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**Part 2: Protocol conformance**
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# Contents

Intellectual Property Rights ..... 2
Legal Notice ..... 2
Modal verbs terminology ..... 2
Foreword ..... 11
Introduction ..... 11
1 Scope ..... 12
2 References ..... 12
3 Definitions and abbreviations ..... 14
3.1 Definitions ..... 14
3.2 Abbreviations ..... 14
4 Default Conditions for UTRAN ..... 14
4.1 Default system information ..... 14
4.2 Simulated A-GPS and A-GNSS environment ..... 15
4.2.1 A-GNSS sub-test cases ..... 15
4.3 A-GPS assistance data sets ..... 15
4.3.1 Adequate assistance data for UE-based A-GPS ..... 16
4.3.2 Inadequate assistance data for UE-based A-GPS ..... 16
4.3.3 Adequate assistance data for UE-assisted A-GPS ..... 16
4.3.4 Inadequate assistance data for UE-assisted A-GPS ..... 17
4.3.5 Response to assistance data requests from UE ..... 17
4.4 A-GNSS assistance data sets ..... 18
4.4.1 Adequate assistance data for UE-based A-GNSS ..... 19
4.4.1.1 Sub-Test 1 ..... 19
4.4.1.2 Sub-Test 2 ..... 20
4.4.1.3 Sub-Test 3 ..... 21
4.4.1.4 Sub-Test 4 ..... 22
4.4.1.4A Sub-Test 8 ..... 23
4.4.1.5 Sub-Test 9 ..... 26
4.4.1.6 Sub-Test 10 ..... 26
4.4.2 Inadequate assistance data for UE-based A-GNSS ..... 27
4.4.3 Adequate assistance data for UE-assisted A-GNSS ..... 27
4.4.3.1 Sub-Test 1 ..... 28
4.4.3.2 Sub-Test 2 ..... 29
4.4.3.3 Sub-Test 3 ..... 30
4.4.3.4 Sub-Test 4 ..... 31
4.4.3.4A Sub-Test 8 ..... 32
4.4.3.5 Sub-Test 9 ..... 33
4.4.3.6 Sub-Test 10 ..... 34
4.4.4 Inadequate assistance data for UE-assisted A-GNSS ..... 34
4.4.5 Response to assistance data requests from UE ..... 34
5 Default Conditions for E-UTRAN ..... 36
5.1 LCS Sub-Test Cases ..... 36
5.2 Default signal conditions ..... 36
5.2.1 Simulated GNSS environment ..... 36
5.2.2 Simulated OTDOA environment ..... 37
5.2.3 Simulated ECID environment ..... 37
5.2.4 Simulated MBS environment ..... 37
5.2.5 Simulated WLAN environment ..... 37
5.2.6 Simulated Bluetooth environment ..... 37
5.2.7 Simulated Sensor environment ..... 37
5.3 Default RRC and NAS message and information elements contents ..... 37
- ATTACH ACCEPT ..... 38

5.4 Default LPP message and information elements contents ..... 38
- LPP REQUEST CAPABILITIES ..... 39
- LPP PROVIDE ASSISTANCE DATA ..... 40
- LPP REQUEST LOCATION INFORMATION ..... 43
- A-GNSS REQUEST LOCATION INFORMATION ..... 44
- OTDOA REQUEST LOCATION INFORMATION ..... 44
- ECID REQUEST LOCATION INFORMATION ..... 44
- TBS REQUEST LOCATION INFORMATION ..... 45
- WLAN REQUEST LOCATION INFORMATION ..... 45
- BT REQUEST LOCATION INFORMATION ..... 45
- SENSOR REQUEST LOCATION INFORMATION ..... 45
5.4.1 Default assistance data information elements ..... 45
5.4.1.1 GNSS Assistance Data Elements ..... 45
5.4.1.2 OTDOA Assistance Data Elements ..... 46
- OTDOA REFERENCE CELL INFO ..... 47
- OTDOA NEIGHBOUR CELL INFO LIST ..... 48
5.4.1.3 MBS Assistance Data Elements ..... 50
5.4.1.4 WLAN Assistance Data Elements ..... 53
5.4.1.5 Sensor Assistance Data Elements ..... 54

6 Protocol Conformance Test Cases for UTRAN ..... 55
6.1 Assisted-GPS Test Cases ..... 55
6.1.1 Assisted GPS Network Induced Tests ..... 55
6.1.1.1 LCS Network Induced location request/ UE-Based GPS/ Emergency Call / with USIM ..... 55
6.1.1.2 LCS Network Induced location request/ UE-Based GPS/ Emergency Call / without USIM ..... 60
6.1.1.3 LCS Network induced location request/ UE-Assisted GPS/ Emergency call/ With USIM ..... 65
6.1.1.4 LCS Network induced location request/ UE-Assisted GPS/ Emergency call/ Without USIM ..... 70
6.1.2 Assisted GPS Mobile Originated Tests ..... 75
6.1.2.1 LCS Mobile originated location request/ UE-Based GPS/ Position estimate request/ Success ..... 75
6.1.2.2 LCS Mobile originated location request/ UE-Based or UE-Assisted GPS/ Assistance data request/ Success ..... 80
6.1.2.3 LCS Mobile originated location request/ UE-Assisted GPS/ Position Estimate/ Success ..... 82
6.1.2.4 LCS Mobile originated location request/ UE-Based GPS/ Transfer to third party/ Success ..... 88
6.1.2.5 LCS Mobile originated location request/ UE-Assisted GPS/ Transfer to third party/ Success ..... 94
6.1.2.6 LCS Mobile originated location request/ UE-Based or UE-Assisted GPS/ Assistance data request/ Failure ..... 99
6.1.2.7 LCS Mobile originated location request/ UE-Based GPS/ Position estimate request/ Failure ..... 102
6.1.3 Assisted GPS Mobile Terminated Tests ..... 108
6.1.3.1 LCS Mobile terminated location request/ UE-Based GPS ..... 108
6.1.3.2 LCS Mobile-terminated location request/UE-Based GPS/ Request for additional assistance data/ Success ..... 114
6.1.3.3 LCS Mobile-terminated location request/UE-Based GPS/ Failure – Not Enough Satellites ..... 121
6.1.3.4 LCS Mobile terminated location request/ UE-Assisted GPS/ Success ..... 128
6.1.3.5 LCS Mobile terminated location request/ UE-Assisted GPS/ Request for additional assistance data/ Success ..... 133
6.1.3.6 LCS Mobile terminated location request/ UE-Based GPS/ Privacy Verification/ Location Allowed if No Response ..... 138
6.1.3.7 LCS Mobile terminated location request/ UE-Based GPS/ Privacy Verification/ Location Not Allowed if No Response ..... 148
6.1.3.8 LCS Mobile terminated location request/ UE-Assisted GPS/ Privacy Verification/ Location Allowed if No Response ..... 155
6.1.3.9 LCS Mobile terminated location request/ UE-Assisted GPS/ Privacy Verification/ Location Not Allowed if No Response ..... 166
6.1.3.10 LCS Mobile terminated location request/ UE-Based or UE-Assisted GPS/ Configuration Incomplete ..... 173
6.2.1 Assisted GNSS Network Induced Tests ..... 177
6.2.1.1 NI-LR Emergency Call: UE-Based A-GNSS ..... 177
6.2.1.2 NI-LR Emergency call: UE-Assisted A-GNSS ..... 184
6.2.2 Assisted GNSS Mobile Originated Tests ..... 190
6.2.2.1 MO-LR Position Estimate: UE-Based A-GNSS ..... 190
6.2.2.2 MO-LR Position Estimate: UE-Assisted A-GNSS ..... 197
6.2.2.3 MO-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites ..... 202

6.2.2.4 MO-LR Assistance Data: UE-Based or UE-Assisted A-GNSS - Success ....211
6.2.2.5 MO-LR Assistance Data: UE-Based or UE-Assisted A-GNSS - Failure....214
6.2.3 Assisted GNSS Mobile Terminated Tests ....217
6.2.3.1 MT-LR: UE-based or UE-Assisted A-GNSS – Request for additional assistance data/ Success ....217
6.2.3.2 MT-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites ....228
6.2.3.3 Location Notification ....237
6.2.3.4 Privacy Verification - Location Allowed if No Response....239
6.2.3.5 Privacy Verification - Location Not Allowed if No Response....242
7 Protocol Conformance Test Cases for E-UTRAN ....246
7.1 NAS Protocol Procedures....246
7.1.1 UE Network Capability ....246
7.2 LCS Procedures....249
7.2.1 Location Notification and Privacy Verification....249
7.2.1.1 Location Notification ....249
7.2.1.2 Privacy Verification – Location Allowed if no Response....252
7.2.1.3 Privacy Verification – Location not Allowed if No Response....258
7.2.2 EPC MO-LR ....263
7.2.2.1 Autonomous Self Location: UE-based....263
7.2.2.2 Basic Self Location: UE-assisted ....269
7.3 LPP Procedures ....283
7.3.1 LPP Common Procedures....283
7.3.1.1 Position Capability Transfer....283
7.3.2 LPP Transport....299
7.3.2.1 LPP Duplicated Message ....299
7.3.2.2 LPP Acknowledgment ....304
7.3.2.3 LPP Retransmission ....309
7.3.3 LPP Error Handling ....314
7.3.3.1 Void....314
7.3.3.1A Void....314
7.3.3.1B LPP Requested Method not Supported – UE-Assisted ....314
7.3.4 LPP Positioning Procedures....322
7.3.4.1 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Based....322
7.3.4.2 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted ....330
7.3.4.3 E-SMLC Initiated Position Measurement without assistance data: UE-Based ....340
7.3.4.4 E-SMLC Initiated Position Measurement without assistance data: UE-Assisted ....349
7.3.5 LPP Abort....361
7.3.5.1 E-SMLC initiated Abort ....361
7.4 Circuit Switched (CS) Fallback....371
7.4.1 MO-LR Procedure ....371
7.4.1.1 CS fallback: Network does not support EPC-MO-LR ....371
7.4.1.2 CS fallback: UE does not support EPC-MO-LR....374
7.5 RRC Protocol Procedures....377
7.5.1 Inter-frequency RSTD measurement indication ....377
7.5.2 Void ....390
8 Default Conditions for NR ....390
8.1 LCS Sub-Test Cases....390
8.1A Test Configurations ....391
8.2 Default signal conditions....391
8.2.1 Simulated GNSS environment....391
8.2.2 Simulated OTDOA (LTE) environment ....391
8.2.3 Simulated ECID (LTE) environment....392
8.2.4 Simulated MBS environment....392
8.2.5 Simulated WLAN environment ....392
8.2.6 Simulated Bluetooth environment ....392
8.2.7 Simulated Sensor environment ....392
8.2.8 Simulated general NR environment....392
8.2.9 Simulated Multi-RTT environment ....393
8.2.10 Simulated DL-AoD environment....393

8.2.11 Simulated DL-TDOA environment ....393
8.2.12 Simulated NR E-CID environment ....393
8.2.13 Simulated UL-TDOA environment ....393
8.2.14 Simulated SL-RTT environment ....394
8.2.15 Simulated SL-AoA environment ....394
8.2.16 Simulated SL-TDOA environment ....394
8.2.17 Simulated SL-TOA environment ....394
8.3 Default RRC and NAS message and information elements contents ....395
8.3.1 RRC message contents for measurement gaps ....395
8.3.2 RRC message contents for UL-SRS Configuration ....398
8.3.3 RRC message contents for SL-PRS Configuration ....401
8.4 Default LPP message and information elements contents ....403
- LPP PROVIDE ASSISTANCE DATA ....404
- LPP REQUEST CAPABILITIES ....408
- LPP REQUEST LOCATION INFORMATION ....410
- NR E-CID REQUEST LOCATION INFORMATION ....412
- NR Multi-RTT REQUEST LOCATION INFORMATION ....412
- NR DL-AoD REQUEST LOCATION INFORMATION ....413
- NR DL-TDOA REQUEST LOCATION INFORMATION ....413
8.4.1 Default assistance data information elements ....413
8.4.1.1 GNSS Assistance Data Elements ....413
8.4.1.2 OTDOA (LTE) Assistance Data Elements ....414
- OTDOA (LTE) REFERENCE CELL INFO ....415
- OTDOA (LTE) NEIGHBOUR CELL INFO LIST ....416
8.4.1.3 MBS Assistance Data Elements ....418
8.4.1.4 WLAN Assistance Data Elements ....418
8.4.1.5 Sensor Assistance Data Elements ....418
8.4.1.6 Multi-RTT Assistance Data Elements ....418
- NR DL-PRS ASSISTANCE DATA ....419
- NR-SelectedDL-PRS-IndexList ....423
- NR-DL-PRS-Info ....424
8.4.1.7 DL-AoD Assistance Data Elements ....426
- NR-PositionCalculationAssistance ....427
8.4.1.8 DL-TDOA Assistance Data Elements ....429
8.5 Default SLPP message and information elements contents ....429
- SLPP REQUEST CAPABILITIES ....430
- SLPP PROVIDE ASSISTANCE DATA ....431
- SLPP REQUEST LOCATION INFORMATION ....433
- CommonIEsRequestLocationInformation ....434
- SL-AoA REQUEST LOCATION INFORMATION ....434
- SL-RTT REQUEST LOCATION INFORMATION ....435
- SL-TDOA REQUEST LOCATION INFORMATION ....435
- SL-TOA REQUEST LOCATION INFORMATION ....435
8.5.1 Default assistance data information elements ....435
*8.5.1.1 Common SL-PRS Assistance Data Elements* ....435
*8.5.1.2 SL-AoA Assistance Data Elements* ....438
*8.5.1.3 SL-RTT Assistance Data Elements* ....439
*8.5.1.4 SL-TDOA Assistance Data Elements* ....439
*8.5.1.5 SL-TOA Assistance Data Elements* ....441

9 Protocol Conformance Test Cases for NR ....442
9.1 FFS ....442
9.2 FFS ....442
9.3 LPP Procedures ....442
9.3.1 LPP Common Procedures ....442
9.3.1.1 Position Capability Transfer ....442
9.3.1.2 LPP Abort ....462
9.3.2 LPP Transport ....469
9.3.2.1 LPP Duplicated Message ....469
9.3.2.2 LPP Acknowledgment ....473
9.3.2.3 LPP Retransmission ....476
9.3.3 LPP Error Handling ....480

9.3.3.1 Void....480
9.3.3.1A Void....480
9.3.3.1B LPP Requested Method not Supported – UE-Assisted....480
9.3.4 LPP Positioning Procedures....488
9.3.4.1 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Based....488
9.3.4.2 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted....494
9.3.4.3 E-SMLC Initiated Position Measurement without assistance data: UE-Based....500
9.3.4.4 E-SMLC Initiated Position Measurement without assistance data: UE-Assisted....508
9.3.4.5 LPP carrier phase positioning procedure....517
9.3.4.5.1 Test Purpose (TP)....517
9.3.4.5.2 Conformance requirements....517
9.3.4.5.3 Test description....519
9.3.4.5.3.1 Pre-test conditions....519
9.3.4.5.3.2 Test procedure sequence....519
9.3.4.5.3.3 Specific message contents....520
9.3.4.6 LPP frequency hopping positioning procedure for Redcap UEs....528
9.3.4.6.1 Test Purpose (TP)....528
9.3.4.6.2 Conformance requirements....528
9.3.4.6.3 Test description....530
9.3.4.6.3.1 Pre-test conditions....530
9.3.4.6.3.2 Test procedure sequence....530
9.3.4.6.3.3 Specific message contents....531
9.3.4.7 PRS aggregation positioning procedure....542
9.3.4.7.1 Test Purpose (TP)....542
9.3.4.7.2 Conformance requirements....542
9.3.4.7.3 Test description....543
9.3.4.7.3.1 Pre-test conditions....543
9.3.4.7.3.2 Test procedure sequence....543
9.3.4.7.3.3 Specific message contents....544
9.4 RRC Protocol Procedures....556
9.4.1 PosSIB broadcasting followed by location information transfer....556
9.4.1.1 Test Purpose (TP)....556
9.4.1.2 Conformance requirements....556
9.4.1.3 Test description....558
9.4.1.3.1 Pre-test conditions....558
9.4.1.3.2 Test procedure sequence....559
9.4.1.3.3 Specific message contents....560
9.4.2 PosSIB broadcasting followed by location information transfer / Positioning SI messages offset....570
9.4.2.1 Test Purpose (TP)....570
9.4.2.2 Conformance requirements....570
9.4.2.3 Test description....571
9.4.2.3.1 Pre-test conditions....571
9.4.2.3.2 Test procedure sequence....571
9.4.2.3.3 Specific message contents....571
9.4.3 On-demand PosSIB followed by location information transfer / RRC_connected state....573
9.4.3.1 Test Purpose (TP)....573
9.4.3.2 Conformance requirements....574
9.4.3.3 Test description....576
9.4.3.3.1 Pre-test conditions....576
9.4.3.3.2 Test procedure sequence....576
9.4.3.3.3 Specific message contents....577
9.4.4 Pre-configured Measurement Gap Procedures....590
9.4.4.1 Test Purpose (TP)....590
9.4.4.2 Conformance requirements....590
9.4.4.3 Test description....594
9.4.4.3.1 Pre-test conditions....594
9.4.4.3.2 Test procedure sequence....594
9.4.4.3.3 Specific message contents....595
9.4.5 Pre-configured PRS processing window procedures....603
9.4.5.1 Test Purpose (TP)....603

9.4.5.2 Conformance requirements ..... 603
9.4.5.3 Test description ..... 606
9.4.5.3.1 Pre-test conditions ..... 606
9.4.5.3.2 Test procedure sequence ..... 607
9.4.5.3.3 Specific message contents ..... 608
9.4.6 UE Positioning Assistance Information ..... 616
9.4.6.1 Test Purpose (TP) ..... 616
9.4.6.2 Conformance requirements ..... 616
9.4.6.3 Test description ..... 617
9.4.6.3.1 Pre-test conditions ..... 617
9.4.6.3.2 Test procedure sequence ..... 618
9.4.6.3.3 Specific message contents ..... 618
9.4.7 LPHAP / SRS configuration enhancements / srs-PosRRC-InactiveValidityAreaNonPreConfig ..... 623
9.4.7.1 Test Purpose (TP) ..... 623
9.4.7.2 Conformance requirements ..... 623
9.4.7.3 Test description ..... 625
9.4.7.3.1 Pre-test conditions ..... 625
9.4.7.3.2 Test procedure sequence ..... 625
9.4.7.3.3 Specific message contents ..... 627
9.5 SLPP Procedures ..... 631
9.5.1 SLPP Common Procedures ..... 631
9.5.1.1 Position Capability Transfer between LMF and UE ..... 631
9.5.1.1.1 Test Purpose (TP) ..... 631
9.5.1.1.2 Conformance requirements ..... 631
9.5.1.1.3 Test description ..... 632
9.5.1.1.3.1 Pre-test conditions ..... 632
9.5.1.1.3.2 Test procedure sequence ..... 632
9.5.1.1.3.3 Specific message contents ..... 633
9.5.1.2 Position Capability Transfer between UEs (Target UE is in coverage) ..... 638
9.5.1.2.1 Test Purpose (TP) ..... 638
9.5.1.2.2 Conformance requirements ..... 638
9.5.1.2.3 Test description ..... 639
9.5.1.2.3.1 Pre-test conditions ..... 639
9.5.1.2.3.2 Test procedure sequence ..... 640
9.5.1.2.3.3 Specific message contents ..... 640
9.5.1.3 Position Capability Transfer between UEs (Target UE is out of coverage) ..... 641
9.5.1.3.1 Test Purpose (TP) ..... 641
9.5.1.3.2 Conformance requirements ..... 641
9.5.1.3.3 Test description ..... 641
9.5.1.3.3.1 Pre-test conditions ..... 641
9.5.1.3.3.2 Test procedure sequence ..... 642
9.5.1.3.3.3 Specific message contents ..... 642
9.5.1.4 SLPP Abort between LMF and UE ..... 642
9.5.1.4.1 Test Purpose (TP) ..... 642
9.5.1.4.2 Conformance requirements ..... 642
9.5.1.4.3 Test description ..... 643
9.5.1.4.3.1 Pre-test conditions ..... 643
9.5.1.4.3.2 Test procedure sequence ..... 644
9.5.1.4.3.3 Specific message contents ..... 644
9.5.1.5 SLPP Abort between UEs (Target UE is in coverage) ..... 647
9.5.1.5.1 Test Purpose (TP) ..... 647
9.5.1.5.2 Conformance requirements ..... 647
9.5.1.5.3 Test description ..... 648
9.5.1.5.3.1 Pre-test conditions ..... 648
9.5.1.5.3.2 Test procedure sequence ..... 648
9.5.1.5.3.3 Specific message contents ..... 649
9.5.1.6 SLPP Abort between UEs (Target UE is out of coverage) ..... 651
9.5.1.6.1 Test Purpose (TP) ..... 651
9.5.1.6.2 Conformance requirements ..... 652
9.5.1.6.3 Test description ..... 652
9.5.1.6.3.1 Pre-test conditions ..... 652
9.5.1.6.3.2 Test procedure sequence ..... 653

9.5.1.6.3.3 Specific message contents....653
9.5.2 SLPP Transport....653
9.5.2.1 SLPP Duplicated Message between LMF and UE....653
9.5.2.1.1 Test Purpose (TP)....653
9.5.2.1.2 Conformance requirements....653
9.5.2.1.3 Test description....654
9.5.2.1.3.1 Pre-test conditions....654
9.5.2.1.3.2 Test procedure sequence....654
9.5.2.1.3.3 Specific message contents....655
9.5.2.2 SLPP Duplicated Message between UEs (Target UE is in coverage)....657
9.5.2.2.1 Test Purpose (TP)....657
9.5.2.2.2 Conformance requirements....657
9.5.2.2.3 Test description....657
9.5.2.2.3.1 Pre-test conditions....657
9.5.2.2.3.2 Test procedure sequence....658
9.5.2.2.3.3 Specific message contents....659
9.5.2.3 SLPP Duplicated Message between UEs (Target UE is out of coverage)....659
9.5.2.3.1 Test Purpose (TP)....659
9.5.2.3.2 Conformance requirements....659
9.5.2.3.3 Test description....660
9.5.2.3.3.1 Pre-test conditions....660
9.5.2.3.3.2 Test procedure sequence....660
9.5.2.3.3.3 Specific message contents....660
9.5.2.4 SLPP Acknowledgment between LMF and UE....660
9.5.2.4.1 Test Purpose (TP)....660
9.5.2.4.2 Conformance requirements....661
9.5.2.4.3 Test description....661
9.5.2.4.3.1 Pre-test conditions....661
9.5.2.4.3.2 Test procedure sequence....662
9.5.2.4.3.3 Specific message contents....662
9.5.2.5 SLPP Acknowledgment between UEs (Target UE is in coverage)....665
9.5.2.5.1 Test Purpose (TP)....665
9.5.2.5.2 Conformance requirements....665
9.5.2.5.3 Test description....665
9.5.2.5.3.1 Pre-test conditions....665
9.5.2.5.3.2 Test procedure sequence....666
9.5.2.5.3.3 Specific message contents....666
9.5.2.6 SLPP Acknowledgment between UEs (Target UE is out of coverage)....668
9.5.2.6.1 Test Purpose (TP)....668
9.5.2.6.2 Conformance requirements....668
9.5.2.6.3 Test description....668
9.5.2.6.3.1 Pre-test conditions....668
9.5.2.6.3.2 Test procedure sequence....669
9.5.2.6.3.3 Specific message contents....669
9.5.2.7 SLPP Retransmission between LMF and UE....669
9.5.2.7.1 Test Purpose (TP)....669
9.5.2.7.2 Conformance requirements....669
9.5.2.7.3 Test description....670
9.5.2.7.3.1 Pre-test conditions....670
9.5.2.7.3.2 Test procedure sequence....671
9.5.2.7.3.3 Specific message contents....671
9.5.2.8 SLPP Retransmission between UEs (Target UE is in coverage)....673
9.5.2.8.1 Test Purpose (TP)....673
9.5.2.8.2 Conformance requirements....674
9.5.2.8.3 Test description....674
9.5.2.8.3.1 Pre-test conditions....674
9.5.2.8.3.2 Test procedure sequence....675
9.5.2.8.3.3 Specific message contents....675
9.5.2.9 SLPP Retransmission between UEs (Target UE is out of coverage)....676
9.5.2.9.1 Test Purpose (TP)....676
9.5.2.9.2 Conformance requirements....676
9.5.2.9.3 Test description....676

9.5.2.9.3.1 Pre-test conditions....676
9.5.2.9.3.2 Test procedure sequence....677
9.5.2.9.3.3 Specific message contents....677
9.5.3 SLPP Error Handling....677
9.5.3.1 SLPP Requested Method not Supported between LMF and UE - UE-Assisted....677
9.5.3.1.1 Test Purpose (TP)....677
9.5.3.1.2 Conformance requirements....677
9.5.3.1.3 Test description....678
9.5.3.1.3.1 Pre-test conditions....678
9.5.3.1.3.2 Test procedure sequence....679
9.5.3.1.3.3 Specific message contents....679
9.5.3.2 SLPP Requested Method not Supported between UEs (Target UE is in coverage) - UE-Assisted....685
9.5.3.2.1 Test Purpose (TP)....685
9.5.3.2.2 Conformance requirements....685
9.5.3.2.3 Test description....685
9.5.3.2.3.1 Pre-test conditions....685
9.5.3.2.3.2 Test procedure sequence....687
9.5.3.2.3.3 Specific message contents....687
9.5.3.3 SLPP Requested Method not Supported between UEs (Target UE is out of coverage) - UE-Assisted....691
9.5.3.3.1 Test Purpose (TP)....691
9.5.3.3.2 Conformance requirements....691
9.5.3.3.3 Test description....691
9.5.3.3.3.1 Pre-test conditions....691
9.5.3.3.3.2 Test procedure sequence....692
9.5.3.3.3.3 Specific message contents....692

**Annex A (informative): Change history....693**

History....698

# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

The present document is part 2 of a multi-part TS:

3GPP TS 37.571-1: User Equipment (UE) conformance specification for UE positioning; Part 1: Conformance test specification.

**3GPP TS 37.571-2: User Equipment (UE) conformance specification for UE positioning; Part 2: Protocol conformance.**

3GPP TS 37.571-3: User Equipment (UE) conformance specification for UE positioning; Part 3: Implementation Conformance Statement (ICS).

3GPP TS 37.571-4: User Equipment (UE) conformance specification for UE positioning; Part 4: Test suites.

3GPP TS 37.571-5: User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data.

# 1 Scope

The present document specifies the protocol conformance testing for UTRAN, E-UTRAN and NR User Equipment (UE) supporting UE positioning.

This is the second part of a multi-part test specification. The following information can be found in this part:

- the overall protocol conformance test structure;
- the protocol conformance test configurations;
- the conformance requirement and reference to the core specifications;
- the test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The Implementation Conformance Statement (ICS) pro-forma could be found in the 3rd part of the present document.

The present document is valid for UE supporting UE positioning implemented according to 3GPP releases starting from Release 99 up to the Release indicated on the cover page of the present document.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document unless the context in which the reference is made suggests a different Release is relevant (information on the applicable release in a particular context can be found in e.g. test case title, description or applicability, message description or content).

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.271: "Functional stage 2 description of Location Services (LCS) ".

[3] 3GPP TS 36.305: "Stage 2 functional specification of User Equipment (UE) positioning in E-UTRAN".

[4] 3GPP TS 36.355: "LTE Positioning Protocol (LPP)".

[5] 3GPP TS 24.171: "Control Plane Location Services (LCS) procedures in the Evolved Packet System (EPS)".

[6] 3GPP TS 24.030: "Location Services (LCS); Supplementary service operations; Stage 3".

[7] 3GPP TS 24.080: "Mobile radio interface layer 3 supplementary services specification; Formats and coding".

[8] 3GPP TS 36.508: "Common test environments for User Equipment (UE)".

[9] 3GPP TS 37.571-1: "User Equipment (UE) conformance specification for UE positioning; Part 1: Conformance test specification".

[10] 3GPP TS 37.571-3: "User Equipment (UE) conformance specification for UE positioning; Part 3: Implementation Conformance Statement (ICS)".

[11] 3GPP TS 37.571-4: "User Equipment (UE) conformance specification for UE positioning; Part 4: Test suites".

[12] 3GPP TS 37.571-5: "User Equipment (UE) conformance specification for UE positioning; Part 5: Test scenarios and assistance data".

[13] 3GPP TS 36.509: "Special conformance testing functions for User Equipment (UE)".

[14] 3GPP TS 34.123-1: "User Equipment (UE) conformance specification; Part 1: Protocol conformance specification".

[15] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".

[16] 3GPP TS 34.108: "Common Test Environments for User Equipment (UE) Conformance Testing".

[17] 3GPP TS 25.331: "RRC Protocol Specification".

[18] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[19] 3GPP TS 24.008: "Mobile radio interface Layer 3 specification; Core network protocols; Stage 3".

[20] 3GPP TS 33.102: "3G security; Security architecture".

[21] ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

[22] 3GPP TS 23.171: "Location Services (LCS); Functional description; Stage 2 (UMTS)".

[23] GSM TS 03.71: "Location Services (LCS); Functional description; Stage 2".

[24] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[25] 3GPP TS 23.272: "Circuit Switched (CS) fallback in Evolved Packet System (EPS); Stage 2".

[26] 3GPP TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

[27] 3GPP TS 22.101: "Service aspects; Service principles".

[28] OMA-TS-LPPe-V2_0: "LPP Extensions Specification", Open Mobile Alliance.

[29] ATIS-0500027: "Recommendations for Establishing Wide Scale Indoor Location Performance", May 2015.

[30] 3GPP TS 38.508-1: "User Equipment (UE) conformance specification; Part 1: Common test environment".

[31] 3GPP TS 38.305: "NG Radio Access Network (NG-RAN); Stage 2 functional specification of User Equipment (UE) positioning in NG-RAN".

[32] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".

[33] 3GPP TS 38.331: "NR Radio Resource Control (RRC) protocol specification".

[34] 3GPP TS 38.509: "Special conformance testing functions for User Equipment (UE)".

[35] 3GPP TS 38.355: "NR; Sidelink Positioning Protocol (SLPP); Protocol specification".

# 3 Definitions and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1], TS 23.271 [2],

TS 36.305 [3], TS 36.355 [4] , TS 38.305 [31] , TS 37.355 [32], TS 38.355 [35] apply.

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| BDS | BeiDou Navigation Satellite System |
| DL | Downlink |
| DL-AoD | Downlink Angle-of-Departure |
| DL-PRS | Downlink Positioning Reference Signal |
| DL-TDOA | Downlink Time Difference Of Arrival |
| LCS | Location Services |
| LPP | LTE Positioning Protocol |
| MBS | Metropolitan Beacon System |
| MO-LR | Mobile Originated Location Request |
| Multi-RTT | Multi-Round Trip Time |
| MT-LR | Mobile Terminated Location Request |
| NAS | Non-Access-Stratum |
| NI-LR | Network Induced Location Request |
| NR | New Radio |
| NR E-CID | NR Enhanced Cell ID (positioning method) |
| OTA | Over The Air |
| RSCP | Reference Signal Carrier Phase |
| RSCPD | Reference Signal Carrier Phase Difference |
| RSRP | Reference Signal Received Power |
| RSRPP | Reference Signal Received Path Power |
| SL-AoA | Sidelink Angle-of-Arrival |
| SL-PRS | Sidelink Positioning Reference Signal |
| SL-PRS-RSRP | Sidelink PRS Reference Signal Received Power |
| SL-PRS-RSRPP | Sidelink PRS Reference Signal Received Path Power |
| SL-RSTD | Sidelink Reference Signal Time Difference |
| SL-RTOA | Sidelink Relative Time of Arrival |
| SL-RTT | Sidelink Round Trip Time |
| SL-TDOA | Sidelink Time Difference Of Arrival |
| SL-TOA | Sidelink Time Of Arrival |
| SLPP | Sidelink Positioning Protocol |
| TBS | Terrestrial Beacon System |
| TEG | Timing Error Group |
| UL | Uplink |
| WLAN | Wireless Local Area Network |
| UL-SRS | Uplink Sounding Reference Signal |

# 4 Default Conditions for UTRAN

## 4.1 Default system information

Default system information, as specified in 3GPP TS 34.108 [16] subclause 6.1, is broadcasted for the A-GPS test cases in subclause 6.1. For the A-GNSS test cases in subclause 6.2, the default system information, as specified in 3GPP TS 36.508 [8], is broadcasted. SIB15/SIB15.x are not broadcasted unless otherwise stated in the specific test cases.

# 4.2 Simulated A-GPS and A-GNSS environment

During A-GPS and A-GNSS tests, where required the SS shall generate satellite signals that are of a sufficient number and strength not to prevent the UE from responding to a positioning request with a valid measurement response. Any assistance data provided during these tests shall be consistent with the satellite signals generated during these tests. Note that some tests require assistance data to be provided even though satellite signals are not required.

It is considered that six satellite signals with the level of the simulated satellites all at -125 dBm ± 6 dB should be suitable, however this does not imply any conformance requirements on the UE.

Suitable GPS and GNSS scenarios together with associated assistance data are defined in TS 37.571-5 [12] subclause 5.1 and 6.1, respectively.

The accuracy of the GPS time-of-week and/or GANSS time-of-day in the provided assistance data shall be within ± 2 seconds relative to the GPS and/or GANSS time in the system simulator. In the case that assistance data is required but satellite signals are not required then this clause does not apply.

During A-GNSS signalling tests where satellite signals are required, the SS shall generate all the UE supported GNSS satellite signals.

## 4.2.1 A-GNSS sub-test cases

The A-GNSS signalling test cases in subclause 6.2 may include several sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in the Table below. The detailed assistance data depend on the particular sub-test case as defined in subclause 4.4.

**Table 4.2.1-1: Sub-Test Case Number Definition for UTRA**

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

# 4.3 A-GPS assistance data sets

This subclause defines the assistance data sets supplied by the SS in A-GPS test cases specified in subclause 6.1.

Throughout this subclause, "adequate assistance data" means the assistance data used in test cases where it is expected that a UE supporting A-GPS will be able to perform the requested positioning operation using the supplied assistance data, and "inadequate assistance data" is the assistance data used in test cases that expect that the UE will be unable to perform the requested operation. The values of all the fields in all cases are defined in TS 37.571-5 [12] subclause 5.1.3.

## 4.3.1 Adequate assistance data for UE-based A-GPS

For UE-based test cases requiring adequate assistance data, the IE "UE positioning GPS assistance data" is spread across two separate MEASUREMENT CONTROL messages, and set as follows:

First MEASUREMENT CONTROL MESSAGE:

| | | |
|---|---|---|
| - UE positioning GPS assistance data | | |
| - UE positioning GPS reference time | | |
| - GPS week | Set according to 4.2 | |
| - GPS Week Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GPS TOW msec | Set according to 4.2 | |
| - UTRAN GPS reference time | Not present | |
| - UE Positioning GPS Reference Time Uncertainty | Set according to 4.2 | Rel-7 UE or later |
| - SFN-TOW uncertainty | Not present | |
| - $T_{UTRAN-GPS}$ drift rate | Not present | |
| - GPS TOW assist | Not present | |
| - UE positioning GPS reference UE position | Set according to 4.2 | |
| - UE positioning GPS DGPS corrections | Not present | |
| - UE positioning GPS navigation model | | |
| - Satellite information | For satellites 1-3 | |
| - SatID | Set according to 4.2 | |
| - Satellite status | NS NN | |
| - GPS ephemeris and clock corr. param. | Set according to 4.2 | |
| - UE positioning GPS ionospheric model | Set according to 4.2 | |
| - UE positioning GPS UTC model | Not present | |
| - UE positioning GPS almanac | Not present | |
| - UE positioning GPS acquisition assistance | Not present | |
| - UE positioning GPS real-time integrity | Not present | |

Second MEASUREMENT CONTROL message:

| | |
|---|---|
| - UE positioning GPS assistance data | |
| - UE positioning GPS reference time | Not present |
| - UE positioning GPS reference UE position | Not present |
| - UE positioning GPS DGPS corrections | Not present |
| - UE positioning GPS navigation model | |
| - Satellite information | For satellites 4-6 |
| - SatID | Set according to 4.2 |
| - Satellite status | NS NN |
| - GPS ephemeris and clock corr. param. | Set according to 4.2 |
| - UE positioning GPS ionospheric model | Not present |
| - UE positioning GPS UTC model | Not present |
| - UE positioning GPS almanac | Not present |
| - UE positioning GPS acquisition assistance | Not present |
| - UE positioning GPS real-time integrity | Not present |

## 4.3.2 Inadequate assistance data for UE-based A-GPS

For UE-based test cases requiring inadequate assistance data, the IE "UE positioning GPS assistance data" is set to "Not present" in the MEASUREMENT CONTROL message.

## 4.3.3 Adequate assistance data for UE-assisted A-GPS

For UE-assisted test cases requiring adequate assistance data, the IE "UE positioning GPS assistance data" is set as follows for the first MEASUREMENT CONTROL message:

| | | |
|---|---|---|
| - UE positioning GPS assistance data | | |
| - UE positioning GPS reference time | | |
| - GPS week | Set according to 4.2 | |
| - GPS Week Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GPS TOW msec | Set according to 4.2 | |
| - UTRAN GPS reference time | Not present | |
| - UE Positioning GPS Reference Time Uncertainty | Set according to 4.2 | Rel-7 UE or later |
| - SFN-TOW uncertainty | Not present | |
| - $T_{UTRAN-GPS}$ drift rate | Not present | |
| - GPS TOW assist | Not present | |
| - UE positioning GPS reference UE position | Not present | |
| - UE positioning GPS DGPS corrections | Not present | |
| - UE positioning GPS navigation model | Not present | |
| - UE positioning GPS ionospheric model | Not present | |
| - UE positioning GPS UTC model | Not present | |
| - UE positioning GPS almanac | Not present | |
| - UE positioning GPS acquisition assistance | | |
| - GPS TOW msec | Set according to 4.2 | |
| - UTRAN GPS reference time | Not present | |
| - UE Positioning GPS Reference Time Uncertainty | Set according to 4.2 | Rel-7 UE or later |
| - Satellite information | Set according to 4.2 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GPS real-time integrity | Not present | |

If the UE requests further assistance data, the SS sends subsequent MEASUREMENT CONTROL messages containing the assistance data fields requested by the UE that are available in the SS as specified in TS 37.571-5 [12] subclause 5.1.3 and in subclause 4.3.5.

## 4.3.4 Inadequate assistance data for UE-assisted A-GPS

For UE-assisted test cases requiring inadequate assistance data, the IE "UE positioning GPS assistance data" is set to "Not present" in the MEASUREMENT CONTROL message.

## 4.3.5 Response to assistance data requests from UE

If the SS needs to send assistance data in response to a request for additional assistance data from the UE, or in response to an MO-LR request for assistance data, the IE "UE positioning GPS assistance data" is set as follows:

| | | |
|---|---|---|
| - UE positioning GPS assistance data | | |
| - UE positioning GPS reference time | Set according to 4.2 if requested by the UE | |
| - GPS week | Set according to 4.2 | |
| - GPS Week Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GPS TOW msec | Set according to 4.2 | |
| - UTRAN GPS reference time | Not present | |
| - UE Positioning GPS Reference Time Uncertainty | Set according to 4.2 | Rel-7 UE or later |
| - SFN-TOW uncertainty | Not present | |
| - $T_{UTRAN-GPS}$ drift rate | Not present | |
| - GPS TOW assist | Not present | |
| - UE positioning GPS reference UE position | Set according to 4.2 if requested by the UE | |
| - UE positioning GPS DGPS corrections | Not sent | |
| - UE positioning GPS navigation model | Set according to 4.2 if requested by the UE | |
| - Satellite information | For satellites 1-6 | |
| - SatID | Set according to 4.2 | |
| - Satellite status | NS NN | |
| - GPS ephemeris and clock corr. param. | Set according to 4.2 | |
| - UE positioning GPS ionospheric model | Set according to 4.2 if requested by the UE | |
| - UE positioning GPS UTC model | Not sent | |
| - UE positioning GPS almanac | Set according to 4.2 if requested by the UE | |
| - WNa | Set according to 4.2 | |
| - Complete Almanac Provided | True | Rel-10 UE or later |
| - Satellite information | Set according to 4.2 | |
| - SV Global Health | Not present | |
| - UE positioning GPS acquisition assistance | Set according to 4.2 if requested by the UE | |
| - GPS TOW msec | Set according to 4.2 | |
| - UTRAN GPS reference time | Not present | |
| - UE Positioning GPS Reference Time Uncertainty | Set according to 4.2 | Rel-7 UE or later |
| - Satellite information | Set according to 4.2 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GPS real-time integrity | Not sent | |

If the UE requests the GPS navigation model then the SS provides navigation model satellite information for at most three satellites in any one MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY message; additional satellites are sent in subsequent MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

If the UE requests the GPS almanac then the SS provides almanac information spread across at least two MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

If the UE requests both GPS navigation model and almanac then the SS provides them in different MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

## 4.4 A-GNSS assistance data sets

This subclause defines the assistance data sets supplied by the SS in A-GNSS test cases specified in subclause 6.2. For A-GNSS sub-test cases which include the GPS L1 C/A signal, the A-GPS assistance data as defined in clause 4.3 apply.

Throughout this subclause, "adequate assistance data" means the assistance data used in test cases where it is expected that a UE supporting A-GNSS will be able to perform the requested positioning operation using the supplied assistance data, and "inadequate assistance data" is the assistance data used in test cases that expect that the UE will be unable to perform the requested operation. The values of all the fields in all cases are defined in 3GPP TS 37.571-5 [12] subclause 6.1.3.

## 4.4.1 Adequate assistance data for UE-based A-GNSS

For UE-based GNSS test cases requiring adequate assistance data, the IE "UE positioning GANSS assistance data" is spread across one or more (dependent on the sub-test) separate MEASUREMENT CONTROL messages, and set as follows:

### 4.4.1.1 Sub-Test 1

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | 2 (GLONASS) | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Set according to 4.2 | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 3 (GLONASS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Set according to 4.2 | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-6 | |
| - GANSS additional clock models | Model-4 | |
| - GANSS additional orbit models | Model-4 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Set according to 4.2 | |
| - GANSS-ID-3 | | |
| - Aux Info List | For satellites 1-6 | |

#### 4.4.1.2 Sub-Test 2

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | Not present | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Set according to 4.2 | |
| - UE positioning GANSS ionospheric model | Set according to 4.2 | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | Not present | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Set according to 4.2 | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-N | |
| - GANSS clock model | Model-1 | |
| - GANSS orbit model | Model-1 | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

Second MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | Not present |
| UE positioning GANSS assistance data | |
| - UE positioning GANSS reference time | Not present |
| - UE positioning GANSS reference UE position | Not present |
| - UE positioning GANSS ionospheric model | Not present |
| - UE positioning GANSS additional ionospheric Model | Not present |
| - UE positioning GANSS Earth orientation Parameters | Not present |
| - GANSS Generic Assistance Data | |
| - GANSS ID | Not present |
| - UE positioning GANSS SBAS ID | Not present |
| - GANSS Time Models | Not present |
| - UE positioning DGANSS corrections | Not present |
| - UE positioning GANSS navigation model | Set according to 4.2 |
| - Non-Broadcast Indication | Not present |
| - Satellite information | For satellites (N+1)-6 |
| - GANSS clock model | Model-1 |
| - GANSS orbit model | Model-1 |
| - UE positioning GANSS additional navigation models | Not present |
| - UE positioning GANSS real-time integrity | Not present |
| - UE positioning GANSS data bit assistance | Not present |
| - UE positioning GANSS reference measurement information | Not present |
| - UE positioning GANSS almanac | Not present |
| - UE positioning GANSS UTC model | Not present |
| - UE positioning GANSS additional UTC models | Not present |
| - UE positioning GANSS auxiliary information | Not present |

### 4.4.1.3 Sub-Test 3

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, First Measurement Control Message |
| UE positioning GANSS assistance data | |
| - UE positioning GANSS reference time | Not present |
| - UE positioning GANSS reference UE position | Not present |
| - UE positioning GANSS ionospheric model | Not present |
| - UE positioning GANSS additional ionospheric Model | Not present |
| - UE positioning GANSS Earth orientation Parameters | Not present |
| - GANSS Generic Assistance Data | |
| - GANSS ID | 1 (Modernized GPS) |
| - UE positioning GANSS SBAS ID | Not present |
| - GANSS Time Models | Not present |
| - UE positioning DGANSS corrections | Not present |
| - UE positioning GANSS navigation model | Not present |
| - UE positioning GANSS additional navigation models | Not present |
| - UE positioning GANSS real-time integrity | Not present |
| - UE positioning GANSS data bit assistance | Not present |
| - UE positioning GANSS reference measurement information | Not present |
| - UE positioning GANSS almanac | Not present |
| - UE positioning GANSS UTC model | Not present |
| - UE positioning GANSS additional UTC models | Not present |
| - UE positioning GANSS auxiliary information | |
| - GANSS-ID-1 | |
| - Aux Info List | For satellites 1-3 |

Second MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, Second Measurement Control Message |
| UE positioning GANSS assistance data | |
| - UE positioning GANSS reference time | Not present |
| - UE positioning GANSS reference UE position | Not present |
| - UE positioning GANSS ionospheric model | Not present |
| - UE positioning GANSS additional ionospheric Model | Not present |
| - UE positioning GANSS Earth orientation Parameters | Not present |
| - GANSS Generic Assistance Data | |
| - GANSS ID | 1 (Modernized GPS) |
| - UE positioning GANSS SBAS ID | Not present |
| - GANSS Time Models | Not present |
| - UE positioning DGANSS corrections | Not present |
| - UE positioning GANSS navigation model | Not present |
| - UE positioning GANSS additional navigation models | Not present |
| - UE positioning GANSS real-time integrity | Not present |
| - UE positioning GANSS data bit assistance | Not present |
| - UE positioning GANSS reference measurement information | Not present |
| - UE positioning GANSS almanac | Not present |
| - UE positioning GANSS UTC model | Not present |
| - UE positioning GANSS additional UTC models | Not present |
| - UE positioning GANSS auxiliary information | |
| - GANSS-ID-1 | |
| - Aux Info List | For satellites 4-6 |

### 4.4.1.4 Sub-Test 4

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, First Measurement Control Message |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, First Measurement Control Message, UE positioning GANSS assistance data. |

Second MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, Second Measurement Control Message |
| - UE positioning GPS UTC model | Set according to 4.2. |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, Second Measurement Control Message, UE positioning GANSS assistance data. |

Third MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 3 (GLONASS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Set according to 4.2 | |
| - GANSS Time Model | | |
| - GANSS Time Model Reference Time | Set according to 4.2 | |
| - $T_{A0}$ | Set according to 4.2 | |
| - $T_{A1}$ | Not present | |
| - $T_{A2}$ | Not present | |
| - GNSS_TO_ID | 0 (GPS) | |
| - Week Number | Not present | |
| - Delta_T | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Set according to 4.2 | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-6 | |
| - GANSS additional clock models | Model-4 | |
| - GANSS additional orbit models | Model-4 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Set according to 4.2 | |
| - GANSS-ID-3 | | |
| - Aux Info List | For satellites 1-6 | |

### 4.4.1.4A Sub-Test 8

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, First Measurement Control Message |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, First Measurement Control Message, UE positioning GANSS assistance data. |

Second MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, Second Measurement Control Message |
| - UE positioning GPS UTC model | Set according to 4.2. |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, Second Measurement Control Message, UE positioning GANSS assistance data. |

Third MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | Not present | (Galileo) |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Set according to 4.2 | |
| - GANSS Time Model | | |
| - GANSS Time Model Reference Time | Set according to 4.2 | |
| - $T_{A0}$ | Set according to 4.2 | |
| - $T_{A1}$ | Not present | |
| - $T_{A2}$ | Not present | |
| - GNSS_TO_ID | 0 (GPS) | |
| - Week Number | Not present | |
| - Delta_T | Set according to 4.2 | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-6 | |
| - GANSS additional clock models | Model 1 | |
| - GANSS additional orbit models | Model 1 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

### 4.4.1.5 Sub-Test 9

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | 3 (BDS) | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Set according to 4.2 | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 4 (BDS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Set according to 4.2 | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-6 | |
| - GANSS additional clock models | Model 6 | |
| - GANSS additional orbit models | Model 6 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

### 4.4.1.6 Sub-Test 10

First MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, First Measurement Control Message |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, First Measurement Control Message, UE positioning GANSS assistance data. |

Second MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.1, Second Measurement Control Message |
| - UE positioning GPS UTC model | Set according to 4.2. |
| UE positioning GANSS assistance data | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.1.3, Second Measurement Control Message, UE positioning GANSS assistance data. |

Third MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 4 (BDS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Set according to 4.2 | |
| - GANSS Time Model | | |
| - GANSS Time Model Reference Time | Set according to 4.2 | |
| - $T_{A0}$ | Set according to 4.2 | |
| - $T_{A1}$ | Not present | |
| - $T_{A2}$ | Not present | |
| - GNSS_TO_ID | 0 (GPS) | |
| - Week Number | Not present | |
| - Delta_T | Set according to 4.2 | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Set according to 4.2 | |
| - Non-Broadcast Indication | Not present | |
| - Satellite information | For satellites 1-6 | |
| - GANSS additional clock models | Model 6 | |
| - GANSS additional orbit models | Model 6 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Not present | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

## 4.4.2 Inadequate assistance data for UE-based A-GNSS

For UE-based test cases requiring inadequate assistance data, the IE "UE positioning GPS assistance data" and "UE positioning GANSS assistance data" is set to "Not present" in the MEASUREMENT CONTROL message.

## 4.4.3 Adequate assistance data for UE-assisted A-GNSS

For UE-assisted test cases requiring adequate assistance data, the IEs "UE positioning GPS assistance data" and "UE positioning GANSS assistance data" are set as follows:

#### 4.4.3.1 Sub-Test 1

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | 2 (GLONASS) | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 3 (GLONASS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Set according to 4.2 | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Set according to 4.2 | |
| - GANSS-ID-3 | | |
| - Aux Info List | For satellites 1-6 | |

### 4.4.3.2 Sub-Test 2

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | Not present | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | Not present | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Set according to 4.2 | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

#### 4.4.3.3 Sub-Test 3

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark |
|---|---|
| UE positioning GPS assistance data | As defined in 4.3.3 |
| UE positioning GANSS assistance data | |
| - UE positioning GANSS reference time | Not present |
| - UE positioning GANSS reference UE position | Not present |
| - UE positioning GANSS ionospheric model | Not present |
| - UE positioning GANSS additional ionospheric Model | Not present |
| - UE positioning GANSS Earth orientation Parameters | Not present |
| - GANSS Generic Assistance Data | |
| - GANSS ID | 1 (Modernized GPS) |
| - UE positioning GANSS SBAS ID | Not present |
| - GANSS Time Models | Not present |
| - UE positioning DGANSS corrections | Not present |
| - UE positioning GANSS navigation model | Not present |
| - UE positioning GANSS additional navigation models | Not present |
| - UE positioning GANSS real-time integrity | Not present |
| - UE positioning GANSS data bit assistance | Not present |
| - UE positioning GANSS reference measurement information | Not present |
| - UE positioning GANSS almanac | Not present |
| - UE positioning GANSS UTC model | Not present |
| - UE positioning GANSS additional UTC models | Not present |
| - UE positioning GANSS auxiliary information | Set according to 4.2 |
| - GANSS-ID-1 | |
| - Aux Info List | For satellites 1-6 |

#### 4.4.3.4 Sub-Test 4

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | Comment |
|---|---|---|
| UE positioning GPS assistance data | As defined in 4.3.3 | |
| - UE positioning GPS UTC model | Set according to 4.2 | |
| UE positioning GANSS assistance data (1) | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.3.3, Measurement Control Message, UE positioning GANSS assistance data. | |
| UE positioning GANSS assistance data (2) | | If UE positioning GANSS assistance data (1) is present, then UE positioning GANSS assistance data (2) is sent in a second MEASUREMENT CONTROL MESSAGE |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 3 (GLONASS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GANSS auxiliary information | Set according to 4.2 | |
| - GANSS-ID-3 | | |
| - Aux Info List | For satellites 1-6 | |

### 4.4.3.4A Sub-Test 8

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | Comment |
|---|---|---|
| UE positioning GPS assistance data | As defined in 4.3.3 | |
| - UE positioning GPS UTC model | Set according to 4.2 | |
| UE positioning GANSS assistance data (1) | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.3.3, Measurement Control Message, UE positioning GANSS assistance data. | |
| UE positioning GANSS assistance data (2) | | If UE positioning GANSS assistance data (1) is present, then UE positioning GANSS assistance data (2) is sent in a second MEASUREMENT CONTROL MESSAGE |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | Not present | (Galileo) |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | |
| - UE positioning GANSS auxiliary information | Not present | |

### 4.4.3.5 Sub-Test 9

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Not present | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | 3 (BDS) | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 4 (BDS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Set according to 4.2 | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | |
| - UE positioning GANSS almanac | Not present | |
| - UE positioning GANSS UTC model | Not present | |
| - UE positioning GANSS additional UTC models | Not present | |
| - UE positioning GANSS auxiliary information | Not present | |

#### 4.4.3.6 Sub-Test 10

MEASUREMENT CONTROL MESSAGE:

| Information Element | Value/Remark | Comment |
|---|---|---|
| UE positioning GPS assistance data | As defined in 4.3.3 | |
| - UE positioning GPS UTC model | Set according to 4.2 | |
| UE positioning GANSS assistance data (1) | If for GPS only L1 C/A supported, not present. If multiple GPS signals supported, as defined in 4.4.3.3, Measurement Control Message, UE positioning GANSS assistance data. | |
| UE positioning GANSS assistance data (2) | | If UE positioning GANSS assistance data (1) is present, then UE positioning GANSS assistance data (2) is sent in a second MEASUREMENT CONTROL MESSAGE |
| - UE positioning GANSS reference time | Not present | |
| - UE positioning GANSS reference UE position | Not present | |
| - UE positioning GANSS ionospheric model | Not present | |
| - UE positioning GANSS additional ionospheric Model | Not present | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | 4 (BDS) | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Not present | |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Not present | |
| - UE positioning GANSS additional navigation models | Not present | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | | |
| - GANSS Signal ID | Not present | |
| - Satellite Information | For satellites 1-6 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | |
| - UE positioning GANSS auxiliary information | Not present | |

If the UE requests further assistance data, the SS sends subsequent MEASUREMENT CONTROL messages containing the assistance data fields requested by the UE that are available in the SS as specified in TS 37.571-5 [12] subclause 6.1.3 and in clause 4.4.5.

### 4.4.4 Inadequate assistance data for UE-assisted A-GNSS

For UE-assisted test cases requiring inadequate assistance data, the IEs "UE positioning GPS assistance data" and "UE positioning GANSS assistance data" are set to "Not present" in the MEASUREMENT CONTROL message.

### 4.4.5 Response to assistance data requests from UE

If the SS needs to send assistance data in response to a request for additional assistance data from the UE, or in response to an MO-LR request for assistance data, the IEs "UE positioning GPS assistance data" and "UE positioning GANSS assistance data" are set as follows:

| Information Element | Value/Remark | |
|---|---|---|
| UE positioning GPS assistance data | Set according to 4.3.5 | |
| - UE positioning GPS UTC model | Set according to 4.2 | |
| UE positioning GANSS assistance data | | |
| - UE positioning GANSS reference time | Set according to 4.2 | |
| - GANSS Day | Set according to 4.2 | |
| - GANSS Day Cycle Number | Set according to 4.2 | Rel-10 UE or later |
| - GANSS TOD | Set according to 4.2 | |
| - GANSS TOD Uncertainty | Set according to 4.2 | |
| - GANSS Time ID | Set according to 4.2 | |
| - UTRAN GANSS reference time | Not present | |
| - $T_{UTRAN-GANSS}$ drift rate | Not present | |
| - UE positioning GANSS reference UE position | Set according to 4.2 | |
| - UE positioning GANSS ionospheric model | Set according to 4.2 | |
| - UE positioning GANSS additional ionospheric Model | Set according to 4.2 | |
| - UE positioning GANSS Earth orientation Parameters | Not present | |
| - GANSS Generic Assistance Data | | |
| - GANSS ID | Set according to 4.2 | |
| - UE positioning GANSS SBAS ID | Not present | |
| - GANSS Time Models | Set according to 4.2 | |
| - GANSS Time Model | Set according to 4.2 | |
| - GANSS Time Model Reference Time | Set according to 4.2 | |
| - $T_{A0}$ | Set according to 4.2 | |
| - $T_{A1}$ | Not present | |
| - $T_{A2}$ | Not present | |
| - GNSS_TO_ID | Set according to 4.2 | |
| - Week Number | Not present | |
| - Delta_T | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning DGANSS corrections | Not present | |
| - UE positioning GANSS navigation model | Set according to 4.2 | |
| - UE positioning GANSS additional navigation models | Set according to 4.2 | |
| - UE positioning GANSS real-time integrity | Not present | |
| - UE positioning GANSS data bit assistance | Not present | |
| - UE positioning GANSS reference measurement information | Set according to 4.2 | |
| - GANSS Signal ID | Set according to 4.2 | |
| - Satellite Information | Set according to 4.2 | |
| - Extra Doppler | Set according to 4.2 | |
| - Azimuth and Elevation | Set according to 4.2 | |
| - Azimuth and Elevation LSB | Set according to 4.2 | Rel-10 UE or later |
| - UE positioning GANSS almanac | Set according to 4.2 | |
| - Complete Almanac Provided | True | Rel-10 UE or later |
| - UE positioning GANSS UTC model | Set according to 4.2 | |
| - UE positioning GANSS additional UTC models | Set according to 4.2 | |
| - UE positioning GANSS auxiliary information | Set according to 4.2 | |

If the UE requests the GPS navigation model or the GANSS navigation model Model-1, Model-2, Model-3 or Model 6 then the SS provides navigation model satellite information for at most three satellites in any one MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY message; additional satellites are sent in subsequent MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

If the UE requests the GPS or GANSS almanac then the SS provides almanac information spread across at least two MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

If the UE requests both navigation model and almanac then the SS provides them in different MEASUREMENT CONTROL or ASSISTANCE DATA DELIVERY messages.

# 5 Default Conditions for E-UTRAN

## 5.1 LCS Sub-Test Cases

Some test cases defined in clause 7 may include several sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 5.1-1. The applicable sub-tests for each test case are specified in the test procedure sequence clause of each test case. If no sub-tests are defined for a specific test case it means that this particular test case is not dependent on a specific positioning method.

**Table 5.1-1: Sub-Test Case Numbers for E-UTRA**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA |
| 6 FDD | UE supporting ECID (FDD) |
| 6 TDD | UE supporting ECID (TDD) |
| 7 | UE supporting GNSS(1) and OTDOA |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS(2) (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 14 | UE supporting Sensor (Rel-13 only) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS(2) (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 23 | UE supporting MBS(2) (Rel-15 onwards) |
| 24 | UE supporting Sensor (Rel-15 onwards) |
| 25 | UE supporting GNSS(1) (Rel-15 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |
| NOTE 2: Metropolitan Beacon System (MBS) is a specific type of Terrestrial Beacon System (TBS) [29] | |

## 5.2 Default signal conditions

### 5.2.1 Simulated GNSS environment

During A-GNSS signalling tests, where required the SS shall generate all UE supported satellite signals. Where required the SS shall provide assistance data dependent on UE capabilities defined in subclause 5.4.1.1 and consistent with the satellite signals generated during these tests if satellite signals are also required.

The levels of the simulated satellites shall be at -125 dBm ± 6 dBm.

GNSS scenarios together with associated assistance data are defined in TS 37.571-5 [12].

The accuracy of the GNSS time in the provided assistance data shall be within ± 2 seconds relative to the GNSS time in the system simulator. In the case that assistance data is required but satellite signals are not required then this clause does not apply.

### 5.2.2 Simulated OTDOA environment

For OTDOA signalling test cases a multi cell environment with Cell 1 and Cell 2 (where required) is used, as defined in 3GPP TS 36.508 [8].

All cells transmit PRS according to the PRS configuration provided in the OTDOA assistance data defined in subclause 5.4.1.2. The positioning subframes are low-interference subframes, i.e. contain no PDSCH transmissions.

Normal propagation condition is used for all cells. Cell 1 is the serving cell and Cell 2 (where required) is a neighbour cell.

Where two cells are required, the two Cells 1 and 2 shall be synchronized, and the timing offset (the RSTD) between the cells, referenced to the UE's antenna input, shall be set equal to the *expectedRSTD* value provided in the OTDOA assistance data, as defined in subclause 5.4.1.2.

The E-UTRA frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

### 5.2.3 Simulated ECID environment

For ECID signalling test cases a multi cell environment with Cell 1 and Cell 2 is used, as defined in 3GPP TS 36.508 [8].

Normal propagation condition is used for all cells. Cell 1 is the serving cell and Cell 2 is a neighbour cell.

The E-UTRA frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

NOTE: If the only ECID measurement supported by the UE is the UE Rx-Tx Time Difference Measurement, Cell 2 does not need to be simulated (see also table 5.4-6).

### 5.2.4 Simulated MBS environment

During MBS signalling tests, where required, the SS shall generate the UE supported MBS signals as defined in the MBS scenarios defined in TS 37.571-5 [12].

### 5.2.5 Simulated WLAN environment

During WLAN signalling tests, where required, the SS shall generate the UE supported WLAN signals as defined in the WLAN scenarios defined in TS 37.571-5 [12].

### 5.2.6 Simulated Bluetooth environment

During Bluetooth signalling tests, where required, the SS shall generate the UE supported Bluetooth signals as defined in the Bluetooth scenarios defined in TS 37.571-5 [12].

### 5.2.7 Simulated Sensor environment

During Sensor signalling tests there is no simulated sensor environment.

## 5.3 Default RRC and NAS message and information elements contents

The default values of common RRC and NAS messages and information elements are used as defined in 3GPP TS 36.508 [8] with the following exceptions.

- ATTACH ACCEPT

**Table 5.3-1: ATTACH ACCEPT**

| Derivation Path: 36.508 Table 4.7.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| EPS network feature support | Set according to Table 5.3-2 | | |

**Table 5.3-2: EPS network feature support**

| Derivation Path: 24.301 clause 9.9.3.12A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| IMS voice over PS session indicator (IMS VoPS) (octet 3, bit 1) | 1 | IMS voice over PS session in S1 mode supported | |
| Emergency bearer services indicator (EMC BS) (octet 3, bit 2) | 1 | emergency bearer services in S1 mode supported | |
| Location services indicator in EPC (EPC-LCS) (octet 3, bit 3) | 1 | location services via EPC supported | |
| Location services indicator in CS (CS-LCS) (octet 3, bit 4 to 5) | 01 | location services via CS domain not supported | |
| octet 3, bit 6 to 8 | 000 | spare | |

## 5.4 Default LPP message and information elements contents

This clause contains the default values of LPP messages and information elements used, unless indicated otherwise in specific clauses of this specification.

- LPP REQUEST CAPABILITIES

**Table 5.4-1: RequestCapabilities**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestCapabilities-r9 SEQUENCE { | | | |
| commonIEsRequestCapabilities SEQUENCE { | Present | Rel-14 onwards | |
| lpp-message-segmentation-req-r14 | 00 | Server is not able to send segmented LPP messages. Server is not able to receive segmented LPP messages. Rel-14 onwards | |
| } | | | |
| a-gnss-RequestCapabilities SEQUENCE { | | | |
| gnss-SupportListReq | TRUE | | |
| assistanceDataSupportListReq | TRUE | | |
| locationVelocityTypesReq | TRUE | | |
| } | | | |
| otdoa-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| epdu-RequestCapabilities | Not present | | |
| sensor-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| tbs-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| wlan-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| bt-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- LPP PROVIDE ASSISTANCE DATA

**Table 5.4-2: ProvideAssistanceData**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | Dependent on test case. | | |
| initiator | | | |
| transactionNumber | | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData SEQUENCE { | | | Sub-tests 7 and 15 only; and as defined in Table 5.4.1.1-1. |
| gnss-CommonAssistData SEQUENCE { | | | |
| gnss-ReferenceTime | As defined in 37.571-5 [12] | | |
| gnss-ReferenceLocation | As defined in 37.571-5 [12] | | |
| gnss-IonosphericModel | As defined in 37.571-5 [12] | | |
| gnss-EarthOrientationParameters | Not present | | |
| } | | | |
| gnss-GenericAssistData(SIZE(1..4))OF{ | SIZE is dependent on the number of GNSSs supported by the UE.<br>If one GNSS supported by the UE, SIZE = 1<br>If two GNSSs supported by the UE, SIZE = 2<br>If three GNSSs supported by the UE, SIZE = 3<br>If four GNSSs supported by the UE, SIZE = 4 | | |
| gnss-ID | For each GNSS supported by the UE. | | |
| sbas-ID | Not present | | |
| gnss-TimeModels | As defined in 37.571-5 [12] | | |
| gnss-DifferentialCorrections | Not present | | |
| gnss-NavigationModel | As defined in 37.571-5 [12] | | |
| gnss-RealTimeIntegrity | Not present | | |
| gnss-DataBitAssistance | Not present | | |
| gnss-AcquisitionAssistance | As defined in 37.571-5 [12] | | |
| gnss-Almanac | As defined in 37.571-5 [12] | | |
| gnss-UTC-Model | As defined in 37.571-5 [12] | | |
| gnss-AuxiliaryInformation | As defined in 37.571-5 [12] | | |
| } | | | |
| gnss-Error | Not present | | |
| } | | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | Sub-test 5 and 7 only |

| | | | |
|---|---|---|---|
| otdoa-ReferenceCellInfo | As defined in Table 5.4.1.2-1 | | |
| otdoa-NeighbourCellInfo | As defined in Table 5.4.1.2-2 | | |
| otdoa-Error | Not present | | |
| } | | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 18 only as defined in clause 5.4.1.5 |
| sensor-AssistanceDataList-r14 | As defined in Table 5.4.1.5-2 | | |
| sensor-Error-r14 | Not present | | |
| } | | | |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 16 only as defined in clause 5.4.1.3 |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE (SIZE(1..n)) OF SEQUENCE{ | | | |
| mbs-AlmanacAssistance-r14 | As defined in Table 5.4.1.3-2 | | |
| mbs-AcquisitonAssistance-r14 | As defined in Table 5.4.1.3-2 | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not present | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 17 only as defined in clause 5.4.1.4 |
| wlan-DataSet-r14 | As defined in Table 5.4.1.4-2 | | |
| wlan-Error-r14 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- LPP REQUEST LOCATION INFORMATION

**Table 5.4-3: RequestLocationInformation**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | Dependent on test case | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 32 | | |
| responseTimeEarlyFix-r12 | Not present | Rel-12 onwards | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | As defined in Table 5.4-4 | | Sub-tests 7 and 15 |
| otdoa-RequestLocationInformation | As defined in Table 5.4-5 | | Sub-test 5 and 7 |
| ecid-RequestLocationInformation | As defined in Table 5.4-6 | | Sub-test 6 |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | As defined in Table 5.4-10 | Rel-13 onwards | Sub-test 14, 18 |
| tbs-RequestLocationInformation-r13 | As defined in Table 5.4-7 | Rel-13 onwards | Sub-tests 12, 16 |
| wlan-RequestLocationInformation-r13 | As defined in Table 5.4-8 | Rel-13 onwards | Sub-test 11, 17 |
| bt-RequestLocationInformation-r13 | As defined in Table 5.4-9 | Rel-13 onwards | Sub-test 13 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- A-GNSS REQUEST LOCATION INFORMATION

**Table 5.4-4: A-GNSS-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.2.7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| A-GNSS-RequestLocationInformation ::= SEQUENCE { | | | |
| gnss-PositioningInstructions SEQUENCE { | | | |
| gnss-Methods | Dependent on the GNSS(s) supported by the UE.<br>If GPS supported bit 0 = 1<br>If Galileo supported bit 3 = 1<br>If GLONASS supported bit 4 = 1<br>If BDS supported bit 5 = 1 | GNSS-ID-Bitmap | |
| fineTimeAssistanceMeasReq | FALSE | | |
| adrMeasReq | FALSE | | |
| multiFreqMeasReq | FALSE | | |
| assistanceAvailability | FALSE | | |
| } | | | |
| } | | | |

- OTDOA REQUEST LOCATION INFORMATION

**Table 5.4-5: OTDOA-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.1.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-RequestLocationInformation ::= SEQUENCE { | | | |
| assistanceAvailability | FALSE | | |
| multipathRSTD-r14 | Not present | | Rel-14 onwards |
| maxNoOfRSTDmeas-r14 | Not present | | Rel-14 onwards |
| } | | | |

- ECID REQUEST LOCATION INFORMATION

**Table 5.4-6: ECID-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.3.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ECID-RequestLocationInformation ::= SEQUENCE { | | | |
| requestedMeasurements | All measurements supported by the UE | | |
| } | | | |

- TBS REQUEST LOCATION INFORMATION

**Table 5.4-7: TBS-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TBS-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| mbsSgnMeasListReq-r13 | TRUE (UE-Assisted MBS) | Rel-13 onwards | |
| mbsAssistanceAvailability-r14 | FALSE | Rel-14 onwards | |
| mbsRequestedMeasurements-r14 | Not present | Rel-14 onwards | |
| } | | | |

- WLAN REQUEST LOCATION INFORMATION

**Table 5.4-8: WLAN-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| WLAN-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| requestedMeasurements-r13 | bit 0 = 1 (rssi) (UE-Assisted WLAN)<br>bit 1 = 1 (rtt) (UE-Assisted WLAN) | Rel-13 onwards | |
| assistanceAvailability-r14 | FALSE | Rel-14 onwards | |
| } | | | |

- BT REQUEST LOCATION INFORMATION

**Table 5.4-9: BT-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.7.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| BT-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| requestedMeasurements-r13 | bit 0 = 1 (rssi) (UE-Assisted BT) | Rel-13 onwards | |
| } | | | |

- SENSOR REQUEST LOCATION INFORMATION

**Table 5.4-10: Sensor-RequestLocationInformation**

| Derivation Path: 36.355 clause 6.5.5.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Sensor-RequestLocationInformation-r13 ::= SEQUENCE { | | | |
| uncompensatedBarometricPressureReq-r13 | TRUE (UE-Assisted Sensor) | Rel-13 onwards | |
| assistanceAvailability-r14 | FALSE | Rel-14 onwards | |
| } | | | |

## 5.4.1 Default assistance data information elements

### 5.4.1.1 GNSS Assistance Data Elements

Table 5.4.1.1-1 defines the GNSS assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message. The GNSS

assistance data provided depends on the mode being used in the test case, the assistance data supported by the UE and the GNSSs supported by the UE. GNSS assistance data IEs not supported by the UE shall not be sent. GNSS assistance data IEs supported by the UE but not listed in Table 5.4.1.1-1 shall not be sent. The content of the assistance data elements is defined in 37.571-5 [12] clause 7.

**Table 5.4.1.1-1: GNSS assistance data to be provided to the UE**

| GNSS Assistance Data IE supported by UE | Mode used in test case | | |
|---|---|---|---|
| | UE-based | UE-assisted, GNSS-AcquisitionAssistance supported by UE | UE-assisted, GNSS-AcquisitionAssistance not supported by UE |
| GNSS-Reference Time | Yes | Yes | Yes |
| GNSS-ReferenceLocation | Yes | No | Yes |
| GNSS-IonosphericModel | Yes | No | No |
| GNSS-TimeModelList | Yes(1) | No | Yes(1) |
| GNSS-NavigationModel | Yes | No | Yes |
| GNSS-AcquisitionAssistance | No | Yes | No |
| GNSS-Almanac | No | No | Yes |
| GNSS-UTC-Model | Yes(3) | Yes(3) | Yes(3) |
| GNSS-AuxiliaryInformation | Yes(2) | Yes(2) | Yes(2) |
| NOTE 1: Only if more than one GNSS supported by the UE.<br>NOTE 2: Only if GLONASS supported by the UE, and/or if the UE supports multiple signals per GNSS, and/or if BDS B1C/B2a signal type supported by the UE.<br>NOTE 3: Only if GLONASS and at least one other GNSS supported by the UE. | | | |

### 5.4.1.2 OTDOA Assistance Data Elements

This subclause defines the OTDOA assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages.

- OTDOA REFERENCE CELL INFO

**Table 5.4.1.2-1: OTDOA-ReferenceCellInfo**

| Derivation Path: 36.355 clause 6.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-ReferenceCellInfo ::= SEQUENCE { | | Cell 1 | |
| physCellId | 0 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 1 in 36.508 [8] | | |
| mnc | As defined for Cell 1 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 1 in 36.508 [8] | | |
| } | | | |
| earfcnRef | Not present | Same as the serving cell | |
| antennaPortConfig | Not present | Same as the serving cell | |
| cpLength | Normal | | |
| prsInfo SEQUENCE { | | | |
| prs-Bandwidth | PRS are transmitted over the used system bandwidth (see subclause 5.2.2) | | |
| prs-ConfigurationIndex | FDD: 2<br>TDD: 4 | | |
| numDL-Frames | sf-1 | | |
| prs-MutingInfo-r9 | Not present | PRS muting is not used. | |
| prsID-r14 | Not present | PRS-ID not used | Rel-14 onwards |
| add-numDL-Frames-r14 | Not present | Not required | Rel-14 onwards |
| prsOccGroupLen-r14 | Not present | No PRS occasion group configured | Rel-14 onwards |
| prsHoppingInfo-r14 | Not present | PRS frequency hopping not used | Rel-14 onwards |
| } | | | |
| earfcnRef-v9a0 | Not present | Same as the serving cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| cpLengthCRS-r14 | Normal | | Rel-14 onwards |
| sameMBSFNconfigRef-r14 | TRUE | Same as the serving cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as the serving cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigRef-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| } | | | |

- OTDOA NEIGHBOUR CELL INFO LIST

**Table 5.4.1.2-2: OTDOA-NeighbourCellInfoList**

| Derivation Path: 36.355 clause 6.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-NeighbourCellInfoList ::= SEQUENCE (SIZE(1)) OF SEQUENCE { | | | |
| SEQUENCE (SIZE(2)) OF SEQUENCE { | | Cell 2 | |
| physCellId | 2 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 2 in 36.508 [8] | | |
| mnc | As defined for Cell 2 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 2 in 36.508 [8] | | |
| } | | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | Not present | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 μs | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| } | | | |
| SEQUENCE { | | Cell 4 | |
| physCellId | 4 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 4 in 36.508 [8] | | |
| mnc | As defined for Cell 4 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 4 in 36.508 [8] | | |
| } | | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |
| antennaPortConfig | Not present | Same as for the reference cell | |

| | | | |
|---|---|---|---|
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | Not present | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 µs | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| } | | | |
| } | | | |

### 5.4.1.3 MBS Assistance Data Elements

Tables 5.4.1.3-1 and 5.4.1.3-2 define the MBS assistance data elements which shall be provided to the UE in sub-test 16 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message.

Table 5.4.1.3-1 defines the MBS assistance data IEs that conditionally depend on the mode being used in the test case and on the assistance data supported by the UE.

Table 5.4.1.3-2 defines the content of the TBS-AssistantDataList.

**Table 5.4.1.3-1: MBS assistance data IEs to be conditionally provided to the UE**

| MBS Assistance Data IE supported by UE | Mode used in test case | |
|---|---|---|
| | UE-based, MBS (Release 14 onwards) | UE-assisted, MBS (Release 14 onwards) |
| mbs-AlmanacAssistance-r14 | Yes | No |
| mbs-AcquisitionAssistance-r14 | Yes | Yes |

**Table 5.4.1.3-2: Content of TBS-AssistanceDataList**

| Derivation Path: TS 36.355 [4] clause 6.5.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb1 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb1 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 3 tb1 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 4 tb1 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 1 tb2 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 2 tb2 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |

| | | | |
|---|---|---|---|
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 3 tb2 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| mbs-AssistanceDataElement-r14 SEQUENCE { | | Beacon 4 tb2 | |
| mbs-AlmanacAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| mbs-AcquisitionAssistance-r14 | According to Table 5.4.1.3-1 and as defined in TS 37.571-5 [12], clause 8 | | |
| } | | | |
| } | | | |
| } | | | |

## 5.4.1.4 WLAN Assistance Data Elements

Tables 5.4.1.4-1 and 5.4.1.4-2 define the WLAN assistance data elements which shall be provided to the UE in sub-test 17 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message.

Table 5.4.1.4-1 defines the WLAN assistance data IE that conditionally depends on the mode being used in the test case and on the assistance data supported by the UE.

Table 5.4.1.4-2 defines the content of the WLAN-DataSet.

**Table 5.4.1.4-1: WLAN assistance data IE to be conditionally provided to the UE**

| WLAN assistance data IE supported by UE | Mode used in test case | |
|---|---|---|
| | UE-based, WLAN (Release 14 onwards) | UE-assisted, WLAN (Release 14 onwards) |
| wlan-AP-Location-r14 (WLAN AP location information) | Yes | No |

**Table 5.4.1.4-2: Content of WLAN-DataSet**

| Derivation Path: TS 36.355 [4] clause 6.5.6.8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| wlan-DataSet-r14::= SEQUENCE (SIZE (1)) OF SEQUENCE { | | | Rel-14 onwards |
| SEQUENCE (SIZE (4)) OF SEQUENCE.{ | | | |
| WLAN-AP-Data-r14 SEQUENCE { | | WLAN AP 1 | |
| wlan-AP-Identifier-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| wlan-AP-Location-r14 | According to Table 5.4.1.4-1 and as defined in TS 37.571-5 [12], clause 9 | | |
| } | | | |
| WLAN-AP-Data-r14 SEQUENCE { | | WLAN AP 2 | |
| wlan-AP-Identifier-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| wlan-AP-Location-r14 | According to Table 5.4.1.4-1 and as defined in TS 37.571-5 [12], clause 9 | | |
| } | | | |
| WLAN-AP-Data-r14 SEQUENCE { | | WLAN AP 3 | |
| wlan-AP-Identifier-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| wlan-AP-Location-r14 | According to Table 5.4.1.4-1 and as defined in TS 37.571-5 [12], clause 9 | | |
| } | | | |
| WLAN-AP-Data-r14 SEQUENCE { | | WLAN AP 4 | |
| wlan-AP-Identifier-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| wlan-AP-Location-r14 | According to Table 5.4.1.4-1 and as defined in TS 37.571-5 [12], clause 9 | | |
| } | | | |
| supportedChannels-11a-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| supportedChannels-11bg-r14 | As defined in TS 37.571-5 [12], clause 9 | | |
| } | | | |
| } | | | |

## 5.4.1.5 Sensor Assistance Data Elements

Tables 5.4.1.5-1 and 5.4.1.5-2 define the Sensor assistance data elements which shall be provided to the UE in sub-test 18 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message.

Table 5.4.1.5-1 defines the Sensor assistance data IE that conditionally depends on the mode being used in the test case.

Table 5.4.1.5-2 defines the content of the Sensor-AssistanceDataList.

**Table 5.4.1.5-1: Sensor assistance data IE to be conditionally provided to the UE**

| **Sensor assistance data to be provided to the UE** | **Mode used in test case** | |
|---|---|---|
| | **UE-based, Sensor (Release 14 onwards)** | **UE-assisted, Sensor (Release 14 onwards)** |
| sensor-AssistanceDataList-r14 | Yes | No |

**Table 5.4.1.5-2: Content of Sensor-AssistanceDataList**

| Derivation Path: TS 36.355 [4] clause 6.5.5.8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Sensor-AssistanceDataList-r14::= SEQUENCE { | According to Table 5.4.1.5-1 | | Rel-14 onwards |
| refPressure-r14 | 0 | 101325 Pa | |
| refPosition-r14 | As defined in TS 37.571-5 [12], clause 6.1.3.4, GNSS-ReferenceLocation | As used in GNSS sub-tests | |
| refTemperature-r14 | 20 | 293K | |
| period-v1520 | Not present | | |
| area-v1520 | Not present | | |
| } | | | |

# 6 Protocol Conformance Test Cases for UTRAN

## 6.1 Assisted-GPS Test Cases

### 6.1.1 Assisted GPS Network Induced Tests

#### 6.1.1.1 LCS Network Induced location request/ UE-Based GPS/ Emergency Call / with USIM

##### 6.1.1.1.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

##### 6.1.1.1.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

   When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

2) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

   - a SETUP message, if the call to be established is a basic call; and
   - an EMERGENCY SETUP message, if the call to be established is an emergency call.

3) If the IE "UE positioning GPS reference time" is included, the UE shall:

   1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

   1> store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

4) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

1> use it as a priori knowledge of the approximate location of the UE.

5) The UE shall when a measurement report is triggered:

2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

4> if the IE "GPS timing of Cell wanted" is set to FALSE:

5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

Reference(s):

- Conformance requirement 1: TS 24.008 clause 4.5.1.5.
- Conformance requirement 2: TS 24.008, clause 5.2.1.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.1b.

#### 6.1.1.1.3 Test Purpose

To verify when an emergency call is initiated by a UE with a USIM, and the network performs a location request using the RRC measurement control procedure by sending Measurement Control message , then the UE respond with a Measurement Report containing UE location.

#### 6.1.1.1.4 Method of Test

Initial Conditions

- System Simulator:
  - 1 cell, default parameters.

- Satellite signals: As specified in 4.2.

- User Equipment:

  - the UE is in state "MM idle" with valid TMSI and CKSN.

### Related PICS/PIXIT Statements

- Emergency speech call yes/no

- UE Based Network Assisted GPS

### Test procedure

The UE is made to initiate an emergency call.

After the call has been through-connected in both directions, the SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages. The last MEASUREMENT CONTROL message orders periodical reporting by sending a MEASUREMENT CONTROL message requesting periodical measurement reporting (1 report, interval 64s).

The UE then performs positioning measurements, calculates "UE Positioning Position Estimate Info" and responds with this in the RRC message MEASUREMENT REPORT.

Finally the SS clears the call.

### Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. Number shall be one programmed in test USIM $EF_{ECC}$ (Emergency Call Codes), ref. 34.108 clause 8.3.2.21. |
| 2 | --> | | | UE establishes RRC procedure for emergency call.<br>Establishment cause: Emergency Call<br>SS checks that the UE capability includes A-GPS UE based positioning measurement |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". |
| 4 | <-- | | AUTHENTICATION REQUEST | IE Authentication Parameter AUTN shall be present in the message. |
| 5 | --> | | AUTHENTICATION RESPONSE | SRES specifies correct value. |
| 6 | | | | SS starts security procedure. |
| 7 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 8 | <-- | | CALL PROCEEDING | |
| 9 | <-- | | ALERTING | |
| 10 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 11 | <-- | | CONNECT | |
| 12 | --> | | CONNECT ACKNOWLEDGE | |
| 13 | UE | | | The DTCH is through connected in both directions. |
| 14 | <- | | MEASUREMENT CONTROL | |
| 15 | <- | | MEASUREMENT CONTROL | |
| 16 | --> | | MEASUREMENT REPORT | |
| 17 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 14):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 15):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 16):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | GPS reference time only |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.1.1.5 Test requirements

After step 12 the UE shall have through connected the DTCH in both directions.

After step 15 the UE shall respond with a MEASUREMENT REPORT message.

### 6.1.1.2 LCS Network Induced location request/ UE-Based GPS/ Emergency Call / without USIM

#### 6.1.1.2.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.1.2.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

   When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

   Normally, the UE will be identified by an IMSI or a TMSI. However, if none of these identifiers is available in the UE, then the UE shall use the IMEI for identification purposes.

2) As a serving network option, emergency calls may be established without the network having to apply the security mode procedure as defined in TS 24.008.

   The following are the only cases where the "security procedure not applied" option may be used:

   a) Authentication is impossible because the USIM is absent.

3) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

   - a SETUP message, if the call to be established is a basic call; and
   - an EMERGENCY SETUP message, if the call to be established is an emergency call.

4) If the IE "UE positioning GPS reference time" is included, the UE shall:

   1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

   1> store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

5) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

   1> use it as a priori knowledge of the approximate location of the UE.

6) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

      3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

         4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

         4> if the IE "GPS timing of Cell wanted" is set to FALSE:

            5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

Reference(s):

- Conformance requirement 1: TS 24.008 clause 4.5.1.5, TS 22.101 clause 8.
- Conformance requirement 2: TS 33.102, clause 6.4.9.2.
- Conformance requirement 3: TS 24.008, clause 5.2.1.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.1b.

#### 6.1.1.2.3 Test Purpose

To verify when an emergency call is initiated by a UE in the "MM idle, no IMSI" state (no USIM inserted) and the network performs a location request using the RRC measurement control procedure by sending Measurement Control message , then the UE respond with a Measurement Report containing UE location.

#### 6.1.1.2.4 Method of Test

Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment:
  - the UE is in MM-state "MM idle, no IMSI", no USIM inserted.

Related PICS/PIXIT Statements

- Emergency speech call yes/no
- UE Based Network Assisted GPS

Test procedure

The UE is made to initiate an emergency call.

After the call has been through-connected in both directions, the SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages. The last MEASUREMENT CONTROL message orders periodical reporting by sending a MEASUREMENT CONTROL message requesting periodical measurement reporting (1 report, interval 64s).

The UE then performs positioning measurements, calculates "UE Positioning Position Estimate Info" and responds with this in the RRC message MEASUREMENT REPORT.

Finally the SS clears the call.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. One of the following emergency numbers shall be used: 000, 08, 112, 110, 118, 119, 911 or 999. |
| 2 | --> | | | UE establishes RRC procedure for emergency call.<br>Establishment cause: Emergency Call<br>SS checks that the UE capability includes A-GPS UE based positioning measurement |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". |
| 4 | <-- | | CM SERVICE ACCEPT | |
| 5 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 6 | <-- | | CALL PROCEEDING | |
| 7 | <-- | | ALERTING | |
| 8 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 9 | <-- | | CONNECT | |
| 10 | --> | | CONNECT ACKNOWLEDGE | |
| 11 | UE | | | The DTCH is through connected in both directions. |
| 12 | <- | | MEASUREMENT CONTROL | |
| 13 | <- | | MEASUREMENT CONTROL | |
| 14 | --> | | MEASUREMENT REPORT | |
| 15 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 12):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 13):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 14):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | GPS reference time only |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.1.2.5 Test requirements

After step 10 the UE shall have through connected the DTCH in both directions.

After step 13 the UE shall respond with a MEASUREMENT REPORT message containing a position estimate.

### 6.1.1.3 LCS Network induced location request/ UE-Assisted GPS/ Emergency call/ With USIM

#### 6.1.1.3.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities.

#### 6.1.1.3.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

2) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

- a SETUP message, if the call to be established is a basic call; and
- an EMERGENCY SETUP message, if the call to be established is an emergency call.

3) if the IE "Measurement command" has the value "setup":

2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

…

2> for any other measurement type:

3> if the measurement is valid in the current RRC state of the UE:

4> begin measurements according to the stored control information for this measurement identity.

4) The UE shall:

1> when a measurement report is triggered:

2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

3> if the IE "Vertical Accuracy" is included:

4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

3> if the IE "Positioning Methods" is set to "GPS":

4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

6> if the IE "GPS timing of Cell wanted" is set to TRUE:

7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

6> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

References

- Conformance requirement 1: TS 24.008 clause 4.5.1.5.
- Conformance requirement 2: TS 24.008, clause 5.2.1.
- Conformance requirement 3: TS 25.331, clause 8.4.1.3.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.1a.

#### 6.1.1.3.3 Test Purpose

To verify when an emergency call is initiated by a UE with a USIM, and the network performs a location request using the RRC measurement control procedure by sending Measurement Control message , then the UE respond with a Measurement Report containing "UE positioning GPS measured results".

#### 6.1.1.3.4 Method of Test

Initial Conditions

System Simulator (SS):

1 cell, default parameters

Satellite signals: As specified in 4.2

UE:

- the UE is in state "MM idle" with valid TMSI and CKSN.

Related PICS/PIXIT Statements

- Emergency speech call yes/no
- UE Assisted Network Assisted GPS

Test Procedure

The UE is made to initiate an emergency call.

After the call has been through-connected in both directions, the SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE then performs positioning measurements and responds with the RRC message MEASUREMENT REPORT.

Finally the SS clears the call.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. Number shall be one programmed in test USIM $EF_{ECC}$ (Emergency Call Codes), ref. 34.108 clause 8.3.2.21. |
| 2 | --> | | | UE establishes RRC procedure for emergency call.<br>Establishment cause: Emergency Call<br>SS checks that the UE capability includes A-GPS UE assisted positioning measurement |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". |
| 4 | <-- | | AUTHENTICATION REQUEST | IE Authentication Parameter AUTN shall be present in the message. |
| 5 | --> | | AUTHENTICATION RESPONSE | SRES specifies correct value. |
| 6 | SS | | | SS starts security procedure. |
| 7 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 8 | <-- | | CALL PROCEEDING | |
| 9 | <-- | | ALERTING | |
| 10 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 11 | <-- | | CONNECT | |
| 12 | --> | | CONNECT ACKNOWLEDGE | |
| 13 | UE | | | The DTCH is through connected in both directions. |
| 14 | <- | | MEASUREMENT CONTROL | |
| 15 | --> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 15a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 15, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 15b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 15, this message contains the IE "UE positioning GPS measured results". |
| 16 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 14):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 15 (Option 1) or 15b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 15 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 15a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

#### 6.1.1.3.5 Test requirements

After step 12 the UE shall have through connected the DTCH in both directions.

After step 14 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

### 6.1.1.4 LCS Network induced location request/ UE-Assisted GPS/ Emergency call/ Without USIM

#### 6.1.1.4.1 Definition

This test case applies to all UEs supporting UE-assisted A-GPS Location Service capabilities.

#### 6.1.1.4.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

   When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

   Normally, the UE will be identified by an IMSI or a TMSI. However, if none of these identifiers is available in the UE, then the UE shall use the IMEI for identification purposes.

2) As a serving network option, emergency calls may be established without the network having to apply the security mode procedure as defined in TS 24.008.

   The following are the only cases where the "security procedure not applied" option may be used:

a) Authentication is impossible because the USIM is absent.

3) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

   - a SETUP message, if the call to be established is a basic call; and
   - an EMERGENCY SETUP message, if the call to be established is an emergency call.

4) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

5) The UE shall:

1> when a measurement report is triggered:

   2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

      3> if the IE "Vertical Accuracy" is included:

         4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

      3> if the IE "Positioning Methods" is set to "GPS":

4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

6> if the IE "GPS timing of Cell wanted" is set to TRUE:

7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

6> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

References

- Conformance requirement 1: TS 24.008 clause 4.5.1.5, TS 22.101 clause 8.
- Conformance requirement 2: TS 33.102, clause 6.4.9.2.
- Conformance requirement 3: TS 24.008, clause 5.2.1.
- Conformance requirement 4: TS 25.331, clause 8.4.1.3.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.1a.

#### 6.1.1.4.3 Test Purpose

To verify that when an emergency call is initiated by a UE with no USIM, and the network performs a network-induced location request using UE-assisted A-GPS, the UE responds with a Measurement Report containing the IE "UE positioning GPS measured results".

#### 6.1.1.4.4 Method of Test

Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment:
  - the UE is in state "MM idle" with no IMSI and no USIM inserted.

Related PICS/PIXIT Statements

- Emergency speech call yes/no
- UE Assisted Network Assisted GPS

Test procedure

The UE is made to initiate an emergency call. The call is established without authentication and security.

After the call has been through-connected in both directions, the SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message including the IE "UE positioning GPS measured results".

Finally the SS clears the call.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. One of the following emergency numbers shall be used: 000, 08, 112, 110, 118, 119, 911 or 999. |
| 2 | --> | | | UE establishes RRC procedure for emergency call. Establishment cause: Emergency Call<br>SS checks that the UE capability includes A-GPS UE-assisted positioning measurement. |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". The mobile identity IE specifies the IMEI of the UE. The cipher key sequence number IE indicates "no key is available". |
| 4 | <-- | | CM SERVICE ACCEPT | |
| 5 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 6 | <-- | | CALL PROCEEDING | |
| 7 | <-- | | ALERTING | |
| 8 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 9 | <-- | | CONNECT | |
| 10 | --> | | CONNECT ACKNOWLEDGE | |
| 11 | UE | | | The DTCH is through connected in both directions. |
| 12 | <- | | MEASUREMENT CONTROL | Assistance data as specified in subclause 4.3.3. |
| 13 | --> | | MEASUREMENT REPORT | UE reports the IE "UE positioning GPS measured results" (Option 1) or requests additional assistance data (Option 2). |
| 13a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 13, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 13b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 13, this message contains the IE "UE positioning GPS measured results". |
| 14 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 12):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in subclause 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 13 (Option 1) or 13b (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 13 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 13a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

#### 6.1.1.4.5 Test requirements

After step 10 the UE shall have through connected the DTCH in both directions.

After step 12 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

## 6.1.2 Assisted GPS Mobile Originated Tests

### 6.1.2.1 LCS Mobile originated location request/ UE-Based GPS/ Position estimate request/ Success

#### 6.1.2.1.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

#### 6.1.2.1.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

3) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

      2> act as specified in subclause 8.6.7.19.3.4.

   If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

4) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

4> if the IE "GPS timing of Cell wanted" is set to FALSE:

5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

5) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

6) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 5 and 6: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 3: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 4: TS 25.331, subclause 8.6.7.19.1b
- Reference [12] in these conformance requirements denotes document ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

#### 6.1.2.1.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-based GPS.

#### 6.1.2.1.4 Method of Test

Initial Conditions

- System Simulator:

- 1 cell, default parameters.
- Satellite signals: As specified in 4.2

- User Equipment:
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI

### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- Method of triggering an MO-LR request for a position estimate.

### Test Procedure

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

Then the UE invokes an MO-LR request of type "locationEstimate". The SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages, including assistance data. The UE then initiates periodic measurement reporting. After the first received MEASUREMENT REPORT message, the SS responds with a FACILITY message containing an MO-LR result. When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

### Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "locationEstimate". |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | <- | | MEASUREMENT CONTROL | |
| 9 | -> | | MEASUREMENT REPORT | |
| 10 | <- | | FACILITY | LCS MO-LR result message containing location estimate |
| 11 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 12 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->locationEstimate |
| SS version indicator | Value 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 9)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

FACILITY (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | FACILITY (0011 1010) |
| Facility | Return result = LCS-MOLR<br>LCS-MOLRRes -> locationEstimate |

RELEASE COMPLETE (Step 11)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |

#### 6.1.2.1.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 8, the UE shall respond with a MEASUREMENT REPORT message.

After step 10, the UE shall send a RELEASE COMPLETE message.

### 6.1.2.2 LCS Mobile originated location request/ UE-Based or UE-Assisted GPS/ Assistance data request/ Success

#### 6.1.2.2.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GPS Location Service capabilities and providing a method to trigger an MO-LR request for assistance data.

#### 6.1.2.2.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

3) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 2 and 3: TS 24.030, subclause 5.1.1

#### 6.1.2.2.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted network assisted GPS.

#### 6.1.2.2.4 Method of Test

Initial Conditions

- System Simulator:
  - 1 cell, default parameters.

- Satellite signals: As specified in 4.2

- User Equipment:

  - The UE shall begin the test with no GPS assistance data stored.

  - The UE is in state "MM idle" with valid TMSI and CKSN.

  - The UE is in state "PMM idle" with valid P-TMSI

### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS

- UE Assisted Network Assisted GPS

- Method of clearing stored GPS assistance data

- Method of triggering an MO-LR request for assistance data.

### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

Then the UE invokes an MO-LR request of type "assistanceData".

The SS transmits an ASSISTANCE DATA delivery message with assistance data. When the assistance data delivery was successful, the SS sends a FACILITY message to the UE.

The UE clears the transaction by sending a RELEASE COMPLETE message.

### Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | | Void | |
| 2 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 3 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 4 | <- | | AUTHENTICATION REQUEST | |
| 5 | -> | | AUTHENTICATION RESPONSE | |
| 6 | SS | | | The SS starts ciphering and integrity protection. |
| 7 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "assistanceData". |
| 8 | <- | | ASSISTANCE DATA DELIVERY | The SS provides the requested data in one or more ASSISTANCE DATA DELIVERY messages as specified in subclause 4.3.5 |
| 9 | <- | | FACILITY | |
| 10 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 11 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 7)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->assistanceData<br>locationMethod -> assistedGPS<br>gpsAssistanceData -> OCTET STRING<br>Octets 1 to 38 are coded in the same way as octets 3 to 7+2n of Requested GPS Data IE in 3GPP TS 49.031 (Contents are not verified, SS will use octet 1 to identify the GPS assistance data requested by the MS) |
| SS version indicator | Value 1 or above |

ASSISTANCE DATA DELIVERY (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| UE positioning OTDOA assistance data for UE-based | Not present |
| UE positioning GPS assistance data | Set as specified in 4.3.5. |

FACILITY (Step 9)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | FACILITY (0011 1010) |
| Facility | Return result = LCS-MOLR<br>LCS-MOLRRes -> EMPTY |

RELEASE COMPLETE (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0x10 1010) |

#### 6.1.2.2.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "assistanceData".

After step 8, the UE shall send a RELEASE COMPLETE message.

### 6.1.2.3 LCS Mobile originated location request/ UE-Assisted GPS/ Position Estimate/ Success

#### 6.1.2.3.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

#### 6.1.2.3.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

3) The UE shall:

   1> when a measurement report is triggered:

   2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

      3> if the IE "Vertical Accuracy" is included:

         4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

      3> if the IE "Positioning Methods" is set to "GPS":

         4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

            5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

               6> if the IE "GPS timing of Cell wanted" is set to TRUE:

                  7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

                  7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

                  7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

               6> if the IE "GPS timing of Cell wanted" is set to FALSE:

                  7> include the IE "GPS TOW msec".

            5> if the UE does not support the capability to provide the GPS timing of the cell:

               6> include the IE "GPS TOW msec".

4) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

5) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

References:

- Conformance requirements 1, 4 and 5: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 25.331, subclause 8.4.1.3

- Conformance requirement 3: TS 25.331, subclause 8.6.7.19b

#### 6.1.2.3.3 Test Purpose

To verify the UE behaviour in the mobile-originated location request procedure using network-assisted UE-assisted GPS to request a position estimate from the network.

#### 6.1.2.3.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment:
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI

##### Related PICS/PIXIT Statements

- UE Assisted Network Assisted GPS
- Method of triggering an MO-LR request for a position estimate.

##### Test Procedure

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request through the Facility IE in a REGISTER message. The MO-LR request is of type "locationEstimate".

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE then initiates periodic measurement reporting. After receiving the MEASUREMENT REPORT message, the SS responds with a FACILITY message containing an MO-LR result. When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request. The IE "MOLR-Type" is set to "locationEstimate". |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | -> | | MEASUREMENT REPORT | UE reports the IE "UE positioning GPS measured results" (Option 1) or requests additional assistance data (Option 2). |
| 8a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 8, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 8b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 8, this message contains the IE "UE positioning GPS measured results". |
| 9 | <- | | FACILITY | LCS MO-LR result message containing location estimate |
| 10 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 11 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type -> locationEstimate<br>Value 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8 (Option 1) or 8b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 8 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 8a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

FACILITY (Step 9)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>FACILITY (0011 1010)<br>Return result = LCS-MOLR<br>LCS-MOLRRes -> locationEstimate<br>locationEstimate ->any values may be used. The SS shall not be required to calculate the value from the returned gps-MeasureInfo values |

RELEASE COMPLETE (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type | Call Independent SS message (1011)<br><br>RELEASE COMPLETE (xx10 1010) |

#### 6.1.2.3.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 7, the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

After step 9, the UE shall send a RELEASE COMPLETE message.

### 6.1.2.4 LCS Mobile originated location request/ UE-Based GPS/ Transfer to third party/ Success

#### 6.1.2.4.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities and providing a method to trigger an MO-LR request for transfer to 3rd party.

#### 6.1.2.4.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) If the UE is requesting that its location be sent to an external LCS client, the message shall include the identity of the LCS client and may include the address of the GMLC through which the LCS client should be accessed.

3) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

4) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

5) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

      2> act as specified in subclause 8.6.7.19.3.4.

   If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

6) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

      3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

         4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

         4> if the IE "GPS timing of Cell wanted" is set to FALSE:

            5> include the IE "GPS TOW msec".

         4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

            5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

               6> if the UE has been able to calculate a 3-dimensional position:

                  7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

               6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

7) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

8) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 7 and 8: TS 24.030, subclause 5.1.1
- Conformance requirement 3: TS 25.331, subclause 8.4.1.3
- Conformance requirement 2: TS 23.171, subclause 8.8.1
- Conformance requirement 4: TS 25.331, subclause 8.4.1.3
- Conformance requirement 5: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4
- Conformance requirement 6: TS 25.331, subclause 8.6.7.19.1b
- Reference [12] in these conformance requirements denotes document ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

#### 6.1.2.4.3 Test Purpose

To verify the UE behaviour in the mobile-originated location request procedure using network-assisted UE-based GPS to request a position estimate from the network for transfer to a third-party LCS client.

#### 6.1.2.4.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment:
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- Method of triggering an MO-LR request for transfer to 3rd party

Test Procedure

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes a MO-LR request through the Facility IE in a REGISTER message. The MO-LR request is of type "locationEstimate". The IE "LCSClientExternalID" is set to the ID of a valid external LCS client.

The SS orders an A-GPS positioning measurement using MEASUREMENT CONTROL messages, including assistance data.

The UE sends a MEASUREMENT REPORT message containing a location estimate.

The SS sends a FACILITY message confirming that the transfer to the external client succeeded. When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request. The IE "MOLR-Type" is set to "locationEstimate". The IE "LCSClientExternalID" is set to a valid ID for an external LCS client. |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | <- | | MEASUREMENT CONTROL | |
| 9 | -> | | MEASUREMENT REPORT | |
| 10 | <- | | FACILITY | LCS MO-LR result message as confirmation that the position estimate was transferred to the requested LCS client. |
| 11 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| | | | | |
| 12 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->locationEstimate<br>lcsClientExternalID -> externalAddress |
| SS version indicator | Value 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 9)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

FACILITY (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | FACILITY (0011 1010) |
| Facility | Return result = LCS-MOLR<br>LCS-MOLRRes -> locationEstimate |

RELEASE COMPLETE (Step 11)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |

#### 6.1.2.4.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate" and the IE "LCSClientExternalID" set to the ID of a valid external LCS client.

After step 8, the UE shall respond with a MEASUREMENT REPORT message containing the IE "Position Estimate".

After step 11, the UE shall send a RELEASE COMPLETE message.

### 6.1.2.5 LCS Mobile originated location request/ UE-Assisted GPS/ Transfer to third party/ Success

#### 6.1.2.5.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities and providing a method to trigger an MO-LR request for transfer to 3rd party.

#### 6.1.2.5.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) If the UE is requesting that its location be sent to an external LCS client, the message shall include the identity of the LCS client and may include the address of the GMLC through which the LCS client should be accessed.

3) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

4) The UE shall:

   1> when a measurement report is triggered:

   2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

3> if the IE "Vertical Accuracy" is included:

4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

3> if the IE "Positioning Methods" is set to "GPS":

4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

6> if the IE "GPS timing of Cell wanted" is set to TRUE:

7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

6> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

5) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

6) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 5 and 6: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 23.171, subclause 8.8.1
- Conformance requirement 3: TS 25.331, subclause 8.4.1.3
- Conformance requirement 4: TS 25.331, subclauses 8.6.7.19.3.3b

#### 6.1.2.5.3 Test Purpose

To verify the UE behaviour in the mobile-originated location request procedure using network-assisted UE-assisted GPS to request a position estimate from the network for transfer to a third-party LCS client.

#### 6.1.2.5.4 Method of Test

Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2.
- User Equipment:
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI.

Related PICS/PIXIT Statements

- UE Assisted Network Assisted GPS
- Method of triggering an MO-LR request for transfer to 3rd party

Test Procedure

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes a MO-LR request through the Facility IE in a REGISTER message. The MO-LR request is of type "locationEstimate". The IE "LCSClientExternalID" is set to the ID of a valid external LCS client.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message containing IE "UE positioning GPS measured results".

The SS sends a FACILITY message confirming that the transfer to the external client succeeded. When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request. The IE "MOLR-Type" is set to "locationEstimate". The IE "LCSClientExternalID" is set to a valid ID for an external LCS client. |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 8a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 8, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 8b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 8, this message contains the IE "UE positioning GPS measured results". |
| 9 | <- | | FACILITY | LCS MO-LR result message as confirmation that the position estimate was transferred to the requested LCS client. |
| 10 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 11 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->locationEstimate<br>lcsClientExternalID -> externalAddress |
| SS version indicator | Value 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Steps 8 (Option 1) or 8b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 8 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 8a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

FACILITY (Step 9)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | FACILITY (0011 1010) |
| Facility | Return result = LCS-MOLR<br>LCS-MOLRRes -> locationEstimate<br>locationEstimate ->any values may be used. The SS shall not be required to calculate the value from the returned gps-MeasureInfo values |

RELEASE COMPLETE (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |

#### 6.1.2.5.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate" and the IE "LCSClientExternalID" set to the ID of a valid external LCS client.

After step 7, the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

After step 9, the UE shall send a RELEASE COMPLETE message.

### 6.1.2.6 LCS Mobile originated location request/ UE-Based or UE-Assisted GPS/ Assistance data request/ Failure

#### 6.1.2.6.1 Definition

This test case applies to all UEs supporting UE-based or UE-assisted GPS Location Service capabilities and providing a method to trigger an MO-LR request for assistance data.

#### 6.1.2.6.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) If the network is unable to successfully fulfil the request received from the MS (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080.

3) PositionMethodFailure: This error is returned by the network when the network is unable to obtain any of the location information requested or none of the information obtained satisfies the requested LCS QoS or if requested LCS assistance data could not be transferred or requested deciphering keys for broadcast assistance data could not be returned.

Reference(s):

- Conformance requirements 1 and 2: TS 24.030, subclause 5.1.1

- Conformance requirement 3: TS 24.080, subclause 4.3.2.29

#### 6.1.2.6.3 Test Purpose

To verify the UE behaviour at a mobile originated location request for GPS assistance data where the network is unable to provide the requested GPS assistance data.

#### 6.1.2.6.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals switched off or not present
- User Equipment:

    The UE shall begin the test with no GPS assistance data stored.
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS, or UE Assisted Network Assisted GPS
- Method of triggering an MO-LR request for assistance data.
- Method of clearing stored GPS assistance data.

##### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "assistanceData".

The SS is unable to provide the requested assistance data.

The SS sends a RELEASE COMPLETE message containing a return error component.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | | Void | |
| 2 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 3 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 4 | <- | | AUTHENTICATION REQUEST | |
| 5 | -> | | AUTHENTICATION RESPONSE | |
| 6 | SS | | | The SS starts ciphering and integrity protection. |
| 7 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "assistanceData". |
| 8 | SS | | | SS is unable to provide the requested assistance data |
| 9 | <- | | RELEASE COMPLETE | SS terminates the dialogue containing a return error component |
| 10 | SS | | | The SS waits for 10 seconds to verify that the UE does not send a RELEASE COMPLETE message. |
| 11 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 7)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->assistanceData<br>locationMethod -> assistedGPS<br>gpsAssistanceData -> OCTET STRING<br>Octets 1 to 38 are coded in the same way as octets 3 to 7+2n of Requested GPS Data IE in 3GPP TS 49.031 (Content is not verified) |
| SS version indicator | Value 1 or above |

RELEASE COMPLETE (Step 9)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |
| Facility | Return error = LCS-MOLR<br>Error -> positionMethodFailure |

### 6.1.2.6.5 Test requirements

After step 6 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "assistanceData".

During step 10 the UE shall not send any RELEASE COMPLETE message.

### 6.1.2.7 LCS Mobile originated location request/ UE-Based GPS/ Position estimate request/ Failure

#### 6.1.2.7.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

#### 6.1.2.7.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

2) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

3) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

      2> act as specified in subclause 8.6.7.19.3.4.

   If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

4) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

      3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

4> if the IE "GPS timing of Cell wanted" is set to FALSE:

5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

5) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS":

2> if there were not enough GPS satellites to be received:

3> set IE "Error reason" to "Not Enough GPS Satellites".

2> if some GPS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE ""Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GPS Additional Assistance Data Request".

6) If the network is unable to successfully fulfil the request received from the MS (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080.

7) PositionMethodFailure: This error is returned by the network when the network is unable to obtain any of the location information requested or none of the information obtained satisfies the requested LCS QoS or if requested LCS assistance data could not be transferred or requested deciphering keys for broadcast assistance data could not be returned.

Reference(s):

- Conformance requirements 1 and 6: TS 24.030, subclause 5.1.1

- Conformance requirement 2: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 3: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 4: TS 25.331, subclause 8.6.7.19.1b
- Conformance requirement 5: TS 25.331, subclause 8.6.7.19.5
- Conformance requirement 7: TS 24.080, subclause 4.3.2.29
- Reference [12] in these conformance requirements denotes document ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

#### 6.1.2.7.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-based GPS when the MO-LR procedure fails due to failure of positioning method.

#### 6.1.2.7.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signal switched off or not present
- User Equipment:
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI
  - The UE shall begin the test with no GPS assistance data stored.

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- Method of triggering an MO-LR request for a position estimate.
- Method of clearing stored GPS assistance data

##### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

Then the UE invokes an MO-LR request of type "locationEstimate". The SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages, including assistance data.

The UE sends a MEASUREMENT REPORT message reporting a positioning error for not enough satellite signals received.

The SS sends a RELEASE COMPLETE message containing a return error component.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with a LCS MO-LR request of type "locationEstimate". |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | <- | | MEASUREMENT CONTROL | |
| 9 | -> | | MEASUREMENT REPORT | Positioning error report "not enough GPS satellites" |
| 10 | SS | | | SS is unable to fulfil the MO-LR request |
| 11 | <- | | RELEASE COMPLETE | SS terminates the dialogue containing a return error component |
| 12 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->locationEstimate |
| SS version indicator | Value 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Not Enough GPS Satellites |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

RELEASE COMPLETE (Step 11)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |
| Facility | Return error = LCS-MOLR<br>Error -> positionMethodFailure |

#### 6.1.2.7.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with a LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 8, the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Not Enough GPS Satellites".

## 6.1.3 Assisted GPS Mobile Terminated Tests

### 6.1.3.1 LCS Mobile terminated location request/ UE-Based GPS

#### 6.1.3.1.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.3.1.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

   In the case of location notification no response is required from the UE, the UE shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

2) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

3) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

2> act as specified in subclause 8.6.7.19.3.4 of TS 25.331.

4) If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

1> update the variable UE_POSITIONING_GPS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in ICD-GPS-200.

5) If the IE "UE positioning GPS reference time" is included, the UE shall:

1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

1> store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

6) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

1> use it as a priori knowledge of the approximate location of the UE.

7) The UE shall when a measurement report is triggered:

2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

4> if the IE "GPS timing of Cell wanted" is set to FALSE:

5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

References

- Conformance requirement 1: TS 24.030, clause 4.1.1.
- Conformance requirement 2: TS 25.331, clause 8.4.1.3.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.3.3a.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.3.4.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 7: TS 25.331, clause 8.6.7.19.1b.

#### 6.1.3.1.3 Test Purpose

To verify that when the UE receives a REGISTER message during an established CS call, containing a LCS Location Notification Invoke component set to NotifyLocationAllowed, the UE displays information about the LCS client correctly and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

To verify that the UE responds with a Measurement Report message containing UE location when the assistance data is divided between several Measurement Control messages using Measurement Command "Modify".

#### 6.1.3.1.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE supporting CS domain services
- UE Based Network Assisted GPS
- UE supporting Mobile Terminated Location Request

Test Procedure

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a DTAP LCS Location Notification Invoke message set to notifyLocationAllowed. The LCS Client Name contained in the USSD text string of the lcs-LocationNotification shall be displayed. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return to terminate the dialogue.

The SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages. The last MEASUREMENT CONTROL message orders periodical reporting.

The UE then initiates periodic measurement reporting and sends a MEASUREMENT REPORT message including a location estimate.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <- | | AUTHENTICATION REQUEST | |
| 2 | -> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE displays information about LCS client |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | <- | | MEASUREMENT CONTROL | Periodical reporting is configured. |
| 9 | -> | | MEASUREMENT REPORT | |
| 10 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -><br>notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -><br>externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = lcs-LocationNotification<br>LocationNotificationRes<br>verificationResponse -><br>permissionGranted |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 9)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.3.1.5 Test requirements

After step 5 the UE shall send a RELEASE COMPLETE message.

After step 8 the UE shall respond with a MEASUREMENT REPORT message.

### 6.1.3.2 LCS Mobile-terminated location request/UE-Based GPS/ Request for additional assistance data/ Success

#### 6.1.3.2.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.3.2.2 Conformance requirements

1) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

   - if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

     - if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

       5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

       5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

   2) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

      1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

      1> if an update has been provided for this satellite:

         2> act as specified in subclause 8.6.7.19.3.4.

3) If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

4) If the IE "UE positioning GPS reference time" is included, the UE shall:

   1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

   - store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

   - if the IE "SFN" and IE "UTRAN GPS timing of cell frames" are included:

     if the UE is able to utilise the IEs:

- store these IEs in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA;
- if the IE "Primary CPICH Info" for FDD or IE "cell parameters id" for TDD is not included:
  - if the UE is not in CELL_DCH state:
    - use IEs "SFN" and "UTRAN GPS timing of cell frames" to estimate the relationship between GPS time and air-interface timing of the NODE B transmission in the serving cell.
  - if the UE is in CELL_DCH state:
    - ignore IEs "SFN" and "UTRAN GPS timing of cell frames".
- if the IE "Primary CPICH Info" for FDD or IE "cell parameters id" for TDD is also included:
  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA;
  - use IEs "SFN" and "UTRAN GPS timing of cell frames" to estimate the relationship between GPS time and air-interface timing of the NODE B transmission in the cell indicated by "Primary CPICH info" or "cell parameters id".

- if the IE "SFN-TOW Uncertainty" is included:
  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it to determine if the relationship between GPS time and air-interface timing of the NODE B transmission is known to within at least 10ms.
- if the IE "$T_{UTRAN-GPS}$ drift rate" is included:
  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA; and
    - may use it as an estimate of the drift rate of the NODE B clock relative to GPS time.
  - if the IE "GPS TOW Assist" is included:
    - for each satellite:

      3> delete all information currently stored in the IE "GPS TOW Assist" in the IE "UE positioning GPS reference time" in the variable UE_POSITIONING_GPS_DATA;

      3> store the received GPS TOW Assist information in the IE "UE positioning GPS reference time" in the variable UE_POSITIONING_GPS_DATA.

5) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

   1> use it as a priori knowledge of the approximate location of the UE.

6) If IE "UE positioning GPS ionospheric model" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS ionospheric model" in variable UE_POSITIONING_GPS_DATA;

   1> act on these GPS ionospheric model parameters in a manner similar to that specified in [12].

7) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

- include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:
  - if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or
  - if the IE "GPS timing of Cell wanted" is set to FALSE:
    - include the IE "GPS TOW msec".
  - if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":
    - if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":
      - if the UE has been able to calculate a 3-dimensional position:
        - include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.
      - if the UE has not been able to calculate a 3-dimensional position:
        - act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".
  - if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":
  - if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

8) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS":

2> if there were not enough GPS satellites to be received:

3> set IE "Error reason" to "Not Enough GPS Satellites".

2> if some GPS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE ""Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GPS Additional Assistance Data Request".

Reference(s):

- Conformance requirement 1: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 2: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.1b.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.3.5.
- Conformance requirement 7: TS 25.331, clause 8.6.7.19.1b.
- Conformance requirement 8: TS 25.331, clause 8.6.7.19.5.

- Reference [12] in these conformance requirements denotes document ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

#### 6.1.3.2.3 Test Purpose

To verify the UE's behaviour in a mobile-terminated location request procedure using UE-based A-GPS with assistance data from the network.

To verify that the UE in CELL_DCH state accepts assistance data received in multiple MEASUREMENT CONTROL messages.

To verify that the UE includes the IE "GPS Additional Assistance Data Request" to request assistance data when it does not have enough assistance data to compute a position.

#### 6.1.3.2.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment:
  - The UE shall begin the test with no GPS assistance data stored.
  - State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- Method of clearing stored GPS assistance data
- UE supporting Mobile Terminated Location Request

##### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a DTAP LCS Location Notification Invoke message set to notifyLocationAllowed. The LCS Client Name contained in the USSD text string of the lcs-LocationNotification shall be displayed. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return to terminate the dialogue.

The SS orders an A-GPS positioning measurement using MEASUREMENT CONTROL including no assistance data.

The UE sends a MEASUREMENT REPORT message to report a positioning error, requesting further assistance data. The SS response with one or more MEASUREMENT CONTROL messages that include the requested assistance data and instructs the UE not to repeat the request for assistance data. The final MEASUREMENT CONTROL message orders periodic reporting.

The UE performs positioning measurements and responds with a MEASUREMENT REPORT message containing a valid position estimate in the IE "UE Positioning Position Estimate Info".

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <- | | AUTHENTICATION REQUEST | |
| 2 | -> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE displays information about LCS client |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | <-- | | MEASUREMENT CONTROL | No assistance data, and "Additional Assistance Data Request" IE set to TRUE. |
| 8 | --> | | MEASUREMENT REPORT | Positioning error report with request for further assistance data. |
| 9 | <-- | | MEASUREMENT CONTROL | The SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5.<br>The final MEASUREMENT CONTROL message contains:<br>Reporting mode: Periodical reporting<br>Amount of reporting: 1<br>Reporting interval: 64000 |
| 10 | --> | | MEASUREMENT REPORT | Measurement report message containing UE position estimate. |
| 11 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>REGISTER (0011 1011)<br>Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -><br>notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -><br>externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>RELEASE COMPLETE (xx10 1010)<br>Return result = lcs-LocationNotification<br><br>LocationNotificationRes<br>verificationResponse -><br>permissionGranted |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Measurement Reporting Mode | Not present |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Inadequate assistance data for UE-based A-GPS" in 4.3.2 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Set as required according to position in sequence of messages |
| - Amount of reporting | Set as required according to position in sequence of messages |
| - Reporting interval | Set as required according to position in sequence of messages |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 10):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.3.2.5 Test Requirements

At step 7 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Assistance Data Missing".

At step 9 the UE shall send a MEASUREMENT REPORT message containing a valid UE position estimate.

### 6.1.3.3 LCS Mobile-terminated location request/UE-Based GPS/ Failure – Not Enough Satellites

#### 6.1.3.3.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.3.3.2 Conformance requirements

1) if the IE "Measurement command" has the value "modify":

2> for all IEs present in the MEASUREMENT CONTROL message:

- if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

- if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

2) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

1> if an update has been provided for this satellite:

2> act as specified in subclause 8.6.7.19.3.4.

3) If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

1> update the variable UE_POSITIONING_GPS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

4) If the IE "UE positioning GPS reference time" is included, the UE shall:

1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

- store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

- if the IE "SFN" and IE "UTRAN GPS timing of cell frames" are included:

- if the UE is able to utilise the IEs:

- store these IEs in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA;

- if the IE "Primary CPICH Info" for FDD or IE "cell parameters id" for TDD is not included:

  - if the UE is not in CELL_DCH state:

    - use IEs "SFN" and "UTRAN GPS timing of cell frames" to estimate the relationship between GPS time and air-interface timing of the NODE B transmission in the serving cell.

  - if the UE is in CELL_DCH state:

    - ignore IEs "SFN" and "UTRAN GPS timing of cell frames".

- if the IE "Primary CPICH Info" for FDD or IE "cell parameters id" for TDD is also included:

  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA;

  - use IEs "SFN" and "UTRAN GPS timing of cell frames" to estimate the relationship between GPS time and air-interface timing of the NODE B transmission in the cell indicated by "Primary CPICH info" or "cell parameters id".

- if the IE "SFN-TOW Uncertainty" is included:

  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it to determine if the relationship between GPS time and air-interface timing of the NODE B transmission is known to within at least 10ms.

- if the IE "$T_{UTRAN-GPS}$ drift rate" is included:

  - store this IE in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA; and

  - may use it as an estimate of the drift rate of the NODE B clock relative to GPS time.

- if the IE "GPS TOW Assist" is included:

  - for each satellite:

    3> delete all information currently stored in the IE "GPS TOW Assist" in the IE "UE positioning GPS reference time" in the variable UE_POSITIONING_GPS_DATA;

    3> store the received GPS TOW Assist information in the IE "UE positioning GPS reference time" in the variable UE_POSITIONING_GPS_DATA.

5) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

   1> use it as a priori knowledge of the approximate location of the UE.

6) If IE "UE positioning GPS ionospheric model" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS ionospheric model" in variable UE_POSITIONING_GPS_DATA;

   1> act on these GPS ionospheric model parameters in a manner similar to that specified in [12].

7) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

- include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

  - if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

  - if the IE "GPS timing of Cell wanted" is set to FALSE:

    - include the IE "GPS TOW msec".

  - if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

    - if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

      - if the UE has been able to calculate a 3-dimensional position:

        - include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

      - if the UE has not been able to calculate a 3-dimensional position:

        - act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

  - if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

  - if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

    6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

8) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS":

2> if there were not enough GPS satellites to be received:

3> set IE "Error reason" to "Not Enough GPS Satellites".

2> if some GPS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE ""Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GPS Additional Assistance Data Request".

Reference(s):

- Conformance requirement 1: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 2: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.1b.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.3.5.
- Conformance requirement 7: TS 25.331, clause 8.6.7.19.1b.

- Conformance requirement 8: TS 25.331, clause 8.6.7.19.5.

- Reference [12] in these conformance requirements denotes document ICD-GPS-200: "Navstar GPS Space Segment/Navigation User Interface".

#### 6.1.3.3.3 Test Purpose

To verify the UE's behaviour in a mobile-terminated location request procedure using UE-based A-GPS with assistance data from the network.

To verify that the UE in CELL_DCH state accepts assistance data received in multiple MEASUREMENT CONTROL messages.

To verify that the UE sets the IE Error Reason in 'UE Positioning Error' to 'Not Enough GPS Satellites' when it does not receive enough satellite signals to compute a position.

#### 6.1.3.3.4 Method of Test

##### Initial Conditions

- System Simulator:
  - 1 cell, default parameters.
  - Satellite signals switched off or not present.
- User Equipment:
  - State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.
  - The UE shall begin the test with no GPS assistance data stored.

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- Method of clearing stored GPS assistance data
- UE supporting Mobile Terminated Location Request

##### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a DTAP LCS Location Notification Invoke message set to notifyLocationAllowed. The LCS Client Name contained in the USSD text string of the lcs-LocationNotification shall be displayed. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return to terminate the dialogue.

The SS orders an A-GPS positioning measurement using two MEASUREMENT CONTROL messages. The last MEASUREMENT CONTROL message orders periodical reporting.

The UE sends a MEASUREMENT REPORT message reporting a positioning error for not enough satellite signal.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE displays information about LCS client |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | <-- | | MEASUREMENT CONTROL | |
| 8 | <-- | | MEASUREMENT CONTROL | Periodical reporting is configured |
| 9 | --> | | MEASUREMENT REPORT | Positioning error report 'not enough GPS satellites' |
| 10 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>REGISTER (0011 1011)<br>Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -><br>notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br><br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>RELEASE COMPLETE (xx10 1010)<br>Return result = lcs-LocationNotification<br>LocationNotificationRes<br>verificationResponse -><br>permissionGranted |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Not Enough GPS Satellites |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.3.3.5 Test Requirements

At step 8 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Not Enough GPS Satellites".

### 6.1.3.4 LCS Mobile terminated location request/ UE-Assisted GPS/ Success

#### 6.1.3.4.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities.

#### 6.1.3.4.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

   In the case of location notification no response is required from the UE, the UE shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

2) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

3) The UE shall:

   1> when a measurement report is triggered:

   2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

      3> if the IE "Vertical Accuracy" is included:

         4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

      3> if the IE "Positioning Methods" is set to "GPS":

         4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

            5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

               6> if the IE "GPS timing of Cell wanted" is set to TRUE:

                  7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

                  7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

                  7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

               6> if the IE "GPS timing of Cell wanted" is set to FALSE:

                  7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

References

- Conformance requirement 1: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 25.331, clause 8.4.1.3.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.1a.

#### 6.1.3.4.3 Test Purpose

To verify the UE behaviour in the mobile-terminated location request procedure using network-assisted UE-assisted GPS to deliver UE positioning measurements to the network.

#### 6.1.3.4.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE supporting CS domain services
- UE Assisted Network Assisted GPS
- UE supporting Mobile Terminated Location Request

Test Procedure

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a DTAP LCS Location Notification Invoke message set to notifyLocationAllowed. The LCS Client Name contained in the USSD text string of the lcs-LocationNotification shall be displayed. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return to terminate the dialogue.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message. The assistance data is as described in subclause 4.3.3 (Adequate assistance data for UE-assisted A-GPS). The MEASUREMENT CONTROL message orders periodical reporting.

The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE then initiates periodic measurement reporting and sends a MEASUREMENT REPORT message including the IE "UE positioning GPS measured results".

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE displays information about LCS client |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | <- | | MEASUREMENT CONTROL | Periodical reporting is configured. |
| 8 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 8a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 8, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 8b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 8, this message contains the IE "UE positioning GPS measured results". |
| 9 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = lcs-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8 (Option 1) or 8b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 8 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 8a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

#### 6.1.3.4.5 Test requirements

After step 5 the UE shall send a RELEASE COMPLETE message.

After step 7 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

### 6.1.3.5 LCS Mobile terminated location request/ UE-Assisted GPS/ Request for additional assistance data/ Success

#### 6.1.3.5.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities.

#### 6.1.3.5.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

   In the case of location notification no response is required from the UE, the UE shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

2) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

3) The UE shall:

   1> when a measurement report is triggered:

   2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

      3> if the IE "Vertical Accuracy" is included:

         4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

      3> if the IE "Positioning Methods" is set to "GPS":

         4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

            5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

               6> if the IE "GPS timing of Cell wanted" is set to TRUE:

7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

6> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

4) 1> if the UE is not able to report the requested measurement results:

2> include IE "UE positioning error" in the MEASUREMENT REPORT and set the contents of this IE as specified in subclause 8.6.7.19.5.

5) if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS":

2> if there were not enough GPS satellites to be received:

3> set IE "Error reason" to "Not Enough GPS Satellites".

2> if some GPS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to FALSE:

4> not include the IE "GPS Additional Assistance Data Request", and use the assistance data available for doing a positioning estimate.

References

- Conformance requirement 1: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 25.331, clause 8.4.1.3.
- Conformance requirements 3 and 4: TS 25.331, clause 8.6.7.19.1a.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.5.

#### 6.1.3.5.3 Test Purpose

To verify the UE behaviour in the mobile-terminated location request procedure using network-assisted UE-assisted GPS to deliver UE positioning measurements to the network.

To verify that the UE includes the IE "GPS Additional Assistance Data Request" to request additional assistance data when it does not have enough assistance data to perform the requested measurements.

#### 6.1.3.5.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- The UE shall begin the test with no GPS assistance data stored.
- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

### Related PICS/PIXIT Statements

- UE supporting CS domain services
- UE Assisted Network Assisted GPS
- Method of clearing stored GPS assistance data
- UE supporting Mobile Terminated Location Request

### Test Procedure

The stored GPS assistance data in the UE shall be cleared.

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a DTAP LCS Location Notification Invoke message set to notifyLocationAllowed. The LCS Client Name contained in the USSD text string of the lcs-LocationNotification shall be displayed. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return to terminate the dialogue.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message. The assistance data is as described in subclause 4.3.2 (Inadequate assistance data for UE-assisted A-GPS). The MEASUREMENT CONTROL message orders periodical reporting.

The UE then initiates periodic measurement reporting and sends a MEASUREMENT REPORT message including a request for additional assistance data. The SS responds with one or more MEASUREMENT CONTROL messages containing assistance data as specified in subclause 4.3.5 (Response to additional assistance data requests from UE). The UE sends a MEASUREMENT REPORT message including the IE "UE positioning GPS measured results".

### Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE displays information about LCS client |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | <- | | MEASUREMENT CONTROL | Periodical reporting is configured. Assistance data set as specified in subclause 4.3.2 (Inadequate assistance data for UE-assisted A-GPS). |
| 8 | -> | | MEASUREMENT REPORT | UE requests additional assistance data. |
| 9 | <- | | MEASUREMENT CONTROL | The SS provides the requested data in one or more MEASUREMENT CONTROL messages as defined in subclause 4.3.5 |
| 10 | -> | | MEASUREMENT REPORT | UE sends the IE "UE positioning GPS measured results". |
| 11 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br><br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = lcs-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Inadequate assistance data for UE-assisted A-GPS" in 4.3.2 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Present, if requested by UE |
| - UTC model | Present, if requested by UE |
| - Ionospheric model | Present, if requested by UE |
| - Navigation model | Present, if requested by UE |
| - DGPS corrections | Present, if requested by UE |
| - Reference location | Present, if requested by UE |
| - Reference time | Present, if requested by UE |
| - Acquisition assistance | Present, if requested by UE |
| - Real-time integrity | Present, if requested by UE |
| - Navigation model additional data | Present, if requested by UE |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 10)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

#### 6.1.3.5.5 Test requirements

After step 5 the UE shall send a RELEASE COMPLETE message.

After step 7 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Assistance data missing".

After step 9 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

### 6.1.3.6 LCS Mobile terminated location request/ UE-Based GPS/ Privacy Verification/ Location Allowed if No Response

#### 6.1.3.6.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.3.6.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

4) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

5) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

6) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

      2> act as specified in subclause 8.6.7.19.3.4 of TS 25.331.

7) If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in ICD-GPS-200.

8) If the IE "UE positioning GPS reference time" is included, the UE shall:

   1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

   1> store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

   NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

9) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

   1> use it as a priori knowledge of the approximate location of the UE.

10) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

4> if the IE "GPS timing of Cell wanted" is set to FALSE:

5> include the IE "GPS TOW msec".

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1.
- Conformance requirements 4 and 5: TS 25.331, subclause 8.4.1.3
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.3.3a.
- Conformance requirement 7: TS 25.331, clause 8.6.7.19.3.4.
- Conformance requirement 8: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 9: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 10: TS 25.331, clause 8.6.7.19.1b.

#### 6.1.3.6.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationAllowedIfNoResponse, the UE notifies the user of the request and indicates that the default response is location allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

#### 6.1.3.6.4 Method of Test

##### Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- px_UeLcsNotification: value for UE LCS Notification timeout timer
- UE supporting Mobile Terminated Location Request

##### Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The LCS Client Name contained in the USSD text string of the lcs-LocationNotification should be displayed with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS orders an A-GPS positioning measurement using MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message including a location estimate.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The user ignores the location request by taking no action.

The SS orders an A-GPS positioning measurement using MEASUREMENT CONTROL messages.

The UE then sends a MEASUREMENT REPORT message including a location estimate.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | MEASUREMENT CONTROL | |
| 10 | <- | | MEASUREMENT CONTROL | |
| 11 | -> | | MEASUREMENT REPORT | |
| 12 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 13 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 14 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 15 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 16 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 17 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 18 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 19 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 20 | UE | | | The user does not reply |
| 21 | SS | | | SS waits until T(LCSN) expires to ensure that the UE does not send a RELEASE COMPLETE message. |
| 22 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 23 | <- | | MEASUREMENT CONTROL | |
| 24 | <- | | MEASUREMENT CONTROL | |
| 25 | -> | | MEASUREMENT REPORT | |
| 26 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 10):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 11)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

REGISTER (Step 12)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 16)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

REGISTER (Step 17)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 22)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |

MEASUREMENT CONTROL (Step 23):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 24):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | Not present |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | Set according to 4.2 (unequal to 0) |
| - Vertical accuracy | Set according to 4.2 (unequal to 0) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 25)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

### 6.1.3.6.5 Test requirements

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 10 the UE shall respond with a MEASUREMENT REPORT message containing a UE position estimate.

After step 15 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

After step 24 the UE shall respond with a MEASUREMENT REPORT message containing a UE position estimate.

### 6.1.3.7 LCS Mobile terminated location request/ UE-Based GPS/ Privacy Verification/ Location Not Allowed if No Response

#### 6.1.3.7.1 Definition

This test case applies to all UEs supporting UE-Based GPS Location Service capabilities.

#### 6.1.3.7.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

4) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

5) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

6) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

   1> if an update has been provided for this satellite:

      2> act as specified in subclause 8.6.7.19.3.4 of TS 25.331.

7) If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

   1> update the variable UE_POSITIONING_GPS_DATA as follows:

      2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

   1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in ICD-GPS-200.

8) If the IE "UE positioning GPS reference time" is included, the UE shall:

   1> store the IE "GPS Week" in "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as the current GPS week;

   1> store the IE "GPS TOW msec" in the IE "UE positioning GPS reference time" in variable UE_POSITIONING_GPS_DATA and use it as an estimate of the GPS Time-of-Week at the time of reception of the complete message containing the IE "GPS TOW msec";

   NOTE: The UE does not need to apply any compensation on the GPS Time-of-Week.

9) If the IE "UE positioning GPS reference UE position" is included, the UE shall:

   1> store this IE in the IE "UE positioning GPS reference UE position" in variable UE_POSITIONING_GPS_DATA; and

   1> use it as a priori knowledge of the approximate location of the UE.

10) The UE shall when a measurement report is triggered:

   2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or on the list of satellites included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning:

      3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

         4> if the UE does not support the capability to perform the UE GPS timing of cell frames measurement; or

         4> if the IE "GPS timing of Cell wanted" is set to FALSE:

            5> include the IE "GPS TOW msec".

         4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

            5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

               6> if the UE has been able to calculate a 3-dimensional position:

                  7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

               6> if the UE has not been able to calculate a 3-dimensional position:

                  7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

         4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

            5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

               6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1.
- Conformance requirements 4 and 5: TS 25.331, clause 8.4.1.3.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.3.3a.
- Conformance requirement 7: TS 25.331, clause 8.6.7.19.3.4.
- Conformance requirement 8: TS 25.331, clause 8.6.7.19.3.7.
- Conformance requirement 9: TS 25.331, clause 8.6.7.19.3.8.
- Conformance requirement 10: TS 25.331, clause 8.6.7.19.1b.

### 6.1.3.7.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationNotAllowedIfNoResponse, the UE notifies the user of the request and indicates that the default response is location not allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

### 6.1.3.7.4 Method of Test

#### Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

#### Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- px_UeLcsNotification: value for UE LCS Notification timeout timer
- UE supporting Mobile Terminated Location Request

#### Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The LCS Client Name contained in the USSD text string of the lcs-LocationNotification should be displayed with the option to accept or deny the request and an indication that location will be not allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS orders an A-GPS positioning measurement using MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message including a location estimate.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The user ignores the location request by taking no action. If the timer expires in the SS before any response from the UE is received, the SS interprets this by applying the default treatment LocationNotAllowed.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | MEASUREMENT CONTROL | |
| 10 | <- | | MEASUREMENT CONTROL | |
| 11 | -> | | MEASUREMENT REPORT | |
| 12 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 13 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 14 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 15 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 16 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 17 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 18 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 19 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 20 | UE | | | The user does not reply |
| 21 | SS | | | SS waits until T(LCSN) expires to verify that the UE does not send a RELEASE COMPLETE message. |
| 22 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 23 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 10):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | Not present |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GPS" in 4.3.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 11)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | |
| - GPS reference time only | |
| - GPS TOW msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

REGISTER (Step 12)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 16)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

REGISTER (Step 17)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 22)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |

### 6.1.3.7.5 Test requirements

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 10 the UE shall respond with a MEASUREMENT REPORT message containing a UE position estimate.

After step 15 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

During step 21 the UE shall not send any RELEASE COMPLETE message.

## 6.1.3.8 LCS Mobile terminated location request/ UE-Assisted GPS/ Privacy Verification/ Location Allowed if No Response

### 6.1.3.8.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities.

### 6.1.3.8.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

4) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

5) The UE shall:

   1> when a measurement report is triggered:

      2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

         3> if the IE "Vertical Accuracy" is included:

            4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

         3> if the IE "Positioning Methods" is set to "GPS":

            4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

               5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

                  6> if the IE "GPS timing of Cell wanted" is set to TRUE:

                     7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

                     7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

                     7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

                  6> if the IE "GPS timing of Cell wanted" is set to FALSE:

                     7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1.
- Conformance requirement 4: TS 25.331, clause 8.4.1.3.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.3b.

#### 6.1.3.8.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationAllowedIfNoResponse, the UE notifies the user of the request and indicates that the default response is location allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

#### 6.1.3.8.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Assisted Network Assisted GPS
- px_UeLcsNotification: value for UE LCS Notification timeout timer
- UE supporting Mobile Terminated Location Request

Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The LCS Client Name contained in the USSD text string of the lcs-LocationNotification should be displayed with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message including IE "UE positioning GPS measured results".

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The user ignores the location request by taking no action. If the timer expires in the SS before any response from the UE is received, the SS interprets this by applying the default treatment LocationAllowed.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE then sends a MEASUREMENT REPORT message including IE "UE positioning GPS measured results".

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | MEASUREMENT CONTROL | Assistance data set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| 10 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 10a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 10, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 10b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 10, this message contains the IE "UE positioning GPS measured results". |
| 11 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 12 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 13 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 14 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 15 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 16 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 17 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 18 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 19 | UE | | | The user does not reply |
| 20 | SS | | | SS waits until T(LCSN) expires to verify that the UE does not send a RELEASE COMPLETE message. |
| 21 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 22 | <- | | MEASUREMENT CONTROL | Assistance data set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.5 |
| 23 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 23a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 23, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 23b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 23, this message contains the IE "UE positioning GPS measured results". |

| | | | |
|---|---|---|---|
| 24 | SS | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>REGISTER (0011 1011)<br>Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>RELEASE COMPLETE (xx10 1010)<br>Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Steps 10 (Option 1) or 10b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 10 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 10a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

REGISTER (Step 11)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 15)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

REGISTER (Step 16)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 21)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |

MEASUREMENT CONTROL (Step 22):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Steps 23 (Option 1) or 23b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 23 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 23a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

#### 6.1.3.8.5 Test requirements

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 9 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

After step 14 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

After step 22 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

## 6.1.3.9 LCS Mobile terminated location request/ UE-Assisted GPS/ Privacy Verification/ Location Not Allowed if No Response

### 6.1.3.9.1 Definition

This test case applies to all UEs supporting UE-Assisted GPS Location Service capabilities.

### 6.1.3.9.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

4) if the IE "Measurement command" has the value "setup":

   2> store this measurement in the variable MEASUREMENT_IDENTITY according to the IE "measurement identity", first releasing any previously stored measurement with that identity if that exists;

   …

   2> for any other measurement type:

      3> if the measurement is valid in the current RRC state of the UE:

         4> begin measurements according to the stored control information for this measurement identity.

5) The UE shall:

   1> when a measurement report is triggered:

      2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

         3> if the IE "Vertical Accuracy" is included:

            4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

         3> if the IE "Positioning Methods" is set to "GPS":

            4> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

               5> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

                  6> if the IE "GPS timing of Cell wanted" is set to TRUE:

7> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GPS timing of cell frames".

6> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec".

5> if the UE does not support the capability to provide the GPS timing of the cell:

6> include the IE "GPS TOW msec".

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1.
- Conformance requirement 4: TS 25.331, clause 8.4.1.3.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.3.3b.

#### 6.1.3.9.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationNotAllowedIfNoResponse, the UE notifies the user of the request and indicates that the default response is location not allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

#### 6.1.3.9.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals: As specified in 4.2

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Assisted Network Assisted GPS
- px_UeLcsNotification: value for UE LCS Notification timeout timer
- UE supporting Mobile Terminated Location Request

Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The LCS Client Name contained in the USSD text string of the lcs-LocationNotification should be displayed with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS orders an A-GPS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.3.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message including IE "UE positioning GPS measured results".

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The user ignores the location request by taking no action. If the timer expires in the SS before any response from the UE is received, the SS interprets this by applying the default treatment LocationNotAllowed.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | MEASUREMENT CONTROL | |
| 10 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 10a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 10, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.3.5. |
| 10b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 10, this message contains the IE "UE positioning GPS measured results". |
| 11 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 12 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 13 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 14 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 15 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 16 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 17 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 18 | UE | | | The UE notifies the user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 19 | UE | | | The user does not reply |
| 20 | SS | | | SS waits until T(LCSN) expires to verify that the UE does not send a RELEASE COMPLETE message. |
| 21 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 22 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 9):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GPS" in 4.3.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Steps 10 (Option 1) or 10b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present |
| - UE positioning error | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT REPORT (Step 10 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | |
| - Almanac | Not checked |
| - UTC model | Not checked |
| - Ionospheric model | Not checked |
| - Navigation model | Not checked |
| - DGPS corrections | Not checked |
| - Reference location | Not checked |
| - Reference time | Not checked |
| - Acquisition assistance | Not checked |
| - Real-time integrity | Not checked |
| - Navigation model additional data | Not checked |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Step 10a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.3.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

REGISTER (Step 11)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-<br>LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 15)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

REGISTER (Step 16)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 21)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0x10 1010) |

#### 6.1.3.9.5 Test requirements

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 9 the UE shall respond with a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results".

After step 14 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

During step 20 the UE shall not send any RELEASE COMPLETE message.

### 6.1.3.10 LCS Mobile terminated location request/ UE-Based or UE-Assisted GPS/ Configuration Incomplete

#### 6.1.3.10.1 Definition

This test case applies to all UEs supporting UE-based or UE-assisted network assisted GPS, but not UE-based OTDOA.

#### 6.1.3.10.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In the case of location notification no response is required from the MS, the MS shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

3) The UE shall perform the following consistency check:

   1> if UE, according to its capabilities, does not support UE-based OTDOA and if IE "Positioning Methods" is set to "OTDOA" and if IE "Method Type" is set to "UE-based":

      2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

   1> if UE, according to its capabilities, does not support UE-based GPS and if IE "Positioning Methods" is set to "GPS" and if IE "Method Type" is set to "UE-based":

      2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

1> if UE, according to its capabilities, does not support UE-assisted GPS and if IE "Positioning Methods" is set to "GPS" and if IE "Method Type" is set to "UE-assisted":

2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

1> if UE, according to its capabilities, does not support UE-based positioning and if IE "Positioning Methods" is set to "OTDOAorGPS" and if IE "Method Type" is set to "UE-based":

2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

1> if UE, according to its capabilities, does not support Rx-Tx time difference type 2 measurement and if IE "Positioning Methods" is set to "Cell ID":

2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

1> if UE, according to its capabilities, does not support UE GPS timing of cell frames measurement and if IE "GPS timing of Cell wanted" is set to TRUE:

2> set the variable CONFIGURATION_INCOMPLETE to TRUE.

4) If the variable CONFIGURATION_INCOMPLETE is set to TRUE, the UE shall:

1> retain the measurement configuration that was valid before the MEASUREMENT CONTROL message was received;

1> set the IE "RRC transaction identifier" in the MEASUREMENT CONTROL FAILURE message to the value of "RRC transaction identifier" in the entry for the MEASUREMENT CONTROL message in the table "Accepted transactions" in the variable TRANSACTIONS and clear that entry;

1> clear the variable CONFIGURATION_INCOMPLETE;

1> set the cause value in IE "failure cause" to "Configuration incomplete";

1> submit the MEASUREMENT CONTROL FAILURE message to lower layers for transmission on the DCCH using AM RLC;

1> continue with any ongoing processes and procedures as if the invalid MEASUREMENT CONTROL message has not been received;

1> and the procedure ends.

5) The UE should set the variable UNSUPPORTED_CONFIGURATION to TRUE if the received message is not according to the UE capabilities.

References

- Conformance requirement 1, 2: TS 24.030, clause 4.1.1.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.1
- Conformance requirement 4: TS 25.331, clause 8.4.1.4a
- Conformance requirement 5: TS 25.331 clause 8.5.20

### 6.1.3.10.3 Test Purpose

To verify that the UE sends a MEASUREMENT CONTROL FAILURE message, after receiving a MEASUREMENT CONTROL message with IE "Method Type" set a value which is inconsistent with the UE positioning capabilities.

To verify that the UE set the "failure cause" IE to value "configuration incomplete" in the uplink MEASUREMENT CONTROL FAILURE message.

6.1.3.10.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals switched off or not present

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Based Network Assisted GPS
- UE Assisted Network Assisted GPS
- UE supporting Mobile Terminated Location Request

Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke component set to notifyLocationAllowed.

The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result.

The SS sends a MEASUREMENT CONTROL message with "Method type" set to a value not supported by the UE as indicated in the "UE positioning capability" contained in the "UE radio access capability" (method not to be supported is UE-based OTDOA).

The UE sends a MEASUREMENT CONTROL FAILURE message with Failure Cause "Configuration Incomplete" or "unsupported configuration.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE notifies the user of the location request |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | SS | | | SS verifies that UE does not support UE-based OTDOA |
| 8 | <- | | MEASUREMENT CONTROL | IE "Method type" is set to a method not supported by the UE (UE-based OTDOA) |
| 9 | -> | | MEASUREMENT CONTROL FAILURE | Failure cause "Configuration Incomplete" or "Unsupported Configuration" |
| 10 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyLocationAllowed<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

MEASUREMENT CONTROL (Step 8):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE-based |
| - Positioning methods | OTDOA |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical reporting criteria |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Not present |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL FAILURE (Step 9)

| Information Element | Value/remark |
|---|---|
| RRC transaction identifier | Set to the same value of the same IE in the MEASUREMENT CONTROL message sent in Step 5 |
| Failure cause | Failure cause "Configuration Incomplete" or "Unsupported Configuration" |

#### 6.1.3.10.5 Test requirements

After step 5 the UE shall send a RELEASE COMPLETE message.

After step 8, the UE shall transmit MEASUREMENT CONTROL FAILURE message, stating the IE "failure cause" as "configuration incomplete". The UE shall not transmit any MEASUREMENT REPORT messages during the execution of this test case.

## 6.2 Assisted-GNSS Test Cases

# 6.2.1 Assisted GNSS Network Induced Tests

## 6.2.1.1 NI-LR Emergency Call: UE-Based A-GNSS

#### 6.2.1.1.1 Definition

This test case applies to all UEs supporting UE-Based GANSS or GNSS Location Service capabilities.

#### 6.2.1.1.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

   When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

2) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

   - a SETUP message, if the call to be established is a basic call; and
   - an EMERGENCY SETUP message, if the call to be established is an emergency call.

3) If the IE "UE positioning GANSS reference time" is included, the UE shall:

   1> if the IE "GANSS Day" is included:

      2> store this IE in "UE positioning GANSS reference time" in variable UE_POSITIONING_GANSS_DATA and use it as the current GANSS day.

   1> store the IE "GANSS TOD" in the IE "UE positioning GANSS reference time" in variable UE_POSITIONING_GANSS_DATA and use it as an estimate of the GANSS Time-of-Day at the time of reception of the complete message containing the IE "GANSS TOD";

   NOTE: The UE does not need to apply any compensation on the GANSS Time-of-Day.

   1> if the IE "GANSS Time ID" is not included:

      2> use Galileo system time as a reference for GANSS-Time-of-Day.

   1> if the IE "GANSS Time ID" is included:

      2> use the system time indicated by this IE as a reference for GANSS-Time-of-Day.

4) If the IE "UE positioning GANSS reference UE position" is included, the UE shall:

1> store this IE in the IE "UE positioning GANSS reference UE position" in variable UE_POSITIONING_GANSS_DATA; and

1> use it as a priori knowledge of the approximate location of the UE.

5) If the IE "UE positioning GANSS time model" is included, the UE shall for each GANSS:

1> store the information in "UE positioning GANSS time model" in variable UE_POSITIONING_GANSS_DATA;

1> use the stored parameters to relate GANSS time for the GANSS indicated by "GANSS ID" to time reference indicated by IE "GNSS_TO_ID".

6) The UE shall when a measurement report is triggered:

1> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or the UE has been able to calculate a position in case of GPS or GANSS positioning or the UE has been able to calculate a position using a standalone positioning method:

2> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

3> if the UE supports the capability to provide the GANSS timing of the cell frames measurement;

3> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero:

4> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid.

3> if the UE does not support the capability to provide the GANSS timing of the cell:

4> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid;

4> include the IE "GANSS Time ID" to identify the GNSS system time.

3> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

4> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

5> if the UE has been able to calculate a 3-dimensional position:

6> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

5> if the UE has not been able to calculate a 3-dimensional position:

6> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

3> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

4> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

5> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

3> if any of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" is reported:

4> should calculate a value of the IE "Confidence", different from "0", as the probability that the UE is located within the uncertainty region of the one of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" that is reported.

NOTE: The value "0" of the IE "Confidence" is interpreted as "no information" by the UTRAN.

Reference(s):

- Conformance requirement 1: TS 24.008 clause 4.5.1.5.
- Conformance requirement 2: TS 24.008, clause 5.2.1.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.7.7.
- Conformance requirement 4: TS 25.331, clause 8.6.7.19.7.8.
- Conformance requirement 5: TS 25.331, clause 8.6.7.19.7.9.
- Conformance requirement 6: TS 25.331, clause 8.6.7.19.1b.

#### 6.2.1.1.3 Test Purpose

To verify when an emergency call is initiated by a UE, and the network performs a GANSS location request using the RRC measurement control procedure, then the UE responds with a Measurement Report containing UE location estimate.

#### 6.2.1.1.4 Method of Test

##### Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment (UE):
  - the UE is in state "MM idle" with valid TMSI and CKSN.

##### Related PICS/PIXIT Statements

- Emergency speech call
- UE Based Network Assisted GANSS
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10)

##### Test procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

The UE is made to initiate an emergency call.

After the call has been through-connected in both directions, the SS orders an A-GNSS positioning measurement using one or more (dependent on the Sub-Test) MEASUREMENT CONTROL messages. The last MEASUREMENT CONTROL message orders periodical reporting by sending a MEASUREMENT CONTROL message requesting periodical measurement reporting (1 report, interval 64s).

The UE then performs positioning measurements, calculates "UE Positioning Position Estimate Info" and responds with this in the RRC message MEASUREMENT REPORT.

Finally the SS clears the call.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. Number shall be one programmed in test USIM $EF_{ECC}$ (Emergency Call Codes), ref. 34.108 clause 8.3.2.21. |
| 2 | --> | | | UE establishes RRC procedure for emergency call.<br>Establishment cause: Emergency Call<br>SS checks that the UE capability includes "Network Assisted GANSS Support List" with "GANSS mode" set to "UE-based", and that the UE includes "Network Assisted GPS Support" for UE-based (Sub-Tests 3, 4, 8 and 10 only). |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". |
| 4 | <-- | | AUTHENTICATION REQUEST | IE Authentication Parameter AUTN shall be present in the message. |
| 5 | --> | | AUTHENTICATION RESPONSE | SRES specifies correct value. |
| 6 | | | | SS starts security procedure. |
| 7 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 8 | <-- | | CALL PROCEEDING | |
| 9 | <-- | | ALERTING | |
| 10 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 11 | <-- | | CONNECT | |
| 12 | --> | | CONNECT ACKNOWLEDGE | |
| 13 | UE | | | The DTCH is through connected in both directions. |
| 14 | <-- | | MEASUREMENT CONTROL | All Sub-Tests |
| 14a | <-- | | MEASUREMENT CONTROL | Sub-Tests 2, 3, 4, 8, 10 only |
| 14b | <-- | | MEASUREMENT CONTROL | Sub-Tests 4, 8, 10 only |
| 15 | --> | | MEASUREMENT REPORT | |
| 16 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 14):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 1, 9 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 2, 3, 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 14a):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 2, 3 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 14b):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 15):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | GPS or GANSS reference time only |
| - GPS TOW msec | Not checked |
| - GANSS TOD msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - Position Data | Not checked |
| - CHOICE *Velocity estimate* | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

#### 6.2.1.1.5 Test requirements

After step 12 the UE shall have through connected the DTCH in both directions.

After step 14, 14a, or 14b (dependent on Sub-Test) the UE shall respond with a MEASUREMENT REPORT message.

### 6.2.1.2 NI-LR Emergency call: UE-Assisted A-GNSS

#### 6.2.1.2.1 Definition

This test case applies to all UEs supporting UE-Assisted GANSS or GNSS Location Service capabilities.

#### 6.2.1.2.2 Conformance requirements

1) A MM connection for an emergency call may be established in all states of the mobility management sublayer which allow MM connection establishment for a normal originating call.

   When a user requests an emergency call establishment the UE will send a CM SERVICE REQUEST message to the network with a CM service type information element indicating emergency call establishment.

2) Having entered the "MM connection pending" state, upon MM connection establishment, the call control entity of the UE sends a setup message to its peer entity. This setup message is

   - a SETUP message, if the call to be established is a basic call; and

   - an EMERGENCY SETUP message, if the call to be established is an emergency call.

3) The a measurement report is triggered:

   1> if the UE was UE shall when able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one satellite included in the

variable UE_POSITIONING_GANSS_DATA in case of GANSS positioning or one cell from the active set in case of CELL ID:

2> if the IE "Vertical Accuracy" is included:

3> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

2> if the IE "Positioning Methods" is set to "GPS" and if the IE "GANSS Positioning Methods" is not present or if the IE "GANSS Positioning Methods" is present indicating GPS allowed:

3> include the IE "UE positioning GPS measured results" in the measurement report and set the contents of the IE as follows:

4> if the UE supports the capability to provide the GPS timing of the cell frames measurement:

5> if the IE "GPS timing of Cell wanted" is set to FALSE:

7> include the IE "GPS TOW msec" and set it to the GPS TOW when the measurements included in the MEASUREMENT REPORT were valid;

7> include the IE "UE Positioning GPS Reference Time Uncertainty" and set it to the uncertainty of the GPS TOW when the measurements included in the MEASUREMENT REPORT were valid.

4> if the UE does not support the capability to provide the GPS timing of the cell:

5> include the IE "GPS TOW msec" and set it to the GPS TOW when the measurements included in the MEASUREMENT REPORT were valid;

5> include the IE "UE Positioning GPS Reference Time Uncertainty" and set it to the uncertainty of the GPS TOW when the measurements included in the MEASUREMENT REPORT were valid.

2> if the IE "Positioning Methods" is set to "GPS" and the IE "GANSS Positioning Methods" is present indicating other GNSS than GPS allowed and if any of these other GNSSs is measured:

3> include the IE "UE positioning GANSS measured results" in the measurement report and set the contents of the IE as follows:

4> if the UE supports the capability to provide the GANSS timing of the cell frames measurement:

5> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero and if IE "UE positioning GPS measured results" is not present:

6> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the measurements included in the MEASUREMENT REPORT were valid.

4> if the UE does not support the capability to provide the GANSS timing of the cell and if IE "UE positioning GPS measured results" is not present:

5> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the measurements included in the MEASUREMENT REPORT were valid.

4> if the UE supports the capability to perform GANSS measurements on multiple GANSS frequencies:

5> if the IE "GANSS Multi-frequency Measurement Requested" is included with one bit set to value one for a supported GANSS, and if any of these GANSS signals are measured:

6> include the IE "GANSS Signal Measurement Information" for each measured GANSS signal.

References

- Conformance requirement 1: TS 24.008 clause 4.5.1.5.

- Conformance requirement 2: TS 24.008, clause 5.2.1.
- Conformance requirement 3: TS 25.331, clause 8.6.7.19.1a.

#### 6.2.1.2.3 Test Purpose

To verify when an emergency call is initiated by a UE, and the network performs a GANSS location request using the RRC measurement control procedure, then the UE responds with a Measurement Report containing "UE positioning GANSS measured results" and "UE positioning GPS measured results" (Sub-tests 3, 4, 8 and 10).

#### 6.2.1.2.4 Method of Test

##### Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2.
- User Equipment (UE):
  - the UE is in state "MM idle" with valid TMSI and CKSN.

##### Related PICS/PIXIT Statements

- Emergency speech call
- UE Assisted Network Assisted GANSS
- UE Assisted Network Assisted GPS (Sub-tests 3, 4, 8 and 10)

##### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

The UE is made to initiate an emergency call.

After the call has been through-connected in both directions, the SS orders an A-GNSS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.4.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The UE then performs positioning measurements and responds with the RRC message MEASUREMENT REPORT.

Finally the SS clears the call.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | UE | | | The "emergency number" is entered. Number shall be one programmed in test USIM $EF_{ECC}$ (Emergency Call Codes), ref. 34.108 clause 8.3.2.21. |
| 2 | --> | | | UE establishes RRC procedure for emergency call.<br>Establishment cause: Emergency Call<br>SS checks that the UE capability includes "Network Assisted GANSS Support List" with "GANSS mode" set to "UE-assisted", and that the UE includes "Network Assisted GPS Support" for UE-assisted (Sub-Tests 3, 4, 8 and 10 only). |
| 3 | --> | | CM SERVICE REQUEST | The CM service type IE indicates "emergency call establishment". |
| 4 | <-- | | AUTHENTICATION REQUEST | IE Authentication Parameter AUTN shall be present in the message. |
| 5 | --> | | AUTHENTICATION RESPONSE | SRES specifies correct value. |
| 6 | SS | | | SS starts security procedure. |
| 7 | --> | | EMERGENCY SETUP | If the Bearer capability IE is not included the default UMTS AMR speech version shall be assumed. |
| 8 | <-- | | CALL PROCEEDING | |
| 9 | <-- | | ALERTING | |
| 10 | <-- | | | SS sets up the radio bearer with the rate indicated by the EMERGENCY SETUP message. |
| 11 | <-- | | CONNECT | |
| 12 | --> | | CONNECT ACKNOWLEDGE | |
| 13 | UE | | | The DTCH is through connected in both directions. |
| 14 | <-- | | MEASUREMENT CONTROL | |
| 15 | --> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 15a | <-- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 15, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.4.5. |
| 15b | --> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 15, this message contains the IE "UE positioning GANSS measured results" and "UE positioning GPS measured results" (Sub-tests 3, 4, 8 and 10). |
| 16 | <-- | | DISCONNECT | SS disconnects the call and associated radio bearer. |

Specific Message Contents

MEASUREMENT CONTROL (Step 14):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Set according to UE capabilities |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GNSS" in 4.4.3 |
| - UE positioning GANSS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GNSS" in 4.4.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 15 (Option 1) or 15b (Option 2))

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present for Sub-Tests 3, 4, 8 and 10 |
| - UE positioning error | Not present |
| - UE positioning GANSS measured results | Present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

MEASUREMENT REPORT (Step 15 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| - Error reason | Assistance Data Missing |
| - GPS Additional Assistance Data Request | Not checked |
| - GANSS Additional Assistance Data Request | Not checked |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

MEASUREMENT CONTROL (Step 15a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Set according to UE capabilities |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.4.5 |
| - UE positioning GANSS assistance data | Set as specified in 4.4.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

#### 6.2.1.2.5 Test requirements

After step 12 the UE shall have through connected the DTCH in both directions.

After step 14 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results" and/or "UE positioning GANSS measured results".

## 6.2.2 Assisted GNSS Mobile Originated Tests

### 6.2.2.1 MO-LR Position Estimate: UE-Based A-GNSS

#### 6.2.2.1.1 Definition

This test case applies to all UEs supporting UE-Based GANSS or GNSS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

#### 6.2.2.1.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

2) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

3) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 2 and 3: TS 24.030, subclause 5.1.1

#### 6.2.2.1.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-based GNSS.

#### 6.2.2.1.4 Method of Test

Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment (UE):
  - The UE is in state “MM idle” with valid TMSI and CKSN.
  - The UE is in state “PMM idle” with valid P-TMSI.

Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of triggering an MO-LR request for a position estimate.

Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: “A-GPS” includes Modernized GPS if supported by the UE. | |

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "locationEstimate".

The SS orders an A-GNSS positioning measurement using one or more (dependent on the Sub-Test) MEASUREMENT CONTROL messages.

The SS responds with a FACILITY message containing an MO-LR result.

When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "locationEstimate". |
| 7 | <-- | | MEASUREMENT CONTROL | All Sub-Tests |
| 7a | <-- | | MEASUREMENT CONTROL | Sub-Tests 2, 3, 4, 8, 10 only |
| 7b | <-- | | MEASUREMENT CONTROL | Sub-Tests 4, 8, 10 only |
| 8 | -> | | MEASUREMENT REPORT | |
| 9 | <- | | FACILITY | LCS MO-LR result message containing a location estimate |
| 10 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 11 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Register message type | xx11 1011 (REGISTER) |
| Facility | Invoke=LCS-MOLR<br>molr-Type ->locationEstimate |
| SS version | Version 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 1, 9 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 2, 3, 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7a):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 2, 3 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7b):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| - CHOICE *Reference time* | GPS or GANSS reference time only |
| - GPS TOW msec | Not checked |
| - GANSS TOD msec | Not checked |
| - CHOICE *Position estimate* | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' |
| - Position Data | Not checked |
| - CHOICE *Velocity estimate* | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

FACILITY (Step 9)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Facility message type | xx11 1010 (FACILITY) |
| Facility | Return Result=LCS-MOLRRes<br>→ locationEstimate |

RELEASE COMPLETE (Step 10)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Release Complete message type | xx10 1010 (RELEASE COMPLETE) |

#### 6.2.2.1.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 7, the UE shall respond with a MEASUREMENT REPORT message.

After step 9, the UE shall send a RELEASE COMPLETE message.

## 6.2.2.2 MO-LR Position Estimate: UE-Assisted A-GNSS

### 6.2.2.2.1 Definition

This test case applies to all UEs supporting UE-Assisted GANSS or GNSS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

### 6.2.2.2.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

2) The network shall pass the result of the location procedure to the MS by sending a FACILITY message to the MS containing a LCS-MOLR return result component.

3) After the last location request operation the MS shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 2 and 3: TS 24.030, subclause 5.1.1

### 6.2.2.2.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-assisted GNSS.

### 6.2.2.2.4 Method of Test

#### Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2
- User Equipment (UE):
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI.

#### Related PICS/PIXIT Statements

- UE Assisted Network Assisted GANSS.
- UE Assisted Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of triggering an MO-LR request for a position estimate.

#### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "locationEstimate".

The SS orders an A-GNSS positioning measurement using a MEASUREMENT CONTROL message, including assistance data as specified in subclause 4.4.3. The UE may request additional assistance data by sending a MEASUREMENT REPORT message containing a positioning error indication with the IE "Error reason" set to "Assistance Data Missing". If the UE requests additional assistance data, the SS provides the requested assistance data in one or more MEASUREMENT CONTROL messages.

The SS responds with a FACILITY message containing an MO-LR result.

When UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "locationEstimate". |
| 7 | <- | | MEASUREMENT CONTROL | |
| 8 | -> | | MEASUREMENT REPORT | UE reports positioning measurement results (Option 1) or requests additional assistance data (Option 2). |
| 8a | <- | | MEASUREMENT CONTROL | If UE requested additional assistance data in step 8, SS provides the requested data in one or more MEASUREMENT CONTROL messages as specified in subclause 4.4.5. |
| 8b | -> | | MEASUREMENT REPORT | If UE requested additional assistance data in step 8, this message contains the IE "UE positioning GANSS measured results" and "UE positioning GPS measured results" (Sub-tests 3, 4, 8 and 10). |
| 9 | <- | | FACILITY | LCS MO-LR result message containing a location estimate |
| 10 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 11 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 6):

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Register message type | xx11 1011 (REGISTER) |
| Facility | Invoke=LCS-MOLR<br>molr-Type ->locationEstimate |
| SS version | Version 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Set according to UE capabilities |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GNSS" in 4.4.3 |
| - UE positioning GANSS assistance data | Set as specified for "Adequate assistance data for UE-assisted A-GNSS" in 4.4.3 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8 (Option 1) or 8b (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Present for Sub-Tests 3, 4, 8 and 10 |
| - UE positioning error | Not present |
| - UE positioning GANSS measured results | Present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

MEASUREMENT REPORT (Step 8 (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS Additional Assistance Data Request | Not checked |
| - GANSS Additional Assistance Data Request | Not checked |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

MEASUREMENT CONTROL (Step 8a (Option 2)):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Set according to UE capabilities |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.4.5 |
| - UE positioning GANSS assistance data | Set as specified in 4.4.5 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

FACILITY (Step 9):

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Facility message type | xx11 1010 (FACILITY) |
| Facility | Return Result=LCS-MOLRRes<br>→ locationEstimate<br>Any values may be used. The SS shall not be required to calculate the value from the returned measurements. |

RELEASE COMPLETE (Step 10):

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Release Complete message type | xx10 1010 (RELEASE COMPLETE) |

#### 6.2.2.2.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 7, the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning GPS measured results" and/or "UE positioning GANSS measured results".

After step 9, the UE shall send a RELEASE COMPLETE message.

### 6.2.2.3 MO-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites

#### 6.2.2.3.1 Definition

This test case applies to all UEs supporting UE-Based GANSS or GNSS Location Service capabilities and providing a method to trigger an MO-LR request for a position estimate.

#### 6.2.2.3.2 Conformance requirements

1) The MS invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

2) if the IE "Measurement command" has the value "modify":

2> for all IEs present in the MEASUREMENT CONTROL message:

3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GANSS assistance data" is present, for any of the optional IEs "UE positioning GANSS reference time", "UE positioning GANSS reference UE position", "UE positioning DGANSS corrections", "UE positioning GANSS ionospheric model", "UE positioning GANSS additional ionospheric model", "UE positioning GANSS UTC model", "UE positioning GANSS additional UTC models", "UE positioning GANSS reference measurement information", "UE positioning GANSS data bit assistance", "UE positioning GANSS Time model", "UE positioning GANSS real-time integrity", "UE positioning GANSS Earth orientation parameters", "UE positioning GANSS auxiliary information" that are present in the MEASUREMENT CONTROL message:

5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

3) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

1> if an update has been provided for this satellite:

2> act as specified in subclause 8.6.7.19.3.4.

If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

1> update the variable UE_POSITIONING_GPS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

4) If the IE "UE positioning GANSS Navigation Model" is included, the UE shall:

1> for each GANSS:

2> for each satellite, the UE shall:

3> for IE "UE positioning GANSS clock model":

4> act as specified in subclause 8.6.7.19.7.4a.

3> for IE "UE positioning GANSS orbit model":

4> act as specified in subclause 8.6.7.19.7.4b.

5) If the IE "UE positioning GANSS clock model" is included, the UE shall:

1> for each GANSS:

2> update the variable UE_POSITIONING_GANSS_DATA as follows:

3> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position.

2> act on these GANSS clock models in a manner similar to that specified in a relevant ICD.

6) If the IE "UE positioning GANSS orbit model" is included, for each satellite of each supported GNSS, the UE shall:

1> update the variable UE_POSITIONING_GANSS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position..

1> act on these GANSS orbit models in a manner similar to that specified in a relevant ICD.

7) The UE shall when a measurement report is triggered:

2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or the UE has been able to calculate a position in case of GPS or GANSS positioning or the UE has been able to calculate a position using a standalone positioning method:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE supports the capability to perform the UE GPS timing of cell frames measurement:

5> if the IE "GPS timing of Cell wanted" is set to TRUE:

6> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD;

6> include the SFN when the position was determined;

6> include the IE "UE GPS timing of cell frames";

6> include the IE "UE Positioning GPS Reference Time Uncertainty".

5> if the IE "GPS timing of Cell wanted" is set to FALSE:

6> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the position was calculated with GPS; and

4> the UE does not support the capability to provide the GPS timing of the cell:

5> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the UE supports the capability to provide the GANSS timing of the cell frames measurement:

5> if the IE "GANSS timing of Cell wanted" is included with one bit set to value one for a supported GANSS:

6> perform the UE GANSS timing of cell frames measurement on the serving cell or on one cell of the active set;

6> include the IE "GANSS Time ID" to identify the GNSS system time;

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

6> include the IE "Reference SFN" and the IE "UE GANSS timing of cell frames".

5> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero:

6> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid.

4> if the UE does not support the capability to provide the GANSS timing of the cell:

5> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid;

5> include the IE "GANSS Time ID" to identify the GNSS system time.

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to value "0":

6> if the IE "Horizontal Accuracy" has been assigned a value "0":

7> may include IE "Ellipsoid point with altitude".

6> if the IE "Horizontal Accuracy" has been assigned a value unequal to "0"; and

6> if the UE has been able to calculate a 3-dimensional position

7> include IE "Ellipsoid point with altitude" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> may act as if IE "Vertical Accuracy" was not included in IE "UE positioning reporting quantity".

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

4> if any of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" is reported:

5> should calculate a value of the IE "Confidence", different from "0", as the probability that the UE is located within the uncertainty region of the one of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" that is reported.

NOTE: The value "0" of the IE "Confidence" is interpreted as "no information" by the UTRAN [57].

4> if IE "Velocity Requested" has been included in IE "UE positioning reporting quantity":

5> include IE "Velocity estimate" if supported and available.

2> if the UE was not able to calculate a position:

3> include IE "UE positioning error" in the MEASUREMENT REPORT and set the contents of this IE as specified in subclause 8.6.7.19.5.

8) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS" and the IE "GANSS Positioning Methods" is present:

2> if there were not enough GANSS satellites to be received:

3> set IE "Error reason" to "Not Enough GANSS Satellites".

2> if some GANSS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GANSS Additional Assistance Data Request".

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to FALSE:

4> not include the IE "GANSS Additional Assistance Data Request", and use the assistance data available for doing a positioning estimate.

9) If the network is unable to successfully fulfil the request received from the MS (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080.

10) PositionMethodFailure: This error is returned by the network when the network is unable to obtain any of the location information requested or none of the information obtained satisfies the requested LCS QoS or if requested LCS assistance data could not be transferred or requested deciphering keys for broadcast assistance data could not be returned.

Reference(s):

- Conformance requirements 1 and 9: TS 24.030, subclause 5.1.1
- Conformance requirement 2: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 3: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 4: TS 25.331, subclause 8.6.7.19.7.4
- Conformance requirement 5: TS 25.331, subclause 8.6..7.19.4a
- Conformance requirement 6: TS 25.331, subclause 8.6.7.19.4b
- Conformance requirement 7: TS 25.331, subclause 8.6.7.19.1b
- Conformance requirement 8: TS 25.331, subclause 8.6.7.19.1b
- Conformance requirement 9: TS 25.331, subclause 8.6.7.19.5
- Conformance requirement 10: TS 24.080, subclause 4.3.2.29

#### 6.2.2.3.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-based GNSS when the MO-LR procedure fails due to failure of positioning method.

#### 6.2.2.3.4 Method of Test

##### Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals switched off or not present.
- User Equipment (UE):
  - The UE is in state “MM idle” with valid TMSI and CKSN.
  - The UE is in state “PMM idle” with valid P-TMSI

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of triggering an MO-LR request for a position estimate.

##### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "locationEstimate".

The SS orders an A-GNSS positioning measurement using one or more (dependent on the Sub-Test) MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message with a positioning error indication.

The SS sends a RELEASE COMPLETE message containing a return error component.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "locationEstimate". |
| 7 | <-- | | MEASUREMENT CONTROL | All Sub-Tests |
| 7a | <-- | | MEASUREMENT CONTROL | Sub-Tests 2, 3, 4, 8, 10 only |
| 7b | <-- | | MEASUREMENT CONTROL | Sub-Tests 4, 8, 10 only |
| 8 | -> | | MEASUREMENT REPORT | Positioning error report "not enough GANSS satellites" or "not enough GPS satellites" (sub-tests 3, 4, 8 and 10 only) |
| 9 | SS | | | SS is unable to fulfil the MO-LR request |
| 10 | <- | | RELEASE COMPLETE | SS terminates the dialogue containing a return error component |
| 11 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Register message type | xx11 1011 (REGISTER) |
| Facility | Invoke=LCS-MOLR<br>molr-Type ->locationEstimate |
| SS version | Version 1 or above |

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 1, 9 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 2, 3, 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7a):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | For Sub-Tests 2, 3 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7b):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | notEnoughGANSS-Satellites or notEnoughGPS-Satellites (sub-tests 3, 4, 8 and 10 only) |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

RELEASE COMPLETE (Step 10)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |
| Facility | Return error = LCS-MOLR<br>Error -> positionMethodFailure |

#### 6.2.2.3.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "locationEstimate".

After step 7b, the UE shall respond with a MEASUREMENT REPORT message.

### 6.2.2.4 MO-LR Assistance Data: UE-Based or UE-Assisted A-GNSS - Success

#### 6.2.2.4.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities and providing a method to trigger an MO-LR request for a assistance data.

#### 6.2.2.4.2 Conformance requirements

1) The UE invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

2) The network shall send a FACILITY message to the UE containing a LCS-MOLR return result component.

3) After the last location request operation the UE shall terminate the dialogue by sending a RELEASE COMPLETE message.

Reference(s):

- Conformance requirements 1, 2 and 3: TS 24.030, subclause 5.1.1

#### 6.2.2.4.3 Test Purpose

To verify the UE behaviour at a mobile originated location request procedure using network-assisted UE-based or UE-assisted GNSS.

#### 6.2.2.4.4 Method of Test

##### Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signal s: As specified in 4.2
- User Equipment (UE):
  - The UE is in state “MM idle” with valid TMSI and CKSN.
  - The UE is in state “PMM idle” with valid P-TMSI.

##### Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- UE Assisted Network Assisted GANSS.
- UE Assisted Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of triggering an MO-LR request for assistance data.

##### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: “A-GPS” includes Modernized GPS if supported by the UE. | |

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "AssistanceData".

The SS transmits an ASSISTANCE DATA delivery message with assistance data. When the assistance data delivery was successful, the SS sends a FACILITY message to the UE.

The SS responds with a FACILITY message containing an MO-LR result.

When the UE receives the FACILITY message, it clears the transaction by sending a RELEASE COMPLETE message.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "AssistanceData". |
| 7 | <- | | ASSISTANCE DATA DELIVERY | The SS provides the requested data in one or more ASSISTANCE DATA DELIVERY messages as specified in subclause 4.4.5. |
| 8 | <- | | FACILITY | |
| 9 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 10 | SS | | | The SS releases the RRC connection and the test case ends. |

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | For sub-tests 1, 2, 9:<br>Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->assistanceData<br>locationMethod -> assistedGANSS<br>ganssAssistanceData -> OCTET STRING<br>Octets 1 to 40 are coded in the same way as the octets 3 to 9+2n of Requested GANSS Data IE in 3GPP TS 49.031<br><br>For sub-tests 3, 4, 8 and 10:<br>Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->assistanceData<br>locationMethod ->assistedGPSandGANSS<br>gpsAssistanceData -> OCTET STRING<br>Octets 1 to 38 are coded in the same way as octets 3 to 7+2n of Requested GPS Data IE in 3GPP TS 49.031<br>ganssAssistanceData -> OCTET STRING<br>Octets 1 to 40 are coded in the same way as the octets 3 to 9+2n of requested GANSS Data IE in 3GPP TS 49.031 |
| SS Version | Value 1 or above |

ASSISTANCE DATA DELIVERY (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| UE positioning OTDOA assistance data for UE-based | Not present |
| UE positioning GPS assistance data | Not present for sub-tests 1, 2 and 9.<br>For sub-tests 3, 4, 8 and 10 set as specified in 4.4.5 |
| UE positioning GANSS assistance data | Set as specified in 4.4.5 |

FACILITY (Step 8)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Facility message type | xx11 1010 (FACILITY) |
| Facility | Return Result=LCS-MOLRRes<br>→ EMPTY |

RELEASE COMPLETE (Step 9)

| Information element | Value/remark |
|---|---|
| Supplementary service protocol discriminator | 1011 (supplementary services (call independent)) |
| Transaction identifier | |
| Release Complete message type | xx10 1010 (RELEASE COMPLETE) |

#### 6.2.2.4.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "assistanceData".

After step 8, the UE shall send a RELEASE COMPLETE message.

### 6.2.2.5 MO-LR Assistance Data: UE-Based or UE-Assisted A-GNSS - Failure

#### 6.2.2.5.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities and providing a method to trigger an MO-LR request for assistance data.

#### 6.2.2.5.2 Conformance requirements

1) The UE invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component.

2) If the network is unable to successfully fulfil the request received from the UE (e.g. to provide a location estimate or location assistance information), it shall clear the transaction by sending a RELEASE COMPLETE message containing a return error component. Error values are specified in 3GPP TS 24.080.

3) PositionMethodFailure: This error is returned by the network when the network is unable to obtain any of the location information requested or none of the information obtained satisfies the requested LCS QoS or if requested LCS assistance data could not be transferred or requested deciphering keys for broadcast assistance data could not be returned.

Reference(s):

- Conformance requirements 1 and 2: TS 24.030, subclause 5.1.1

- Conformance requirement 3: TS 24.080, subclause 4.3.2.29

#### 6.2.2.5.3 Test Purpose

To verify the UE behaviour at a mobile originated location request for assistance data where the network is unable to provide the requested assistance data.

#### 6.2.2.5.4 Method of Test

Initial Conditions

- System Simulator (SS):

- 1 cell, default parameters.
- Satellite signals: As specified in 4.2

- User Equipment (UE):

  The UE shall begin the test with neither GPS nor GANSS assistance data stored.

  - The UE is in state “MM idle” with valid TMSI and CKSN.
  - The UE is in state “PMM idle” with valid P-TMSI

### Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- UE Assisted Network Assisted GANSS.
- UE Assisted Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of triggering an MO-LR request for assistance data.

### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: "A-GPS" includes Modernized GPS if supported by the UE. | |

The UE invokes call independent supplementary service through a CM SERVICE REQUEST. The SS initiates authentication and ciphering.

The UE invokes an MO-LR request of type "assistanceData".

The SS is unable to provide the requested assistance data.

The SS sends a RELEASE COMPLETE message containing a return error component.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | -> | | | The UE establishes an RRC connection for location service. The SS verifies that the IE "Establishment cause" in the received RRC CONNECTION REQUEST message is set to "Originated High Priority Signalling". |
| 2 | -> | | CM SERVICE REQUEST | The CM service type IE indicates "call independent supplementary service" |
| 3 | <- | | AUTHENTICATION REQUEST | |
| 4 | -> | | AUTHENTICATION RESPONSE | |
| 5 | SS | | | The SS starts ciphering and integrity protection. |
| 6 | -> | | REGISTER | Call Independent SS containing Facility IE with an LCS MO-LR request of type "AssistanceData". |
| 7 | SS | | | SS is unable to provide the requested assistance data |
| 8 | <- | | RELEASE COMPLETE | SS terminates the dialogue containing a return error component |
| 9 | SS | | | The SS waits for 10 seconds to verify that the UE does not send a RELEASE COMPLETE message. |
| 10 | SS | | | The SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (xx11 1011) |
| Facility | Sub-tests 1, 2 and 9:<br>Invoke = LCS-MOLR<br>LCS-MOLRArg<br>molr-Type ->assistanceData<br>locationMethod -> assistedGANSS<br>ganssAssistanceData -> OCTET STRING<br>Octets 1 to 40 are coded in the same way as octets 3 to 9+2n of Requested GANSS Data IE in 3GPP TS 49.031<br><br>Sub-tests 3, 4, 8 and 10:<br>Invoke = LCS-MOLR<br><br>LCS-MOLRArg<br><br>molr-Type ->assistanceData<br><br>LocationMethod ->assistedGPSandGANSS<br><br>gpsAssistanceData -> OCTET STRING<br>Octets 1 to 38 are coded in the same way as octets 3 to 7+2n of Requested GPS Data IE in 3GPP TS 49.031<br><br>ganssAssistanceData -> OCTET STRING<br>Octets 1 to 40 are coded in the same way as the octets 3 to 9+2n of requested GANSS Data IE in 3GPP TS 49.031 |
| SS version indicator | Value 1 or above |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |
| Facility | Return error = LCS-MOLR<br>Error -> positionMethodFailure |

#### 6.2.2.5.5 Test requirements

After step 5 the UE shall transmit a REGISTER message with an LCS MO-LR request with the IE "MOLR-Type" set to "assistanceData".

During step 9 the UE shall not send any RELEASE COMPLETE message.

## 6.2.3 Assisted GNSS Mobile Terminated Tests

### 6.2.3.1 MT-LR: UE-based or UE-Assisted A-GNSS – Request for additional assistance data/ Success

#### 6.2.3.1.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities.

#### 6.2.3.1.2 Conformance requirements

1) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GANSS assistance data" is present, for any of the optional IEs "UE positioning GANSS reference time", "UE positioning GANSS reference UE position", "UE positioning DGANSS corrections", "UE positioning GANSS ionospheric model", "UE positioning GANSS additional ionospheric model", "UE positioning GANSS UTC model", "UE positioning GANSS additional UTC models", "UE positioning GANSS reference measurement information", "UE positioning GANSS data bit assistance", "UE positioning GANSS Time model", "UE positioning GANSS real-time integrity", "UE positioning GANSS Earth orientation parameters", "UE positioning GANSS auxiliary information" that are present in the MEASUREMENT CONTROL message:

            5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

            5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

2) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

   1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

1> if an update has been provided for this satellite:

2> act as specified in subclause 8.6.7.19.3.4.

If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

1> update the variable UE_POSITIONING_GPS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

3) If the IE "UE positioning GANSS Navigation Model" is included, the UE shall:

1> for each GANSS:

2> for each satellite, the UE shall:

3> for IE "UE positioning GANSS clock model":

4> act as specified in subclause 8.6.7.19.7.4a.

3> for IE "UE positioning GANSS orbit model":

4> act as specified in subclause 8.6.7.19.7.4b.

4) If the IE "UE positioning GANSS clock model" is included, the UE shall:

1> for each GANSS:

2> update the variable UE_POSITIONING_GANSS_DATA as follows:

3> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position.

2> act on these GANSS clock models in a manner similar to that specified in a relevant ICD.

5) If the IE "UE positioning GANSS orbit model" is included, for each satellite of each supported GNSS, the UE shall:

1> update the variable UE_POSITIONING_GANSS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position..

1> act on these GANSS orbit models in a manner similar to that specified in a relevant ICD.

6) The UE shall when a measurement report is triggered:

2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or the UE has been able to calculate a position in case of GPS or GANSS positioning or the UE has been able to calculate a position using a standalone positioning method:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE supports the capability to perform the UE GPS timing of cell frames measurement:

5> if the IE "GPS timing of Cell wanted" is set to TRUE:

6> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD;

6> include the SFN when the position was determined;

6> include the IE "UE GPS timing of cell frames";

6> include the IE "UE Positioning GPS Reference Time Uncertainty".

5> if the IE "GPS timing of Cell wanted" is set to FALSE:

6> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the position was calculated with GPS; and

4> the UE does not support the capability to provide the GPS timing of the cell:

5> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the UE supports the capability to provide the GANSS timing of the cell frames measurement:

5> if the IE "GANSS timing of Cell wanted" is included with one bit set to value one for a supported GANSS:

6> perform the UE GANSS timing of cell frames measurement on the serving cell or on one cell of the active set;

6> include the IE "GANSS Time ID" to identify the GNSS system time;

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

6> include the IE "Reference SFN" and the IE "UE GANSS timing of cell frames".

5> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero:

6> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid.

4> if the UE does not support the capability to provide the GANSS timing of the cell:

5> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid;

5> include the IE "GANSS Time ID" to identify the GNSS system time.

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to value "0":

6> if the IE "Horizontal Accuracy" has been assigned a value "0":

7> may include IE "Ellipsoid point with altitude".

6> if the IE "Horizontal Accuracy" has been assigned a value unequal to "0"; and

6> if the UE has been able to calculate a 3-dimensional position

7> include IE "Ellipsoid point with altitude" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> may act as if IE "Vertical Accuracy" was not included in IE "UE positioning reporting quantity".

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

4> if any of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" is reported:

5> should calculate a value of the IE "Confidence", different from "0", as the probability that the UE is located within the uncertainty region of the one of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" that is reported.

NOTE: The value "0" of the IE "Confidence" is interpreted as "no information" by the UTRAN [57].

4> if IE "Velocity Requested" has been included in IE "UE positioning reporting quantity":

5> include IE "Velocity estimate" if supported and available.

2> if the UE was not able to calculate a position:

3> include IE "UE positioning error" in the MEASUREMENT REPORT and set the contents of this IE as specified in subclause 8.6.7.19.5.

7) The UE shall:

1> when a measurement report is triggered:

2> if the UE was able to perform measurements on at least one neighbour cell included in the variable UE_POSITIONING_OTDOA_DATA_UE_ASSISTED in case of OTDOA or one satellite included in the variable UE_POSITIONING_GPS_DATA in case of GPS positioning or one cell from the active set in case of CELL ID:

3> if the IE "Vertical Accuracy" is included:

4> interpret the presence of this IE to indicate that the UTRAN desires to compute a 3-dimensional position estimate.

if the IE "Positioning Methods" is set to "GPS" and the IE "GANSS Positioning Methods" is present indicating other GNSS than GPS allowed and if any of these other GNSSs is measured:

4> include the IE "UE positioning GANSS measured results" in the measurement report and set the contents of the IE as follows:

5> if the UE supports the capability to provide the GANSS timing of the cell frames measurement:

6> if the IE "GANSS timing of Cell wanted" is included with one bit set to value one for a supported GANSS and if IE "UE GPS timing of cell frames" is not present:

7> perform the UE GANSS timing of cell frames measurement on the serving cell or on one cell of the active set;

7> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

7> include the IE "Reference SFN" and the IE "UE GANSS timing of cell frames".

6> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero and if IE "UE positioning GPS measured results" is not present:

7> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the measurements included in the MEASUREMENT REPORT were valid.

5> if the UE does not support the capability to provide the GANSS timing of the cell and if IE "UE positioning GPS measured results" is not present:

6> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the measurements included in the MEASUREMENT REPORT were valid.

5> if the UE supports the capability to provide the GANSS carrier-phase measurements:

6> if the IE "GANSS Carrier-Phase Measurement Requested" is included with one bit set to value one for a supported GANSS:

7> include the IE "Carrier Quality Indication" and include the IE "ADR".

5> if the UE supports the capability to perform GANSS measurements on multiple GANSS frequencies:

6> if the IE "GANSS Multi-frequency Measurement Requested" is included with one bit set to value one for a supported GANSS, and if any of these GANSS signals are measured:

7> include the IE "GANSS Signal Measurement Information" for each measured GANSS signal.

8) 1> if the UE is unable to report the requested measurement results due to missing GPS assistance data:

2> the UE may at anytime send a measurement report containing the IE "UE positioning error" and set the contents of this IE as specified in subclause 8.6.7.19.5.

2> after sending the measurement report, the UE shall not send another measurement report to request the same GPS assistance data for at least 20s. This requirement does not apply after release of the current RRC connection.

1> if the UE is unable to report the requested measurement results due to missing GANSS assistance data:

2> the UE may at anytime send a measurement report containing the IE "UE positioning error" and set the contents of this IE as specified in subclause 8.6.7.19.5;

2> after sending the measurement report, the UE shall not send another measurement report to request the same GANSS assistance data for at least 20s. This requirement does not apply after release of the current RRC connection.

9) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS" and the IE "GANSS Positioning Methods" is present:

2> if there were not enough GANSS satellites to be received:

3> set IE "Error reason" to "Not Enough GANSS Satellites".

2> if some GANSS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GANSS Additional Assistance Data Request".

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to FALSE:

4> not include the IE "GANSS Additional Assistance Data Request", and use the assistance data available for doing a positioning estimate.

Reference(s):

- Conformance requirement 1: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 2: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 3: TS 25.331, subclause 8.6.7.19.7.4
- Conformance requirement 4: TS 25.331, subclause 8.6.7.19.7.4a
- Conformance requirement 5: TS 25.331, subclause 8.6.7.19.7.4b
- Conformance requirement 6: TS 25.331, subclause 8.6.7.19.1b
- Conformance requirement 7: TS 25.331, subclause 8.6.7.19.1a
- Conformance requirement 8: TS 25.331, subclause 8.6.7.19.1a, 8.6.7.19.1b
- Conformance requirement 9: TS 25.331, subclause 8.6.7.19.5

#### 6.2.3.1.3 Test Purpose

To verify the UE's behaviour in a mobile-terminated location request procedure using UE-based or UE-assisted A-GNSS with assistance data from the network.

To verify that the UE in CELL_DCH state accepts assistance data received in multiple MEASUREMENT CONTROL messages.

To verify that the UE includes the IE "GPS Additional Assistance Data Request" or "GANSS Additional Assistance Data Request" to request assistance data when it does not have enough assistance data.

#### 6.2.3.1.4 Method of Test

Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals: As specified in 4.2.
- User Equipment (UE):
  - The UE is in state "MM idle" with valid TMSI and CKSN.
  - The UE is in state "PMM idle" with valid P-TMSI.
  - The UE shall begin the test with no GPS and GANSS assistance data stored.

Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.

- UE Assisted Network Assisted GANSS
- UE Based Network Assisted GPS (Sub-tests 3. 4, 8 and 10).
- UE Assisted Network Assisted GPS (Sub-tests 3, 4, 8 and 10).
- Method of clearing stored GPS/GANSS assistance data.

### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: “A-GPS” includes Modernized GPS if supported by the UE. | |

The stored GPS/GANSS assistance data in the UE shall be cleared.

The SS initiates authentication and ciphering and orders a positioning measurement using a MEASUREMENT CONTROL message including no assistance data.
The UE sends a MEASUREMENT REPORT message to report a positioning error, requesting further assistance data.
The SS response with one or more MEASUREMENT CONTROL messages that include the requested assistance data and instructs the UE not to repeat the request for assistance data. The final MEASUREMENT CONTROL message orders periodic reporting.

The UE performs positioning measurements and responds with a MEASUREMENT REPORT message including the IE "UE Positioning Position Estimate Info" in case of UE-based, or including the IE "UE positioning GANSS measured results" and/or "UE positioning GPS measured results" in case of UE-assisted.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <- | | AUTHENTICATION REQUEST | |
| 2 | -> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | The SS starts ciphering and integrity protection. |
| 4 | | | Void | |
| 5 | | | Void | |
| 6 | | | Void | |
| 7 | <-- | | MEASUREMENT CONTROL | No assistance data, and "Additional Assistance Data Request" IE set to TRUE. |
| 8 | -> | | MEASUREMENT REPORT | Positioning error report ‘Assistance Data Missing’ |
| 9 | <- | | MEASUREMENT CONTROL | The SS provides the requested data in one or more MEASUREMENT CONTROL messages. The last message contains:<br>Reporting mode: Periodical reporting<br>Amount of reporting: 1<br>Reporting interval: 64000 |
| 10 | --> | | MEASUREMENT REPORT | Measurement report message containing UE position estimate (UE-based), or IE "UE positioning GANSS measured results" and/or "UE positioning GPS measured results’’ (UE-assisted). |
| 11 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| Measurement Information Elements | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based or UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | UE assisted: Set according to UE capabilities<br>UE based: Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for "Inadequate assistance data for UE-based A-GNSS" in 4.4.2<br>or<br>"Inadequate assistance data for UE-assisted A-GNSS" in 4.4.4 |
| - UE positioning GANSS assistance data | Set as specified for "Inadequate assistance data for UE-based A-GNSS" in 4.4.2<br>or<br>"Inadequate assistance data for UE-assisted A-GNSS" in 4.4.4 |
| Physical Channel Information Elements | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8)

| Information element | Value/remark |
|---|---|
| Measurement Information Elements | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| - Error reason | Assistance Data Missing |
| -GPS additional assistance data requested | Sub-tests 3, 4, 8 and 10 |
| -Almanac | Present, if requested by UE |
| -UTC model | Present, if requested by UE |
| -Ionospheric model | Present, if requested by UE |
| -Navigation model | Present, if requested by UE |
| -DGPS corrections | Present, if requested by UE |
| -Reference location | Present, if requested by UE |
| -Reference time | Present, if requested by UE |
| -Acquisition assistance | Present, if requested by UE |
| -Real-time integrity | Present, if requested by UE |
| -Navigation model additional data | Present, if requested by UE |
| -GANSS additional assistance data requested | Sub-tests 1, 2, 4, 9 |
| -GANSS Reference time | Present, if requested by UE |
| -Reference location | Present, if requested by UE |
| -GANSS Ionospheric model | Present, if requested by UE |
| -GANSS Additional Ionospheric Model for Data ID = ‘00’ | Present, if requested by UE |
| -GANSS Additional Ionospheric Model for Data ID = ‘11’ | Present, if requested by UE |
| -GANSS Earth orientation parameters | Present, if requested by UE |
| -GANSS Real-time integrity | Present, if requested by UE |
| -GANSS Almanac | Present, if requested by UE |
| -GANSS Time Model | Present, if requested by UE |
| -GANSS Ephemeris Extension Check | Present, if requested by UE |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

MEASUREMENT CONTROL (Step 9)

| Information element | Value/remark |
|---|---|
| Measurement Information Elements | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based or UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-test 1: bit 5 = 1<br>Sub-test 2: bit 1 = 1<br>Sub-test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | UE assisted: Set according to UE capabilities<br>UE based: Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Set as required according to position in sequence of messages |
| - Periodical reporting criteria | Set as required according to position in sequence of messages |
| - Amount of reporting | Set as required according to position in sequence of messages |
| - Reporting interval | Set as required according to position in sequence of messages |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified in 4.4.5 |
| - UE positioning GANSS assistance data | Set as specified in 4.4.5 |
| Physical Channel Information Elements | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 10)

| Information element | Value/remark |
|---|---|
| Measurement Information Elements | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | |
| -CHOICE Position estimate | One of 'Ellipsoid point with uncertainty Circle' or 'Ellipsoid point with uncertainty Ellipse' or 'Ellipsoid point with altitude and uncertainty Ellipsoid' (UE-based) |
| - UE positioning GPS measured results | Present for UE-assisted (Sub-Tests 3, 4, 8 and 10) |
| - UE positioning error | Not present |
| - UE positioning GANSS measured results | Present for UE-assisted. |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

#### 6.2.3.1.5 Test requirements

After step 7 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Assistance Data Missing".

After step 9 the UE shall send a MEASUREMENT REPORT message containing a valid UE position estimate (UE-based) or GANSS and/or GPS measurements (UE-assisted).

### 6.2.3.2 MT-LR Position Estimate: UE-Based A-GNSS – Failure Not Enough Satellites

#### 6.2.3.2.1 Definition

This test case applies to all UEs supporting UE-Based GANSS or GNSS Location Service capabilities.

#### 6.2.3.2.2 Conformance requirements

1) if the IE "Measurement command" has the value "modify":

   2> for all IEs present in the MEASUREMENT CONTROL message:

      3> if a measurement was stored in the variable MEASUREMENT_IDENTITY associated to the identity by the IE "measurement identity":

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GPS assistance data" is present, for any of the optional IEs "UE positioning GPS reference time", "UE positioning GPS reference UE position", "UE positioning GPS DGPS corrections", "UE positioning GPS ionospheric model", "UE positioning GPS UTC model", "UE positioning GPS acquisition assistance", "UE positioning GPS real-time integrity" that are present in the MEASUREMENT CONTROL message:

         4> if measurement type is set to "UE positioning measurement" and the IE "UE positioning GANSS assistance data" is present, for any of the optional IEs "UE positioning GANSS reference time", "UE positioning GANSS reference UE position", "UE positioning DGANSS corrections", "UE positioning

GANSS ionospheric model", "UE positioning GANSS additional ionospheric model", "UE positioning GANSS UTC model", "UE positioning GANSS additional UTC models", "UE positioning GANSS reference measurement information", "UE positioning GANSS data bit assistance", "UE positioning GANSS Time model", "UE positioning GANSS real-time integrity", "UE positioning GANSS Earth orientation parameters", "UE positioning GANSS auxiliary information" that are present in the MEASUREMENT CONTROL message:

5> replace all instances of the IEs listed above (and all their children) stored in variable MEASUREMENT_IDENTITY associated to the identity indicated by the IE "measurement identity" with the IEs received in the MEASUREMENT CONTROL message;

5> leave all other stored information elements unchanged in the variable MEASUREMENT_IDENTITY.

2) If the IE "UE positioning GPS Navigation Model" is included, for each satellite, the UE shall:

1> use IE "Satellite Status" to determine if an update of IE "UE positioning GPS Ephemeris and Clock Correction parameters" has been provided for the satellite indicated by the IE "SatID";

1> if an update has been provided for this satellite:

2> act as specified in subclause 8.6.7.19.3.4.

If the IE "UE positioning GPS Ephemeris and Clock Correction parameters" is included, for each satellite, the UE shall:

1> update the variable UE_POSITIONING_GPS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GPS Navigation Model" in the variable UE_POSITIONING_GPS_DATA, possibly overwriting any existing information in this position.

1> act on these GPS ephemeris and clock correction parameters in a manner similar to that specified in [12].

3) If the IE "UE positioning GANSS Navigation Model" is included, the UE shall:

1> for each GANSS:

2> for each satellite, the UE shall:

3> for IE "UE positioning GANSS clock model":

4> act as specified in subclause 8.6.7.19.7.4a.

3> for IE "UE positioning GANSS orbit model":

4> act as specified in subclause 8.6.7.19.7.4b.

4) If the IE "UE positioning GANSS clock model" is included, the UE shall:

1> for each GANSS:

2> update the variable UE_POSITIONING_GANSS_DATA as follows:

3> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position.

2> act on these GANSS clock models in a manner similar to that specified in a relevant ICD.

5) If the IE "UE positioning GANSS orbit model" is included, for each satellite of each supported GNSS, the UE shall:

1> update the variable UE_POSITIONING_GANSS_DATA as follows:

2> store this IE at the position indicated by the IE "Sat ID" in the IE "UE positioning GANSS Navigation Model" in the variable UE_POSITIONING_GANSS_DATA, possibly overwriting any existing information in this position..

1> act on these GANSS orbit models in a manner similar to that specified in a relevant ICD.

6) The UE shall when a measurement report is triggered:

2> if the UE has been able to calculate a position after performing measurements on the cells included in the variable UE_POSITIONING_OTDOA_DATA_UE_BASED in case of OTDOA or the UE has been able to calculate a position in case of GPS or GANSS positioning or the UE has been able to calculate a position using a standalone positioning method:

3> include IE "UE positioning Position Estimate Info" in the MEASUREMENT REPORT and set the contents of the IE as follows:

4> if the UE supports the capability to perform the UE GPS timing of cell frames measurement:

5> if the IE "GPS timing of Cell wanted" is set to TRUE:

6> perform the UE GPS timing of cell frames measurement on the serving cell or on one cell of the active set.

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD;

6> include the SFN when the position was determined;

6> include the IE "UE GPS timing of cell frames";

6> include the IE "UE Positioning GPS Reference Time Uncertainty".

5> if the IE "GPS timing of Cell wanted" is set to FALSE:

6> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the position was calculated with GPS; and

4> the UE does not support the capability to provide the GPS timing of the cell:

5> include the IE "GPS TOW msec" and set it to the GPS TOW when the position estimate was valid.

4> if the UE supports the capability to provide the GANSS timing of the cell frames measurement:

5> if the IE "GANSS timing of Cell wanted" is included with one bit set to value one for a supported GANSS:

6> perform the UE GANSS timing of cell frames measurement on the serving cell or on one cell of the active set;

6> include the IE "GANSS Time ID" to identify the GNSS system time;

6> include the IE "Primary CPICH Info" for FDD or the IE "cell parameters id" for TDD; and

6> include the IE "Reference SFN" and the IE "UE GANSS timing of cell frames".

5> if the IE "GANSS timing of Cell wanted" is not included, or included with each bit set to value zero:

6> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid.

4> if the UE does not support the capability to provide the GANSS timing of the cell:

5> include the IE "GANSS TOD msec" and set it to the GANSS TOD when the position estimate was valid;

5> include the IE "GANSS Time ID" to identify the GNSS system time.

4> if IE "Vertical Accuracy" has been included in IE "UE positioning reporting quantity":

5> if the IE "Vertical Accuracy" has been assigned to value "0":

6> if the IE "Horizontal Accuracy" has been assigned a value "0":

7> may include IE "Ellipsoid point with altitude".

6> if the IE "Horizontal Accuracy" has been assigned a value unequal to "0"; and

6> if the UE has been able to calculate a 3-dimensional position

7> include IE "Ellipsoid point with altitude" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> may act as if IE "Vertical Accuracy" was not included in IE "UE positioning reporting quantity".

5> if the IE "Vertical Accuracy" has been assigned to a value unequal to "0":

6> if the UE has been able to calculate a 3-dimensional position:

7> include IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

6> if the UE has not been able to calculate a 3-dimensional position:

7> act as if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity".

4> if IE "Vertical Accuracy" has not been included in IE "UE positioning reporting quantity":

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to value "0":

6> may include IE "Ellipsoid point".

5> if IE "Horizontal Accuracy" in IE "UE positioning reporting quantity" has been assigned to a value unequal to 0:

6> include either IE "Ellipsoid point with uncertainty circle" or IE "Ellipsoid point with uncertainty ellipse" or IE "Ellipsoid point with altitude and uncertainty ellipsoid" as the position estimate.

4> if any of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" is reported:

5> should calculate a value of the IE "Confidence", different from "0", as the probability that the UE is located within the uncertainty region of the one of the IEs "Ellipsoid point with uncertainty ellipse" or "Ellipsoid point with altitude and uncertainty ellipsoid" that is reported.

NOTE: The value "0" of the IE "Confidence" is interpreted as "no information" by the UTRAN [57].

4> if IE "Velocity Requested" has been included in IE "UE positioning reporting quantity":

5> include IE "Velocity estimate" if supported and available.

2> if the UE was not able to calculate a position:

3> include IE "UE positioning error" in the MEASUREMENT REPORT and set the contents of this IE as specified in subclause 8.6.7.19.5.

7) The UE shall set the contents of the IE "UE positioning Error" as follows:

…

1> if the IE "Positioning Methods" in IE "UE positioning reporting quantity" has been assigned to value "GPS" and the IE "GANSS Positioning Methods" is present:

2> if there were not enough GANSS satellites to be received:

3> set IE "Error reason" to "Not Enough GANSS Satellites".

2> if some GANSS assistance data was missing:

3> set IE "Error reason" to "Assistance Data Missing"; and

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to TRUE:

4> include the IE "GANSS Additional Assistance Data Request".

3> if the IE "Additional Assistance Data Request" included in the IE "UE positioning reporting quantity" stored in the variable MEASUREMENT_IDENTITY is set to FALSE:

4> not include the IE "GANSS Additional Assistance Data Request", and use the assistance data available for doing a positioning estimate.

Reference(s):

- Conformance requirement 1: TS 25.331, subclause 8.4.1.3.
- Conformance requirement 2: TS 25.331, subclauses 8.6.7.19.3.3a, 8.6.7.19.3.4.
- Conformance requirement 3: TS 25.331, subclause 8.6.7.19.7.4
- Conformance requirement 4: TS 25.331, subclause 8.6..7.19.4a
- Conformance requirement 5: TS 25.331, subclause 8.6.7.19.4b
- Conformance requirement 6: TS 25.331, subclause 8.6.7.19.1b
- Conformance requirement 7: TS 25.331, subclause 8.6.7.19.5

#### 6.2.3.2.3 Test Purpose

To verify the UE behaviour at a mobile terminated location request procedure using network-assisted UE-based GNSS when the MT-LR procedure fails due to failure of positioning method.

To verify that the UE in CELL_DCH state accepts assistance data received in multiple MEASUREMENT CONTROL messages.

To verify that the UE sets the IE Error Reason in ‘UE Positioning Error’ to ‘Not Enough GANSS Satellites’ when it does not receive enough satellite signals to compute a position.

#### 6.2.3.2.4 Method of Test

Initial Conditions

- System Simulator (SS):
  - 1 cell, default parameters.
  - Satellite signals switched off or not present.
- User Equipment (UE):
  - State “CS-CELL DCH Initial (State 6-1) ” as specified in clause 7.4.1 of TS 34.108.

### Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Based Network Assisted GPS (Sub-tests 3, 4, 8 and 10).

### Test Procedure

This test case includes sub-test cases dependent on the GNSS supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined below:

| Sub-Test Case Number | Supported GNSS |
|---|---|
| 1 | UE supporting A-GLONASS only |
| 2 | UE supporting A-Galileo only |
| 3 | UE supporting A-GPS and Modernized GPS only |
| 4 | UE supporting A-GPS(1) and A-GLONASS only |
| 8 | UE supporting A-GPS(1) and A-Galileo only |
| 9 | UE supporting A-BDS only |
| 10 | UE supporting A-GPS(1) and A-BDS only |
| NOTE 1: “A-GPS” includes Modernized GPS if supported by the UE. | |

The SS initiates authentication and ciphering and orders an A-GNSS positioning measurement using one or more (dependent on the sub-test) MEASUREMENT CONTROL messages.

The UE sends a MEASUREMENT REPORT message reporting a positioning error for not enough satellite signal.

### Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <- | | AUTHENTICATION REQUEST | |
| 2 | -> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | | | Void | |
| 5 | | | Void | |
| 6 | | | Void | |
| 7 | <-- | | MEASUREMENT CONTROL | All Sub-Tests |
| 7a | <-- | | MEASUREMENT CONTROL | Sub-Tests 2, 3, 4, 8, 10 only |
| 7b | <-- | | MEASUREMENT CONTROL | Sub-Tests 4, 8, 10 only |
| 8 | -> | | MEASUREMENT REPORT | Positioning error report ‘not enough satellites’ |
| 9 | SS | | | The SS releases the RRC connection and the test case ends. |

Specific Message Contents

MEASUREMENT CONTROL (Step 7):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE Measurement type | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 1: bit 5 = 1<br>Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE Reporting criteria | |
| - Periodical reporting criteria | For Sub-Tests 1, 9 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 2, 3, 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the first MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7a):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE Measurement type | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 2: bit 1 = 1<br>Sub-Test 3: bit 0 and 3 = 1<br>Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 9: bit 6 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE Reporting criteria | |
| - Periodical reporting criteria | For Sub-Tests 2, 3 only |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - No reporting | For Sub-Tests 4, 8, 10 only |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the second MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Step 7b):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE Measurement type | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 127 |
| - Vertical accuracy | 127 |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Velocity Requested | Not present |
| - GANSS Positioning Method | Sub-Test 4: bit 0 and 3 and 5 = 1<br>Sub-Test 8: bit 0 and 1 and 3 = 1<br>Sub-Test 10: bit 0 and 3 and 6 = 1 |
| - GANSS timing of cell wanted | Not present |
| - GANSS Carrier-Phase Measurement Requested | Not present |
| - GANSS Multi-frequency Measurement Requested | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE Reporting criteria | |
| - Periodical reporting criteria | |
| - Amount of reporting | 1 |
| - Reporting interval | 64000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| - UE positioning GANSS assistance data | Set as specified for the third MEASUREMENT CONTROL message for "Adequate assistance data for UE-based A-GNSS" in 4.4.1 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Step 8)

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE Measurement | UE positioning measured results |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | Not present |
| - Error reason | notEnoughGANSS-Satellites or notEnoughGPS-Satellites (sub-tests 3, 4, 8 and 10 only) |
| - UE positioning GANSS measured results | Not present |
| Measured Results on secondary UL frequency | Not present |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Additional Measured results on secondary UL frequency | Not present |
| Event Results | Not present |
| Event results on secondary UL frequency | Not present |
| Inter-RAT cell info indication | Not present |
| E-UTRA Measured Results | Not present |
| E-UTRA Event Results | Not present |
| CSG Proximity Indication | Not present |

#### 6.2.3.2.5 Test requirements

After step 7 the UE shall send a MEASUREMENT REPORT message containing the IE "UE positioning error", with "Error reason" set to "Not Enough Satellites".

### 6.2.3.3 Location Notification

#### 6.2.3.3.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities and LCS value added location request notification capability.

#### 6.2.3.3.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

In the case of location notification no response is required from the UE, the UE shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

References

- Conformance requirement 1: TS 24.030, clause 4.1.1

#### 6.2.3.3.3 Test Purpose

To verify that when the UE receives a REGISTER message during an established CS call, containing a LCS Location Notification Invoke component set to NotifyLocationAllowed, the UE notifies the UE user of the location request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

6.2.3.3.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals switched off or not present

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Assisted Network Assisted GANSS.
- UE supporting LCS value added location request notification capability.

Test Procedure

The SS initiates authentication and ciphering and sends an SS REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyLocationAllowed. The UE notifies the UE user of the location request. The UE then responds with a RELEASE COMPLETE message containing a LocationNotification return result to terminate the dialogue.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <- | | AUTHENTICATION REQUEST | |
| 2 | -> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyLocationAllowed |
| 5 | UE | | | The UE notifies the UE user of the location request |
| 6 | -> | | RELEASE COMPLETE | The UE terminates the dialogue |
| 7 | SS | | | SS releases the RRC connection and the test case ends |

Specific Message Contents

REGISTER (Step 4)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator<br>Transaction identifier<br>Message type<br>Facility | Call Independent SS message (1011)<br><br>REGISTER (0011 1011)<br>Invoke = lcs-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyLocationAllowed,<br>locationType -> current Location ,<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 6)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = lcs-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

#### 6.2.3.3.5 Test requirements

After step 4 the UE shall notify the UE user of the location request.

After step 5 the UE shall send a RELEASE COMPLETE message.

### 6.2.3.4 Privacy Verification - Location Allowed if No Response

#### 6.2.3.4.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities and LCS value added location request notification capability.

#### 6.2.3.4.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1

#### 6.2.3.4.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationAllowedIfNoResponse, the UE notifies the UE user of the location request and indicates that the default response is location allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

#### 6.2.3.4.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters

- Satellite signals switched off or not present

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Assisted Network Assisted GANSS.
- UE supporting LCS value added location request notification capability.
- px_UeLcsNotification: value for UE LCS Notification timeout timer.

Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be allowed if no user response is received.

The user ignores the location request by taking no action, allowing the verification process to time-out.

The SS send a RELEASE COMPLETE.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 10 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 11 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 12 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 13 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 14 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationAllowedIfNoResponse |
| 15 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 16 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be allowed in the absence of a response |
| 17 | UE | | | The user does not reply |
| 18 | SS | | | SS waits until T(LCSN) expires to ensure that the UE does not send a RELEASE COMPLETE message. |
| 19 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 20 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Steps 4, 9 and 14)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

RELEASE COMPLETE (Step 13)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

RELEASE COMPLETE (Step 19)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |

#### 6.2.3.4.5 Test requirements

After steps 4, 9 and 14 the UE shall notify the UE user of the location request and indicate to the user that location will be allowed in the absence of a response.

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 12 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

During step 18 the UE shall not send any RELEASE COMPLETE message.

### 6.2.3.5 Privacy Verification - Location Not Allowed if No Response

#### 6.2.3.5.1 Definition

This test case applies to all UEs supporting UE-Based or UE-Assisted GANSS or GNSS Location Service capabilities and LCS value added location request notification capability.

#### 6.2.3.5.2 Conformance requirements

1) The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

2) In case of privacy verification the MS shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

3) If the timer expires in the network before any response from the MS (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in GSM 03.71 for GSM and TS 23.171 for UMTS (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

References

- Conformance requirement 1, 2 and 3: TS 24.030, clause 4.1.1

#### 6.2.3.5.3 Test Purpose

To verify that when the UE receives a REGISTER message, containing a LCS Location Notification Invoke component set to notifyAndVerify-LocationNotAllowedIfNoResponse, the UE notifies the UE user of the location request and indicates that the default response is location not allowed if no response and providing the opportunity to accept or deny the request and sends a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied or permissionGranted as appropriate.

#### 6.2.3.5.4 Method of Test

Initial Conditions

System Simulator (SS):

- 1 cell, default parameters
- Satellite signals switched off or not present

UE:

- State "CS-CELL DCH Initial (State 6-1)" as specified in clause 7.4.1 of TS 34.108.

Related PICS/PIXIT Statements

- UE Based Network Assisted GANSS.
- UE Assisted Network Assisted GANSS.
- UE supporting LCS value added location request notification capability.
- px_UeLcsNotification: value for UE LCS Notification timeout timer.

Test Procedure

The SS initiates authentication and ciphering and sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be not allowed if no user response is received.

The user accepts the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionGranted.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be not allowed if no user response is received.

The user denies the location request. The UE responds with a RELEASE COMPLETE message containing a LocationNotification return result with verificationResponse set to permissionDenied.

The SS sends a REGISTER message containing a Facility IE containing a LCS Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse.

The UE notifies the UE user of the location request with the option to accept or deny the request and an indication that location will be not allowed if no user response is received.

The user ignores the location request by taking no action, allowing the verification process to time-out.

The SS send a RELEASE COMPLETE.

Expected Sequence

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | <-- | | AUTHENTICATION REQUEST | |
| 2 | --> | | AUTHENTICATION RESPONSE | |
| 3 | SS | | | SS starts security procedure |
| 4 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 5 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 6 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 7 | UE | | | The user accepts the location request before timer T(LCSN) expires |
| 8 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionGranted |
| 9 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 10 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 11 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 12 | UE | | | The user denies the location request before timer T(LCSN) expires |
| 13 | -> | | RELEASE COMPLETE | Containing a LocationNotification return result with verificationResponse set to permissionDenied |
| 14 | <- | | REGISTER | Call Independent SS containing Facility IE Location Notification Invoke message set to notifyAndVerify-LocationNotAllowedIfNoResponse |
| 15 | SS | | | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification |
| 16 | UE | | | The UE notifies the UE user of the location request and indicates to the user that location will be not allowed in the absence of a response |
| 17 | UE | | | The user does not reply |
| 18 | SS | | | SS waits until T(LCSN) expires to verify that the UE does not send a RELEASE COMPLETE message. |
| 19 | <- | | RELEASE COMPLETE | SS terminates the dialogue |
| 20 | SS | | | SS releases the connection and the test case ends |

Specific Message Contents

REGISTER (Steps 4, 9 and 14)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | REGISTER (0011 1011) |
| Facility | Invoke = LCS-LocationNotification<br>LocationNotificationArg<br>notificationType -> notifyAndVerify-LocationNotAllowedIfNoResponse<br>locationType -> current Location<br>lcsClientExternalID -> externalAddress<br>lcsClientName ->dataCodingScheme<br>nameString |

RELEASE COMPLETE (Step 8)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionGranted |

RELEASE COMPLETE (Step 13)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (xx10 1010) |
| Facility | Return result = LCS-LocationNotification<br>LocationNotificationRes<br>verificationResponse -> permissionDenied |

RELEASE COMPLETE (Step 19)

| Information element | Value/remark |
|---|---|
| Protocol Discriminator | Call Independent SS message (1011) |
| Transaction identifier | |
| Message type | RELEASE COMPLETE (0010 1010) |

#### 6.2.3.5.5 Test requirements

After steps 4, 9 and 14 the UE shall notify the UE user of the location request and indicate to the user that location will be not allowed in the absence of a response.

After step 7 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionGranted.

After step 12 the UE shall send a RELEASE COMPLETE message with verificationResponse set to permissionDenied.

During step 18 the UE shall not send any RELEASE COMPLETE message.

# 7 Protocol Conformance Test Cases for E-UTRAN

## 7.1 NAS Protocol Procedures

### 7.1.1 UE Network Capability

#### 7.1.1.1 Test Purpose (TP)

(1)

```
with { the UE having received an RRCConnectionSetup message. }
ensure that {
  when {    the UE transmits ATTACH REQUEST }
    then {  the UE correctly sets UE Network Capability IE values for LCS and LPP }
            }
```

#### 7.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171 clause 4.2.1 and TS 24.301 clause 9.9.3.3.

[TS 24.171, clause 4.2.1]

The UE announces to the network its ability to support LCS notification mechanism and/or LPP messages using the UE Network Capability IE defined in 3GPP TS 24.301.

[TS 24.301, clause 9.9.3.3]

The purpose of the UE network capability information element is to provide the network with information concerning aspects of the UE related to EPS or interworking with GPRS. The contents might affect the manner in which the network handles the operation of the UE. The UE network capability information indicates general UE characteristics and it shall therefore, except for fields explicitly indicated, be independent of the frequency band of the channel it is sent on.

…

#### 7.1.1.3 Test description

##### 7.1.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.
- Satellite signals switched off or not present

UE:

-

Preamble:

- the UE is in state Switched OFF (state 1) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

-

7.1.1.3.2 Test procedure sequence

**Table 7.1.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | | <-- | RRC: SYSTEM INFORMATION (BCCH) | - | - |
| 1a | UE is switched on. | | | | |
| 2 | UE transmits an *RRCConnectionRequest* message. | --> | RRC: *RRCConnectionRequest* | - | - |
| 3 | SS transmits an *RRCConnectionSetup* message. | <-- | RRC: *RRCConnectionSetup* | - | - |
| 4 | The UE transmits an *RRCConnectionSetupComplete* message to confirm the successful completion of the connection establishment and to initiate the Attach procedure by including the ATTACH REQUEST message. The PDN CONNECTIVITY REQUEST message is piggybacked in ATTACH REQUEST | --> | RRC: *RRCConnectionSetupComplete*<br>NAS: ATTACH REQUEST<br>NAS: PDN CONNECTIVITY REQUEST | 1 | P |
| 5 to 17 | Steps 5 to 17 of the registration procedure described in TS 36.508 subclause 4.5.2.3 are performed.<br>NOTE: The UE performs registration and the RRC connection is released. | | | | |

7.1.1.3.3 Specific message contents

**Table 7.1.1.3.3-1: ATTACH REQUEST (step 4, Table 7.1.1.3.2-1)**

| Derivation Path: 24.301 clause 8.2.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | EMM | | |
| Security header type | '0000'B | Plain NAS message, not security protected | |
| Attach request message identity | '0100 0001'B | Attach request | |
| EPS attach type | '0001'B | EPS attach | EPS_only |
| | '0010'B | combined EPS/IMSI attach | combined_EPS_IMSI |
| NAS key set identifier | Any allowed value | | |
| Old GUTI or IMSI | Any allowed value | | |
| UE network capability | Set according to Table 7.1.1.3.3-2 | | |
| ESM message container | PDN CONNECTIVITY REQUEST message to request PDN connectivity to the default PDN | | |
| Old P-TMSI signature | Not present or any allowed value | | |
| Additional GUTI | Not present or any allowed value | | |
| Last visited registered TAI | Not present or any allowed value | | |
| DRX parameter | Not present or any allowed value | | |
| MS network capability | Not present or any allowed value | | |
| Old location area identification | Not present or any allowed value | | |
| TMSI status | Not present or any allowed value | | |
| Mobile station classmark 2 | Not present or any allowed value | | |
| Mobile station classmark 3 | Not present or any allowed value | | |
| Supported Codecs | Not present or any allowed value | | |
| Additional update type | Not present | | EPS_only |
| Additional update type | Not present or any allowed value | | combined_EPS_IMSI |

| Condition | Explanation |
|---|---|
| EPS_only | See the definition below table 4.7.2-1 in TS 36.508. |
| combined_EPS_IMSI | See the definition below table 4.7.2-1 in TS 36.508. |

NOTE: This message is sent integrity protected when a valid security context exists and without integrity protection otherwise.

**Table 7.1.1.3.3-2: UE network capability (step 4, Table 7.1.1.3.2-1)**

| Derivation Path: 24.301 clause 9.9.3.34 | | | |
|---|---|---|---|
| **Information Element Contents** | **Value/remark** | **Comment** | **Condition** |
| Octet 7, bit 3 | Set according to pc_MT_LR_loc_notif | Location services (LCS) notification mechanisms capability | |
| Octet 7, bit 4 | 1 (LPP supported) | LTE Positioning Protocol (LPP) capability | |
| All other octets/bits | Any allowed value | | |

# 7.2 LCS Procedures

## 7.2.1 Location Notification and Privacy Verification

### 7.2.1.1 Location Notification

#### 7.2.1.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE receives a REGISTER message containing the LCS-LocationNotification Invoke component
            set to NotifyLocationAllowed }
    then {  UE notifies the user of the location procedure and terminates the dialogue by sending a
            RELEASE COMPLETE message }
            }
```

#### 7.2.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171, clause 5.2.1.1.1.

[TS 24.171, clause 5.2.1.1.1]

The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

…

In the case of location notification no response is required from the UE, the UE shall terminate the dialogue by sending a RELEASE COMPLETE message containing a LocationNotification return result.

…

#### 7.2.1.1.3 Test description

##### 7.2.1.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- UE supporting LCS value added location request notification capability.

7.2.1.1.3.2 Test procedure sequence

**Table 7.2.1.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 2 | The UE notifies the user of the location procedure | | | 1 | P |
| 3 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 1 | P |

7.2.1.1.3.3 Specific message contents

**Table 7.2.1.1.3.3-1: *DLInformationTransfer* (step 1, Table 7.2.1.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.1.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.1.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (step 1, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | Set according to Table 7.2.1.1.3.3-3 | REGISTER | |
| Additional information | Not present. | | |

**Table 7.2.1.1.3.3-3: REGISTER (step 1, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Register message type | 0011 1011 | REGISTER | |
| Facility | Invoke = lcs-LocationNotification | Set according to Table 7.2.1.1.3.3-4 | |

**Table 7.2.1.1.3.3-4: LCS-LocationNotification (step 1, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationArg ::= SEQUENCE { | | | |
| notificationType | notifyLocationAllowed | | |
| locationType | currentLocation | | |
| lcsClientExternalID SEQUENCE { | | | |
| externalAddress | ISDN-AddressString | | |
| } | | | |
| lcsClientName SEQUENCE { | | | |
| dataCodingScheme | USSD-DataCodingScheme | | |
| nameString | NameString | | |
| } | | | |
| } | | | |

**Table 7.2.1.1.3.3-5: *ULInformationTransfer* (step 3, Table 7.2.1.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.1.3.3-6 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.1.3.3-6: UPLINK GENERIC NAS TRANSPORT (step 3, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | Set according to Table 7.2.1.1.3.3-7 | RELEASE COMPLETE | |
| Additional information | Not present | | |

**Table 7.2.1.1.3.3-7: RELEASE COMPLETE (step 3, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Facility | Return result = LocationNotificationRes | Set according to Table 7.2.1.1.3.3-8 | |

**Table 7.2.1.1.3.3-8: LocationNotificationRes (step 3, Table 7.2.1.1.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationRes ::= SEQUENCE { | | | |
| verificationResponse | permissionGranted | | |
| } | | | |

## 7.2.1.2 Privacy Verification – Location Allowed if no Response

### 7.2.1.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE receives a REGISTER message containing the LCS-LocationNotification Invoke component
            set to NotifyAndVerify-LocationAllowedIfNoResponse }
    then {  UE notifies the user of the location procedure and indicates that the default response
            is location allowed, allows the user to accept or deny the request and terminates the
            dialogue by sending a RELEASE COMPLETE message with verificationResponse set as
            appropriate }
            }
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE receives a REGISTER message containing the LCS-LocationNotification Invoke component
            set to NotifyAndVerify-LocationAllowedIfNoResponse }
    then {  UE notifies the user of the location procedure and indicates that the default response
            is location allowed, allows the user to accept or deny the request and waits for the
            user to respond }
            }
```

#### 7.2.1.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171, clause 5.2.1.1.1.

[TS 24.171, clause 5.2.1.1.1]

The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

In case of privacy verification the UE shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

If the timer T(LCSN) expires in the network before any response from the UE (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in 3GPP TS 23.271 (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

…

#### 7.2.1.2.3 Test description

##### 7.2.1.2.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- UE supporting LCS value added location request notification capability.
- px_UeLcsNotification: value for UE LCS Notification timeout timer.

7.2.1.2.3.2 Test procedure sequence

**Table 7.2.1.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 2 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 3 | The UE notifies the user of the location procedure and indicates that location will be allowed in the absence of a response | | | 1 | P |
| 4 | The user accepts the location request before timer T(LCSN) expires | | | - | - |
| 5 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 1 | P |
| 6 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 7 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 8 | The UE notifies the user of the location procedure and indicates that location will be allowed in the absence of a response | | | 1 | P |
| 9 | The user denies the location request before timer T(LCSN) expires | | | - | - |
| 10 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 1 | P |
| 11 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 12 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 13 | The UE notifies the user of the location procedure and indicates that location will be allowed in the absence of a response | | | 2 | P |
| 14 | The user does not reply | | | - | - |
| 15 | SS waits until T(LCSN) expires to ensure that the UE does not send a RELEASE COMPLETE message. | | | 2 | P |
| 16 | The SS terminates the dialogue by sending a RELEASE COMPLETE message. | <-- | *DLInformationTransfer* (RELEASE COMPLETE) | - | - |

7.2.1.2.3.3 Specific message contents

**Table 7.2.1.2.3.3-1: *DLInformationTransfer* (steps 1, 6, 11 and 16, Table 7.2.1.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.2.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.2.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 1, 6, 11 and 16, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | **Step 1, 6, 11:** Set according to Table 7.2.1.2.3.3-3 | REGISTER | |
| | **Step 16:** Set according to Table 7.2.1.2.3.3-11 | RELEASE COMPLETE | |
| Additional information | Not present. | | |

**Table 7.2.1.2.3.3-3: REGISTER (steps 1, 6, and 11, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Register message type | 0011 1011 | REGISTER | |
| Facility | Invoke = lcs-LocationNotification | Set according to Table 7.2.1.2.3.3-4 | |

**Table 7.2.1.2.3.3-4: LCS-LocationNotification (step 1, 6 and 11, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationArg ::= SEQUENCE { | | | |
| notificationType | notifyAndVerify-LocationAllowedIfNoResponse | | |
| locationType | currentLocation | | |
| lcsClientExternalID SEQUENCE { | | | |
| externalAddress | ISDN-AddressString | | |
| } | | | |
| lcsClientName SEQUENCE { | | | |
| dataCodingScheme | USSD-DataCodingScheme | | |
| nameString | NameString | | |
| } | | | |
| } | | | |

**Table 7.2.1.2.3.3-5: *ULInformationTransfer* (steps 5 and 10, Table 7.2.1.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.2.3.3-6 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.2.3.3-6: UPLINK GENERIC NAS TRANSPORT (steps 5 and 10, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | **Step 5:** Set according to Table 7.2.1.2.3.3-7 | RELEASE COMPLETE | |
| | **Step 10:** Set according to Table 7.2.1.2.3.3-9 | RELEASE COMPLETE | |
| Additional information | Not present | | |

**Table 7.2.1.2.3.3-7: RELEASE COMPLETE (step 5, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Facility | Return result = LocationNotificationRes | Set according to Table 7.2.1.2.3.3-8 | |

**Table 7.2.1.2.3.3-8: LocationNotificationRes (step 5, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationRes ::= SEQUENCE { | | | |
| verificationResponse | permissionGranted | | |
| } | | | |

**Table 7.2.1.2.3.3-9: RELEASE COMPLETE (step 10, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Facility | Return result = LocationNotificationRes | Set according to Table 7.2.1.2.3.3-10 | |

**Table 7.2.1.2.3.3-10: LocationNotificationRes (step 10, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationRes ::= SEQUENCE { | | | |
| verificationResponse | permissionDenied | | |
| } | | | |

**Table 7.2.1.2.3.3-11: RELEASE COMPLETE (step 16, Table 7.2.1.2.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Cause | 31 = Normal Unspecified | Set according to TS 24.008 | |

### 7.2.1.3 Privacy Verification – Location not Allowed if No Response

#### 7.2.1.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE receives a REGISTER message containing the LCS-LocationNotification Invoke component
            set to NotifyAndVerify-LocationNotAllowedIfNoResponse }
    then {  UE notifies the user of the location procedure and indicates that the default response
            is location not allowed, allows the user to accept or deny the request and terminates
            the dialogue by sending a RELEASE COMPLETE message with verificationResponse set as
            appropriate }
            }
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE receives a REGISTER message containing the LCS-LocationNotification Invoke component
            set to NotifyAndVerify-LocationNotAllowedIfNoResponse }
    then {  UE notifies the user of the location procedure and indicates that the default response
            is location not allowed, allows the user to accept or deny the request and waits for the
            user to respond }
            }
```

#### 7.2.1.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171, clause 5.2.1.1.1.

[TS 24.171, clause 5.2.1.1.1]

The network invokes a location notification procedure by sending a REGISTER message containing a LCS-LocationNotification invoke component to the UE. This may be sent either to request verification for MT-LR or to notify about already authorized MT-LR.

In case of privacy verification the UE shall respond to the request by sending a RELEASE COMPLETE message containing the mobile subscriber's response in a return result component.

If the timer T(LCSN) expires in the network before any response from the UE (e.g. due to no response from the user), the network shall interpret this by applying the default treatment defined in 3GPP TS 23.271 (i.e. disallow location if barred by subscription and allow location if allowed by subscription).

…

#### 7.2.1.3.3 Test description

##### 7.2.1.3.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- UE supporting LCS value added location request notification capability.

- px_UeLcsNotification: value for UE LCS Notification timeout timer.

7.2.1.3.3.2 Test procedure sequence

**Table 7.2.1.3.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 2 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 3 | The UE notifies the user of the location procedure and indicates that location will be not allowed in the absence of a response | | | 1 | P |
| 4 | The user accepts the location request before timer T(LCSN) expires | | | - | - |
| 5 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 1 | P |
| 6 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 7 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 8 | The UE notifies the user of the location procedure and indicates that location will be not allowed in the absence of a response | | | 1 | P |
| 9 | The user denies the location request before timer T(LCSN) expires | | | - | - |
| 10 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 1 | P |
| 11 | The SS sends a REGISTER message containing a LCS-LocationNotification Invoke component. | <-- | *DLInformationTransfer* (REGISTER) | - | - |
| 12 | SS starts timer T(LCSN) set to 90% of px_UeLcsNotification | | | - | - |
| 13 | The UE notifies the user of the location procedure and indicates that location will be not allowed in the absence of a response | | | 2 | P |
| 14 | The user does not reply | | | - | - |
| 15 | SS waits until T(LCSN) expires to ensure that the UE does not send a RELEASE COMPLETE message. | | | 2 | P |
| 16 | The SS terminates the dialogue by sending a RELEASE COMPLETE message. | <-- | *DLInformationTransfer* (RELEASE COMPLETE) | - | - |

7.2.1.3.3.3 Specific message contents

**Table 7.2.1.3.3.3-1: *DLInformationTransfer* (steps 1, 6, 11 and 16, Table 7.2.1.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.3.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.3.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 1, 6, 11 and 16, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | **Step 1, 6, 11:** Set according to Table 7.2.1.3.3.3-3 | REGISTER | |
| | **Step 16:** Set according to Table 7.2.1.3.3.3-11 | RELEASE COMPLETE | |
| Additional information | Not present. | | |

**Table 7.2.1.3.3.3-3: REGISTER (steps 1, 6, and 11, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Register message type | 0011 1011 | REGISTER | |
| Facility | Invoke = lcs-LocationNotification | Set according to Table 7.2.1.3.3.3-4 | |

**Table 7.2.1.3.3.3-4: LCS-LocationNotification (step 1, 6 and 11, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationArg ::= SEQUENCE { | | | |
| notificationType | notifyAndVerify-LocationNotAllowedIfNoResponse | | |
| locationType | currentLocation | | |
| lcsClientExternalID SEQUENCE { | | | |
| externalAddress | ISDN-AddressString | | |
| } | | | |
| lcsClientName SEQUENCE { | | | |
| dataCodingScheme | USSD-DataCodingScheme | | |
| nameString | NameString | | |
| } | | | |
| } | | | |

**Table 7.2.1.3.3.3-5: *ULInformationTransfer* (steps 5 and 10, Table 7.2.1.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.1.3.3.3-6 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.1.3.3.3-6: UPLINK GENERIC NAS TRANSPORT (steps 5 and 10, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | **Step 5:** Set according to Table 7.2.1.3.3.3-7 | RELEASE COMPLETE | |
| | **Step 10:** Set according to Table 7.2.1.3.3.3-9 | RELEASE COMPLETE | |
| Additional information | Not present | | |

**Table 7.2.1.3.3.3-7: RELEASE COMPLETE (step 5, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Facility | Return result = LocationNotificationRes | Set according to Table 7.2.1.3.3.3-8 | |

**Table 7.2.1.3.3.3-8: LocationNotificationRes (step 5, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationRes ::= SEQUENCE { | | | |
| verificationResponse | permissionGranted | | |
| } | | | |

**Table 7.2.1.3.3.3-9: RELEASE COMPLETE (step 10, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Facility | Return result = LocationNotificationRes | Set according to Table 7.2.1.3.3.3-10 | |

**Table 7.2.1.3.3.3-10: LocationNotificationRes (step 10, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LocationNotificationRes ::= SEQUENCE { | | | |
| verificationResponse | permissionDenied | | |
| } | | | |

**Table 7.2.1.3.3.3-11: RELEASE COMPLETE (step 16, Table 7.2.1.3.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |
| Cause | 31 = Normal Unspecified | Set according to TS 24.008 | |

## 7.2.2 EPC MO-LR

### 7.2.2.1 Autonomous Self Location: UE-based

#### 7.2.2.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { an EPC-MO-LR location session is initiated at the UE of type "assistanceData"  }
    then { UE sends a REGISTER message containing a LCS-MOLR invoke component }
}
```

(2)

```
with { UE having performed the last location request operation }
ensure that {
  when { UE has received a FACILITY message containing the LCS-MOLR return result component }
    then { UE terminates the dialogue by sending a RELEASE COMPLETE message }
}
```

#### 7.2.2.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171, clause 5.2.2.1.

[TS 24.171, clause 5.2.2.1.1]

The UE invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

...

The network shall pass the result of the location procedure to the UE by sending a FACILITY message to the UE containing a LCS-MOLR return result component.

…

After the last location request operation the UE shall terminate the dialogue by sending a RELEASE COMPLETE message.

…

#### 7.2.2.1.3 Test description

##### 7.2.2.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.
- Satellite signals (Sub-test 15): As specified in 5.2.1.
- MBS signals (Sub-test 16): as specified in 5.2.4.
- WLAN signals (Sub-test 17): as specified in 5.2.5.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- Method of triggering an EPC-MO-LR request for assistance data.

#### 7.2.2.1.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.2.2.1.3.2-0 below:

**Table 7.2.2.1.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.2.2.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 0 | The SS sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 0A | Cause the UE to initiate MO-LR procedure | - | - | - | - |
| 1 | The UE sends a NAS PDU containing an MO-LR Request of type "assistanceData" inside an RRC UL Information Transfer message. The embedded LPP message specifies the type of assistance data. | --> | *ULInformationTransfer* (REGISTER) | 1 | P |
| 2 | The SS provides the requested assistance data in an LPP message of type "Assistance Data". | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 3 | The SS sends a FACILITY message containing a LCS-MOLR return result component. | <-- | *DLInformationTransfer* (FACILITY) | - | - |
| 4 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer* (RELEASE COMPLETE) | 2 | P |

#### 7.2.2.1.3.3 Specific message contents

**Table 7.2.2.1.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 0, Table 7.2.2.1.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 15: 0 0 0 0 0 0 0 0<br>Sub-test 16: 0 0 0 0 0 0 1 0<br>Sub-test 17: 0 0 0 0 0 0 1 1<br>Sub-test 18: 0 0 0 0 0 1 0 1 | Sub-test 15: AGNSS<br>Sub-test 16: MBS<br>Sub-test 17: WLAN<br>Sub-test 18: Sensor | |

**Table 7.2.2.1.3.3-1: *ULInformationTransfer* (steps 1 and 4, Table 7.2.2.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.2.1.3.3-2 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.1.3.3-2: UPLINK GENERIC NAS TRANSPORT (steps 1 and 4, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000010 | Location services message container | |
| Generic message container | **Step 1:** Set according to Table 7.2.2.1.3.3-3 | REGISTER | |
| | **Step 4:** Set according to Table 7.2.2.1.3.3-11 | RELEASE COMPLETE | |
| Additional information | Not present | | |

**Table 7.2.2.1.3.3-3: REGISTER (step 1, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.080 Table 2.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Register message type | xx11 1011 | REGISTER | |
| Facility | Invoke=LCS-MOLR | Set according to Table 7.2.2.1.3.3-4 | |
| SS version | Version 1 or above | | |

**Table 7.2.2.1.3.3-4: LCS-MOLRArg (step 1, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-MOLRArg ::= SEQUENCE { | | | |
| molr-Type | assistanceData | | |
| multiplePositioningProtocolPDUs SEQUENCE (SIZE (1..3)) OF OCTET STRING | At least one LPP message of type Request Assistance Data (UE may include additional LPP messages) | Set according to Table 7.2.2.1.3.3-5 | |
| } | | | |

**Table 7.2.2.1.3.3-5: LPP Request Assistance Data (step 1, Table 7.2.2.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| commonIEsRequestAssistanceData | Present or not present | | |
| a-gnss-RequestAssistanceData | Present for sub-test 15. May be present for other sub-tests | | |
| otdoa-RequestAssistanceData | Not present | | |
| epdu-RequestAssistanceData | Not present | | |
| sensor-RequestAssistanceData-r14 | Present for sub-test 18. May be present for other sub-tests | Rel-14 onwards | |
| tbs-RequestAssistanceData-r14 | Present for sub-test 16. May be present for other sub-tests | Rel-14 onwards | |
| wlan-RequestAssistanceData-r14 | Present for sub-test 17. May be present for other sub-tests | Rel-14 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.1.3.3-6: *DLInformationTransfer* (steps 2 and 3, Table 7.2.2.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.2.1.3.3-7 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.1.3.3-7: DOWNLINK GENERIC NAS TRANSPORT (steps 2 and 3, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | **Step 2:** 00000001 | LTE Positioning Protocol (LPP) message container | |
| | **Step 3:** 00000010 | Location services message container | |
| Generic message container | **Step 2:** Set according to Table 7.2.2.1.3.3-8 | LPP Provide Assistance Data | |
| | **Step 3:** Set according to Table 7.2.2.1.3.3-9 | FACILITY | |
| Additional information | **Step 2:** Present | Routing Identifier/ Correlation ID | |
| | **Step 3:** Not present. | | |

**Table 7.2.2.1.3.3-8: LPP Provide Assistance Data (step 2, Table 7.2.2.1.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Assistance Data message in step 1 Table 7.2.2.1.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| a-gnss-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 1, Table 7.2.2.1.3.2-1 which are available according to TS 37.571-5 [12]. | | |
| sensor-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 1, Table 7.2.2.1.3.2-1 which are available according to subclause 5.4.1.5. | Rel-14 onwards | |
| tbs-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 1, Table 7.2.2.1.3.2-1 which are available according to subclause 5.4.1.3. | Rel-14 onwards | |
| wlan-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 1, Table 7.2.2.1.3.2-1 which are available according to subclause 5.4.1.4. | Rel-14 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.1.3.3-9: FACILITY (step 3, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Facility message type | 0011 1010 | FACILITY | |
| Facility | Return Result=LCS-MOLRRes | Set according to Table 7.2.2.1.3.3-10 | |

**Table 7.2.2.1.3.3-10: LCS-MOLRRes (step 3, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-MOLRRes::= SEQUENCE { | empty | | |
| } | | | |

**Table 7.2.2.1.3.3-11: RELEASE COMPLETE (step 4, Table 7.2.2.1.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |

## 7.2.2.2 Basic Self Location: UE-assisted

### 7.2.2.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { an EPC-MO-LR location session is initiated at the UE of type "locationEstimate"  }
    then { UE sends a REGISTER message containing a LCS-MOLR invoke component }
            }
```

(2)

```
with { UE having performed the last location request operation }
ensure that {
  when { UE has received a FACILITY message containing the LCS-MOLR return result component }
    then { UE terminates the dialogue by sending a RELEASE COMPLETE message }
            }
```

### 7.2.2.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 24.171, clause 5.2.2.1.

[TS 24.171, clause 5.2.2.1.1]

The UE invokes a MO-LR by sending a REGISTER message to the network containing a LCS-MOLR invoke component. SS Version Indicator value 1 or above shall be used.

...

The network shall pass the result of the location procedure to the UE by sending a FACILITY message to the UE containing a LCS-MOLR return result component.

…

After the last location request operation the UE shall terminate the dialogue by sending a RELEASE COMPLETE message.

…

#### 7.2.2.2.3 Test description

##### 7.2.2.2.3.1 Pre-test conditions

System Simulator:

- Sub-tests 11, 12, 13, 14, 15, 16, 17, 18: Cell 1.
- Sub-test 5: Cell 1, Cell 2 as specified in 5.2.2.
- Sub-tests 6 FDD, 6 TDD: Cell 1, Cell 2 as specified in 5.2.3.
- Satellite signals (Sub-test 15): As specified in 5.2.1.
- WLAN signals (Sub-test 11, 17): as specified in 5.2.5.
- MBS signals (Sub-tests 12, 16): as specified in 5.2.4.
- Bluetooth signals (Sub-test 13): as specified in 5.2.6.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

- Method of triggering an EPC-MO-LR request for a location estimate.

##### 7.2.2.2.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.2.2.2.3.2-0 below:

**Table 7.2.2.2.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA |
| 6 FDD | UE supporting ECID (FDD) |
| 6 TDD | UE supporting ECID (TDD) |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 14 | UE supporting Sensor (Rel-13 only) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.2.2.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 0 | IF NOT<br>sub-test 6 FDD or sub-test 6 TDD<br>THEN<br>The SS sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 0A | Cause the UE to initiate MO-LR procedure | - | - | - | - |
| 1 | The UE sends a NAS PDU containing an MO-LR Request of type "locationEstimate" inside an RRC UL Information Transfer message. The MO-LR message may optionally include up to three LPP positioning messages. | --> | *ULInformationTransfer*<br>(REGISTER) | 1 | P |
| 2a | IF<br>the UE does not include a LPP Provide Capabilities message in step 1<br>THEN<br>the SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer*<br>(LPP REQUEST CAPABILITIES) | - | - |
| 2b | IF<br>the SS performed step 2a<br>THEN<br>the UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer*<br>(LPP PROVIDE CAPABILITIES) | - | - |
| 2c | IF<br>the UE LPP message at step 2b includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 2d | IF<br>the UE included a LPP message of type Request Assistance Data in step 1<br>THEN<br>SS sends a LPP message of type Provide Assistance Data including an error indication without assistance data. | <-- | *DLInformationTransfer*<br>(LPP PROVIDE ASSISTANCE DATA) | - | - |
| 3 | IF NOT<br>sub-test 6 FDD or sub-test 6 TDD or sub-test-11 or sub-test 12 or sub-test 13 or sub-test 14<br>THEN<br>The SS sends a LPP message of type Provide Assistance Data including the assistance data as defined in subclause 5.4.1, dependent on UE capabilities. | <-- | *DLInformationTransfer*<br>(LPP PROVIDE ASSISTANCE DATA) | - | - |
| 4 | The SS sends a LPP message of type Request Location Information. | <-- | *DLInformationTransfer*<br>(LPP REQUEST LOCATION INFORMATION) | - | - |
| 5 | The UE sends a LPP message of type Provide Location Information including measurements as requested at step 4. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | - | - |
| 5a | IF<br>the UE LPP message at step 5 includes an acknowledgement request<br>THEN<br>the SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 6 | The SS sends a FACILITY message containing a LCS-MOLR return result component. | <-- | *DLInformationTransfer*<br>(FACILITY) | - | - |
| 7 | The UE terminates the dialogue by sending a RELEASE COMPLETE message. | --> | *ULInformationTransfer*<br>(RELEASE COMPLETE) | 2 | P |

7.2.2.2.3.3 Specific message contents

**Table 7.2.2.2.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 0, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 15: 0 0 0 0 0 0 0 0<br>Sub-test 5: 0 0 0 0 0 0 0 1<br>Sub-test 11, 17: 0 0 0 0 0 0 1 1<br>Sub-tests 12, 16: 0 0 0 0 0 0 1 0<br>Sub-test 13: 0 0 0 0 0 1 0 0<br>Sub-test 14, 18: 0 0 0 0 0 1 0 1 | Sub-test 15: AGNSS<br>Sub-test 5: OTDOA<br>Sub-test 11, 17: WLAN<br>Sub-tests 12, 16: MBS<br>Sub-test 13: Bluetooth<br>Sub-test 14, 18: Sensor | |

**Table 7.2.2.2.3.3-1: *ULInformationTransfer* (steps 1, 2b, 5 and 7, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.2.2.3.3-2 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.3.3-2: UPLINK GENERIC NAS TRANSPORT (steps 1, 2b, 5 and 7, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | **Steps 1 and 7:** 00000010 | Location services message container | |
| | **Step 2b, and 5:** 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1:** Set according to Table 7.2.2.2.3.3-3 | REGISTER | |
| | **Step 2b:** Set according to Table 7.2.2.2.3.3-8 | LPP Provide Capabilities | |
| | **Step 5:** Set according to Table 7.2.2.2.3.3-13 | LPP Provide Location Information | |
| | **Step 7:** Set according to Table 7.2.2.2.3.3-16 | RELEASE COMPLETE | |
| Additional information | **Steps 1 and 7:** Not present | | |
| | **Step 2b:** Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 2a Table 7.2.2.2.3.2-1) | |
| | **Step 5:** | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 4 Table 7.2.2.2.3.2-1) | |

**Table 7.2.2.2.3.3-3: REGISTER (step 1, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 Table 2.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Register message type | xx11 1011 | REGISTER | |
| Facility | Invoke=LCS-MOLR | Set according to Table 7.2.2.2.3.3-4 | |
| SS version | Version 1 or above | | |

**Table 7.2.2.2.3.3-4: LCS-MOLRArg (step 1, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-MOLRArg ::= SEQUENCE { | | | |
| molr-Type | locationEstimate | | |
| multiplePositioningProtocolPDUs SEQUENCE (SIZE (1..3)) OF OCTET STRING | May include up to three LPP messages | | |
| } | | | |

**Table 7.2.2.2.3.3-5: *DLInformationTransfer* (steps 2a, 2c, 2d, 3, 4, 5a and 6, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.2.2.2.3.3-6 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.3.3-6: DOWNLINK GENERIC NAS TRANSPORT (steps 2a, 2c, 2d, 3, 4, 5a and 6, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | **Step 2a, 2c, 2d, 3, 4, 5a:** 00000001 | LTE Positioning Protocol (LPP) message container | |
| | **Step 6:** 00000010 | Location services message container | |
| Generic message container | **Step 2a:** Set according to Table 7.2.2.2.3.3-7 | LPP Request Capabilities | |
| | **Step 2c, 5a:** Set according to Table 7.2.2.2.3.3-9 | LPP Acknowledgement | |
| | **Step 2d:** Set according to Table 7.2.2.2.3.3-10 | LPP Provide Assistance Data | |
| | **Step 3:** Set according to Table 7.2.2.2.3.3-11 | LPP Provide Assistance Data | |
| | **Step 4:** Set according to Table 7.2.2.2.3.3-12 | LPP Request Location Information | |
| | **Step 6:** Set according to Table 7.2.2.2.3.3-14 | FACILITY | |
| Additional information | **Steps 2a, 2c, 2d, 3, 4, 5a:** Present | Routing Identifier/ Correlation ID | |
| | **Step 6:** Not present. | | |

**Table 7.2.2.2.3.3-7: LPP Request Capabilities (step 2a, Table 7.2.2.2.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.2.2.2.3.3-8: LPP Provide Capabilities (step 2b, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 2a Table 7.2.2.2.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | Contains a different value compared to any other UL message already sent by the UE. | |
| acknowledgement SEQUENCE { | Present, or not present. | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities SEQUENCE{ | Dependent on UE capabilities | | |
| ueRxTxSupTDD-r13 | Present (TRUE) for sub-test 6 TDD | Rel-13 onwards | |
| } | | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.3.3-9: LPP Acknowledgement (steps 2c and 5a, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | **Step 2c:**<br>(0..255) | Contains the same value of the sequenceNumber field as received by the SS in the LPP Provide Capabilities message in step 2b, Table 7.2.2.2.3.2-1. | |
| | **Step 5a:**<br>(0..255) | Contains the same value of the sequenceNumber field as received by the SS in the LPP Provide Location Information message in step 5, Table 7.2.2.2.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.2.2.2.3.3-10: LPP Provide Assistance Data (step 2d, Table 7.2.2.2.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | Contains the same value as any potential LPP Request Assistance Data message included by the UE at step 1, Table 7.2.2.2.3.2-1. | |
| Initiator | targetDevice | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present. | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData SEQUENCE { | Present, if UE requested GNSS assistance data at step 1, Table 7.2.2.2.3.2-1. | | |
| gnss-CommonAssistData | Not present | | |
| gnss-GenericAssistData | Not present | | |
| gnss-Error CHOICE { | | | |
| locationServerErrorCauses SEQUENCE { | | | |
| cause | undefined | | |
| } | | | |
| } | | | |
| } | | | |
| otdoa-ProvideAssistanceData SEQUENCE { | Present, if UE requested OTDOA assistance data at step 1, Table 7.2.2.2.3.2-1. | | |
| otdoa-ReferenceCellInfo | Not present | | |
| otdoa-NeighbourCellInfo | Not present | | |
| otdoa-Error CHOICE { | | | |
| locationServerErrorCauses SEQUENCE { | | | |
| cause | undefined | | |
| } | | | |
| } | | | |
| } | | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 SEQUENCE { | Present, if UE requested Sensor assistance data at step 1, Table 7.2.2.2.3.2-1. | Rel-14 onwards | |
| sensor-AssistanceDataList-r14 | Not present | | |
| sensor-Error-r14 CHOICE{ | | | |
| locationServerErrorCauses-r13 SEQUENCE { | | | |
| cause-r13 | undefined | | |
| } | | | |
| } | | | |
| } | | | |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | Present, if UE requested MBS assistance data at step 1, Table 7.2.2.2.3.2-1. | Rel-14 onwards | |

| | | | |
|---|---|---|---|
| tbs-AssistanceDataList-r14 | Not present | | |
| tbs-Error-r14 CHOICE{ | | | |
| locationServerErrorCauses-r13 SEQUENCE { | | | |
| cause-r13 | undefined | | |
| } | | | |
| } | | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 SEQUENCE { | Present, if UE requested WLAN assistance data at step 1, Table 7.2.2.2.3.2-1. | Rel-14 onwards | |
| wlan-AssistanceDataList-r14 | Not present | | |
| wlan-Error-r14 CHOICE{ | | | |
| locationServerErrorCauses-r13 SEQUENCE { | | | |
| cause-r13 | undefined | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.3.3-11: LPP Provide Assistance Data (step 3, Table 7.2.2.2.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 7.2.2.2.3.3-12: LPP Request Location Information (step 4, Table 7.2.2.2.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |

**Table 7.2.2.2.3.3-13: LPP ProvideLocation Information (step 5, Table 7.2.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Location Information message in step 4 Table 7.2.2.2.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | Contains a different value compared to any other UL message already sent by the UE. | |
| acknowledgement SEQUENCE { | Present, or not present. | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | May be present | | |
| locationEstimate | Not present | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | Not present | Rel-12 onwards | |
| } | | | |
| a-gnss-ProvideLocationInformation | Present for sub-test 15. Any value acceptable | | |
| otdoa-ProvideLocationInformation | Present for sub-test 5. Any value acceptable | | |
| ecid-ProvideLocationInformation | Present for sub-tests 6 FDD, 6 TDD. Any value acceptable | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Present for sub-test 14, 18 Any value acceptable | Rel-13 onwards | |
| tbs-ProvideLocationInformation-r13 | Present for sub-tests 12, 16 Any value acceptable | Rel-13 onwards | |
| wlan-ProvideLocationInformation-r13 | Present for sub-test 11, 17 Any value acceptable | Rel-13 onwards | |
| bt-ProvideLocationInformation-r13 | Present for sub-test 13 Any value acceptable | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.3.3-14: FACILITY (step 6, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Facility message type | 0011 1010 | FACILITY | |
| Facility | Return Result=LCS-MOLRRes | Set according to Table 7.2.2.2.3.3-15 | |

**Table 7.2.2.2.3.3-15: LCS-MOLRRes (step 6, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-MOLRRes::= SEQUENCE { | | | |
| locationEstimate | Any value. The SS shall not be required to calculate the value from the returned measurements. | | |
| } | | | |

**Table 7.2.2.2.3.3-16: RELEASE COMPLETE (step 7, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 Table 2.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Release Complete message type | xx10 1010 | RELEASE COMPLETE | |

# 7.3 LPP Procedures

## 7.3.1 LPP Common Procedures

### 7.3.1.1 Position Capability Transfer

#### 7.3.1.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives a LPP message of type REQUEST CAPABILITIES }
    then { UE sends a LPP message of type PROVIDE CAPABILITIES with the correct supported
capabilities }
}
```

#### 7.3.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.1.

[TS 36.355, clause 5.1.3]

Upon receiving a *RequestCapabilities* message, the target device shall generate a *ProvideCapabilities* message as a response.

The target device shall:

1> for each positioning method for which a request for capabilities is included in the message:

2> if the target device supports this positioning method:

3> include the capabilities of the device for that supported positioning method in the response message;

1> set the IE *LPP-TransactionID* in the response message to the same value as the IE *LPP-TransactionID* in the received message;

…

[TS 36.355, clause 5.1.4]

When triggered to transmit a *ProvideCapabilities* message, the target device shall:

1> for each positioning method whose capabilities are to be indicated:

2> set the corresponding IE to include the device's capabilities;

2> if OTDOA capabilities are to be indicated:

3> include the IE *supportedBandListEUTRA*;

…

#### 7.3.1.1.3 Test description

##### 7.3.1.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

-

7.3.1.1.3.2 Test procedure sequence

**Table 7.3.1.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | 1 | P |
| 2a | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

7.3.1.1.3.3 Specific message contents

**Table 7.3.1.1.3.3-1: *DLInformationTransfer* (steps 1 and 2a, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.1.1.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 1 and 2a, Table 7.3.1.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1:** Set according to Table 7.3.1.1.3.3-3 | LPP Request Capabilities | |
| | **Step 2a:** Set according to Table 7.3.1.1.3.3-14 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 7.3.1.1.3.3-3: LPP Request Capabilities (step 1, Table 7.3.1.1.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 with the following exception: | | | |
| epdu-RequestCapabilities SEQUENCE (SIZE (1)) OF SEQUENCE{ | | | |
| ePDU-Identifier SEQUENCE { | | | |
| ePDU-ID | 1 | OMA LPPe | |
| ePDU-Name | Not present | | |
| } | | | |
| ePDU-Body | Set according to Table 7.3.1.1.3.3-15 | | |
| } | | | |

**Table 7.3.1.1.3.3-4: *ULInformationTransfer* (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.1.1.3.3-5 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-5: UPLINK GENERIC NAS TRANSPORT (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | Set according to Table 7.3.1.1.3.3-6 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.1.1.3.2-1) | |

**Table 7.3.1.1.3.3-6: LPP Provide Capabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 7.3.1.1.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities SEQUENCE { | Dependent on UE capabilities | Rel-14 onwards | |
| segmentationInfo-r14 | Not present | | |
| lpp-message-segmentation-r14 | Present or not present and value dependent on pc_LPP_MsgSegmentation | Rel-14 onwards | |
| } | | | |
| a-gnss-ProvideCapabilities SEQUENCE { | Present or not present dependent on (pc_UEB_AGNSS OR pc_UEA_AGNSS) | | |
| gnss-SupportList | Set according to Table 7.3.1.1.3.3-7 | | |
| assistanceDataSupportList | Set according to Table 7.3.1.1.3.3-8 | | |
| locationCoordinateTypes | Present or not present dependent on pc_UEB_AGNSS. Set according to Table 7.3.1.1.3.3-9 | | |
| velocityTypes | Present or not present dependent on pc_UEB_AGNSS. Set according to Table 7.3.1.1.3.3-10 | | |
| periodicalReportingNotSupported-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| } | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA. Set according to Table 7.3.1.1.3.3-11 | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID. Set according to Table 7.3.1.1.3.3-12 | | |

| | | | |
|---|---|---|---|
| epdu-ProvideCapabilities | Present or not present dependent on UE capabilities. Set according to Table 7.3.1.1.3.3-13 | | |
| sensor-ProvideCapabilities-r13 | Present or not present dependent on UE capabilities. Set according to Table 7.3.1.1.3.3-20 | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Present or not present dependent on pc_UA_MBS. Set according to Table 7.3.1.1.3.3-17 | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Present or not present dependent on UE capabilities. Set according to Table 7.3.1.1.3.3-18 | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Present or not present dependent on UE capabilities. Set according to Table 7.3.1.1.3.3-19 | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-7: gnss-SupportList (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.2.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| gnss-SupportList SEQUENCE (SIZE(1..n)) OF SEQUENCE{ | | Size n of SEQUENCE is dependent on UE capabilities | |
| gnss-ID | Dependent on UE capabilities | | |
| sbas-IDs | Dependent on UE capabilities | Present only if gnss-ID = sbas | |
| agnss-Modes | Dependent on UE capabilities | | |
| gnss-Signals | Dependent on UE capabilities | | |
| fta-MeasSupport SEQUENCE { | Present or not present dependent on pc_GNSS_FTA | | |
| cellTime | Dependent on UE capabilities | | |
| mode | Dependent on UE capabilities | | |
| } | | | |
| adr-Support | Dependent on UE capabilities | | |
| velocityMeasurementSupport | Dependent on UE capabilities | | |
| adrEnhancementsSupport-r15 | Dependent on UE capabilities | Rel-15 onwards | |
| ha-gnss-Modes-r15 | Dependent on UE capabilities | Rel-15 onwards | |
| } | | | |

**Table 7.3.1.1.3.3-8: assistanceDataSupportList (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.2.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| assistanceDataSupportList SEQUENCE{ | | | |
| gnss-CommonAssistanceDataSupport SEQUENCE{ | | | |
| gnss-ReferenceTimeSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-ReferenceLocationSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-IonosphericModelSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-EarthOrientationParametersSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-RTK-ReferenceStationInfoSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-RTK-AuxiliaryStationDataSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| } | | | |
| gnss-GenericAssistanceDataSupport SEQUENCE (SIZE (1..n)) OF SEQUENCE{ | | Size n of SEQUENCE is dependent on UE capabilities | |
| gnss-ID | Dependent on UE capabilities | | |
| sbas-ID | Dependent on UE capabilities | Present only if gnss-ID = sbas | |
| gnss-TimeModelsSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-DifferentialCorrectionsSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-NavigationModelSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-RealTimeIntegritySupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-DataBitAssistanceSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-AcquisitionAssistanceSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-AlmanacSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-UTC-ModelSupport | Present or not present and value dependent on UE capabilities. | | |
| gnss-AuxiliaryInformationSupport | Present or not present and value dependent on UE capabilities. | | |
| bds-DifferentialCorrectionsSupport-r12 | Present or not present and value dependent on UE capabilities. | Rel-12 onwards | |
| bds-GridModelSupport-r12 | Present or not present and value dependent on UE capabilities. | Rel-12 onwards | |
| gnss-RTK-ObservationsSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |

| | | | |
|---|---|---|---|
| glo-RTK-BiasInformationSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-RTK-MAC-CorrectionDifferencesSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-RTK-ResidualsSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-RTK-FKP-GradientsSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-SSR-OrbitCorrectionsSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-SSR-ClockCorrectionsSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-SSR-CodeBiasSupport-r15 | Present or not present and value dependent on UE capabilities. | Rel-15 onwards | |
| gnss-SSR-URA-Support-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| gnss-SSR-PhaseBiasSupport-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| gnss-SSR-STEC-CorrectionSupport-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| gnss-SSR-GriddedCorrectionSupport-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| navic-DifferentialCorrectionsSupport-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| navic-GridModelSupport-r16 | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-9: locationCoordinateTypes (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.4.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| locationCoordinateTypes SEQUENCE { | | | |
| ellipsoidPoint | Dependent on UE capabilities | | |
| ellipsoidPointWithUncertaintyCircle | Dependent on UE capabilities | | |
| ellipsoidPointWithUncertaintyEllipse | Dependent on UE capabilities | | |
| polygon | Dependent on UE capabilities | | |
| ellipsoidPointWithAltitude | Dependent on UE capabilities | | |
| ellipsoidPointWithAltitudeAndUncertaintyEllipsoid | Dependent on UE capabilities | | |
| ellipsoidArc | Dependent on UE capabilities | | |
| highAccuracyEllipsoidPointWithUncertaintyEllipse-r15 | Dependent on UE capabilities | Rel-15 onwards | |
| highAccuracyEllipsoidPointWithAltitudeAndUncertaintyEllipsoid-r15 | Dependent on UE capabilities | Rel-15 onwards | |
| } | | | |

**Table 7.3.1.1.3.3-10: velocityTypes (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.4.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| velocityTypes SEQUENCE { | | | |
| horizontalVelocity | Dependent on UE capabilities | | |
| horizontalWithVerticalVelocity | Dependent on UE capabilities | | |
| horizontalVelocityWithUncertainty | Dependent on UE capabilities | | |
| horizontalWithVerticalVelocityAndUncertainty | Dependent on UE capabilities | | |
| } | | | |

**Table 7.3.1.1.3.3-11: otdoa-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.1.7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| otdoa-ProvideCapabilities SEQUENCE { | | | |
| otdoa-Mode | Dependent on UE capabilities | | |
| supportedBandListEUTRA SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Shall be present if otdoa-ProvideCapabilities is present | Size n of SEQUENCE is dependent on UE capabilities | |
| bandEUTRA | Dependent on UE capabilities.<br>(NOTE: The reported OTDOA supported bands can be just a subset of the EUTRA supported bands) | If bandEUTRA-v9a0 is included, then the corresponding entry of this IE shall be set to maxFBI. | |
| } | | | |
| supportedBandListEUTRA-v9a0 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Dependent on UE capabilities | Size n of SEQUENCE is dependent on UE capabilities | |
| bandEUTRA-v9a0 | Dependent on UE capabilities.<br>(NOTE: The reported OTDOA supported bands can be just a subset of the EUTRA supported bands) | | |
| } | | | |
| interFreqRSTDmeasurement-r10 | Dependent on UE capabilities | | Rel-10 onwards |
| additionalNeighbourCellInfoList-r10 | Dependent on UE capabilities | | Rel-10 onwards |
| prs-id-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| tp-separation-via-muting-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| additional-prs-config-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| prs-based-tbs-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| additionalPathsReport-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| densePrsConfig-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| maxSupportedPrsBandwidth-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| prsOccGroup-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| prsFrequencyHopping-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| maxSupportedPrsConfigs-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| periodicalReporting-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| multiPrbNprs-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| numberOfRXantennas-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| motionMeasurements-r15 | Dependent on UE capabilities | | Rel-15 onwards |
| interRAT-RSTDmeasurement-r15 | Dependent on UE capabilities | | Rel-15 onwards |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-12: ecid-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.3.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ecid-ProvideCapabilities SEQUENCE { | | | |
| ecid-MeasSupported | Dependent on UE capabilities | | |
| ueRxTxSupTDD-r13 | Dependent on UE capabilities | | Rel-13 onwards |
| periodicalReporting-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| triggeredReporting-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | | Rel-14 onwards |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-13: epdu-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.4.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| epdu-ProvideCapabilities SEQUENCE (SIZE (1)) OF SEQUENCE{ | | | |
| ePDU-Identifier SEQUENCE { | | | |
| ePDU-ID | 1 | OMA LPPe | |
| ePDU-Name | Present or not present. | | |
| } | | | |
| ePDU-Body | Set according to Table 7.3.1.1.3.3-16 | | |
| } | | | |

**Table 7.3.1.1.3.3-14: LPP Acknowledgement (step 2a, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 7.3.1.1.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.3.1.1.3.3-15: ePDU-Body OCTET STRING (step 1, Table 7.3.1.1.3.2-1)**

| Derivation Path: OMA-TS-LPPe-V1_0 [28] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OMA-LPPe-MessageExtension ::= SEQUENCE { | | | |
| lppeCompatibilityLevel | 0 | | |
| lppeVersion SEQUENCE { | | | |
| majorVersion | 1 | | |
| minorVersion | 0 | | |

| | | | |
|---|---|---|---|
| } | | | |
| lppeMode | normal | | |
| messageExtensionBody CHOICE { | | | |
| requestCapabilities SEQUENCE { | | | |
| commonIEsRequestCapabilities SEQUENCE { | | | |
| iP-Address-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| assistanceContainerSupportReq SEQUENCE { | Present | | |
| vendorOrOperatorIDList | Not present | | |
| } | | | |
| locationInformationContainerSupportReq SEQUENCE { | Present | | |
| vendorOrOperatorIDList | Not present | | |
| } | | | |
| relativeLocationChange-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| highAccuracyFormatCapabilitiesReq SEQUENCE { | Present | | |
| } | | | |
| segmentedAssistanceData-ReqCapabilities SEQUENCE { | Present | | |
| } | | | |
| referencePointCapabilitiesReq SEQUENCE { | Present | | |
| referencePointProviderSupportListReq | Not present | | |
| } | | | |
| scheduledLocation-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| accessCapabilitiesReq SEQUENCE { | Present | | |
| } | | | |
| segmentedLocationInformation-ReqCapabilities SEQUENCE { | Present | | |
| } | | | |
| } | | | |
| agnss-RequestCapabilities SEQUENCE { | | | |
| assistanceDataSupportListReq | Present | | |
| environmentObservationSupportListReq | Present | | |
| haGNSSsupportReq | Present | | |
| } | | | |
| otdoa-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| eotd-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| otdoa-utra-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-lte-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-gsm-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-utra-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| wlan-ap-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-wimax-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| sensor-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| srn-RequestCapabilities SEQUENCE { | Present | | |
| capabilitiesRequestedFor | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-16: ePDU-Body OCTET STRING (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: OMA-TS-LPPe-V1_0 [28] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OMA-LPPe-MessageExtension ::= SEQUENCE { | | | |
| lppeCompatibilityLevel | 0 | | |
| lppeVersion SEQUENCE { | | | |
| majorVersion | 1 | | |
| minorVersion | 0 | | |
| } | | | |
| lppeMode | normal | | |
| messageExtensionBody CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| commonIEsProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| agnss-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| otdoa-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| eotd-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| otdoa-utra-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| ecid-lte-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| ecid-gsm-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| ecid-utra-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| wlan-ap-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| ecid-wimax-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| sensor-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| srn-ProvideCapabilities | Present or not present and value dependent on UE capabilities. | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-17: tbs-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.4.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| tbs-ProvideCapabilities-r13 SEQUENCE { | | Rel-13 onwards | |
| tbs-Modes-r13 | Dependent on UE capabilities | | |
| mbs-AssistanceDataSupportList-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| periodicalReportingSupported-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| mbs-ConfigSupport-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| mbs-IdleStateForMeasurements-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-18: wlan-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| wlan-ProvideCapabilities-r13 SEQUENCE { | | Rel-13 onwards | |
| wlan-Modes-r13 | Dependent on UE capabilities | | |
| wlan-MeasSupported-r13 | Dependent on UE capabilities | | |
| wlan-AP-AD-Supported-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| periodicalReportingSupported-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-19: bt-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.7.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| bt-ProvideCapabilities-r13 SEQUENCE { | | Rel-13 onwards | |
| bt-Modes-r13 | Dependent on UE capabilities | | |
| bt-MeasSupported-r13 | Dependent on UE capabilities | | |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| periodicalReportingSupported-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| } | | | |
| } | | | |

**Table 7.3.1.1.3.3-20: sensor-ProvideCapabilities (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.5.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| sensor-ProvideCapabilities-r13 SEQUENCE { | | Rel-13 onwards | |
| sensor-Modes-r13 | Dependent on UE capabilities | | |
| sensor-AssistanceDataSupportList-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| periodicalReportingSupported-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| idleStateForMeasurements-r14 | Dependent on UE capabilities | Rel-14 onwards | |
| sensor-MotionInformationSup-r15 | Dependent on UE capabilities | Rel-15 onwards | |
| adjustmentSupported-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |

## 7.3.2 LPP Transport

### 7.3.2.1 LPP Duplicated Message

#### 7.3.2.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives a LPP message carrying the same sequence number as that last received for the
         associated location session }
    then { UE discards the LPP message }
}
```

#### 7.3.2.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 4.3.

[TS 36.355, clause 4.3.1]

A UE implementing LPP for the control plane solution shall support LPP reliable transport (including all three of duplicate detection, acknowledgement, and retransmission).

The following requirements in subclauses 4.3.2, 4.3.3, and 4.3.4 [LPP] for LPP reliable transport apply only when the capability is supported.

[TS 36.355, clause 4.3.2]

A sender shall include a sequence number in all LPP messages sent for a particular location session. The sequence number shall be distinct for different LPP messages sent in the same direction in the same location session.

…

A receiver shall record the most recent received sequence number for each location session. If a message is received carrying the same sequence number as that last received for the associated location session, it shall be discarded.

7.3.2.1.3 Test description

7.3.2.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

-

7.3.2.1.3.2 Test procedure sequence

**Table 7.3.2.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a LPP message of type Request Capabilities including a sequence number. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | Immediately after step 1, the SS sends the same LPP message as in step 1. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 3 | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3a | IF the UE LPP message at step 3 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS waits for 10 seconds to ensure the UE does not send another LPP message of type Provide Capabilities with the same transaction ID as received in step 1 or 2. | | | 1 | P |

7.3.2.1.3.3 Specific message contents

**Table 7.3.2.1.3.3-1: *DLInformationTransfer* (steps 1, 2, and 3a, Table 7.3.2.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.1.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.1.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 1, 2 and 3a, Table 7.3.2.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Steps 1 and 2:** Set according to Table 7.3.2.1.3.3-3 | LPP Request Capabilities | |
| | **Step 3a:** Set according to Table 7.3.2.1.3.3-7 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.2.1.3.3-3: LPP Request Capabilities (steps 1 and 2, Table 7.3.2.1.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 with the following exceptions: | | | |
| sequenceNumber | 0 | | |

**Table 7.3.2.1.3.3-4: *ULInformationTransfer* (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.1.3.3-5 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.1.3.3-5: UPLINK GENERIC NAS TRANSPORT (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | Set according to Table 7.3.2.1.3.3-6 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.2.1.3.2-1) | |

**Table 7.3.2.1.3.3-6: LPP Provide Capabilities (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 7.3.2.1.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.1.3.3-7: LPP Acknowledgement (step 3a, Table 7.3.2.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 3, Table 7.3.2.1.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

## 7.3.2.2 LPP Acknowledgment

### 7.3.2.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives a LPP message carrying an acknowledgement request indicator  }
    then { UE returns an acknowledgement response }
           }
```

### 7.3.2.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 4.3.

[TS 36.355, clause 4.3.1]

A UE implementing LPP for the control plane solution shall support LPP reliable transport (including all three of duplicate detection, acknowledgement, and retransmission).

The following requirements in subclauses 4.3.2, 4.3.3, and 4.3.4 [LPP] for LPP reliable transport apply only when the capability is supported.

[TS 36.355, clause 4.3.3.1]

Upon reception of an LPP message which includes the IE *ackRequested* set to TRUE, a receiver returns an LPP message with an acknowledgement response, i.e., that includes the *ackIndicator* IE set to the same sequence number of the message being acknowledged.

An acknowledgment response may contain no LPP message body (in which case only the sequence number being acknowledged is significant); alternatively, the acknowledgment may be sent in an LPP message along with an LPP message body.

### 7.3.2.2.3 Test description

#### 7.3.2.2.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

-

7.3.2.2.3.2 Test procedure sequence

**Table 7.3.2.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a LPP message of type Request Capabilities including a request for acknowledgement. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 Option 1 | Option 1: The UE sends an acknowledgement along with an LPP message of type Provide Capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES, incl. acknowledgement response) | 1 | P |
| 2 Option 2 | Option 2: The UE sends a LPP Acknowledgement response, followed by a LPP message of type Provide Capabilities. | --><br>--> | *ULInformationTransfer* (LPP ACKNOWLEDGEMENT)<br>*ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | 1 | P |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

7.3.2.2.3.3 Specific message contents

**Table 7.3.2.2.3.3-1: *DLInformationTransfer* (steps 1, and 3, Table 7.3.2.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.2.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 1, and 3, Table 7.3.2.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1:** Set according to Table 7.3.2.2.3.3-3 | LPP Request Capabilities | |
| | **Step 3:** Set according to Table 7.3.2.2.3.3-8 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.2.2.3.3-3: LPP Request Capabilities (step 1, Table 7.3.2.2.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 with the following exceptions: | | | |
| sequenceNumber | 0 | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |

**Table 7.3.2.2.3.3-4: *ULInformationTransfer* (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.2.3.3-5 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.3-5: UPLINK GENERIC NAS TRANSPORT (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 2:**<br>Set according to Table 7.3.2.2.3.3-6 | LPP Provide Capabilities | |
| | **Step 2 (Option 2),**<br>Set according to Table 7.3.2.2.3.3-7 | LPP Acknowledgement | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message | |

**Table 7.3.2.2.3.3-6: LPP Provide Capabilities (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 7.3.2.2.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present. Present for Option 1. | | |
| ackRequested | TRUE or FALSE | | |
| ackIndicator | 0 (Option 1)<br>Not present (Option 2) | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.3-7: LPP Acknowledgement (step 2 – Option 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | 0 | | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.3.2.2.3.3-8: LPP Acknowledgement (step 3, Table 7.3.2.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 7.3.2.2.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

## 7.3.2.3 LPP Retransmission

### 7.3.2.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing}
ensure that {
  when { UE does not receive an LPP acknowledgement for an LPP message which requires
         acknowledgement }
    then { UE retransmits the LPP message up to three times. If still unacknowledged after that, the
           UE aborts all LPP activity for the associated session}
}
```

### 7.3.2.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 4.3.

[TS 36.355, clause 4.3.1]

A UE implementing LPP for the control plane solution shall support LPP reliable transport (including all three of duplicate detection, acknowledgement, and retransmission).

The following requirements in subclauses 4.3.2, 4.3.3, and 4.3.4 [LPP] for LPP reliable transport apply only when the capability is supported.

[TS 36.355, clause 4.3.4.1]

When an LPP message which requires acknowledgement is sent and not acknowledged, it is resent by the sender following a timeout period up to three times. If still unacknowledged after that, the sender aborts all LPP activity for the associated session.

#### 7.3.2.3.3 Test description

##### 7.3.2.3.3.1 Pre-test conditions

System Simulator:

- Cell 1.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

- UE supporting sending of acknowledgement request in LPP Provide Capabilities message.

##### 7.3.2.3.3.2 Test procedure sequence

**Table 7.3.2.3.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends a LPP message of type Provide Capabilities including a request for acknowledgement along with a sequence number. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | SS does not send an acknowledgement | | | - | - |
| 4 | After an implementation specific timeout period, the UE retransmits the LPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | 1 | P |
| 5 | SS does not send an acknowledgement | | | - | - |
| 6 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the LPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 7 | SS does not send an acknowledgement | | | - | - |
| 8 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the LPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 9 | SS does not send an acknowledgement | | | | |
| 10 | UE aborts all procedures and activity associated with LPP support for the location session.<br>SS waits for 10 seconds to ensure the UE does not send another LPP message. | | | 1 | P |

7.3.2.3.3.3 Specific message contents

**Table 7.3.2.3.3.3-1: *DLInformationTransfer* (step 1, Table 7.3.2.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.3.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.3.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (step 1, Table 7.3.2.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | Set according to Table 7.3.2.3.3.3-3 | LPP Request Capabilities | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.2.3.3.3-3: LPP Request Capabilities (step 1, Table 7.3.2.3.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1. | | | |

**Table 7.3.2.3.3.3-4: *ULInformationTransfer* (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.2.3.3.3-5 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.3.3.3-5: UPLINK GENERIC NAS TRANSPORT (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | Set according to Table 7.3.2.3.3.3-6 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message | |

**Table 7.3.2.3.3.3-6: LPP Provide Capabilities (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1 Table 7.3.2.3.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.3 LPP Error Handling

### 7.3.3.1 Void

### 7.3.3.1A Void

### 7.3.3.1B LPP Requested Method not Supported – UE-Assisted

#### 7.3.3.1B.1 Test Purpose (TP)

(1)

```
with {  a UE supporting at least one of UE-assisted GNSS, UE-assisted OTDOA, UE-assisted ECID, UE-
assisted WLAN, UE-assisted Bluetooth, UE-assisted Sensor or UE-assisted MBS but not all of them }
and with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives a LPP message requesting at least one location method not supported }
    then { the UE provides location information for the supported methods}
}
```

#### 7.3.3.1B.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clauses 5.3.3 and 5.4.3.

[TS 36.355, clause 5.3.3]

Upon receiving a *RequestLocationInformation* message, the target device shall:

1> if the requested information is compatible with the target device capabilities and configuration:

[…]

1> otherwise:

2> if one or more positioning methods are included that the target device does not support:

3> continue to process the message as if it contained only information for the supported positioning methods;

3> handle the signalling content of the unsupported positioning methods by LPP error detection as in 5.4.3.

[TS 36.355, clause 5.4.3]

Upon receiving any LPP message, the receiving entity shall attempt to decode the message and verify the presence of any errors and:

1> if the message type is an LPP *RequestAssistanceData* or *RequestLocationInformation* and some or all of the requested information is not supported:

2> return any information that can be provided in a normal response, which includes indications on other information that is not supported.

#### 7.3.3.1B.3 Test description

##### 7.3.3.1B.3.1 Pre-test conditions

System Simulator:

- If OTDOA is supported by the UE: Cells 1 and 2, as specified in 5.2.2.
- If ECID is supported by the UE: Cells 1 and 2, as specified in 5.2.3. If OTDOA is also supported then Cells 1 and 2 are as specified in 5.2.2.

- If GNSS is supported by the UE: Cell 1 and satellite signals, as specified in 5.2.1. If OTDOA is also supported then Cell 1 is as specified in 5.2.2.
- If WLAN is supported by the UE: Cell 1 and WLAN signals, as specified in 5.2.5.
- If MBS is supported by the UE: Cell 1 and MBS signals, as specified in 5.2.4.
- If Bluetooth is supported by the UE: Cell 1 and Bluetooth signals, as specified in 5.2.6.
- If Sensor is supported by the UE: Cell 1.

UE:

- -

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

- -

7.3.3.1B.3.2 Test procedure sequence

**Table 7.3.3.1B.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 0 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 0a | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 0b | IF the UE LPP message at step 0a includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 1 | IF the UE supports any positioning method other than ECID or Bluetooth, THEN the SS sends a LPP message of type Provide Assistance Data containing the data for all supported positioning methods. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 2 | The SS sends a LPP message of type Request Location Information including all specified positioning methods. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 3 | The UE sends a LPP message of type Provide Location Information including information for the supported method(s). | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 3a | IF the UE LPP message at step 3 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

7.3.3.1B.3.3 Specific message contents

**Table 7.3.3.1B.3.3-1: *DLInformationTransfer* (steps 0, 0b, 1, 2 and 3a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.3.1B.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.3.1B.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 0, 0b, 1, 2 and 3a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 0:** Set according to Table 7.3.3.1B.3.3-2a | LPP Request Capabilities. | |
| | **Step 1:** Set according to Table 7.3.3.1B.3.3-3 | LPP Provide Assistance Data | |
| | **Step 2:** Set according to Table 7.3.3.1B.3.3-4 | LPP Request Location Information | |
| | **Steps 0b and 3a:** Set according to Table 7.3.3.1B.3.3-8 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.3.1B.3.3-2a: LPP Request Capabilities (step 0, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.3.1B.3.3-3: LPP Provide Assistance data (step 1, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| a-gnss-ProvideAssistanceData | Present for all supported GNSSs if UE supports UE-assisted A-GNSS. As defined in clause 5.4 | | |
| otdoa-ProvideAssistanceData | Present if UE supports UE-assisted OTDOA. As defined in clause 5.4 | | |
| sensor-ProvideAssistanceData-r14 | Present if UE supports UE-assisted Sensor. As defined in clause 5.4 | Rel-14 onwards | |
| tbs-ProvideAssistanceData-r14 | Present if UE supports UE-assisted MBS. As defined in clause 5.4 | Rel-14 onwards | |
| wlan-ProvideAssistanceData-r14 | Present if UE supports UE-assisted WLAN. As defined in clause 5.4 | Rel-14 onwards | |

**Table 7.3.3.1B.3.3-4: LPP Request Location Information (step 2, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |
| a-gnss-RequestLocationInformation | Present. As defined in Table 5.4-4 | | |
| gnss-Methods | GNSS-ID-Bitmap: bits 0, 3, 4, 5 = 1 | | |
| otdoa-RequestLocationInformation | Present. As defined in Table 5.4-5 | | |
| ecid-RequestLocationInformation | Present. As defined in Table 5.4-6 | | |
| requestedMeasurements | bits 0, 1, 2 = 1 | | |
| tbs-RequestLocationInformation-r13 | Present. As defined in Table 5.4-7 | Rel-13 onwards | |
| sensor-RequestLocationInformation-r13 | Present. As defined in Table 5.4-10 | Rel-13 onwards | |
| wlan-RequestLocationInformation-r13 | Present. As defined in Table 5.4-8 | Rel-13 onwards | |
| bt-RequestLocationInformation-r13 | Present. As defined in Table 5.4-9 | Rel-13 onwards | |

**Table 7.3.3.1B.3.3-5: *ULInformationTransfer* (steps 0a and 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.3.1B.3.3-6 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.3.1B.3.3-6: UPLINK GENERIC NAS TRANSPORT (steps 0a and 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 0a:** Set according to Table 7.3.3.1B.3.3-6a | LPP Provide Capabilities | |
| | **Step 3:** Set according to Table 7.3.3.1B.3.3-7 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 0 or 2 Table 7.3.3.1B.3.2-1) | |

**Table 7.3.3.1B.3.3-6a: LPP Provide Capabilities (step 0a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 0, Table 7.3.3.1B.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.3.1B.3.3-7: LPP Provide Location Information (step 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 2, Table 7.3.3.1B.3.1-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | May be present | | |
| locationEstimate | Not present | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | Not present | Rel-12 onwards | |
| } | | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | Present if UE supports UE-assisted A-GNSS. | | |
| gnss-SignalMeasurementInformation | Present for each supported GNSS. Any value acceptable | | |
| gnss-LocationInformation | Not present | | |
| gnss-Error | May be present if UE only supports one GNSS | | |
| } | | | |
| otdoa-ProvideLocationInformation SEQUENCE { | Present if UE supports UE-assisted OTDOA. | | |
| otdoa-SignalMeasurementInformation | Present. Any value acceptable | | |
| otdoa-Error | May be present | | |
| } | | | |
| ecid-ProvideLocationInformation SEQUENCE { | Present if UE supports UE-assisted ECID. | | |
| ecid-SignalMeasurementInformation | Present. Any value acceptable | | |
| ecid-Error | May be present | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 SEQUENCE { | Present if UE supports UE-assisted Sensor. | Rel-13 onwards | |
| sensor-MeasurementInformation-r13 | Present. Any value acceptable | | |
| sensor-Error-r13 | May be present | | |
| } | | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | Present if UE supports UE-assisted MBS | Rel-13 onwards | |

| | | | |
|---|---|---|---|
| tbs-MeasurementInformation-r13 SEQUENCE { | | | |
| measurementReferenceTime-r13 | Present. Any value acceptable | | |
| mbs-SgnMeasList-r13 | Present. Any value acceptable | | |
| } | | | |
| tbs-Error-r13 | May be present | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 SEQUENCE { | Present if UE supports UE-assisted WLAN. | Rel-13 onwards | |
| wlan-MeasurementInformation-r13 | Present. Any value acceptable | | |
| wlan-Error-r13 | May be present | | |
| } | | | |
| bt-ProvideLocationInformation-r13 SEQUENCE { | Present if UE supports UE-assisted Bluetooth. | Rel-13 onwards | |
| bt-MeasurementInformation-r13 | Present. Any value acceptable | | |
| bt-Error-r13 | May be present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.3.1B.3.3-8: LPP Acknowledgement (steps 0b and 3a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 0a or 3, Table 7.3.3.1B.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

## 7.3.4 LPP Positioning Procedures

### 7.3.4.1 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Based

#### 7.3.4.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives assistance data and a location request for UE-based  }
    then { UE sends a PROVIDE LOCATION INFORMATION  message containing a location estimate }
            }
```

#### 7.3.4.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.2.4, 5.3.3 and 5.3.4.

[TS 36.355, clause 5.2.4]

Upon receiving a *ProvideAssistanceData* message, the target device shall:

1> for each positioning method contained in the message:

2> deliver the related assistance data to upper layers.

[TS 36.355, clause 5.3.3]

Upon receiving a *RequestLocationInformation* message, the target device shall:

1> if the requested information is compatible with the target device capabilities and configuration:

2> include the requested information in a *ProvideLocationInformation* message;

2> set the IE *LPP-TransactionID* in the response to the same value as the IE *LPP-TransactionID* in the received message;

2> deliver the *ProvideLocationInformation* message to lower layers for transmission.

1> otherwise:

[…]

[TS 36.355, clause 5.3.4]

When triggered to transmit *ProvideLocationInformation* message, the target device shall:

1> for each positioning method contained in the message:

2> set the corresponding IE to include the available location information;

1> deliver the response to lower layers for transmission.

#### 7.3.4.1.3 Test description

##### 7.3.4.1.3.1 Pre-test conditions

System Simulator:

- Cell 1.
- Satellite signals (sub-test case 15): As specified in 5.2.1.
- MBS signals (Sub-test 16): As specified in 5.2.4 .
- WLAN signals (Sub-test 17): as specified in 5.2.5.

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

-

7.3.4.1.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.3.4.1.3.2-0 below:

**Table 7.3.4.1.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.3.4.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The stored assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1a | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 1b | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 1c | IF the UE LPP message at step 1b includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 2 | The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 3 | The SS sends a LPP message of type Request Location Information including a request for a location estimate. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 4 | The UE sends a LPP message of type Provide Location Information including a location estimate. | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 4a | IF the UE LPP message at step 4 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

7.3.4.1.3.3 Specific message contents

**Table 7.3.4.1.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 15: 0 0 0 0 0 0 0 0<br>Sub-test 16: 0 0 0 0 0 0 1 0<br>Sub-test 17: 0 0 0 0 0 0 1 1<br>Sub-test 18: 0 0 0 0 0 1 0 1 | Sub-test 15: GNSS<br>Sub-test 16: MBS<br>Sub-test 17: WLAN<br>Sub-test 18: Sensor | |

**Table 7.3.4.1.3.3-2: *DLInformationTransfer* (steps 1a, 1c, 2, 3 and 4a, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.1.3.3-3 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.1.3.3-3: DOWNLINK GENERIC NAS TRANSPORT (steps 1a, 1c, 2, 3 and 4a, Table 7.3.4.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1a:** Set according to Table 7.3.4.1.3.3-3a | LPP Request Capabilities. | |
| | **Step 2:** Set according to Table 7.3.4.1.3.3-4 | LPP Provide Assistance Data | |
| | **Step 3:** Set according to Table 7.3.4.1.3.3-5 | LPP Request Location Information | |
| | **Steps 1c and 4a:** Set according to Table 7.3.4.1.3.3-9 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.4.1.3.3-3a: LPP Request Capabilities (step 1a, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.4.1.3.3-4: LPP Provide Assistance data (step 2, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 7.3.4.1.3.3-5: LPP Request Location Information (step 3, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationEstimateRequired | | |

**Table 7.3.4.1.3.3-6: *ULInformationTransfer* (steps 1b and 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |

| | | | |
|---|---|---|---|
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.1.3.3-7 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.1.3.3-7: UPLINK GENERIC NAS TRANSPORT (steps 1b and 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobilitymanagement messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1b:** Set according to Table 7.3.4.1.3.3-7a | LPP Provide Capabilities | |
| | **Step 4:** Set according to Table 7.3.4.1.3.3-8 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1a or 3 Table 7.3.4.1.3.2-1) | |

**Table 7.3.4.1.3.3-7a: LPP Provide Capabilities. (step 1b, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1a, Table 7.3.4.1.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.1.3.3-8: LPP Provide Location Information (step 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location | |

| | | | |
|---|---|---|---|
| | | Information message in step 3, Table 7.3.4.1.3.1-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | Present | | |
| locationEstimate | Present. Any value acceptable | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | Not present | Rel-12 onwards | |
| } | | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | Present for sub-test 15 | | |
| gnss-SignalMeasurementInformation | Not present | | |
| gnss-LocationInformation SEQUENCE { | Present | | |
| measurementReferenceTime | Any value acceptable | | |
| agnss-List | Any value acceptable | | |
| } | | | |
| gnss-Error | Not present | | |
| } | | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 18 | Rel-13 onwards | |
| sensor-MeasurementInformation-r13 | Present. Any value acceptable | | |
| sensor-Error-r13 | Not present | | |
| } | | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 16 | Rel-13 onwards | |
| tbs-MeasurementInformation-r13 SEQUENCE { | Present | Rel-13 onwards | |
| measurementReferenceTime-r13 | Any value acceptable | | |
| mbs-SgnMeasList-r13 | Any value acceptable | | |
| } | | | |
| tbs-Error-r13 | Not present | Rel-13 onwards | |
| } | | | |
| wlan-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 17 | Rel-13 onwards | |
| wlan-MeasurementInformation-r13 | Present. Any value acceptable | | |
| wlan-Error-r13 | Not present | | |
| } | | | |
| bt-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 7.3.4.1.3.3-9: LPP Acknowledgement (steps 1c and 4a, Table 7.3.4.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1b or 4, Table 7.3.4.1.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

## 7.3.4.2 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted

### 7.3.4.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives assistance data and a location request for UE-assisted  }
    then { UE sends a PROVIDE LOCATION INFORMATION  message containing location measurements }
           }
```

### 7.3.4.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.2.4, 5.3.3 and 5.3.4.

[TS 36.355, clause 5.2.4]

Upon receiving a *ProvideAssistanceData* message, the target device shall:

1> for each positioning method contained in the message:

2> deliver the related assistance data to upper layers.

[TS 36.355, clause 5.3.3]

Upon receiving a *RequestLocationInformation* message, the target device shall:

1> if the requested information is compatible with the target device capabilities and configuration:

2> include the requested information in a *ProvideLocationInformation* message;

2> set the IE *LPP-TransactionID* in the response to the same value as the IE *LPP-TransactionID* in the received message;

2> deliver the *ProvideLocationInformation* message to lower layers for transmission.

1> otherwise:

[…]

[TS 36.355, clause 5.3.4]

When triggered to transmit *ProvideLocationInformation* message, the target device shall:

1> for each positioning method contained in the message:

2> set the corresponding IE to include the available location information;

1> deliver the response to lower layers for transmission.

#### 7.3.4.2.3 Test description

##### 7.3.4.2.3.1 Pre-test conditions

System Simulator:

- Sub-tests 11, 12, 13, 14, 15, 16, 17, 18: Cell 1.
- Sub-test 5 and 7: Cells 1 and 2, as specified in 5.2.2.
- Sub-tests 6 FDD, 6 TDD: Cells 1 and 2, as specified in 5.2.3.
- Satellite signals (Sub-test 15): As specified in 5.2.1.
- WLAN signals (Sub-test 11, 17): as specified in 5.2.5.
- MBS signals (Sub-tests 12, 16): as specified in 5.2.4
- Bluetooth signals (Sub-test 13): as specified in 5.2.6.

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

-

##### 7.3.4.2.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.3.4.2.3.2-0 below:

**Table 7.3.4.2.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA |
| 6 FDD | UE supporting ECID (FDD) |
| 6 TDD | UE supporting ECID (TDD) |
| 7 | UE supporting GNSS(1) and OTDOA |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 14 | UE supporting Sensor (Rel-13 only) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.3.4.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | IF<br>sub-test 7 or 15 or 16 or 17 or 18<br>THEN<br>The stored assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1a | IF<br>Sub-test 5 or 7<br>THEN<br>The stored OTDOA assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1b | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer*<br>(LPP REQUEST CAPABILITIES) | - | - |
| 1c | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer*<br>(LPP PROVIDE CAPABILITIES) | - | - |
| 1d | IF<br>the UE LPP message at step 1c includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 2 | IF NOT<br>sub-test 6 FDD or 6 TDD or 11 or 12 or 13 or 14<br>THEN<br>The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer*<br>(LPP PROVIDE ASSISTANCE DATA) | - | - |
| 3 | The SS sends a LPP message of type Request Location Information including a request for location measurements. | <-- | *DLInformationTransfer*<br>(LPP REQUEST LOCATION INFORMATION) | - | - |
| - | Steps 4a1-4a2 and 4b1-4b4 represent alternative UE behaviours depending on the UE implementation | - | - | - | - |
| 4a1 (Note 1) | All sub-tests:<br>The UE sends a LPP message of type Provide Location Information including location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 4a2 | IF<br>the UE LPP message at step 4 a1 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 4b1 (Note 2) | IF sub-test 7<br>THEN<br>The UE sends a LPP message of type Provide Location Information including "early fix" location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | - | - |
| 4b2 | IF<br>the UE LPP message at step 4b1 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 4b3 | The UE sends a LPP message of type Provide Location Information including location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 4b4 | IF<br>the UE LPP message at step 4b3 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |

Note 1: This alternative is applicable to all releases of LPP and may be followed even in the case of sub-test 7 and LPP release 12 onwards.

Note 2: This alternative is applicable only to LPP release 12 onwards.

7.3.4.2.3.3 Specific message contents

**Table 7.3.4.2.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-tests 7, 15: 0 0 0 0 0 0 0 0<br>Sub-test 16: 0 0 0 0 0 0 1 0<br>Sub-test 17: 0 0 0 0 0 0 1 1<br>Sub-test 18: 0 0 0 0 0 1 0 1 | Sub-tests 7, 15: GNSS<br>Sub-test 16: MBS<br>Sub-test 17: WLAN<br>Sub-test 18: Sensor | |

**Table 7.3.4.2.3.3-2: RESET UE POSITIONING STORED INFORMATION (step 1a, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 7.3.4.2.3.3-3: *DLInformationTransfer* (steps 1b, 1d, 2, 3, 4a2, 4b2 and 4b4, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.2.3.3-4 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.2.3.3-4: DOWNLINK GENERIC NAS TRANSPORT (steps 1b, 1d, 2, 3, 4a2, 4b2 and 4b4, Table 7.3.4.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1b:**<br>Set according to Table 7.3.4.2.3.3-4a | LPP Request Capabilities. | |
| | **Step 2:**<br>Set according to Table 7.3.4.2.3.3-5 | LPP Provide Assistance Data | |

| | | | |
|---|---|---|---|
| | **Step 3:** Set according to Table 7.3.4.2.3.3-6 | LPP Request Location Information | |
| | **Steps 1d, 4a2, 4b2 and 4b4:** Set according to Table 7.3.4.2.3.3-10 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.4.2.3.3-4a: LPP Request Capabilities (step 1b, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.4.2.3.3-5: LPP Provide Assistance data (step 2, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 7.3.4.2.3.3-6: LPP Request Location Information (step 3, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 32 | | |
| responseTimeEarlyFix-r12 | Sub-tests 5, 6 FDD, 6 TDD, 11, 12, 13, 14, 15, 16, 17, 18: not present<br>Sub-test 7: 10 | Rel-12 onwards | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| a-gnss-RequestLocationInformation | Set according to Table 7.3.4.2.3.3-11 | | |

**Table 7.3.4.2.3.3-7: *ULInformationTransfer* (steps 1c, 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.2.3.3-8 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |

| | | | |
|---|---|---|---|
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.2.3.3-8: UPLINK GENERIC NAS TRANSPORT (steps 1c, 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1c:** Set according to Table 7.3.4.2.3.3-8a | LPP Provide Capabilities | |
| | **Steps 4 a1, 4b1 and 4b3:** Set according to Table 7.3.4.2.3.3-9 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1b or 3 Table 7.3.4.2.3.2-1) | |

**Table 7.3.4.2.3.3-8a: LPP Provide Capabilities. (step 1c, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1b, Table 7.3.4.2.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities SEQUENCE{ | Dependent on UE capabilities | | |
| ueRxTxSupTDD-r13 | Present (TRUE) for sub-test 6 TDD | Rel-13 onwards | |
| } | | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.2.3.3-9: LPP Provide Location Information (steps 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 3, Table 7.3.4.2.3.1-1 | |
| } | | | |
| endTransaction | **Step 4a1, 4b3:** TRUE | | |
| | **Step 4b1:** FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | **Step 4a1, 4b3:** May be present | | |
| | **Step 4b1:** Present | | |
| locationEstimate | Not present | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | **Step 4a1, 4b3:** Not present | Rel-12 onwards | |
| | **Step 4b1:** Any value acceptable | | |
| } | | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | **Step 4a1:** Present for sub-tests 7, 15 | | |
| | **Step 4b1, 4b3:** May be present | One of a-gnss-ProvideLocationInformation or otdoa-ProvideLocationInformation shall be present | |
| gnss-SignalMeasurementInformation SEQUENCE { | Present | | |
| measurementReferenceTime | Present. Any value acceptable | | |
| gnss-MeasurementList SEQUENCE (SIZE(1..n)) OF SEQUENCE { | Present. SIZE n is the number of GNSSs supported by the UE, one instance for each GNSS supported by the UE | | |
| gnss-ID | Present | | |
| gnss-SgnMeasList | Present, one instance for each frequency within the GNSS supported by the UE. Any value acceptable | | |
| } | | | |
| } | | | |
| } | | | |
| gnss-LocationInformation | Not present | | |
| gnss-Error | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideLocationInformation SEQUENCE { | **Step 4a1:** Present for sub-tests 5, 7 | | |
| | **Step 4b1, 4b3:** May be present | One of a-gnss-ProvideLocationInformation or otdoa-ProvideLocationInformation shall be present | |
| otdoa-SignalMeasurementInformation | Present. Any value acceptable | | |
| otdoa-Error | May be present | | |
| } | | | |
| ecid-ProvideLocationInformation SEQUENCE { | Present for sub-test 6 FDD, 6 TDD. | | |
| ecid-SignalMeasurementInformation | Present. Any value acceptable | | |
| ecid-Error | Not present | | |
| } | | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 14, 18 | Rel-13 onwards | |
| sensor-MeasurementInformation-r13 | Present. Any value acceptable | | |
| sensor-Error-r13 | May be present | | |
| } | | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-tests 12, 16 | Rel-13 onwards | |
| tbs-MeasurementInformation-r13 SEQUENCE { | Present | | |
| measurementReferenceTime-r13 | Any value acceptable | | |
| mbs-SgnMeasList-r13 | Any value acceptable | | |
| } | | | |
| tbs-Error-r13 | Not present | | |
| } | | | |
| wlan-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 11, 17 | Rel-13 onwards | |
| wlan-MeasurementInformation-r13 | Present. Any value acceptable | | |
| wlan-Error-r13 | May be present | | |
| } | | | |
| bt-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 13 | Rel-13 onwards | |
| bt-MeasurementInformation-r13 | Present. Any value acceptable | | |
| bt-Error-r13 | May be present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.2.3.3-10: LPP Acknowledgement (steps 1d, 4a2, 4b2 and 4b4, Table 7.3.4.2.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1c or 4 a1 or 4b1 or 4b3, Table 7.3.4.2.3.2-1 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.3.4.2.3.3-11: A-GNSS Request Location Information (step 3, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 5.4-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-4 with the following exceptions: | | | |
| multiFreqMeasReq | TRUE, if UE supports multi frequency GNSS | | MultiFreqAGNSS |

| Condition | Description |
|---|---|
| MultiFreqAGNSS | The UE supports at least one of pc_A_GPS_L2C, pc_A_GPS_L5, pc_QZSS_QZS_L1C, pc_QZSS_QZS_L2C, pc_QZSS_QZS_L5, pc_GLONASS_G2, pc_GLONASS_G3, pc_GALILEO_E5a, pc_GALILEO_E5b, pc_GALILEO_E6, pc_GALILEO_E5aE5b, pc_BDS_B1C, pc_BDS_B2a or pc_BDS_B3I |

## 7.3.4.3 E-SMLC Initiated Position Measurement without assistance data: UE-Based

### 7.3.4.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when {    UE has no assistance data stored and receives a location request for UE-based and the UE
            requires assistance data in order to fulfill the location request }
    then {  UE sends a REQUEST ASSISTANCE DATA message followed by a PROVIDE LOCATION INFORMATION
            message containing a location estimate }
            }
```

### 7.3.4.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.2.3, 5.2.4, 5.3.3 and 5.3.4.

[TS 36.355, clause 5.2.3]

When triggered to transmit a *RequestAssistanceData* message, the target device shall:

1> set the IEs for the positioning-method-specific request for assistance data to request the data indicated by upper layers.

[TS 36.355, clause 5.2.4]

Upon receiving a *ProvideAssistanceData* message, the target device shall:

1> for each positioning method contained in the message:

2> deliver the related assistance data to upper layers.

[TS 36.355, clause 5.3.3]

Upon receiving a *RequestLocationInformation* message, the target device shall:

1> if the requested information is compatible with the target device capabilities and configuration:

2> include the requested information in a *ProvideLocationInformation* message;

2> set the IE *LPP-TransactionID* in the response to the same value as the IE *LPP-TransactionID* in the received message;

2> deliver the *ProvideLocationInformation* message to lower layers for transmission.

1> otherwise:

[…]

[TS 36.355, clause 5.3.4]

When triggered to transmit *ProvideLocationInformation* message, the target device shall:

1> for each positioning method contained in the message:

2> set the corresponding IE to include the available location information;

1> deliver the response to lower layers for transmission.

#### 7.3.4.3.3 Test description

##### 7.3.4.3.3.1 Pre-test conditions

System Simulator:

- Cell 1.
- Satellite signals (sub-test 15): As specified in 5.2.1.
- MBS signals (sub-test 16): As specified in 5.2.4.
- WLAN signals (Sub-test 17): as specified in 5.2.5.

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- Method of triggering an LPP Request Assistance Data message.

##### 7.3.4.3.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.3.4.3.3.2-0 below:

**Table 7.3.4.3.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.3.4.3.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The stored assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1a | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 1b | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 1c | IF the UE LPP message at step 1b includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 2 | The SS sends a LPP message of type Request Location Information including a request for a location estimate. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 3 | The UE sends a LPP message of type Request Assistance Data including a request for assistance data. NOTE: This requires a method of triggering an Request Assistance Data message. | --> | *ULInformationTransfer* (LPP REQUEST ASSISTANCE DATA) | 1 | P |
| 4 | The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The UE sends a LPP message of type Provide Location Information including a location estimate. | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 5a | IF the UE LPP message at step 5 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

7.3.4.3.3.3 Specific message contents

**Table 7.3.4.3.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 15: 0 0 0 0 0 0 0 0 | Sub-test 15: GNSS<br>Sub-test 16: MBS<br>Sub-test 17: WLAN<br>Sub-test 18: | |

| | | | |
|---|---|---|---|
| | Sub-test 16: 0 0 0 0 0 0 1 0<br>Sub-test 17: 0 0 0 0 0 0 1 1<br>Sub-test 18: 0 0 0 0 0 1 0 1 | Sensor | |

**Table 7.3.4.3.3.3-2: *DLInformationTransfer* (steps 1a, 1c, 2, 4 and 5a, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.3.3.3-3 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-3: DOWNLINK GENERIC NAS TRANSPORT (steps 1a, 1c, 2, 4 and 5a, Table 7.3.4.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1a:** Set according to Table 7.3.4.3.3.3-3a | LPP Request Capabilities. | |
| | **Step 2:** Set according to Table 7.3.4.3.3.3-4 | LPP Request Location Information | |
| | **Step 4:** Set according to Table 7.3.4.3.3.3-9 | LPP Provide Assistance Data | |
| | **Steps 1c and 5a:** Set according to Table 7.3.4.3.3.3-11 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.4.3.3.3-3a: LPP Request Capabilities (step 1a, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.4.3.3.3-4: LPP Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationEstimateRequired | | |
| a-gnss-RequestLocationInformation | Set according to Table 7.3.4.3.3.3-5 | | Sub-test 15 |
| sensor-RequestLocationInformation-r14 | Set according to Table 7.3.4.3.3.3-5B | Rel-14 onwards | Sub-test 18 |
| tbs-RequestLocationInformation-r13 | Set according to Table 7.3.4.3.3.3-5A | Rel-13 onwards | Sub-test 16 |
| wlan-RequestLocationInformation-r14 | Set according to Table 7.3.4.3.3.3-5C | Rel-14 onwards | Sub-test 17 |

**Table 7.3.4.3.3.3-5: A-GNSS Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-4 with the following exceptions: | | | |
| assistanceAvailability | TRUE | | |

**Table 7.3.4.3.3.3-5A: TBS Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-7 with the following exceptions: | | | |
| mbsSgnMeasListReq-r13 | FALSE (UE-based MBS) | Rel-13 onwards | |
| mbsAssistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.3.3.3-5B: Sensor Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-10 with the following exceptions: | | | |
| uncompensatedBarometricPressureReq-r13 | FALSE (UE-based Sensor) | Rel-13 onwards | |
| assistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.3.3.3-5C: WLAN Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-8 with the following exceptions: | | | |
| requestedMeasurements-r13 | bit 0 = 0 (rssi) (UE-based WLAN)<br>bit 1 = 0 (rtt) (UE-based WLAN) | Rel-13 onwards | |
| assistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.3.3.3-6: *ULInformationTransfer* (steps 1b, 3 and 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table | UPLINK GENERIC | |

| | | | |
|---|---|---|---|
| | 7.3.4.3.3.3-7 | NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-7: UPLINK GENERIC NAS TRANSPORT (steps 1b, 3 and 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1b:** Set according to Table 7.3.4.3.3.3-7a | LPP Provide Capabilities | |
| | **Step 3:** Set according to Table 7.3.4.3.3.3-8 | LPP Request Assistance Data | |
| | **Step 5:** Set according to Table 7.3.4.3.3.3-10 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1a or 2 or 4 Table 7.3.4.3.3.2-1) | |

**Table 7.3.4.3.3.3-7a: LPP Provide Capabilities. (step 1b, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1a, Table 7.3.4.3.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-8: LPP Request Assistance Data (step 3, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present or not present | | |

| | | | |
|---|---|---|---|
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| commonIEsRequestAssistanceData | Present or not present | | |
| a-gnss-RequestAssistanceData | Present for sub-test 15 | | |
| otdoa-RequestAssistanceData | Not present | | |
| epdu-RequestAssistanceData | Not present | | |
| sensor-RequestAssistanceData-r14 | Present for sub-test 18 | Rel-14 onwards | |
| tbs-RequestAssistanceData-r14 | Present for sub-test 16 | | |
| wlan-RequestAssistanceData-r14 | Present for sub-test 17 | Rel-14 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-9: LPP Provide Assistance Data (step 4, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Assistance Data message in step 3 Table 7.3.4.3.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | Present if acknowledgement field is included by the UE at step 3, Table 7.3.4.3.3.2-1. | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value as the sequenceNumber in step 3, Table 7.3.4.3.3.2-1 | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| a-gnss-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to TS 37.571-5 [12]. | | |
| sensor-ProvideAssistanceData-r14 | The SS provides the assistance data requested | Rel-14 onwards | |

| | | | |
|---|---|---|---|
| | by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 5.4.1.5. | | |
| tbs-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 5.4.1.3. | Rel-14 onwards | |
| wlan-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 5.4.1.4. | Rel-14 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-10: LPP Provide Location Information (step 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 2, Table 7.3.4.3.3.1-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | Present | | |
| locationEstimate | Present.<br>Any value acceptable | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | Not present | Rel-12 onwards | |
| } | | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | Present for sub-test 15 | | |
| gnss-SignalMeasurementInformation | Not present | | |
| gnss-LocationInformation | Present | | |

| | | | |
|---|---|---|---|
| SEQUENCE { | | | |
| measurementReferenceTime | Any value acceptable | | |
| agnss-List | Any value acceptable | | |
| } | | | |
| gnss-Error | Not present | | |
| } | | | |
| otdoa-ProvideLocationInformation | Not present | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Present for sub-test 18 | Rel-13 onwards | |
| tbs-ProvideLocationInformation-r13 | Present for sub-test 16 | Rel-13 onwards | |
| wlan-ProvideLocationInformation-r13 | Present for sub-test 17 | Rel-13 onwards | |
| bt-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.3.3.3-11: LPP Acknowledgement (steps 1c and 5a, Table 7.3.4.3.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1b or 5, Table 7.3.4.3.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

## 7.3.4.4 E-SMLC Initiated Position Measurement without assistance data: UE-Assisted

### 7.3.4.4.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when {    UE has no assistance data stored and receives a location request for UE-assisted and the
            UE requires assistance data in order to fulfill the location request }
    then {  UE sends a REQUEST ASSISTANCE DATA message followed by a PROVIDE LOCATION INFORMATION
            message containing location measurements }
            }
```

### 7.3.4.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.2.3, 5.2.4, 5.3.3 and 5.3.4.

[TS 36.355, clause 5.2.3]

When triggered to transmit a *RequestAssistanceData* message, the target device shall:

1> set the IEs for the positioning-method-specific request for assistance data to request the data indicated by upper layers.

[TS 36.355, clause 5.2.4]

Upon receiving a *ProvideAssistanceData* message, the target device shall:

1> for each positioning method contained in the message:

2> deliver the related assistance data to upper layers.

[TS 36.355, clause 5.3.3]

Upon receiving a *RequestLocationInformation* message, the target device shall:

1> if the requested information is compatible with the target device capabilities and configuration:

2> include the requested information in a *ProvideLocationInformation* message;

2> set the IE *LPP-TransactionID* in the response to the same value as the IE *LPP-TransactionID* in the received message;

2> deliver the *ProvideLocationInformation* message to lower layers for transmission.

1> otherwise:

[…]

[TS 36.355, clause 5.3.4]

When triggered to transmit *ProvideLocationInformation* message, the target device shall:

1> for each positioning method contained in the message:

2> set the corresponding IE to include the available location information;

1> deliver the response to lower layers for transmission.

#### 7.3.4.4.3 Test description

##### 7.3.4.4.3.1 Pre-test conditions

System Simulator:

- Sub-tests 15, 16, 17, 18: Cell 1.
- Sub-test 5 and 7: Cells 1 and 2, as specified in 5.2.2.
- Satellite signals (Sub-test 15): As specified in 5.2.1.
- MBS signals (Sub-test 16): As specified in 5.2.4 .
- WLAN signals (Sub-test 17): As specified in 5.2.5 .

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8].

Related PICS/PIXIT Statements:

- Method of triggering an LPP Request Assistance Data message.

#### 7.3.4.4.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.3.4.4.3.2-0 below:

**Table 7.3.4.4.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA |
| 7 | UE supporting GNSS(1) and OTDOA |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 7.3.4.4.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | IF<br>sub-test 7 or 15 or 16 or 17 or 18<br>THEN<br>The stored assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1a | IF<br>Sub-test 5 or 7<br>THEN<br>The stored OTDOA assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1b | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 1c | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 1d | IF<br>the UE LPP message at step 1c includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 2 | The SS sends a LPP message of type Request Location Information including a request for location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 3 | The UE sends a LPP message of type Request Assistance.NOTE: This requires a method of triggering a Request Assistance Data message. | --> | *ULInformationTransfer* (LPP REQUEST ASSISTANCE DATA) | 1 | P |
| 4 | The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 4a | IF sub-test 7 THEN the UE may send a second LPP message of type Request Assistance Data including a request for GNSS assistance data or OTDOA assistance data. | --> | *ULInformationTransfer* (LPP REQUEST ASSISTANCE DATA) | 1 | P |
| 4b | IF in step 4a the UE sends a second LPP message of type Request Assistance Data THEN the SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |

| | | | | | |
|---|---|---|---|---|---|
| - | Steps 5a1-5a2 and 5b1-5b4 represent alternative UE behaviours depending on the UE implementation | - | - | - | - |
| 5a1 (Note 1) | All sub-tests:<br>The UE sends a LPP message of type Provide Location Information including location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 5a2 | IF<br>the UE LPP message at step 5a1 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 5b1 (Note 2) | IF sub-test 7<br>THEN<br>The UE sends a LPP message of type Provide Location Information including "early fix" location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | - | - |
| 5b2 | IF<br>the UE LPP message at step 5b1 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 5b3 | The UE sends a LPP message of type Provide Location Information including location measurements. | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 5b4 | IF<br>the UE LPP message at step 5b3 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |

Note 1: This alternative is applicable to all releases of LPP and may be followed even in the case of sub-test 7 and LPP release 12 onwards.

Note 2: This alternative is applicable only to LPP release 12 onwards.

7.3.4.4.3.3 Specific message contents

**Table 7.3.4.4.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-tests 7, 15: 0 0 0 0 0 0 0 0<br>Sub-test 16: 0 0 0 0 0 0 0 1 0<br>Sub-test 17: 0 0 0 0 0 0 0 1 1<br>Sub-test 18: 0 0 0 0 0 1 0 1 | Sub-tests 7, 15: GNSS<br>Sub-test 16: MBS<br>Sub-test 17: WLAN<br>Sub-test 18: Sensor | |

**Table 7.3.4.4.3.3-2: RESET UE POSITIONING STORED INFORMATION (step 1a, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 7.3.4.4.3.3-3: *DLInformationTransfer* (steps 1b, 1d, 2, 4, 4b, 5a2, 5b2 and 5b4, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.4.3.3-4 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-4: DOWNLINK GENERIC NAS TRANSPORT (steps 1b, 1d, 2, 4, 4b, 5a2, 5b2 and 5b4, Table 7.3.4.4.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1b:** Set according to Table 7.3.4.4.3.3-4a | LPP Request Capabilities | |
| | **Step 2:** Set according to Table 7.3.4.4.3.3-5 | LPP Request Location Information | |
| | **Steps 4 and 4b:** Set according to Table 7.3.4.4.3.3-10 | LPP Provide Assistance Data | |
| | **Steps 1d, 5a2, 5b2 and 5b4:** Set according to Table 7.3.4.4.3.3-12 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.4.4.3.3-4a: LPP Request Capabilities (step 1b, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.4.4.3.3-5: LPP Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |
| a-gnss-RequestLocationInformation | Set according to Table 7.3.4.4.3.3-6 | | Sub-tests 7, 15 |
| otdoa-RequestLocationInformation | Set according to Table 7.3.4.4.3.3-6a | | Sub-test 5, 7 |

| | | | |
|---|---|---|---|
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 32 | | |
| responseTimeEarlyFix-r12 | Sub-tests 5, 15, 16: not present<br>Sub-test 7: 10 | Rel-12 onwards | |
| } | | | |
| velocityRequest | FALSE | | |
| } | | | |
| sensor-RequestLocationInformation-r14 | Set according to Table 7.3.4.4.3.3-6c | Rel-14 onwards | Sub-test 18 |
| tbs-RequestLocationInformation-r13 | Set according to Table 7.3.4.4.3.3-6b | Rel-13 onwards | Sub-test 16 |
| wlan-RequestLocationInformation-r14 | Set according to Table 7.3.4.4.3.3-6d | Rel-14 onwards | Sub-test 17 |

**Table 7.3.4.4.3.3-6: A-GNSS Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-4 with the following exceptions: | | | |
| assistanceAvailability | TRUE | | |
| multiFreqMeasReq | TRUE, if UE supports multi frequency GNSS | | MultiFreqAGNSS |

| Condition | Description |
|---|---|
| MultiFreqAGNSS | The UE supports at least one of pc_A_GPS_L2C, pc_A_GPS_L5, pc_QZSS_QZS_L1C, pc_QZSS_QZS_L2C, pc_QZSS_QZS_L5, pc_GLONASS_G2, pc_GLONASS_G3, pc_GALILEO_E5a, pc_GALILEO_E5b, pc_GALILEO_E6, pc_GALILEO_E5aE5b, pc_BDS_B1C, pc_BDS_B2a or pc_BDS_B3I |

**Table 7.3.4.4.3.3-6a: OTDOA Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-5 with the following exceptions: | | | |
| assistanceAvailability | TRUE | | |

**Table 7.3.4.4.3.3-6b: TBS Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-7 with the following exceptions: | | | |
| mbsAssistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.4.3.3-6c: Sensor Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-10 with the following exceptions: | | | |
| assistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.4.3.3-6d: WLAN Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-8 with the following exceptions: | | | |
| assistanceAvailability-r14 | TRUE | Rel-14 onwards | |

**Table 7.3.4.4.3.3-7: *ULInformationTransfer* (steps 1c, 3, 4a, 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.4.4.3.3-8 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-8: UPLINK GENERIC NAS TRANSPORT (steps 1c, 3, 4a, 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1c:** Set according to Table 7.3.4.4.3.3-8a | LPP Provide Capabilities | |
| | **Steps 3 and 4a:** Set according to Table 7.3.4.4.3.3-9 | LPP Request Assistance Data | |
| | **Steps 5a1, 5b1 and 5b3:** Set according to Table 7.3.4.4.3.3-11 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1b, 2 or 4 Table 7.3.4.4.3.2-1) | |

**Table 7.3.4.4.3.3-8a: LPP Provide Capabilities. (step 1c, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1b, Table 7.3.4.4.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-9: LPP Request Assistance Data (steps 3 and 4a, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present or not present | | |

| | | | |
|---|---|---|---|
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| commonIEsRequestAssistanceData | Present or not present | | |
| a-gnss-RequestAssistanceData | Present for sub-tests 7, 15 | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then one contains a-gnss-RequestAssistance Data and the other contains otdoa-RequestAssistance Data | |
| otdoa-RequestAssistanceData | Present for sub-test 5,7 | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then one contains a-gnss-RequestAssistance Data and the other contains otdoa-RequestAssistance Data | |
| epdu-RequestAssistanceData | Not present | | |
| sensor-RequestAssistanceData-r14 | Present for sub-test 18 | Rel-14 onwards | |
| tbs-RequestAssistanceData-r14 | Present for sub-test 16 | Release 14 onwards | |
| wlan-RequestAssistanceData-r14 | Present for sub-test 17 | Rel-14 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-10: LPP Provide Assistance Data (steps 4 and 4b, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Assistance Data message in step 3 or 4a Table 7.3.4.4.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |

| | | | |
|---|---|---|---|
| acknowledgement SEQUENCE { | Present if acknowledgement field is included by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1. | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value as the sequenceNumber in step 3 or 4a, Table 7.3.4.4.3.2-1. | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| a-gnss-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1 which are available according to TS 37.571-5 [12]. | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then the SS sends two seperate LPP Provide Assistance Data messages in steps 4 and 4b each containing the relevant assistance data. | Sub-tests 7, 15 |
| otdoa-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1 according to subclause 5.4.1.2. | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then the SS sends two seperate LPP Provide Assistance Data messages in steps 4 and 4b each containing the relevant assistance data. | Sub-tests 5,7 |
| sensor-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 5.4.1.5. | Release 14 onwards | Sub-test 18 |
| tbs-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 5.4.1.3. | Release 14 onwards | Sub-test 16 |
| wlan-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 5.4.1.4. | Release 14 onwards | Sub-test 17 |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-11: LPP Provide Location Information (steps 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 2, Table 7.3.4.4.3.1-1 | |
| } | | | |
| endTransaction | **Step 5a1, 5b3:** TRUE | | |
| | **Step 5b1:** FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation SEQUENCE { | **Step 5a1, 5b3:** May be present | | |
| | **Step 5b1:** Present | | |
| locationEstimate | Not present | | |
| velocityEstimate | Not present | | |
| locationError | Not present | | |
| earlyFixReport-r12 | **Step 5a1, 5b3:** Not present | Rel-12 onwards | |
| | **Step 5b1:** Any value acceptable | | |
| } | | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | **Step 5a1:** Present for sub-tests 7, 15 | | |
| | **Step 5b1, 5b3:** May be present | One of a-gnss-ProvideLocationInformation or otdoa-ProvideLocationInformation shall be present | |
| gnss-SignalMeasurementInformation SEQUENCE { | Present | | |
| measurementReferenceTime | Present. Any value acceptable | | |
| gnss-MeasurementList SEQUENCE (SIZE(1..n)) OF SEQUENCE { | Present. SIZE n is the number of GNSSs supported by the UE, one instance for each GNSS supported by the UE | | |
| gnss-ID | Present | | |
| gnss-SgnMeasList | Present, one instance for each frequency within the GNSS supported by the UE. Any value acceptable | | |
| } | | | |
| } | | | |
| } | | | |
| gnss-LocationInformation | Not present | | |
| gnss-Error | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| otdoa-ProvideLocationInformation SEQUENCE { | **Step 5a1:** Present for sub-tests 5, 7 | | |
| | **Step 5b1, 5b3:** May be present | One of a-gnss-ProvideLocationInformation or otdoa-ProvideLocationInformation shall be present | |
| otdoa-SignalMeasurementInformation | Present. Any value acceptable | | |
| otdoa-Error | May be present | | |
| } | | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Present for sub-test 18 | Rel-13 onwards | |
| tbs-ProvideLocationInformation-r13 | Present for sub-test 16 | Rel-13 onwards | |
| wlan-ProvideLocationInformation-r13 | Present for sub-test 17 | Rel-13 onwards | |
| bt-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.4.4.3.3-12: LPP Acknowledgement (steps 1d, 5a2, 5b2 and 5b4, Table 7.3.4.4.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1c or 5a1 or 5b1 or 5b3, Table 7.3.4.4.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

# 7.3.5 LPP Abort

## 7.3.5.1 E-SMLC initiated Abort

### 7.3.5.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing }
ensure that {
  when { UE receives a LPP Abort message carrying the transaction ID of an on-going procedure }
    then { UE aborts the on-going procedure }
          }
```

#### 7.3.5.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355, clause 5.5.3.

[TS 36.355, clause 5.5.3]

Upon receiving an *Abort* message, a device shall:

1> abort any ongoing procedure associated with the transaction ID indicated in the message.

#### 7.3.5.1.3 Test description

##### 7.3.5.1.3.1 Pre-test conditions

System Simulator:

- Sub-tests 11, 12, 13, 15, 16, 17: Cell 1.
- Sub-test 5: Cell 1 as specified in 5.2.2.

UE:

- The UE shall begin the tests with no assistance data stored.

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP TS 36.508 [8].

Related PICS/PIXIT Statements:

-

##### 7.3.5.1.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 7.3.5.1.3.2-0 below:

**Table 7.3.5.1.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

Note that this test case does not include a sub-test for the case where ECID or Sensor is supported by the UE as the behaviour required cannot be guaranteed in these cases.

**Table 7.3.5.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 00 | The SS sends a RESET UE POSITIONING STORED INFORMATION message. | **<--** | RESET UE POSITIONING STORED INFORMATION | **-** | **-** |
| 0 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 0a | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 0b | IF the UE LPP message at step 0a includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 0c | IF NOT Sub-test 11 or 12 or 13 THEN The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | | |
| 1 | The SS sends a LPP message of type Request Location Information including a transaction ID. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 2 | Immediately after step 1, the SS sends a LPP message of type Abort using the same transaction ID chosen in step 1. | <-- | *DLInformationTransfer* (LPP ABORT) | - | - |
| 3 | The SS waits for 10 seconds to ensure the UE does not send a LPP message of type Provide Location Information with the same transaction ID as in step 1. | | | 1 | P |

7.3.5.1.3.3 Specific message contents

**Table 7.3.5.1.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 00, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-tests 15: 0 0 0 0 0 0 0 0<br>Sub-test 5: 0 0 0 0 0 0 0 1<br>Sub-test 11, 17: 0 0 0 0 0 0 1 1<br>Sub-tests 12, 16: 0 0 0 0 0 0 1 0<br>Sub-test 13: 0 0 0 0 0 1 0 0 | Sub-tests 15: AGNSS<br>Sub-test 5: OTDOA<br>Sub-test 11, 17: WLAN<br>Sub-tests 12, 16: MBS<br>Sub-test 13: Bluetooth | |

**Table 7.3.5.1.3.3-1: *DLInformationTransfer* (steps 0, 0b, 0c, 1 and 2, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.5.1.3.3-2 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.5.1.3.3-2: DOWNLINK GENERIC NAS TRANSPORT (steps 0, 0b, 0c, 1 and 2, Table 7.3.5.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 0:** Set according to Table 7.3.5.1.3.3-2a | LPP Request Capabilities. | |
| | **Step 0b:** Set according to Table 7.3.5.1.3.3-2b | LPP Acknowledgement | |
| | **Step 0c:** Set according to Table 7.3.5.1.3.3-2c | LPP Provide Assistance Data | |
| | **Step 1:** Set according to Table 7.3.5.1.3.3-3 | LPP Request Location Information | |
| | **Step 2:** Set according to Table 7.3.5.1.3.3-4 | LPP Abort | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.3.5.1.3.3-2a: LPP Request Capabilities (step 0, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.3.5.1.3.3-2b: LPP Acknowledgement (step 0b, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 0a, Table 7.3.5.1.3.2-1. | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.3.5.1.3.3-2c: LPP Provide Assistance Data (step 0c, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| OTDOA-NeighbourCellInfoList ::= SEQUENCE (SIZE(1)) OF SEQUENCE { | | | Sub-test 5 |
| SEQUENCE (SIZE(18)) OF SEQUENCE { | Sequence contains 18 instances of the following data. | | |
| physCellId | Refer to Sequence data values in Table 7.3.5.1.3.3-2d | | |
| cellGlobalId | For values of cellidentity refer to Sequence data values in Table 7.3.5.1.3.3-2d | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Same as for the reference cell | |
| prs-SubframeOffset | Not present | Same as for the reference cell | |
| expectedRSTD | Refer to Sequence data values in Table 7.3.5.1.3.3-2d | | |
| expectedRSTD-Uncertainty | Refer to Sequence data values in Table 7.3.5.1.3.3-2d | | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| } | | | |
| } | | | |

**Table 7.3.5.1.3.3-2d: Sequence data values for 18 instances of sequence for Table 7.3.5.1.3.3-2c**

| Cell | Value physCellId | Value cellidentity (E-UTRAN Cell Identity) | | Value expectedRSTD | Value expectedRSTD-Uncertainty | Comment |
|---|---|---|---|---|---|---|
| | | Value eNB ID | Value Cell Identity | | | |
| Cell 2 | 2 | '0000 0000 0000 0000 0001'B | '0000 0010'B | 8192 | 10 | As defined for Cell 2 in 36.508 [8] |
| Cell 4 | 4 | '0000 0000 0000 0000 0011'B | ‘0000 0100’B | 8192 | 10 | As defined for Cell 4 in 36.508 [8] |
| Dummy cell | 1 | '0000 0000 0000 0000 0001'B | '0000 0001'B | 8253 | 51 | |
| Dummy cell | 3 | '0000 0000 0000 0000 0010'B | '0000 0011'B | 8211 | 51 | |
| Dummy cell | 6 | '0000 0000 0000 0000 0100'B | '0000 0110'B | 8221 | 51 | |
| Dummy cell | 7 | '0000 0000 0000 0000 0110'B | ‘0000 0111’B | 8192 | 51 | |
| Dummy cell | 8 | '0000 0000 0000 0000 0010'B | '0000 1000'B | 8233 | 51 | |
| Dummy cell | 9 | '0000 0000 0000 0000 0100'B | ‘0000 1001’B | 8161 | 51 | |
| Dummy cell | 10 | '0000 0000 0000 0000 0101'B | '0000 1010'B | 8226 | 51 | |
| Dummy cell | 11 | '0000 0000 0000 0000 0110'B | '0000 1011'B | 8232 | 51 | |
| Dummy cell | 16 | '0000 0000 0000 0000 0010'B | '0001 0000'B | 8223 | 51 | |
| Dummy cell | 111 | '0000 0000 0000 0000 1100'B | '0110 1111'B | 8236 | 51 | |
| Dummy cell | 118 | '0000 0000 0000 0000 1111'B | ‘0111 0110’B | 8223 | 51 | |
| Dummy cell | 119 | '0000 0000 0000 0000 1110'B | ‘0111 0111’B | 8221 | 51 | |
| Dummy cell | 120 | '0000 0000 0000 0000 1111'B | ‘0111 1000’B | 8223 | 51 | |
| Dummy cell | 122 | '0000 0000 0000 0000 1010'B | ‘0111 1010’B | 8243 | 51 | |
| Dummy cell | 125 | '0000 0000 0000 0000 1011'B | ‘0111 1101’B | 8253 | 51 | |
| Dummy cell | 126 | '0000 0000 0000 0000 1100'B | ‘0111 1110’B | 8257 | 51 | |

**Table 7.3.5.1.3.3-3: LPP Request Location Information (step 1, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 0 | | |
| } | | | |
| locationInformationType | locationEstimateRequired | In case of "UE-based" Location method supported by the UE | |
| | locationMeasurementsRequired | In case of "UE-assisted" Location method supported by the UE | |
| time | 10 | | |

**Table 7.3.5.1.3.3-4: LPP Abort (step 2, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 0 | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present. | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| abort SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| abort-r9 SEQUENCE { | | | |
| commonIEsAbort SEQUENCE { | | | |
| abortCause | networkAbort | | |
| } | | | |
| epdu-Abort | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.5.1.3.3-5: *ULInformationTransfer* (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.3.5.1.3.3-6 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.5.1.3.3-6: UPLINK GENERIC NAS TRANSPORT (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | Set according to Table 7.3.5.1.3.3-7 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 0 Table 7.3.5.1.3.2-1) | |

**Table 7.3.5.1.3.3-7: LPP Provide Capabilities. (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 0, Table 7.3.5.1.3.2-1. | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

# 7.4 Circuit Switched (CS) Fallback

## 7.4.1 MO-LR Procedure

### 7.4.1.1 CS fallback: Network does not support EPC-MO-LR

#### 7.4.1.1.1 Test Purpose (TP)

(1)

```
with { UE in E-UTRA RRC_IDLE state having received an ATTACH ACCEPT message indicating location
       services via EPC not supported and location services via CS domain supported }
ensure that {
  when { UE initiates MO-LR procedure }
    then { UE transmits an EXTENDED SERVICE REQUEST message }
           }
```

#### 7.4.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 23.272, clause 8.3.1.

[TS 23.272, clause 8.3.1]

MO-LR procedure in the CS fallback in EPS is performed as specified in TS 23.271 [8].

When the MO-LR procedure is triggered by the UE's application, UE will check the LCS Support Indication provided by the Attach and TAU procedures as specified in TS 23.401 [2]:

- If the LCS Support Indication indicates EPC-MO-LR is supported, and if the UE supports EPC-MO-LR, the UE stays in LTE and initiates the EPC-MO-LR procedure.
- If EPC-MO-LR is not supported by either the network or the UE and if the LCS Support Indication indicates CS-MO-LR is supported, and the UE supports CS-MO-LR, the UE assumes CS-MO-LR is provided. Also, if EPC-MO-LR is not supported by either the network or the UE and if network does not provide information on whether CS-MO-LR is supported, then UE assumes CS-MO-LR may be provided. In these cases, if the previous combined EPS/IMSI Attach or Combined TA/LA Update is accepted with no "SMS only" indication, then the UE initiates CS Fallback to perform CS-MO-LR.

NOTE: Based on UE implementation, UE may avoid initiating CS-MO-LR when an IMS VoIP session is active.

- Otherwise, the UE shall not attempt the EPC-MO-LR procedure, i.e. neither EPC-MO-LR nor CS-MO-LR with CS Fallback.

If the UE decided to initiate the CS Fallback for the LCS based on LCS Support Indication check, then, the following is applied:

- When UE is in active mode, UE and the network follows the procedure in clause 6.2 "Mobile Originating Call in Active-Mode". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedures as specified in TS 23.271 [8].
- When UE is in active mode but there's no need for PS-Handover, then UE and the network follows the procedure in clause 6.3 "Mobile Originating Call in Active Mode - No PS HO Support in GERAN". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedure as specified in TS 23.271 [8].
- When UE is in idle mode, UE follows the procedure in clause 6.4 "Mobile Originating Call in Idle Mode". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedure as specified in TS 23.271 [8].

7.4.1.1.3 Test description

7.4.1.1.3.1 Pre-test conditions

System Simulator:

- Cell 1 (E-UTRA) and Cell 5 (UTRA)
- System information combination 4 as defined in TS 36.508 [8] clause 4.4.3.1 is applied to cell 1.

**Table 7.4.1.1.3.1-1: Cell power levels**

| Parameter | Unit | Cell1 | Cell 5 | Remark |
|---|---|---|---|---|
| Cell-specific RS EPRE | dBm/15kHz | -75 | - | The power levels are such that the UE camps on cell 1. |
| CPICH_Ec | dBm/3.84 MHz | - | -70 | |

UE:

- The UE is configured to initiate combined EPS/IMSI attach.

Preamble:

- The UE is in state Registered, Idle Mode (state 2) on cell 1 according to TS 36.508 [8] During the registration procedure, the LAI of cell 5 is allocated to the UE.

Related PICS/PIXIT Statements:

- Method of triggering a CS-MO-LR request for a location estimate.

7.4.1.1.3.2 Test procedure sequence

**Table 7.4.1.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | Cause the UE to initiate MO-LR procedure. | - | - | - | - |
| 2 | The UE transmits an *RRCConnectionRequest* message on Cell 1. | --> | *RRCConnectionRequest* | - | - |
| 3 | The SS transmits an *RRCConnectionSetup* message on Cell 1. | <-- | *RRCConnectionSetup* | - | - |
| 4 | The UE transmits an *RRCConnectionSetupComplete* message on Cell 1.<br>This message includes an EXTENDED SERVICE REQUEST message. | --> | *RRCConnectionSetupComplete* | 1 | P |
| 5 | The SS transmits an *RRCConnectionRelease* message for redirection to UTRAN carrier on Cell 5. | <-- | *RRCConnectionRelease* | - | - |
| 6-14 | Steps 1 to 6 and steps 10 to 12 of the test procedure in subclause 6.1.2.1 are performed on Cell 5.<br>Note: RRC connection establishment procedure and LCS procedure are performed in UTRAN cell. | - | - | - | - |

7.4.1.1.3.3 Specific message contents

**Table 7.4.1.1.3.3-1: SystemInformationBlockType6 for cell 1 (preamble, table 7.4.1.1.3.2-1)**

| Derivation path: 36.508 table 4.4.3.3-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| SystemInformationBlockType6 ::= SEQUENCE { | | | |
| carrierFreqListUTRA-FDD SEQUENCE (SIZE (1..maxUTRA-FDD-Carrier)) OF SEQUENCE { | | | FDD |
| carrierFreq[n] | Same as cell 5 | | |
| cellReselectionPriority[n] | 3 | Lower than cell 1 priority (priority = 4) | |
| } | | | |
| carrierFreqListUTRA-TDD SEQUENCE (SIZE (1..maxUTRA-TDD-Carrier)) OF SEQUENCE { | 1 entry | | TDD |
| carrierFreq[n] | Same as cell 5 | | |
| cellReselectionPriority[n] | 3 | Lower than cell 1 priority (priority = 4) | |
| } | | | |
| } | | | |

**Table 7.4.1.1.3.3-2: Message ATTACH ACCEPT (preamble, Table 7.4.1.1.3.2-1)**

| Derivation Path: TS 36.508 Table 4.7.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Location services indicator in EPC (EPC-LCS) | 0 | location services via EPC not supported | |
| Location services indicator in CS (CS-LCS) | 01 | location services via CS domain supported | |

**Table 7.4.1.1.3.3-3: EXTENDED SERVICE REQUEST (step 4, table 13.1.2.3.2-1)**

| Derivation path: 36.508 table 4.7.2-14A | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| Service type | 0000 'mobile originating CS fallback or 1xCS fallback' | | |
| CSFB response | Not present | | |
| EPS bearer context status | Not present or any allowed value | | |

**Table 7.4.1.1.3.3-4: Message *RRCConnectionRelease* (step 5, Table 7.4.1.1.3.2-1)**

| Derivation Path: TS 36.508 Table 4.6.1-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionRelease ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Downlink UARFCN of cell 5 | | UTRA-FDD |
| utra-TDD | Downlink UARFCN of cell 5 | | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|

| | |
|---|---|
| UTRA-FDD | UTRA FDD cell environment |
| UTRA-TDD | UTRA TDD cell environment |

**Table 7.4.1.1.3.3-5: Message FACILITY (step 6, Table 7.4.1.1.3.2-1)**

| Derivation Path: 24.080 Table 2.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Supplementary service protocol discriminator | 1011 | supplementary services (call independent) | |
| Transaction identifier | | | |
| Facility message type | 0011 1010 | FACILITY | |
| Facility | Return Result=LCS-MOLRRes | Set according to Table 7.4.1.1.3.3-6 | |

**Table 7.4.1.1.3.3-6: LCS-MOLRRes (step 6, Table 7.2.2.2.3.2-1)**

| Derivation Path: 24.080 clause 4.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LCS-MOLRRes::= SEQUENCE { | | | |
| locationEstimate | '90D6B9D6B860B800300 220430144'O | Ellipsoid point with altitude and uncertainty<br>Ellipsoid = 9<br>Spare = 0<br>Degrees of latitude = D6B9D6<br>Degrees of longitude = B860B8<br>Altitude =0030<br>Uncertainty semi-major= 02<br>Uncertainty semi-minor =20<br>Orientation of major axis =43<br>Uncertainty Altitude= 01<br>Confidence=44 | |
| } | | | |

## 7.4.1.2 CS fallback: UE does not support EPC-MO-LR

### 7.4.1.2.1 Test Purpose (TP)

(1)

```
with {  UE in E-UTRA RRC_IDLE state having received an ATTACH ACCEPT message indicating location
        services via EPC supported and location services via CS domain supported }
ensure that {
  when { UE initiates MO-LR procedure }
    then { UE transmits an EXTENDED SERVICE REQUEST message }
           }
```

### 7.4.1.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 23.272, clause 8.3.1.

[TS 23.272, clause 8.3.1]

MO-LR procedure in the CS fallback in EPS is performed as specified in TS 23.271 [8].

When the MO-LR procedure is triggered by the UE's application, UE will check the LCS Support Indication provided by the Attach and TAU procedures as specified in TS 23.401 [2]:

- If the LCS Support Indication indicates EPC-MO-LR is supported, and if the UE supports EPC-MO-LR, the UE stays in LTE and initiates the EPC-MO-LR procedure.

- If EPC-MO-LR is not supported by either the network or the UE and if the LCS Support Indication indicates CS-MO-LR is supported, and the UE supports CS-MO-LR, the UE assumes CS-MO-LR is provided. Also, if EPC-MO-LR is not supported by either the network or the UE and if network does not provide information on whether CS-MO-LR is supported, then UE assumes CS-MO-LR may be provided. In these cases, if the previous combined EPS/IMSI Attach or Combined TA/LA Update is accepted with no "SMS only" indication, then the UE initiates CS Fallback to perform CS-MO-LR.

NOTE: Based on UE implementation, UE may avoid initiating CS-MO-LR when an IMS VoIP session is active.

- Otherwise, the UE shall not attempt the EPC-MO-LR procedure, i.e. neither EPC-MO-LR nor CS-MO-LR with CS Fallback.

If the UE decided to initiate the CS Fallback for the LCS based on LCS Support Indication check, then, the following is applied:

- When UE is in active mode, UE and the network follows the procedure in clause 6.2 "Mobile Originating Call in Active-Mode". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedures as specified in TS 23.271 [8].

- When UE is in active mode but there's no need for PS-Handover, then UE and the network follows the procedure in clause 6.3 "Mobile Originating Call in Active Mode - No PS HO Support in GERAN". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedure as specified in TS 23.271 [8].

- When UE is in idle mode, UE follows the procedure in clause 6.4 "Mobile Originating Call in Idle Mode". After UE changes its RAT from E-UTRAN to UTRAN/GERAN, it performs CS-MO-LR procedure as specified in TS 23.271 [8].

#### 7.4.1.2.3 Test description

##### 7.4.1.2.3.1 Pre-test conditions

System Simulator:

- Cell 1 (E-UTRA) and Cell 5 (UTRA)

- System information combination 4 as defined in TS 36.508 [8] clause 4.4.3.1 is applied to cell 1.

**Table 7.4.1.2.3.1-1: Cell power levels**

| Parameter | Unit | Cell1 | Cell 5 | Remark |
|---|---|---|---|---|
| Cell-specific RS EPRE | dBm/15kHz | -75 | - | The power levels are such that the UE camps on cell 1. |
| CPICH_Ec | dBm/3.84 MHz | - | -70 | |

UE:

- The UE is configured to initiate combined EPS/IMSI attach.

Preamble:

- The UE is in state Registered, Idle Mode (state 2) on cell 1 according to TS 36.508 [8] During the registration procedure, the LAI of cell 5 is allocated to the UE related PICS/PIXIT Statements:

- Method of triggering a CS-MO-LR request for a location estimate.

7.4.1.2.3.2 Test procedure sequence

**Table 7.4.1.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | Cause the UE to initiate MO-LR procedure. | - | - | - | - |
| 2 | The UE transmits an *RRCConnectionRequest* message on Cell 1. | --> | *RRCConnectionRequest* | - | - |
| 3 | The SS transmits an *RRCConnectionSetup* message on Cell 1. | <-- | *RRCConnectionSetup* | - | - |
| 4 | The UE transmits an *RRCConnectionSetupComplete* message on Cell 1.<br>This message includes an EXTENDED SERVICE REQUEST message. | --> | *RRCConnectionSetupComplete* | 1 | P |
| 5 | The SS transmits an *RRCConnectionRelease* message for redirection to UTRAN carrier on Cell 5. | <-- | *RRCConnectionRelease* | - | - |
| 6-14 | Steps 1 to 6 and steps 10 to 12 of the test procedure in subclause 6.1.2.1 are performed on Cell 5.<br>Note: RRC connection establishment procedure and LCS procedure are performed in UTRAN cell. | - | - | - | - |

7.4.1.2.3.3 Specific message contents

**Table 7.4.1.2.3.3-1: SystemInformationBlockType6 for cell 1 (preamble, table 7.4.1.2.3.2-1)**

| Same content as Table 7.4.1.1.3.3-1 |
|---|

**Table 7.4.1.2.3.3-2: Message ATTACH ACCEPT (preamble, Table 7.4.1.2.3.2-1)**

| Derivation Path: TS 36.508 Table 4.7.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Location services indicator in EPC (EPC-LCS) | 1 | location services via EPC supported | |
| Location services indicator in CS (CS-LCS) | 01 | location services via CS domain supported | |

**Table 7.4.1.2.3.3-3: EXTENDED SERVICE REQUEST (step 4, table 13.1.2.3.2-1)**

| Derivation path: 36.508 table 4.7.2-14A | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| Service type | 0000 'mobile originating CS fallback or 1xCS fallback' | | |
| CSFB response | Not present | | |
| EPS bearer context status | Not present or any allowed value | | |

**Table 7.4.1.2.3.3-4: Message *RRCConnectionRelease* (step 5, Table 7.4.1.2.3.2-1)**

| Derivation Path: TS 36.508 Table 4.6.1-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionRelease ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Downlink UARFCN of cell 5 | | UTRA-FDD |
| utra-TDD | Downlink UARFCN of cell 5 | | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| UTRA-FDD | UTRA FDD cell environment |
| UTRA-TDD | UTRA TDD cell environment |

**Table 7.4.1.2.3.3-5: Message *FACILITY* (step 6, Table 7.4.1.2.3.2-1)**

| same content as Table 7.4.1.1.3.3-5 |
|---|

# 7.5 RRC Protocol Procedures

## 7.5.1 Inter-frequency RSTD measurement indication

### 7.5.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when {    the UE receives OTDOA assistance data including inter-frequency cells together with
            a location request for OTDOA }
    then { the UE initiates the RRC Inter-frequency RSTD measurement procedure to indicate
 “start”.}
}
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
  when { the UE stops reporting inter-frequency RSTD measurements }
    then { the UE initiates the RRC Inter-frequency RSTD measurement procedure to indicate
 “stop”. }
}
```

### 7.5.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 36.355 clauses 6.5.1.2, 6.5.1.5, and TS 36.331 clauses 5.5.7.2, 5.7.7.3.

[TS 36.355, clause 6.5.1.2]

If inter-frequency neighbour cells are included in *OTDOA-NeighbourCellInfoList*, where an inter-frequency is a E-UTRA frequency which is different from the E-UTRA serving cell frequency, the LPP layer shall inform lower layers to start performing inter-frequency RSTD measurements for these neighbour cells and also provide to lower layers the information about these neighbour cells, e.g. EARFCN and PRS positioning occasion information.

[TS 36.355, clause 6.5.1.5]

If the target device stops reporting inter-frequency RSTD measurements, where the inter-frequency RSTD measurement is an OTDOA RSTD measurement with at least one cell on a frequency different from the serving cell frequency, the LPP layer shall inform lower layers that inter-frequency RSTD measurements are stopped.

[TS 36.331, clause 5.5.7.2]

The UE shall:

1> if and only if upper layers indicate to start performing inter-frequency RSTD measurements; and the UE requires measurement gaps for these measurements while measurement gaps are either not configured or not sufficient:

2> initiate the procedure to indicate start;

NOTE 1: The UE verifies the measurement gap situation only upon receiving the indication from upper layers. If at this point in time sufficient gaps are available, the UE does not initiate the procedure. Unless it receives a new indication from upper layers, the UE is only allowed to further repeat the procedure in the same PCell once per frequency if the provided measurement gaps are insufficient.

1> if and only if upper layers indicate to stop performing inter-frequency RSTD measurements:

2> initiate the procedure to indicate stop;

NOTE 2: The UE may initiate the procedure to indicate stop even if it did not previously initiate the procedure to indicate start.

[TS 36.331, clause 5.5.7.3]

The UE shall set the contents of *InterFreqRSTDMeasurementIndication* message as follows:

1> set *the rstd-InterFreqIndication* as follows:

2> if the procedure is initiated to indicate start of inter-frequency RSTD measurements:

3> set the *rstd-InterFreqInfoList* according to the information received from upper layers;

2> else if the procedure is initiated to indicate stop of inter-frequency RSTD measurements:

3> set the rstd-InterFreqIndication to the value stop;

1> submit the *InterFreqRSTDMeasurementIndication* message to lower layers for transmission, upon which the procedure ends;

### 7.5.1.3 Test description

#### 7.5.1.3.1 Pre-test conditions

System Simulator:

- Cell 1 as specified in 5.2.2.

UE:

-

Preamble:

- The UE is in state Generic RB Established (state 3) according to 3GPP 36.508 [8] (no measurement gaps are configured).

Related PICS/PIXIT Statements:

- Support for inter-frequency RSTD measurements that require measurement gaps for these measurements.

7.5.1.3.2 Test procedure sequence

**Table 7.5.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The stored OTDOA assistance data in the UE are cleared. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 1a | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 1b | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 1c | IF the UE LPP message at step 1b includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| - | EXCEPTION: In parallel to the events described in Steps 2 to 6, the steps specified in Table 7.5.1.3.2-2 take place | | | | |
| 2 | The SS sends a LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 3-5 | Void | | | | |
| 6 | The SS sends a LPP message of type Request Location Information including a request for OTDOA location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| - | EXCEPTION: In parallel with the events described in Steps 7 to 10, the steps specified in Table 7.5.1.3.2-3 take place. | | | | |
| 7 | The UE sends a LPP message of type Provide Location Information. | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | - | - |
| 7a | IF the UE LPP message at step 7 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 8-10 | Void | | | | |

**Table 7.5.1.3.2-2: Parallel behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The UE transmits an RRC *InterFreqRSTDMeasurementIndication* message to indicate “start” | --> | *InterFreqRSTDMeasurementIndication* | 1 | P |
| 2 | The SS transmits an *RRCConnectionReconfiguration* message to set up the measurement gap configuration | <-- | *RRCConnectionReconfiguration* | - | - |
| 3 | The UE transmits an *RRCConnectionReconfigurationComplete* message to confirm the set up of the measurement gap configuration | --> | *RRCConnectionReconfigurationComplete* | - | - |

**Table 7.5.1.3.2-3: Parallel behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The UE transmits an RRC *InterFreqRSTDMeasurementIndication* message to indicate “stop” | --> | *InterFreqRSTDMeasurementIndication* | 2 | P |
| 2 | The SS transmits an *RRCConnectionReconfiguration* message to release the measurement gap configuration | <-- | *RRCConnectionReconfiguration* | - | - |
| 3 | The UE transmits an *RRCConnectionReconfigurationComplete* message to confirm the release of the measurement gap configuration | --> | *RRCConnectionReconfigurationComplete* | - | - |

### 7.5.1.3.3 Specific message contents

For the default message content as specified in subclause 5.4, the values for sub-test 5 are used.

**Table 7.5.1.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.5.1.3.2-1)**

| Derivation Path: 36.509 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | 0 0 0 0 0 0 0 1 | OTDOA | |

**Table 7.5.1.3.3-2: *DLInformationTransfer* (steps 1a, 1c, 2, 6 and 7a, Table 7.5.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| dlInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.5.1.3.3-3 | DOWNLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-3: DOWNLINK GENERIC NAS TRANSPORT (steps 1a, 1c, 2, 6 and 7a, Table 7.5.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.31.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Downlink generic NAS transport message identity | 01101000 | Downlink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1a:**<br>Set according to Table 7.5.1.3.3-3a | LPP Request Capabilities. | |
| | Step 2:<br>Set according to Table 7.5.1.3.3-4 | LPP Provide Assistance Data | |
| | Step 6:<br>Set according to Table 7.5.1.3.3-8 | LPP Request Location Information | |
| | Step**s 1c and** 7a:<br>Set according to Table 7.5.1.3.3-12 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/ Correlation ID | |

**Table 7.5.1.3.3-3a: LPP Request Capabilities (step 1a, Table 7.5.1.3.2-1)**

| Derivation Path: Table 5.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-1 | | | |

**Table 7.5.1.3.3-4: LPP Provide Assistance data (step 2, Table 7.5.1.3.2-1)**

| Derivation Path: Table 5.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-2 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | |
| otdoa-NeighbourCellInfo | Set according to Table 7.5.1.3.3-5. | | |
| } | | | |

**Table 7.5.1.3.3-5: OTDOA-NeighbourCellInfoList (step 2, Table 7.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-NeighbourCellInfoList ::= SEQUENCE (SIZE(2)) OF SEQUENCE { | | | |
| SEQUENCE (SIZE(1)) OF SEQUENCE { | | Cell 3 | |
| physCellId | 3 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 3 in 36.508 [8] | | |
| mnc | As defined for Cell 3 in 36.508 [8] | | |
| cellidentity | As defined for Cell 3 in 36.508 [8] | | |
| } | | | |
| earfcn | For E-UTRA band < 65: as defined for Cell 3 in 36.508 [8]<br>For E-UTRA band > 64: not present | | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo SEQUENCE { | | | |
| prs-Bandwidth | PRS are transmitted over the used system bandwidth (see subclause 5.2.2) | | |
| prs-ConfigurationIndex | 12 | | |
| numDL-Frames | sf-1 | | |
| prs-MutingInfo-r9 | Not present | PRS muting is not used. | |
| prsID-r14 | Not present | PRS-ID not used | Rel-14 onwards |
| add-numDL-Frames-r14 | Not present | Not required | Rel-14 onwards |
| prsOccGroupLen-r14 | Not present | No PRS occasion group configured | Rel-14 onwards |
| prsHoppingInfo-r14 | Not present | PRS frequency hopping not used | Rel-14 onwards |
| } | | | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | 10 | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 µs | |
| earfcn-v9a0 | For E-UTRA band < 65: not present<br>For E-UTRA Band > 64: as defined for Cell 3 in 36.508 [8] | | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Normal | | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| } | | | |

| | | | |
|---|---|---|---|
| SEQUENCE (SIZE(1)) OF SEQUENCE { | | Cell 6 | Assumes that earfcn for Cell 6 is different from earfcn for Cell 3. |
| physCellId | 6 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 6 in 36.508 [8] | | |
| mnc | As defined for Cell 6 in 36.508 [8] | | |
| cellidentity | As defined for Cell 6 in 36.508 [8] | | |
| } | | | |
| earfcn | For E-UTRA band < 65: as defined for Cell 6 in 36.508 [8]<br>For E-UTRA band > 64: not present | | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo SEQUENCE { | | | |
| prs-Bandwidth | PRS are transmitted over the used system bandwidth (see subclause 5.2.2) | | |
| prs-ConfigurationIndex | 12 | | |
| numDL-Frames | sf-1 | | |
| prs-MutingInfo-r9 | Not present | PRS muting is not used. | |
| prsID-r14 | Not present | PRS-ID not used | Rel-14 onwards |
| add-numDL-Frames-r14 | Not present | Not required | Rel-14 onwards |
| prsOccGroupLen-r14 | Not present | No PRS occasion group configured | Rel-14 onwards |
| prsHoppingInfo-r14 | Not present | PRS frequency hopping not used | Rel-14 onwards |
| } | | | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | 10 | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 μs | |
| earfcn-v9a0 | For E-UTRA band < 65: not present<br>For E-UTRA band > 64: as defined for Cell 6 in 36.508 [8] | | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Normal | | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 7.5.1.3.3-6: RRC InterFreqRSTDMeasurementIndication (step 1, Table 7.5.1.3.2-2)**

| Derivation Path: 36.331, clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| InterFreqRSTDMeasurementIndication-r10 ::= SEQUENCE{ | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| interFreqRSTDMeasurementIndication-r10 SEQUENCE { | | | |
| rstd-InterFreqIndication-r10 CHOICE { | | | |
| start SEQUENCE { | | | |
| rstd-InterFreqInfoList-r10 SEQUENCE { | 2 entries | | |
| carrierFreq-r10[1] | For E-UTRA band < 65: as defined for Cell 3 in 36.508 [8]<br>For E-UTRA band > 64: 65535 | | |
| measPRS-Offset-r10[1] | (0..39) | | |
| carrierFreq-v1090[1] | For E-UTRA band < 65: not present<br>For E-UTRA band > 64: as defined for Cell 3 in 36.508 [8] | | |
| carrierFreq-r10[2] | For E-UTRA band < 65: as defined for Cell 6 in 36.508 [8]<br>For E-UTRA band > 64: 65535 | | |
| measPRS-Offset-r10[2] | (0..39) | | |
| carrierFreq-v1090[2] | For E-UTRA band < 65: Not present<br>For E-UTRA band > 64: as defined for Cell 6 in 36.508 [8] | | |
| } | | | |
| } | | | |
| } | | | |
| lateNonCriticalExtension OCTET STRING | Not present | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| criticalExtensionsFuture SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-7: RRCConnectionReconfiguration (step 2, Table 7.5.1.3.2-2)**

| Derivation Path: 36.508, Table 4.6.1-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in 36.508, Table 4.6.1-8 with the following exceptions: | | | |
| measConfig ::= SEQUENCE { | | | |
| measGapConfig CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | Value of measPRS-Offset-r10 as provided by the UE in Table 7.5.1.3.3-6. | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 7.5.1.3.3-8: LPP Request Location Information (step 6, Table 7.5.1.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |
| time | 10 | | |

**Table 7.5.1.3.3-9: ULInformationTransfer (steps 1b and 7, Table 7.5.1.3.2-1)**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulInformationTransfer-r8 SEQUENCE { | | | |
| dedicatedInfoType CHOICE { | | | |
| dedicatedInfoNAS OCTET STRING | Set according to Table 7.5.1.3.3-10 | UPLINK GENERIC NAS TRANSPORT | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-10: UPLINK GENERIC NAS TRANSPORT (steps 1b and 7, Table 7.5.1.3.2-1)**

| Derivation Path: 24.301 Table 8.2.32.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 0111 | EPS mobility management messages | |
| Security header type | 0000 | Plain NAS message | |
| Uplink generic NAS transport message identity | 01101001 | Uplink generic NAS transport | |
| Generic message container type | 00000001 | LTE Positioning Protocol (LPP) message container | |
| Generic message container | **Step 1b:** Set according to Table 7.5.1.3.3-10a | LPP Provide Capabilities | |
| | **Step 7:** Set according to Table 7.5.1.3.3-11 | LPP Provide Location Information | |
| Additional information | present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1a or 6 Table 7.5.1.3.2-1) | |

**Table 7.5.1.3.3-10a: LPP Provide Capabilities. (step 1b, Table 7.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1a, Table 7.5.1.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| otdoa-ProvideCapabilities | Dependent on UE capabilities | | |
| ecid-ProvideCapabilities | Dependent on UE capabilities | | |
| epdu-ProvideCapabilities | Not present | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| wlan-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| bt-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-11: LPP Provide Location Information (step 7, Table 7.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 6, Table 7.5.1.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| a-gnss-ProvideLocationInformation | Not present | | |
| otdoa-ProvideLocationInformation | May be present. Any value acceptable | | |
| ecid-ProvideLocationInformation | Not present | | |
| epdu-ProvideLocationInformation | Not present | | |
| sensor-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| tbs-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| wlan-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| bt-ProvideLocationInformation-r13 | Not present | Rel-13 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-12: LPP Acknowledgement (steps 1c and 7a, Table 7.5.1.3.2-1)**

| Derivation Path: 36.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1b or 7, Table 7.5.1.3.2-1 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 7.5.1.3.3-13: RRC InterFreqRSTDMeasurementIndication (step 1, Table 7.5.1.3.2-3)**

| Derivation Path: 36.331, clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| InterFreqRSTDMeasurementIndication-r10 ::= SEQUENCE{ | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| interFreqRSTDMeasurementIndication-r10 SEQUENCE { | | | |
| rstd-InterFreqIndication-r10 CHOICE { | | | |
| stop | NULL | | |
| } | | | |
| lateNonCriticalExtension OCTET STRING | Not present | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| criticalExtensionsFuture SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.5.1.3.3-14: RRCConnectionReconfiguration (step 2, Table 7.5.1.3.2-3)**

| Derivation Path: 36.508, Table 4.6.1-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in 36.508, Table 4.6.1-8 with the following exceptions: | | | |
| measConfig ::= SEQUENCE { | | | |
| measGapConfig CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |

### 7.5.2 Void

# 8 Default Conditions for NR

## 8.1 LCS Sub-Test Cases

Some test cases defined in clause 9 may include several sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 8.1-1. The applicable sub-tests for each test case are specified in the test procedure sequence clause of each test case. If no sub-tests are defined for a specific test case it means that this particular test case is not dependent on a specific positioning method and is applicable in all cases, independently of the positioning method(s) supported by the UE.

**Table 8.1-1: Sub-Test Case Numbers for NR**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 1 | Void |
| 2 | Void |
| 3 | Void |
| 4 | Void |
| 5 | UE supporting OTDOA (LTE) |
| 6 | UE supporting ECID (LTE) |
| 7 | UE supporting GNSS(1) and OTDOA (LTE) |
| 8 | Void |
| 9 | Void |
| 10 | Void |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS(2) (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 14 | UE supporting Sensor (Rel-13 only) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS(2) (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 19 | UE supporting Multi-RTT (Rel-16 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| 22 | UE supporting NR E-CID (Rel-16 onwards) |
| 23 | UE supporting MBS(2) (Rel-16 onwards) |
| 24 | UE supporting Sensor (Rel-16 onwards) |
| 25 | UE supporting GNSS(1) (Rel-15 onwards) |
| 26 | UE supporting Multi-RTT (Rel-17 onwards) |
| 27 | UE supporting DL-AoD (Rel-17 onwards) |
| 28 | UE supporting DL-TDOA (Rel-17 onwards) |
| 29 | UE supporting UL-TDOA (Rel-17 onwards) |
| 30 | UE supporting Multi-RTT (Rel-18 onwards) |
| 31 | UE supporting DL-AoD (Rel-18 onwards) |
| 32 | UE supporting DL-TDOA (Rel-18 onwards) |
| 33 | UE supporting SL-RTT (Rel-18 onwards) |
| 34 | UE supporting SL-AoA (Rel-18 onwards) |
| 35 | UE supporting SL-TDOA (Rel-18 onwards) |
| 36 | UE supporting SL-TOA (Rel-18 onwards) |
| NOTE 1: The GNSS combination of BDS, Galileo, GLONASS, GPS supported by the UE | |
| NOTE 2: Metropolitan Beacon System (MBS) is a specific type of Terrestrial Beacon System (TBS) [29] | |

# 8.1A Test Configurations

For ease of use of this document a number of Test Configurations corresponding to Network Deployment Types are defined in Table 8.1A-1.

**Table 8.1A-1: Test Configuration**

| Test Configuration | Network Deployment Type |
|---|---|
| A | EN-DC |
| B | NG-RAN NR |
| C | NE-DC |
| D | NG-RAN E-UTRA |
| E | NGEN-DC |

For ease of use of this document a number of sidelink positioning coverage scenarios are defined in Table 8.1A-2.

**Table 8.1A-2: Sidelink positioning coverage scenarios**

| Test Configuration | Sidelink Positioning Coverage Scenarios |
|---|---|
| A | In-coverage |
| B | Partial-coverage |
| C | out-of-coverage |

# 8.2 Default signal conditions

## 8.2.1 Simulated GNSS environment

Same as defined in clause 5.2.1.

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration D (Table 8.1A-1) the LTE Cell 1 frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.2 Simulated OTDOA (LTE) environment

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration B (Table 8.1A-1) an additional independent multi cell LTE environment is used with LTE Cell 1 and LTE Cell 2 (where required) . The E-UTRA frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

For Test Configuration D (Table 8.1A-1), the LTE Cell 1 and LTE Cell 2 (where required) E-UTRA frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

All LTE cells transmit PRS according to the PRS configuration provided in the OTDOA (LTE) assistance data defined in subclause 8.4.1.2. The positioning subframes are low-interference subframes, i.e. contain no PDSCH transmissions.

LTE Cell 1 is the OTDOA reference cell, LTE Cell 2 (where required) is an OTDOA neighbour cell.

Where two LTE cells are required, the two LTE Cells 1 and 2 shall be synchronized, and the timing offset (the RSTD) between the cells, referenced to the UE's antenna input, shall be set equal to the *expectedRSTD* value provided in the OTDOA (LTE) assistance data, as defined in subclause 8.4.1.2.

Normal propagation condition is used for all cells.

## 8.2.3 Simulated ECID (LTE) environment

Same as defined in clause 5.2.3.

For Test Configuration B (Table 8.1A-1) up to and including LPP Rel-15, UE Rx-Tx measurement is not possible and therefore there is little value in testing ECID (LTE) for positioning purposes and ECID (LTE) shall not be tested.

For Test Configuration D (Table 8.1A-1), the LTE Cell 1 and LTE Cell 2 E-UTRA frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.4 Simulated MBS environment

Same as defined in clause 5.2.4.

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration D (Table 8.1A-1) the LTE Cell 1 frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.5 Simulated WLAN environment

Same as defined in clause 5.2.5.

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration D (Table 8.1A-1) the LTE Cell 1 frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.6 Simulated Bluetooth environment

Same as defined in clause 5.2.6.

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration D (Table 8.1A-1) the LTE Cell 1 frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.7 Simulated Sensor environment

Same as defined in clause 5.2.7.

For Test Configuration B (Table 8.1A-1) the NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

For Test Configuration D (Table 8.1A-1) the LTE Cell 1 frequency to be tested and other default conditions are as specified for signalling test cases in 3GPP TS 36.508 [8].

## 8.2.8 Simulated general NR environment

For NR FR2, the connection between the SS and the DUT shall be OTA. The SS shall ensure that the NR cell is suitable throughout the test. The OTA link shall be sufficient to provide stable LPP/SLPP message transmissions between the SS and the DUT.

For NR FR1 and the other technologies used for the test cases in Chapter 9 (e.g. LTE, GNSS, WLAN…) the connection shall be conducted, following the settings described in this section.

## 8.2.9 Simulated Multi-RTT environment

For Multi-RTT signalling test cases NR Cell 1 is used, as defined in 3GPP TS 38.508-1 [30].

NR cell transmits DL-PRS according to the DL-PRS configuration provided in the Multi-RTT assistance data defined in subclause 8.4.1.6. The UE transmits UL-SRS according to the UL-SRS configuration provided in the RRC message as defined in subclause 8.3.2.

Normal propagation condition is used for all cells.

The simulated general NR environment as specified in 8.2.8 also applies.

Multi-RTT tests are only applicable for Test Configuration B (Table 8.1A-1). The NR Cell 1 frequency to be used for testing and other default conditions is as specified for signalling test cases in 3GPP TS 38.508-1 [30].

## 8.2.10 Simulated DL-AoD environment

For DL-AoD signalling test cases NR Cell 1 and NR Cell 2 (if required) are used, as defined in 3GPP TS 38.508-1 [30].

All NR cells transmit DL-PRS according to the DL-PRS configuration provided in the DL-AoD assistance data defined in subclause 8.4.1.7.

Normal propagation condition is used for all cells.

The simulated general NR environment as specified in 8.2.8 also applies.

DL-AoD tests are only applicable for Test Configuration B (Table 8.1A-1). The NR Cell 1 and NR Cell 2 frequencies to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

## 8.2.11 Simulated DL-TDOA environment

For DL-TDOA signalling test cases a multi cell environment with NR Cell 1, NR Cell 2 and NR Cell 3 (if required) are used, as defined in 3GPP TS 38.508-1 [30].

All NR cells transmit DL-PRS according to the DL-PRS configuration provided in the DL-TDOA assistance data defined in subclause 8.4.1.8.

Normal propagation condition is used for all cells.

NR Cell 1 is the DL-TDOA reference cell. NR Cell 2 and NR Cell 3 are DL-TDOA neighbour cells.

The simulated general NR environment as specified in 8.2.8 also applies.

DL-TDOA tests are only applicable for Test Configuration B (Table 8.1A-1). The NR Cell 1, NR Cell 2 and NR Cell 3 frequencies to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

## 8.2.12 Simulated NR E-CID environment

For NR E-CID signalling test cases NR Cell 1 is used, as defined in 3GPP TS 38.508-1 [30].

## 8.2.13 Simulated UL-TDOA environment

For UL-TDOA signalling test cases NR Cell 1 is used, as defined in 3GPP TS 38.508-1 [30].

The simulated general NR environment as specified in 8.2.8 also applies.

UL-TDOA test is only applicable for Test Configuration B (Table 8.1A-1). The NR Cell 1 frequency to be used for testing and other default conditions is as specified for signalling test cases in 3GPP TS 38.508-1 [30].

The simulated general NR environment as specified in 8.2.8 also applies.

NR E-CID tests are only applicable for Test Configuration B (Table 8.1A-1). The NR Cell 1 frequency to be used for testing and other default conditions is as specified for signalling test cases in 3GPP TS 38.508-1 [30].

## 8.2.14 Simulated SL-RTT environment

For SL-RTT signalling test cases NR Cell 1 is used for Test Configuration A (Table 8.1A-2) and Test Configuration B (Table 8.1A-2), as defined in 3GPP TS 38.508-1 [30]. The NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

In the test, there is one target UE receiving SL-PRS and performing SL Rx-TX time difference measurements and two anchor UEs NR-SS-UE1 and anchor NR-SS-UE2 (if the UE supports SL-Target UE-based SL-RTT). The synchronization source type is GNSS for all anchor UEs.

The NR-SS-UEs transmit SL-PRS according to the sl-PRS-TxInfo provided in the SLPP PROVIDE ASSISTANCE DATA as defined in subclause 8.5.

The simulated general NR environment as specified in 8.2.8 also applies.

## 8.2.15 Simulated SL-AoA environment

For SL-AoA signalling test cases NR Cell 1 is used for Test Configuration A (Table 8.1A-2) and Test Configuration B (Table 8.1A-2), as defined in 3GPP TS 38.508-1 [30]. The NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

In the test, there is one target UE receiving SL-PRS and performing SL-PRS-RSRP measurements and two anchor UEs NR-SS-UE1 and anchor NR-SS-UE2 (if the UE supports SL-Target UE-based SL-AoA). The synchronization source type is GNSS for all anchor UEs.

The NR-SS-UEs transmit SL-PRS according to the sl-PRS-TxInfo provided in the SLPP PROVIDE ASSISTANCE DATA as defined in subclause 8.5.

The simulated general NR environment as specified in 8.2.8 also applies.

## 8.2.16 Simulated SL-TDOA environment

For SL-TDOA signalling test cases NR Cell 1 is used for Test Configuration A (Table 8.1A-2) and Test Configuration B (Table 8.1A-2), as defined in 3GPP TS 38.508-1 [30]. The NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

In the test, there is one target UE receiving SL-PRS and performing SL Rx-Tx time difference measurements and three anchor UEs NR-SS-UE1, NR-SS-UE2 and anchor NR-SS-UE3 (if the UE supports SL-Target UE-based SL-TDOA). The synchronization source type is GNSS for all anchor UEs.

The NR-SS-UEs transmit SL-PRS according to the sl-PRS-TxInfo provided in the SLPP PROVIDE ASSISTANCE DATA as defined in subclause 8.5.

The simulated general NR environment as specified in 8.2.8 also applies.

## 8.2.17 Simulated SL-TOA environment

For SL-TOA signalling test cases NR Cell 1 is used for Test Configuration A (Table 8.1A-2) and Test Configuration B (Table 8.1A-2), as defined in 3GPP TS 38.508-1 [30]. The NR Cell 1 frequency to be used for testing and other default conditions are as specified for signalling test cases in 3GPP TS 38.508-1 [30].

In the test, there is one target UE receiving SL-PRS and performing SL-RTOA measurements and two anchor UEs NR-SS-UE1 and anchor NR-SS-UE2 (if the UE supports SL-Target UE-based SL-TOA). The synchronization source type is GNSS for all anchor UEs.

The NR-SS-UEs transmit SL-PRS according to the sl-PRS-TxInfo provided in the SLPP PROVIDE ASSISTANCE DATA as defined in subclause 8.5.

The simulated general NR environment as specified in 8.2.8 also applies.

# 8.3 Default RRC and NAS message and information elements contents

The default values of common RRC and NAS messages and information elements are used as defined in 3GPP TS 38.508-1 [30] with the following exceptions:

**Table 8.3-1: Void**

**Table 8.3-2: Void**

- REGISTRATION ACCEPT

**Table 8.3-3: REGISTRATION ACCEPT**

| Derivation Path: 38.508-1 Table 4.7.1-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| 5GS network feature support | Set according to Table 8.3-4 | | |

**Table 8.3-4: 5GS network feature support**

| Derivation Path: 24.501 clause 9.11.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in 38.508-1 Table 4.7.1-7 with the following exceptions: | | | |
| Emergency service support indicator for 3GPP access (EMC) (octet 3, bit 3 and bit 4) | 01 | Emergency services supported in NR connected to 5GCN only | |

## 8.3.1 RRC message contents for measurement gaps

**Table 8.3.1-1: RRCReconfiguration for meas gaps**

| Derivation Path: TS 38.331 [6], clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | Not present | | |
| | Not present | | |
| measConfig | MeasConfig | Table 8.3.1-2 | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.1-2: MeasConfig**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList | MeasObjectToAddModList | Table 8.3.1-4 | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList | Not present | | |
| s-MeasureConfig | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | MeasGapConfig | Table 8.3.1-3 | |
| measGapSharingConfig | Not present | | |
| } | | | |

**Table 8.3.1-3: MeasGapConfig**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= SEQUENCE { | | | |
| gapFR2 | Not present | | |
| gapFR1 | Not present | | |
| gapUE CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset | 10 | | |
| mgl | ms3 | | |
| mgrp | ms160 | | |
| mgta | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.1-4: MeasObjectToAddModList**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectToAddModList ::=SEQUENCE (SIZE (1..maxNrofObjectId)) OF MeasObjectToAddMod { | 1 entry | | |
| MeasObjectToAddMod[1] SEQUENCE { | | entry 1 | |
| measObjectId | 1 | | |
| measObject CHOICE { | | | |
| measObjectNR | MeasObjectNR | Table 8.3.1-5 | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.1-5: MeasObjectNR**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectNR ::=SEQUENCE { | | | |
| ssbFrequency | ARFCN-ValueNR for NR Cell 1 | | |
| ssbSubcarrierSpacing | Subcarrier spacing of SSB for NR Cell 1 | | |
| smtc1 | Not present | | |
| smtc2 | Not present | | |
| refFreqCSI-RS | Not present | | |
| referenceSignalConfig SEQUENCE { | | | |
| ssb-ConfigMobility | Not present | | |
| csi-rs-ResourceConfigMobility | Not present | | |
| } | | | |
| absThreshSS-BlocksConsolidation | Not present | | |
| absThreshCSI-RS-Consolidation | Not present | | |
| nrofSS-BlocksToAverage | Not present | | |
| nrofCSI-RS-ResourcesToAverage | Not present | | |
| quantityConfigIndex | 1 | | |
| offsetMO SEQUENCE { | | | |
| rsrpOffsetSSB | dB0 | | |
| rsrqOffsetSSB | dB0 | | |
| sinrOffsetSSB | dB0 | | |
| rsrpOffsetCSI-RS | dB0 | | |
| rsrqOffsetCSI-RS | dB0 | | |
| sinrOffsetCSI-RS | dB0 | | |
| } | | | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList | Not present | | |
| excludedCellsToRemoveList | Not present | | |
| excludedCellsToAddModList | Not present | | |
| allowedCellsToRemoveList | Not present | | |
| allowedCellsToAddModList | Not present | | |
| freqBandIndicatorNR | Not present | | |
| measCycleSCell | Not present | | |
| smtc3list-r16 | Not present | | |
| rmtc-Config-r16 | Not present | | |
| t312-r16 | Not present | | |
| associatedMeasGapSSB-r17 | Not present | | |
| associatedMeasGapCSIRS-r17 | Not present | | |
| smtc4list-r17 | Not present | | |
| measCyclePSCell-r17 | Not present | | |
| cellsToAddModListExt-v1710 | Not present | | |
| associatedMeasGapSSB2-v1720 | Not present | | |
| associatedMeasGapCSIRS2-v1720 | Not present | | |
| } | | | |

## 8.3.2 RRC message contents for UL-SRS Configuration

**Table 8.3.2-1: RRCReconfiguration**

| Derivation Path: TS 38.508-1, table 4.6.1-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.2-2: CellGroupConfig**

| Derivation Path: 38.508-1 [30], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| uplinkConfig SEQUENCE { | | | |
| initialUplinkBWP SEQUENCE { | | | |
| srs-Config CHOICE { | | | |
| setup | srs-Config | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.2-3: SRS-Config**

| Derivation Path: TS 38.331, clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-Config ::= SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r16 | Not present | | |
| srs-PosResourceSetToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceSet-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16[1] | 0 | entry 1 | |
| } | | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| } | | | |
| alpha-r16 | Not present | | |
| p0-r16 | -100 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 1 | Set according to Table 4.4.2-2 in TS 38.508-1. | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r16 | Not present | | |
| srs-PosResourceToAddModList-r16 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] | SRS-PosResource-r16 | entry 1 | |
| } | | | |
| } | | | |

**Table 8.3.2-4: SRS-PosResource-r16**

| Derivation Path: TS 38.331, clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosResource-r16::=SEQUENCE { | | | |
| srs-PosResourceId-r16 | 0 | | |
| transmissionComb-r16 CHOICE { | | | |
| n4-r16 SEQUENCE { | | | |
| combOffset-n4-r16 | 0 | | |
| cyclicShift-n4-r16 | 0 | | |
| } | | | |
| } | | | |
| resourceMapping-r16 SEQUENCE { | | | |
| startPosition-r16 | 1 | | |
| nrofSymbols-r16 | n2 | | |
| } | | | |
| freqDomainShift-r16 | 0 | | |
| freqHopping-r16 SEQUENCE { | | | |
| c-SRS-r16 | 0 | | |
| } | | | |
| groupOrSequenceHopping-r16 | neither | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| periodicityAndOffset-p-r16 CHOICE { | | | |
| sl160 | 20 | | FR1 |
| | 23 | | FR1 AND (HD_FDD OR TDD) AND SCS15Khz |
| | 27 | | FR1 AND TDD AND SCS30Khz |
| sl1280 | 163 | | FR2 |
| } | | | |
| } | | | |
| } | | | |
| sequenceId-r16 | 0 | | |
| spatialRelationInfoPos-r16 CHOICE { | | | |
| servingRS-r16 SEQUENCE { | | | |
| servingCellId | 0 | | |
| referenceSignal-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 1 | Set according to Table 4.4.2-2 in TS 38.508-1. | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| **Condition** | **Explanation** |
|---|---|
| HD_FDD | pc_halfDuplexFDD_TypeA_RedCap_r17 (i.e HD_FDD UE are performing test on FDD band) |

## 8.3.3 RRC message contents for SL-PRS Configuration

**Table 8.3.3-1: RRCReconfiguration**

| Derivation Path: TS 38.508-1 [30], Table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| sl-ConfigDedicatedNR-r16 CHOICE { | | | |
| setup | SL-ConfigDedicatedNR | Table 8.3.3-2 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.3-2: SL-ConfigDedicatedNR**

| Derivation Path: TS 38.508-1 [30], Table 4.6.6-7 with condition SCHEDULING | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-ConfigDedicatedNR-r16 ::= SEQUENCE { | | | |
| sl-PHY-MAC-RLC-Config-r16 SEQUENCE { | | | |
| sl-FreqInfoToAddModList-r16 SEQUENCE (SIZE (1..maxNrofFreqSL-r16)) OF SL-FreqConfig-r16 { | 1 entry | | |
| SL-FreqConfig-r16[1] | SL-FreqConfig | Table 8.3.3-3 | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.3-3: SL-FreqConfig**

| Derivation Path: TS 38.508-1 [30], Table 4.6.6-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-FreqConfig-r16 ::= SEQUENCE { | | | |
| sl-BWP-ToAddModList-r16 SEQUENCE (SIZE (1..maxNrofSL-BWPs-r16)) OF SL-BWP-Config-r16 { | 1 entry | | |
| SL-BWP-Config-r16[1] | SL-BWP-Config | Table 8.3.3-4 | |
| } | | | |
| } | | | |

**Table 8.3.3-4: SL-BWP-Config**

| Derivation Path: TS 38.508-1 [30], Table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-BWP-Config-r16 ::= SEQUENCE { | | | |
| sl-BWP-Id | BWP-Id | As defined in TS 38.508-1 [30] Table 4.6.3-12 | |
| sl-BWP-Generic-r16 SEQUENCE { | | | |
| sl-BWP-r16 | BWP | As defined in TS 38.508-1 [30] Table 4.6.3-8 | |
| sl-LengthSymbols-r16 | sym14 | | |
| sl-StartSymbol-r16 | sym0 | | |
| sl-PSBCH-Config-r16 | Not present | | |
| sl-TxDirectCurrentLocation-r16 | Not present | | |
| } | | | |
| sl-BWP-PoolConfig-r16 | SL-BWP-PoolConfig | Table 8.3.3-5 | pc_sl_PRS_RxInSharedResourcePool |
| | Not present | | |
| sl-BWP-PoolConfigA2X-r18 | Not present | | |
| sl-BWP-PRS-PoolConfig-r18 | Not present | | |
| sl-BWP-PRS-PoolConfig-r18 CHOICE { | | | |
| setup | SL-BWP-PRS-PoolConfig-r18 | As defined in TS 38.508-1 [30] Table 4.6.6-4A | pc_sl_PRS_RxInDedicatedResourcePool |
| } | | | |
| } | | | |

**Table 8.3.3-5: SL-BWP-PoolConfig**

| Derivation Path: TS 38.331 [6], clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-BWP-PoolConfig-r16 ::= SEQUENCE { | | | |
| sl-RxPool-r16 SEQUENCE (SIZE (1..maxNrofRXPool-r16)) OF SL-ResourcePool-r16 { | 1 entry | | |
| SL-ResourcePool-r16[1] | SL-ResourcePool | As defined in TS 38.508-1 [30] Table 4.6.6-25 with condition SLPP | |
| } | | | |
| sl-TxPoolSelectedNormal-r16 | Not present | | |
| sl-TxPoolScheduling-r16 SEQUENCE { | | | |
| sl-PoolToReleaseList-r16 | Not present | | |
| sl-PoolToAddModList-r16 SEQUENCE (SIZE (1..maxNrofTXPool-r16)) OF SL-ResourcePoolConfig-r16 { | 1 entry | | |
| SL-ResourcePoolConfig-r16[1] SEQUENCE { | | entry 1 | |
| sl-ResourcePoolID-r16 | 1 | | |
| sl-ResourcePool-r16 | SL-ResourcePool | As defined in TS 38.508-1 [30] Table 4.6.6-25 with condition SLPP | |
| } | | | |
| } | | | |
| } | | | |
| sl-TxPoolExceptional-r16 | Not present | | |
| } | | | |

## 8.4 Default LPP message and information elements contents

The default values of LPP messages and information elements used, unless indicated otherwise in specific clauses of this specification are as defined in clause 5.4 with the following exceptions:

- LPP PROVIDE ASSISTANCE DATA

**Table 8.4-1: ProvideAssistanceData**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | Dependent on test case. | | |
| initiator | | | |
| transactionNumber | | | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| a-gnss-ProvideAssistanceData SEQUENCE { | | | Sub-tests 7 and 15 only; and as defined in Table 5.4.1.1-1. |
| gnss-CommonAssistData SEQUENCE { | | | |
| gnss-ReferenceTime | As defined in 37.571-5 [12] | | |
| gnss-ReferenceLocation | As defined in 37.571-5 [12] | | |
| gnss-IonosphericModel | As defined in 37.571-5 [12] | | |
| gnss-EarthOrientationParameters | Not present | | |
| gnss-RTK-ReferenceStationInfo-r15 | Not present | Rel-15 onwards | |
| gnss-RTK-CommonObservationInfo-r15 | Not present | Rel-15 onwards | |
| gnss-RTK-AuxiliaryStationData-r15 | Not present | Rel-15 onwards | |
| gnss-SSR-CorrectionPoints-r16 | Not present | Rel-16 onwards | |
| gnss-Integrity-ServiceParameters-r17 | Not present | Rel-17 onwards | |
| gnss-Integrity-ServiceAlert-r17 | Not present | Rel-17 onwards | |
| } | | | |
| gnss-GenericAssistData(SIZE(1..4))OF{ | SIZE is dependent on the number of GNSSs supported by the UE.<br>If one GNSS supported by the UE, SIZE = 1<br>If two GNSSs supported by the UE, SIZE = 2<br>If three GNSSs supported by the UE, SIZE = 3<br>If four GNSSs supported by the UE, SIZE = 4 | | |
| gnss-ID | For each GNSS supported by the UE. | | |
| sbas-ID | Not present | | |
| gnss-TimeModels | As defined in 37.571-5 [12] | | |
| gnss-DifferentialCorrections | Not present | | |
| gnss-NavigationModel | As defined in 37.571-5 [12] | | |
| gnss-RealTimeIntegrity | Not present | | |
| gnss-DataBitAssistance | Not present | | |
| gnss-AcquisitionAssistance | As defined in 37.571-5 [12] | | |
| gnss-Almanac | As defined in 37.571-5 [12] | | |
| gnss-UTC-Model | As defined in 37.571-5 [12] | | |

| | | | |
|---|---|---|---|
| gnss-AuxiliaryInformation | As defined in 37.571-5 [12] | | |
| bds-DifferentialCorrections-r12 | Not present | Rel-12 onwards | |
| bds-GridModel-r12 | Not present | Rel-12 onwards | |
| gnss-RTK-Observations-r15 | Not present | Rel-15 onwards | |
| glo-RTK-BiasInformation-r15 | Not present | Rel-15 onwards | |
| gnss-RTK-MAC-CorrectionDifferences-r15 | Not present | Rel-15 onwards | |
| gnss-RTK-Residuals-r15 | Not present | Rel-15 onwards | |
| gnss-RTK-FKP-Gradients-r15 | Not present | Rel-15 onwards | |
| gnss-SSR-OrbitCorrections-r15 | Not present | Rel-15 onwards | |
| gnss-SSR-ClockCorrections-r15 | Not present | Rel-15 onwards | |
| gnss-SSR-CodeBias-r15 | Not present | Rel-15 onwards | |
| gnss-SSR-URA-r16 | Not present | Rel-16 onwards | |
| gnss-SSR-PhaseBias-r16 | Not present | Rel-16 onwards | |
| gnss-SSR-STEC-Correction-r16 | Not present | Rel-16 onwards | |
| gnss-SSR-GriddedCorrection-r16 | Not present | Rel-16 onwards | |
| navic-DifferentialCorrections-r16 | Not present | Rel-16 onwards | |
| navic-GridModel-r16 | Not present | Rel-16 onwards | |
| } | | | |
| gnss-Error | Not present | | |
| } | | | |
| otdoa-ProvideAssistanceData SEQUENCE { | | | Sub-tests 5 and 7 only |
| otdoa-ReferenceCellInfo | As defined in Table 8.4.1.2-1 | | |
| otdoa-NeighbourCellInfo | As defined in Table 8.4.1.2-2 | | |
| otdoa-Error | Not present | | |
| otdoa-ReferenceCellInfoNB-r14 | Not present | Rel-14 onwards | |
| otdoa-NeighbourCellInfoNB-r14 | Not present | Rel-14 onwards | |
| } | | | |
| epdu-Provide-AssistanceData | Not present | | |
| sensor-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 18 only as defined in clause 5.4.1.5 |
| sensor-AssistanceDataList-r14 | As defined in Table 5.4.1.5-2 | | |
| sensor-Error-r14 | Not present | | |
| } | | | |
| tbs-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 16 only as defined in clause5.4.1.3 |
| tbs-AssistanceDataList-r14 SEQUENCE { | | | |
| mbs-AssistanceDataList-r14 SEQUENCE (SIZE(1..n)) OF SEQUENCE{ | | | |
| mbs-AlmanacAssistance-r14 | As defined in Table 5.4.1.3-2 | | |
| mbs-AcquisitonAssistance-r14 | As defined in Table 5.4.1.3-2 | | |
| } | | | |
| } | | | |
| tbs-Error-r14 | Not present | | |
| } | | | |
| wlan-ProvideAssistanceData-r14 SEQUENCE { | | Rel-14 onwards | Sub-test 17 only as defined in clause 5.4.1.4 |
| wlan-DataSet-r14 | As defined in Table 5.4.1.4-2 | | |
| wlan-Error-r14 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | Rel-16 onwards | Sub-test 19, Sub-test 26 and Sub-test 30 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 8.4.1.6-1 | | |
| nr-SelectedDL-PRS-IndexList-r16 | As defined in Table 8.4.1.6-2 | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | Rel-17 onwards | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | Rel-17 onwards | |
| assistanceDataValidityArea-r17 | Not present | Rel-17 onwards | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | Rel-16 onwards | Sub-test 20, Sub-test 27 and Sub-test 31 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 8.4.1.6-1 | | |
| nr-SelectedDL-PRS-IndexList-r16 | As defined in Table 8.4.1.6-2 | | |
| nr-PositionCalculationAssistance-r16 | Present or not present dependent on pc_UEB_DL_AoD. As defined in Table 8.4.1.7-1 | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-PRS-BeamInfo-r17 | Not present | Rel-17 onwards | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | Rel-17 onwards | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | Rel-17 onwards | |
| assistanceDataValidityArea-r17 | Not present | Rel-17 onwards | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | Rel-16 onwards | Sub-test 21, Sub-test 28 and Sub-test 32 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 8.4.1.6-1 | | |
| nr-SelectedDL-PRS-IndexList-r16 | As defined in Table 8.4.1.6-2 | | |
| nr-PositionCalculationAssistance-r16 | Present or not present dependent on pc_UEB_DL_TDOA. As defined in Table 8.4.1.7-1 | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-On-Demand-DL-PRS-Configurations-r17 | Not present | Rel-17 onwards | |
| nr-On-Demand-DL-PRS-Configurations-Selected-IndexList-r17 | Not present | Rel-17 onwards | |
| assistanceDataValidityArea-r17 | Not present | Rel-17 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- LPP REQUEST CAPABILITIES

**Table 8.4-2: RequestCapabilities**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestCapabilities-r9 SEQUENCE { | | | |
| commonIEsRequestCapabilities SEQUENCE { | Present | Rel-14 onwards | |
| lpp-message-segmentation-req-r14 | 00 | Server is not able to send segmented LPP messages. Server is not able to receive segmented LPP messages. Rel-14 onwards | |
| } | | | |
| a-gnss-RequestCapabilities SEQUENCE { | | | |
| gnss-SupportListReq | TRUE | | |
| assistanceDataSupportListReq | TRUE | | |
| locationVelocityTypesReq | TRUE | | |
| } | | | |
| otdoa-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| ecid-RequestCapabilities SEQUENCE { | Present | | |
| } | | | |
| epdu-RequestCapabilities | Not present | | |
| sensor-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| tbs-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| wlan-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| bt-RequestCapabilities-r13 SEQUENCE { | Present | Rel-13 onwards | |
| } | | | |
| nr-ECID-RequestCapabilities-r16 SEQUENCE { | Present | Rel-16 onwards | |
| } | | | |
| nr-Multi-RTT-RequestCapabilities-r16 SEQUENCE { | Present | Rel-16 onwards | |
| } | | | |
| nr-DL-AoD-RequestCapabilities-r16 SEQUENCE { | Present | Rel-16 onwards | |
| } | | | |
| nr-DL-TDOA-RequestCapabilities-r16 SEQUENCE { | Present | Rel-16 onwards | |
| } | | | |
| nr-UL-RequestCapabilities-r16 SEQUENCE { | Present | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- LPP REQUEST LOCATION INFORMATION

**Table 8.4-3: RequestLocationInformation**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| acknowledgement | Not present | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | Dependent on test case | | |
| triggeredReporting | Not present | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 32 | | |
| responseTimeEarlyFix-r12 | Not present | Rel-12 onwards | |
| } | | | |
| velocityRequest | FALSE | | |
| messageSizeLimitNB-r14 | Not present | Rel-14 onwards | |
| segmentationInfo-r14 | Not present | Rel-14 onwards | |
| scheduledLocationTime-r17 | Not present | Rel-17 onwards | |
| targetIntegrityRisk-r17 | Not present | Rel-17 onwards | |
| } | | | |
| environment | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |
| a-gnss-RequestLocationInformation | As defined in Table 5.4-4 | | Sub-tests 7 and 15 |
| otdoa-RequestLocationInformation | As defined in Table 5.4-5 | | Sub-test 5 and 7 |
| ecid-RequestLocationInformation | As defined in Table 5.4-6 | | Sub-test 6 |
| epdu-RequestLocationInformation | Not Present | | |
| sensor-RequestLocationInformation-r13 | As defined in Table 5.4-10 | Rel-13 onwards | Sub-test 14, 18 |
| tbs-RequestLocationInformation-r13 | As defined in Table 5.4-7 | Rel-13 onwards | Sub-tests 12, 16 |
| wlan-RequestLocationInformation-r13 | As defined in Table 5.4-8 | Rel-13 onwards | Sub-test 11, 17 |
| bt-RequestLocationInformation-r13 | As defined in Table 5.4-9 | Rel-13 onwards | Sub-test 13 |
| nr-ECID-RequestLocationInformation-r16 | As defined in Table 8.4-4 | Rel-16 onwards | Sub-test 22 |
| nr-Multi-RTT-RequestLocationInformation-r16 | As defined in Table 8.4-5 | Rel-16 onwards | Sub-test 19, Sub-test 26 and Sub-test 30 |
| nr-DL-AoD-RequestLocationInformation-r16 | As defined in Table 8.4-6 | Rel-16 onwards | Sub-test 20, Sub-test 27 and Sub-test 31 |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-RequestLocationInformation-r16 | As defined in Table 8.4-7 | Rel-16 onwards | Sub-test 21, Sub-test 28 and Sub-test 32 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

- NR E-CID REQUEST LOCATION INFORMATION

**Table 8.4-4: NR-ECID-RequestLocationInformation**

| Derivation Path: 37.355 clause 6.5.9.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-ECID-RequestLocationInformation-r16 ::= SEQUENCE { | | | |
| requestedMeasurements-r16 | All measurements supported by the UE | | |
| } | | | |

- NR Multi-RTT REQUEST LOCATION INFORMATION

**Table 8.4-5: NR-Multi-RTT-RequestLocationInformation**

| Derivation Path: 37.355 clause 6.5.12.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-Multi-RTT-RequestLocationInformation-r16 ::= SEQUENCE { | | | |
| nr-UE-RxTxTimeDiffMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-Multi-RTT-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RxTxTimeDiffMeasPerTRP-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTxTEG-Request-r17 | Not present | Rel-17 onwards | |
| measureSameDL-PRS-ResourceWithDifferentRxTxTEGs-r17 | Not present | Rel-17 onwards | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | Rel-17 onwards | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | Rel-17 onwards | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | Rel-17 onwards | |
| additionalPathsExt-r17 | Not present | Rel-17 onwards | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | Rel-17 onwards | |
| multiMeasInSameReport-r17 | Not present | Rel-17 onwards | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | Rel-17 onwards | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | Rel-18 onwards | |
| timingReportingGranularityFactorExt-r18 | Not present | Rel-18 onwards | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | Rel-18 onwards | |
| nr-DL-PRS-MeasurementTimeWindowsConfig-r18 | Not present | Rel-18 onwards | |
| nr-NTN-UE-RxTxMeasurementsRequest-r18 | Not present | Rel-18 onwards | |
| } | | | |

- NR DL-AoD REQUEST LOCATION INFORMATION

**Table 8.4-6: NR-DL-AoD-RequestLocationInformation**

| Derivation Path: 37.355 clause 6.5.11.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-AoD-RequestLocationInformation-r16 ::= SEQUENCE { | | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-AoD-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r16 | Not present | | |
| maxDL-PRS-RSRP-MeasurementsPerTRP-r17 | Not present | Rel-17 onwards | |
| maxDL-PRS-RSRPP-MeasurementsPerTRP-r17 | Not present | Rel-17 onwards | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | Rel-17 onwards | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | Rel-17 onwards | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | Rel-17 onwards | |
| } | | | |
| multiMeasInSameReport-r17 | Not present | Rel-17 onwards | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | Rel-18 onwards | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | Rel-18 onwards | |
| } | | | |

- NR DL-TDOA REQUEST LOCATION INFORMATION

**Table 8.4-7: NR-DL-TDOA-RequestLocationInformation**

| Derivation Path: 37.355 clause 6.5.10.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-TDOA-RequestLocationInformation-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-RstdMeasurementInfoRequest-r16 | Not present | | |
| nr-RequestedMeasurements-r16 | bit 0 = 1 (prsrsrpReq) | | |
| nr-AssistanceAvailability-r16 | FALSE | | |
| nr-DL-TDOA-ReportConfig-r16 SEQUENCE { | | | |
| maxDL-PRS-RSTD-MeasurementsPerTRPPair-r16 | Not present | | |
| timingReportingGranularityFactor-r16 | Not present | | |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Not present | Rel-17 onwards | |
| reducedDL-PRS-ProcessingSamples-r17 | Not present | Rel-17 onwards | |
| lowerRxBeamSweepingFactor-FR2-r17 | Not present | Rel-17 onwards | |
| } | | | |
| additionalPaths-r16 | Not present | | |
| nr-UE-RxTEG-Request-r17 | Not present | Rel-17 onwards | |
| nr-los-nlos-IndicatorRequest-r17 | Not present | Rel-17 onwards | |
| additionalPathsExt-r17 | Not present | Rel-17 onwards | |
| additionalPathsDL-PRS-RSRP-Request-r17 | Not present | Rel-17 onwards | |
| multiMeasInSameReport-r17 | Not present | Rel-17 onwards | |
| nr-DL-PRS-JointMeasurementRequest-r18 | Not present | Rel-18 onwards | |
| nr-DL-PRS-RxHoppingRequest-r18 | Not present | Rel-18 onwards | |
| } | | | |

## 8.4.1 Default assistance data information elements

### 8.4.1.1 GNSS Assistance Data Elements

The GNSS assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message are as defined in clause 5.4.1.1.

#### 8.4.1.2 OTDOA (LTE) Assistance Data Elements

This clause defines the OTDOA (LTE) assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages.

- OTDOA (LTE) REFERENCE CELL INFO

**Table 8.4.1.2-1: OTDOA-ReferenceCellInfo**

| Derivation Path: 37.355 clause 6.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-ReferenceCellInfo ::= SEQUENCE { | | LTE Cell 1 | |
| physCellId | 0 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 1 in 36.508 [8] | | |
| mnc | As defined for Cell 1 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 1 in 36.508 [8] | | |
| } | | | |
| earfcnRef | As defined for Cell 1 in 36.508 [8] | | |
| antennaPortConfig | As defined for Cell 1 in 36.508 [8] | | |
| cpLength | Normal | | |
| prsInfo SEQUENCE { | | | |
| prs-Bandwidth | PRS are transmitted over the used system bandwidth (see clause 5.2.2) | | |
| prs-ConfigurationIndex | FDD: 2<br>TDD: 4 | | |
| numDL-Frames | sf-1 | | |
| prs-MutingInfo-r9 | Not present | PRS muting is not used. | |
| prsID-r14 | Not present | PRS-ID not used | Rel-14 onwards |
| add-numDL-Frames-r14 | Not present | Not required | Rel-14 onwards |
| prsOccGroupLen-r14 | Not present | No PRS occasion group configured | Rel-14 onwards |
| prsHoppingInfo-r14 | Not present | PRS frequency hopping not used | Rel-14 onwards |
| } | | | |
| earfcnRef-v9a0 | As defined for Cell 1 in 36.508 [8] | | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| cpLengthCRS-r14 | Normal | | Rel-14 onwards |
| sameMBSFNconfigRef-r14 | FALSE | Not the same as the serving cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigRef-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| nr-LTE-SFN-Offset-r15 | Not present | | Rel-15 onwards |
| tdd-config-v1520 | Not present | | Rel-15 onwards |
| nr-LTE-fineTiming-Offset-r15 | Not present | | Rel-15 onwards |
| } | | | |

- OTDOA (LTE) NEIGHBOUR CELL INFO LIST

**Table 8.4.1.2-2: OTDOA-NeighbourCellInfoList**

| Derivation Path: 37.355 clause 6.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| OTDOA-NeighbourCellInfoList ::= SEQUENCE (SIZE(1)) OF SEQUENCE { | | | |
| SEQUENCE (SIZE(2)) OF SEQUENCE { | | Cell 2 | |
| physCellId | 2 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 2 in 36.508 [8] | | |
| mnc | As defined for Cell 2 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 2 in 36.508 [8] | | |
| } | | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | Not present | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 μs | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| tdd-config-v1520 | Not present | | Rel-15 onwards |
| } | | | |
| SEQUENCE { | | Cell 4 | |
| physCellId | 4 | | |
| cellGlobalId SEQUENCE { | | | |
| mcc | As defined for Cell 4 in 36.508 [8] | | |
| mnc | As defined for Cell 4 in 36.508 [8] | | |
| cellidentity | As defined for E-UTRAN Cell Identifier for Cell 4 in 36.508 [8] | | |
| } | | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |

| | | | |
|---|---|---|---|
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Slot timing is the same as for reference cell | |
| prs-SubframeOffset | Not present | | |
| expectedRSTD | 8192 | Value 0 | |
| expectedRSTD-Uncertainty | 10 | About 1 μs | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| tdd-config-v1520 | Not present | | Rel-15 onwards |
| } | | | |
| } | | | |

### 8.4.1.3 MBS Assistance Data Elements

The MBS assistance data elements which shall be provided to the UE in sub-test 16 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message are as defined in clause 5.4.1.3.

### 8.4.1.4 WLAN Assistance Data Elements

The WLAN assistance data elements which shall be provided to the UE in sub-test 17 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message are as defined in clause 5.4.1.4.

### 8.4.1.5 Sensor Assistance Data Elements

The Sensor assistance data elements which shall be provided to the UE in sub-test 18 via LPP Provide Assistance Data messages in the absence of a corresponding LPP Request Assistance Data message are as defined in clause 5.4.1.5.

### 8.4.1.6 Multi-RTT Assistance Data Elements

This clause defines the Multi-RTT assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages.

- NR DL-PRS ASSISTANCE DATA

**Table 8.4.1.6-1: NR-DL-PRS-AssistanceData**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ReferenceInfo-r16 SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | |
| nr-DL-PRS-ResourceID-List-r16 | Not present | | |
| nr-DL-PRS-ResourceSetID-r16 | Not present | | |
| } | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | SubcarrierSpacing | 38.508-1 [30] Table 4.6.3-188 | |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | same value as 'offsetToCarrier' as defined for the DL frequency of NR Cell 1 | | |
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the NR Cell 1 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | As defined in TS 38.508-1 table 4.4.2-2, Physical layer cell Identity for SIG for NR Cell 1 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 1 | | |
| mnc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 1 | | |
| nr-cellidentity-r15 | Set to NR Cell 1 Identifier defined in TS 38.508-1 Table 4.4.2-2 | | |
| } | | | |
| nr-ARFCN-r16 | ARFCN-ValueNR for NR Cell 1 frequency | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 4 | About 1 μs | |
| nr-DL-PRS-Info-r16 | NR-DL-PRS-Info-r16 as specified in Table 8.4.1.6-3 | | |
| prs-OnlyTP-r16 | Not present | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-r17 | Not present | Rel-17 onwards | |
| } | | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | In case of sub-test 20/27/31 UE-based DL-AoD or sub-test 21/28/32 DL-TDOA method supported by the UE as defined in clause 8.2.10 and clause 8.2.11. |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | As defined in TS 38.508-1 table 4.4.2-2, Physical layer cell Identity for SIG for NR Cell 2 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 2 | | |
| mnc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 2 | | |
| nr-cellidentity-r15 | Set to NR Cell 2 Identifier defined in TS 38.508-1 Table 4.4.2-2 | | |
| } | | | |
| nr-ARFCN-r16 | ARFCN-ValueNR for NR Cell 2 | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 4 | About 1 μs | |
| nr-DL-PRS-Info-r16 | NR-DL-PRS-Info-r16 as specified in Table 8.4.1.6-3 | | |
| prs-OnlyTP-r16 | Not present | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-r17 | Not present | Rel-17 onwards | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | In case of sub-test 21/28/32 UE-based DL-TDOA method supported by the UE as defined in clause 8.2.11 |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-SubcarrierSpacing-r16 | SubcarrierSpacing | 38.508-1 [30] Table 4.6.3-188 | |
| dl-PRS-ResourceBandwidth-r16 | 1 | 24 PRBs | |
| dl-PRS-StartPRB-r16 | same value as 'offsetToCarrier' as defined for the DL frequency of NR Cell 3 | | |

| | | | |
|---|---|---|---|
| dl-PRS-PointA-r16 | absoluteFrequencyPointA as defined for the DL frequency of the NR Cell 3 | | |
| dl-PRS-CombSizeN-r16 | n2 | | |
| dl-PRS-CyclicPrefix-r16 | normal | | |
| } | | | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | As defined in TS 38.508-1 table 4.4.2-2, Physical layer cell Identity for SIG for NR Cell 3 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | As defined in TS 38.508-1 [30] Table 4.4.2-3 for NR Cell 3 | | |
| mnc-r15 | As defined in TS 38.508-1 [30] Table 4.4.2-3 for NR Cell 3 | | |
| nr-cellidentity-r15 | Set to NR Cell 3 Identifier defined in TS 38.508-1 [30] Table 4.4.2-2 | | |
| } | | | |
| nr-ARFCN-r16 | ARFCN-ValueNR for NR Cell 3 frequency | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 4 | About 1 μs | |
| nr-DL-PRS-Info-r16 | NR-DL-PRS-Info-r16 as specified in Table 8.4.1.6-3 | | |
| prs-OnlyTP-r16 | Not present | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-r17 | Not present | Rel-17 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-SSB-Config-r16 SEQUENCE (SIZE (1..nrMaxTRPs-r16)) OF NR-SSB-Config-r16 { | 1 entry | | In case of sub-test 21/28/32 UE-based DL-TDOA method supported by the UE as defined in clause 8.2.11 |
| NR-SSB-Config-r16[1] SEQUENCE { | | entry 1 | |
| nr-PhysCellID-r16 | As defined in TS 38.508-1 table 4.4.2-2, Physical layer cell Identity for SIG for NR Cell 3 | | |
| nr-ARFCN-r16 | ARFCN-ValueNR for NR Cell 3 frequency | As defined in 38.508 -1 [30] subclause 6.2.3. | |
| ss-PBCH-BlockPower-r16 | 0 | | |
| halfFrameIndex-r16 | 0 | | |
| ssb-periodicity-r16 | ms20 | | |
| ssb-PositionsInBurst-r16 | Not present | | |

| | | | |
|---|---|---|---|
| ssb-SubcarrierSpacing-r16 | Subcarrier spacing of SSB for NR Cell 3 | | |
| sfn-SSB-Offset-r16 | 0 | | |
| } | | | |
| } | | | |
| } | | | |

- NR-SelectedDL-PRS-IndexList

**Table 8.4.1.6-2: NR-SelectedDL-PRS-IndexList**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-SelectedDL-PRS-IndexList-r16 ::= SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-SelectedDL-PRS-PerFreq-r16 { | 2 entries | | |
| NR-SelectedDL-PRS-PerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-SelectedDL-PRS-FrequencyLayerIndex-r16 | 0 | | |
| nr-SelectedDL-PRS-IndexListPerFreq-r16 | Not present | | |
| } | | | |
| NR-SelectedDL-PRS-PerFreq-r16[2] SEQUENCE { | | entry 2 | Sub-test 21/28/32 UE-Based only as defined in clause 8.2.11 |
| nr-SelectedDL-PRS-FrequencyLayerIndex-r16 | 1 | | |
| nr-SelectedDL-PRS-IndexListPerFreq-r16 | Not present | | |
| } | | | |
| } | | | |

- NR-DL-PRS-Info

**Table 8.4.1.6-3: NR-DL-PRS-Info**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceSetID-r16 | 0 | | |
| dl-PRS-Periodicity-and-ResourceSetSlotOffset-r16 | The periodicity is 160ms and the resource set slot offset is 11 ms for any SCS configuration | | |
| dl-PRS-ResourceRepetitionFactor-r16 | Not present | | |
| dl-PRS-ResourceTimeGap-r16 | Not present | | |
| dl-PRS-NumSymbols-r16 | n4 | | |
| dl-PRS-MutingOption1-r16 | Not present | | |
| dl-PRS-MutingOption2-r16 | Not present | | |
| dl-PRS-ResourcePower-r16 | 0 | | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | |
| dl-PRS-SequenceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[1] Table 8.4.1.6-1 |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[2], Table 8.4.1.6-1 |
| | 2 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[1] Table 8.4.1.6-1 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |

| | | | |
|---|---|---|---|
| n2-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[1] Table 8.4.1.6-1 |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[2] Table 8.4.1.6-1 |
| | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2]<br><br>NR-DL-PRS-AssistanceDataPerTRP-r16[1] Table 8.4.1.6-1 |
| } | | | |
| dl-PRS-ResourceSlotOffset-r16 | 0 | | |
| dl-PRS-ResourceSymbolOffset-r16 | 0 | | |
| dl-PRS-QCL-Info-r16 | Not present | | |
| dl-PRS-ResourcePrioritySubset-r17 | Not present | Rel-17 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.4.1.7 DL-AoD Assistance Data Elements

This clause defines the DL-AoD assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages in addition to those defined in Table 8.4.1.6-1 and Table 8.4.1.6-2 and Table 8.4.1.6-3.

- NR-PositionCalculationAssistance

**Table 8.4.1.7-1: NR-PositionCalculationAssistance**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-PositionCalculationAssistance-r16 ::= SEQUENCE { | | | |
| nr-TRP-LocationInfo-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-TRP-LocationInfoPerFreqLayer-r16 { | 2 entries | | |
| NR-TRP-LocationInfoPerFreqLayer-r16[1] SEQUENCE { | | entry1 | |
| referencePoint-r16 | | | |
| trp-LocationInfoList-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF TRP-LocationInfoElement-r16 { | 2 entries | | |
| TRP-LocationInfoElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | 0 | | |
| nr-PhysCellID-r16 | Not present | | |
| nr-CellGlobalID-r16 | Not present | | |
| nr-ARFCN-r16 | Not present | | |
| associated-DL-PRS-ID-r16 | Not present | | |
| trp-Location-r16 | Not present | Same as the reference point location | |
| trp-DL-PRS-ResourceSets-r16 | Not present | | |
| } | | | |
| TRP-LocationInfoElement-r16[2] SEQUENCE { | | entry 2 | In case of sub-test 20/27/31 UE-based DL-AoD or sub-test 21/28/32 DL-TDOA method supported by the UE as defined in clause 8.2.10 and clause 8.2.11. |
| dl-PRS-ID-r16 | 1 | | |
| nr-PhysCellID-r16 | Not present | | |
| nr-CellGlobalID-r16 | Not present | | |
| nr-ARFCN-r16 | Not present | | |
| associated-DL-PRS-ID-r16 | Not present | | |
| trp-Location-r16 | trp-Location-r16 for NR cell 2 as defined in 37.571-5 [12] | | |
| trp-DL-PRS-ResourceSets-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| NR-TRP-LocationInfoPerFreqLayer-r16[2] SEQUENCE { | | entry 2 | |
| referencePoint-r16 | Not present | Same as in the previous entry of the NR-TRP-LocationInfoPerFreqLayer list. | |
| trp-LocationInfoList-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF TRP-LocationInfoElement-r16 { | 1 entry | | |
| TRP-LocationInfoElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | 2 | | |
| nr-PhysCellID-r16 | Not present | | |
| nr-CellGlobalID-r16 | Not present | | |
| nr-ARFCN-r16 | Not present | | |

| | | | |
|---|---|---|---|
| associated-DL-PRS-ID-r16 | Not present | | |
| trp-Location-r16 | trp-Location-r16 for NR cell 3 as defined in TS 37.571-5 [12] | | |
| trp-DL-PRS-ResourceSets-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nr-DL-PRS-BeamInfo-r16 | Not present | | |
| nr-RTD-Info-r16 | Not present | | |
| nr-TRP-BeamAntennaInfo-r17 | Not present | Rel-17 onwards | |
| nr-DL-PRS-Expected-LOS-NLOS-Assistance-r17 | Not present | Rel-17 onwards | |
| nr-DL-PRS-TRP-TEG-Info-r17 | Not present | Rel-17 onwards | |
| } | | | |

### 8.4.1.8 DL-TDOA Assistance Data Elements

The DL-TDOA assistance data elements which shall be provided to the UE in the tests in LPP Provide Assistance Data messages are defined in Table 8.4.1.6-1, Table 8.4.1.6-2, Table 8.4.1.6-3 and Table 8.4.1.7-1.

## 8.5 Default SLPP message and information elements contents

This clause contains the default values of SLPP messages and information elements used, unless indicated otherwise in specific clauses of this specification.

- SLPP REQUEST CAPABILITIES

**Table 8.5-1: RequestCapabilities**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| | '00 00 01 00'H | Application Layer ID in initiating UE side | UE-only mode |
| transactionID | (0..255) | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| | Present. Any value acceptable. | OCTET STRING (SIZE (6)) | UE-only mode |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| requestCapabilities SEQUENCE { | | | |
| commonIEsRequestCapabilities | Present | OCTET STRING containing CommonIEsRequestCapabilities | |
| commonSL-PRS-MethodsIEsRequestCapabilities | Present | OCTET STRING containing CommonSL-PRS-MethodsIEsRequestCapabilities | |
| sl-AoA-RequestCapabilities | Present | OCTET STRING containing SL-AoA-RequestCapabilities | |
| sl-RTT-RequestCapabilities | Present | OCTET STRING containing SL-RTT-RequestCapabilities | |
| sl-TDOA-RequestCapabilities | Present | OCTET STRING containing SL-TDOA-RequestCapabilities | |
| sl-TOA-RequestCapabilities | Present | OCTET STRING containing SL-TOA-RequestCapabilities | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |

| Condition | Explanation |
|---|---|
| UE-only mode | The SLPP signalling procedure is between sidelink UEs |

- SLPP PROVIDE ASSISTANCE DATA

**Table 8.5-2: ProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| | '00 00 01 00'H | Application Layer ID in initiating UE side | UE-only mode |
| transactionID | (0..255) | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| | Present. Any value acceptable. | OCTET STRING (SIZE (6)) | UE-only mode |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| commonIEsProvideAssistanceData | Not present | | |
| commonSL-PRS-MethodsIEsProvide AssistanceData | Present | OCTET STRING containing CommonSL-PRS-MethodsIEsProvide AssistanceData as defined in Table 8.5.1.1-1 | |
| sl-AoA-ProvideAssistanceData | Present | OCTET STRING containing SL-AoA-ProvideAssistance Data as defined in Table 8.5.1.2-1 | Sub-test 34 |
| sl-RTT-ProvideAssistanceData | Present | OCTET STRING containing SL-RTT-ProvideAssistance Data as defined in Table 8.5.1.3-1 | Sub-test 33 |
| sl-TDOA-ProvideAssistanceData | Present | OCTET STRING containing SL-TDOA-ProvideAssistance Data as defined in Table 8.5.1.4-1 | Sub-test 35 |
| sl-TOA-ProvideAssistanceData | Present | OCTET STRING containing SL-TOA-ProvideAssistance Data as defined in Table 8.5.1.5-1 | Sub-test 36 |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |

| Condition | Explanation |
|---|---|
| UE-only mode | The SLPP signalling procedure is between sidelink UEs |

- SLPP REQUEST LOCATION INFORMATION

**Table 8.5-3: RequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| | '00 00 01 00'H | Application Layer ID in initiating UE side | UE-only mode |
| transactionID | (0..255) | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| | Present. Any value acceptable. | OCTET STRING (SIZE (6)) | UE-only mode |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| commonIEsRequestLocationInformation | Present | OCTET STRING containing CommonIEsRequestLocationInformation as defined in Table 8.5-4 | |
| commonSL-PRS-MethodsIEsRequestLocationInformation | Not present | | |
| sl-AoA-RequestLocationInformation | Present | OCTET STRING containing SL-AoA-RequestLocationInformation as defined in Table 8.5-5 | Sub-test 34 |
| sl-RTT-RequestLocationInformation | Present | OCTET STRING containing SL-RTT-RequestLocationInformation as defined in Table 8.5-6 | Sub-test 33 |
| sl-TDOA-RequestLocationInformation | Present | OCTET STRING containing SL-TDOA-RequestLocationInformation as defined in Table 8.5-7 | Sub-test 35 |
| sl-TOA-RequestLocationInformation | Present | OCTET STRING containing SL-TOA-RequestLocationInformation as defined in Table 8.5-8 | Sub-test 36 |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |

| Condition | Explanation |
|---|---|
| UE-only mode | The SLPP signalling procedure is between sidelink UEs |

- CommonIEsRequestLocationInformation

**Table 8.5-4: CommonIEsRequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CommonIEsRequestLocationInformation ::= SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| periodicalReporting | Not present | | |
| additionalInformation | onlyReturnInformationRequested | | |
| qos SEQUENCE { | | | |
| horizontalAccuracy | Not present | | |
| verticalCoordinateRequest | FALSE | | |
| verticalAccuracy | Not present | | |
| rangeAccuracy | Not present | | |
| azimuthAccuracy | Not present | | |
| elevationAccuracy | Not present | | |
| responseTime SEQUENCE { | | | |
| time | 128 | | |
| tenMilliSeconds | Not present | | |
| } | | | |
| velocityRequest | Not present | | |
| } | | | |
| environment | Not present | | |
| scheduledLocationTime | Not present | | |
| locationCoordinateTypes | Not present | | |
| velocityTypes | Not present | | |
| } | | | |

- SL-AoA REQUEST LOCATION INFORMATION

**Table 8.5-5: SL-AoA-RequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-AoA-RequestLocationInformation ::= SEQUENCE { | | | |
| sl-AoA-Request | aoa | | |
| measurementReportingType | Not present | | |
| measurementsForMultipleARP-IDs-Rx | Not present | | |
| sl-ARP-InfoRequest | Not present | | |
| sl-LOS-NLOS-IndicatorRequest | Not present | | |
| sl-PRS-RSRP-Request | Not present | | |
| sl-RSRPP-Request | Not present | | |
| sl-AdditionalPathsRequest | Not present | | |
| } | | | |

- SL-RTT REQUEST LOCATION INFORMATION

**Table 8.5-6: SL-RTT-RequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-RequestLocationInformation ::= SEQUENCE { | | | |
| sl-ARP-InfoRequest | Not present | | |
| sl-LOS-NLOS-IndicatorRequest | Not present | | |
| sl-PRS-RSRP-Request | Not present | | |
| sl-RSRPP-Request | Not present | | |
| sl-AdditionalPathsRequest | Not present | | |
| dummy | Not present | | |
| multipleSL-PRS-RxTxTimeDiffRequest | Not present | | |
| measurementsForMultipleARP-IDs-Rx | Not present | | |
| associatedSL-PRS-TxTimeStampRequest | Not present | | |
| } | | | |

- SL-TDOA REQUEST LOCATION INFORMATION

**Table 8.5-7: SL-TDOA-RequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-RequestLocationInformation ::= SEQUENCE { | | | |
| sl-ARP-InfoRequest | Not present | | |
| sl-LOS-NLOS-IndicatorRequest | Not present | | |
| sl-PRS-RSRP-Request | Not present | | |
| sl-RSRPP-Request | Not present | | |
| sl-AdditionalPathsRequest | Not present | | |
| measurementsForMultipleARP-IDs-Rx | Not present | | |
| maxNumberOfMeasurementsPerARP | Not present | | |
| } | | | |

- SL-TOA REQUEST LOCATION INFORMATION

**Table 8.5-8: SL-TOA-RequestLocationInformation**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-RequestLocationInformation ::= SEQUENCE { | | | |
| sl-ARP-InfoRequest | Not present | | |
| sl-LOS-NLOS-IndicatorRequest | Not present | | |
| sl-PRS-RSRP-Request | Not present | | |
| sl-RSRPP-Request | Not present | | |
| sl-AdditionalPathsRequest | Not present | | |
| measurementsForMultipleARP-IDs-Rx | Not present | | |
| maxNumberOfMeasurementsPerARP | Not present | | |
| } | | | |

## 8.5.1 Default assistance data information elements

### *8.5.1.1 Common SL-PRS Assistance Data Elements*

This clause defines the common SL-PRS assistance data elements which shall be provided to the UE in the tests in SLPP Provide Assistance Data messages.

**Table 8.5.1.1-1: CommonSL-PRS-MethodsIEsProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CommonSL-PRS-MethodsIEsProvideAssistance Data ::= SEQUENCE { | | | |
| sl-PRS-AssistanceDataInfo SEQUENCE (SIZE (1..maxNrOfUEs)) OF SL-PRS-AssistanceData { | 3 entries | | |
| SL-PRS-AssistanceData[1] SEQUENCE { | | entry 1 | |
| applicationLayerID | '00 00 01 00'H | NR-SS-UE1 | |
| sl-PRS-SequenceID | 0 | | |
| anchorUE-LocationInformation | AnchorLocationCoordinates for NR-SS-UE1 | As defined in TS 37.571-5 [12] clause 13.1. | |
| arp-LocationInfo SEQUENCE { | | | |
| referencePoint | Not present | | |
| arp-LocationInfoList SEQUENCE (SIZE (1..4)) OF ARP-LocationInfoElement { | 1 entry | | |
| ARP-LocationInfoElement[1] SEQUENCE { | | entry 1 | |
| sl-PRS-ARP-ID | 1 | | |
| arp-LocationInfo | RelativeLocation | As defined in TS 37.571-5 [12] clause 13.2. | |
| } | | | |
| } | | | |
| } | | | |
| sl-PRS-TxInfo SEQUENCE { | | | |
| sl-PRS-Priority | 1 | | |
| sl-PRS-DelayBudget | 1000 | | |
| sl-PRS-Bandwidth | mhz5 | | |
| sl-PRS-Periodicity | Not present | | |
| sl-PRS-Transmission | true | | |
| } | | | |
| sl-POS-ARP-ID-Tx | Not present | | |
| } | | | |
| SL-PRS-AssistanceData [2] SEQUENCE { | | entry 2 | In case of sub-test 33 pc_Target_UEB_SL_RTT, sub-test 34 pc_Target_UEB_SL_AoA, sub-test 35 pc_Target_UEB_SL_TDOA or sub-test 36 pc_Target_UEB_SL_TOA supported by the UE |
| applicationLayerID | '00 00 03 00'H | NR-SS-UE2 | |
| sl-PRS-SequenceID | 1 | | |
| anchorUE-LocationInformation | AnchorLocationCoordinates for NR-SS-UE2 | As defined in TS 37.571-5 [12] clause 13.1. | |
| arp-LocationInfo SEQUENCE { | | | |
| referencePoint | Not present | | |
| arp-LocationInfoList arp-LocationInfoList SEQUENCE (SIZE (1..4)) OF ARP-LocationInfoElement { | 1 entry | | |
| ARP-LocationInfoElement[1] SEQUENCE { | | entry 1 | |
| sl-PRS-ARP-ID | 1 | | |
| arp-LocationInfo | RelativeLocation | As defined in TS 37.571-5 [12] clause 13.2. | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| sl-PRS-TxInfo SEQUENCE { | | | |
| sl-PRS-Priority | 1 | | |
| sl-PRS-DelayBudget | 1000 | | |
| sl-PRS-Bandwidth | mhz5 | | |
| sl-PRS-Periodicity | Not present | | |
| sl-PRS-Transmission | true | | |
| } | | | |
| sl-POS-ARP-ID-Tx | Not present | | |
| } | | | |
| SL-PRS-AssistanceData [3] SEQUENCE { | | entry 3 | In case of sub-test 35 pc_Target_UEB_SL_TDOA supported by the UE |
| applicationLayerID | '00 00 04 00'H | NR-SS-UE3 | |
| sl-PRS-SequenceID | 2 | | |
| anchorUE-LocationInformation | AnchorLocationCoordinates for NR-SS-UE3 | As defined in TS 37.571-5 [12] clause 13.1. | |
| arp-LocationInfo SEQUENCE { | | | |
| referencePoint | Not present | | |
| arp-LocationInfoList arp-LocationInfoList SEQUENCE (SIZE (1..4)) OF ARP-LocationInfoElement { | 1 entry | | |
| ARP-LocationInfoElement[1] SEQUENCE { | | entry 1 | |
| sl-PRS-ARP-ID | 1 | | |
| arp-LocationInfo | RelativeLocation | As defined in TS 37.571-5 [12] clause 13.2. | |
| } | | | |
| } | | | |
| } | | | |
| sl-PRS-TxInfo SEQUENCE { | | | |
| sl-PRS-Priority | 1 | | |
| sl-PRS-DelayBudget | 1000 | | |
| sl-PRS-Bandwidth | mhz5 | | |
| sl-PRS-Periodicity | Not present | | |
| sl-PRS-Transmission | true | | |
| } | | | |
| sl-POS-ARP-ID-Tx | Not present | | |
| } | | | |
| } | | | |
| sl-PRS-Error | Not present | | |
| } | | | |

### 8.5.1.2 SL-AoA Assistance Data Elements

This clause defines the *SL-AoA* assistance data elements which shall be provided to the UE in the tests in SLPP Provide Assistance Data messages.

**Table 8.5.1.2-1: sl-AoA-ProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-AoA-ProvideAssistanceData ::= SEQUENCE { | | | |
| sl-AoA-AssistanceDataInfo | Not present | | |
| sl-AoA-Error | Not present | | |
| } | | | |

### 8.5.1.3 SL-RTT Assistance Data Elements

This clause defines the SL-RTT assistance data elements which shall be provided to the UE in the tests in SLPP Provide Assistance Data messages.

**Table 8.5.1.3-1: SL-RTT-ProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-ProvideAssistanceData ::= SEQUENCE { | | | |
| } | | | |

### 8.5.1.4 SL-TDOA Assistance Data Elements

This clause defines the *SL-TDOA* assistance data elements which shall be provided to the UE in the tests in SLPP Provide Assistance Data messages.

**Table 8.5.1.4-1: SL-TDOA-ProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TDOA-ProvideAssistanceData ::= SEQUENCE { | | | |
| sl-RTD-Info SEQUENCE ( SIZE (1.. maxNrOfUEs)) OF RTD-InfoListPerAnchorUE { | 3 entries | | |
| RTD-InfoListPerAnchorUE[1] SEQUENCE { | | entry 1 | |
| applicationLayerID | '00 00 01 00'H | NR-SS-UE1 | |
| referenceRTD-Info SEQUENCE { | | | |
| syncSourceType | gnss | | |
| applicationLayerID | Not present | | |
| nrCell-Identify | Not present | | |
| } | | | |
| rtd-BetweenAnchorUEs CHOICE { | | | |
| subframeOffset | 0 | | |
| } | | | |
| rtd-Quality | Not present | | |
| syncSourceType | gnss | | |
| } | | | |
| RTD-InfoListPerAnchorUE[2] SEQUENCE { | | entry 2 | In case of sub-test 35 pc_Target_UEB_SL_TDOA supported by the UE |
| applicationLayerID | '00 00 03 00'H | NR-SS-UE2 | |
| referenceRTD-Info SEQUENCE { | | | |
| syncSourceType | gnss | | |
| applicationLayerID | Not present | | |
| nrCell-Identify | Not present | | |
| } | | | |
| rtd-BetweenAnchorUEs CHOICE { | | | |
| subframeOffset | 0 | | |
| } | | | |
| rtd-Quality | Not present | | |
| syncSourceType | gnss | | |
| } | | | |
| RTD-InfoListPerAnchorUE[3] SEQUENCE { | | entry 3 | In case of sub-test 35 pc_Target_UEB_SL_TDOA supported by the UE |
| applicationLayerID | '00 00 04 00'H | NR-SS-UE3 | |
| referenceRTD-Info | | | |
| referenceRTD-Info SEQUENCE { | | | |
| syncSourceType | gnss | | |
| applicationLayerID | Not present | | |
| nrCell-Identify | Not present | | |
| } | | | |
| rtd-BetweenAnchorUEs CHOICE { | | | |
| subframeOffset | 0 | | |
| } | | | |
| rtd-Quality | Not present | | |
| syncSourceType | gnss | | |
| } | | | |
| } | | | |
| sl-TDOA-Error | Not present | | |
| } | | | |

### 8.5.1.5 SL-TOA Assistance Data Elements

This clause defines the ***SL-TOA*** assistance data elements which shall be provided to the UE in the tests in SLPP Provide Assistance Data messages.

**Table 8.5.1.5-1: SL-TOA-ProvideAssistanceData**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TOA-ProvideAssistanceData ::= SEQUENCE { | | | |
| sl-RTD-Info SEQUENCE ( SIZE (1.. maxNrOfUEs)) OF RTD-InfoListPerAnchorUE { | 2 entries | | |
| RTD-InfoListPerAnchorUE[1] SEQUENCE { | | entry 1 | |
| applicationLayerID | '00 00 01 00'H | NR-SS-UE1 | |
| referenceRTD-Info SEQUENCE { | | | |
| syncSourceType | gnss | | |
| applicationLayerID | Not present | | |
| nrCell-Identify | Not present | | |
| } | | | |
| rtd-BetweenAnchorUEs CHOICE { | | | |
| subframeOffset | 0 | | |
| } | | | |
| rtd-Quality | Not present | | |
| syncSourceType | gnss | | |
| } | | | |
| RTD-InfoListPerAnchorUE[2] SEQUENCE { | | entry 2 | In case of sub-test 36 pc_Target_ UEB_SL_T OA supported by the UE |
| applicationLayerID | '00 00 03 00'H | NR-SS-UE2 | |
| referenceRTD-Info SEQUENCE { | | | |
| syncSourceType | gnss | | |
| applicationLayerID | Not present | | |
| nrCell-Identify | Not present | | |
| } | | | |
| rtd-BetweenAnchorUEs CHOICE { | | | |
| subframeOffset | 0 | | |
| } | | | |
| rtd-Quality | Not present | | |
| syncSourceType | gnss | | |
| } | | | |
| } | | | |
| sl-TOA-Error | Not present | | |
| } | | | |

# 9 Protocol Conformance Test Cases for NR

## 9.1 FFS

## 9.2 FFS

## 9.3 LPP Procedures

### 9.3.1 LPP Common Procedures

#### 9.3.1.1 Position Capability Transfer

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

##### 9.3.1.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives a LPP message of type REQUEST CAPABILITIES }
    then { UE sends a LPP message of type PROVIDE CAPABILITIES with the correct supported
capabilities }
}
```

##### 9.3.1.1.2 Conformance requirements

As defined in clause 7.3.1.1.2.

##### 9.3.1.1.3 Test description

###### 9.3.1.1.3.1 Pre-test conditions

System Simulator:

- For Test Configuration B (Table 9.3.1.1.3.2-1): NR Cell 1.
- For Test Configuration D (Table 9.3.1.1.3.2-1): LTE Cell 1.

UE:

-

Preamble:

- For Test Configuration B (Table 9.3.1.1.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.1.1.3.2-1): FFS

Related PICS/PIXIT Statements:

-

9.3.1.1.3.2 Test procedure sequence

**Table 9.3.1.1.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.1.1 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour is as defined in Table 7.3.1.1.3.2-1.

9.3.1.1.3.3 Specific message contents

As defined in clause 7.3.1.1.3.3, with the following exceptions:

Table 9.3.1.1.3.3-1 replaces Table 7.3.1.1.3.3-1, Table 9.3.1.1.3.3-2 replaces Table 7.3.1.1.3.3-2, Table 9.3.1.1.3.3-3 replaces Table 7.3.1.1.3.3-4 and Table 9.3.1.1.3.3-4 replaces Table 7.3.1.1.3.3-5.

**Table 9.3.1.1.3.3-1: *DLInformationTransfer* (steps 1 and 2a, Table 7.3.1.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.1.1.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-2: DL NAS TRANSPORT (steps 1 and 2a, Table 7.3.1.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities | |
| | **Step 2a:** Set according to Table 7.3.1.1.3.3-14 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.1.1.3.3-2A: Void**

**Table 9.3.1.1.3.3-3: *ULInformationTransfer* (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.1.1.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-4: UL NAS TRANSPORT (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.3.1.1.3.3-5 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.1.1.3.2-1) | |

**Table 9.3.1.1.3.3-5: LPP PROVIDE CAPABILITIES (step 2, Table 7.3.1.1.3.2-1)**

| Derivation Path: Table 7.3.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.1.1.3.3-6 with the following exception: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Present or not present dependent on pc_NR_ECID. Set according to Table 9.3.1.1.3.3-6. | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 SEQUENCE { | Present or not present dependent on pc_Multi_RTT. | Rel-16 onwards | |
| nr-Multi-RTT-PRS-Capability-r16 | Set according to Table 9.3.1.1.3.3-7 | | |
| nr-Multi-RTT-MeasurementCapability-r16 | Set according to Table 9.3.1.1.3.3-8 | | |
| nr-DL-PRS-QCL-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-9 | | |
| nr-DL-PRS-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-10 | | |
| nr-UL-SRS-Capability-r16 | Set according to Table 9.3.1.1.3.3-11 | | |
| additionalPathsReport-r16 | Dependent on UE capabilities | | |
| periodicalReporting-r16 | Dependent on UE capabilities | | |
| ten-ms-unit-ResponseTime-r17 | Not checked | Rel-17 onwards | |
| nr-DL-PRS-ExpectedAoD-or-AoA-Sup-r17 | Not checked | Rel-17 onwards | |
| nr-Multi-RTT-On-Demand-DL-PRS-Support-r17 SEQUENCE { | | | |
| nr-on-demand-DL-PRS-InformationSup-r17 | Not checked | Rel-17 onwards | |
| nr-on-demand-DL-PRS-ConfigurationsSup-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| nr-UE-RxTx-TEG-ID-ReportingSupport-r17 | Not checked | Rel-17 onwards | |
| nr-los-nlos-IndicatorSupport-r17 SEQUENCE { | | | |
| type-r17 | Not checked | Rel-17 onwards | |
| granularity-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| additionalPathsExtSupport-r17 | Not checked | Rel-17 onwards | |
| scheduledLocationRequestSupported-r17 | Not checked | Rel-17 onwards | |
| nr-dl-prs-AssistanceDataValidity-r17 | Not checked | Rel-17 onwards | |
| multiMeasInSameMeasReport-r17 | Not checked | Rel-17 onwards | |
| mg-ActivationRequest-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_mg_ActivationRequest_Multi_RTT |
| posMeasGapSupport-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_posMeasGapSupport_Multi_RTT |

| | | | |
|---|---|---|---|
| nr-MultiRTT-OnDemandPRS-ForBWA-Support-r18 | Not checked | Rel-18 onwards | |
| periodicReportingIntervalMsSupport-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| nr-DL-AoD-ProvideCapabilities-r16 SEQUENCE { | Present or not present dependent on (pc_UEA_DL_AoD OR pc_UEB_DL_AoD). | Rel-16 onwards | |
| nr-DL-AoD-Mode-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-PRS-Capability-r16 | Set according to Table 9.3.1.1.3.3-7. | | |
| nr-DL-AoD-MeasurementCapability-r16 | Set according to Table 9.3.1.1.3.3-12. | | |
| nr-DL-PRS-QCL-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-9. | | |
| nr-DL-PRS-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-10. | | |
| periodicalReporting-r16 | Dependent on UE capabilities | | |
| ten-ms-unit-ResponseTime-r17 | Not checked | Rel-17 onwards | |
| nr-PosCalcAssistanceSupport-r17 | Not checked | Rel-17 onwards | |
| nr-los-nlos-AssistanceDataSupport-r17 SEQUENCE { | | | |
| type-r17 | Not checked | Rel-17 onwards | |
| granularity-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-Sup-r17 | Not checked | Rel-17 onwards | |
| dl-PRS-ResourcePrioritySubset-Sup-r17 | Not checked | Rel-17 onwards | |
| nr-DL-PRS-BeamInfoSup-r17 | Not checked | Rel-17 onwards | |
| nr-DL-AoD-On-Demand-DL-PRS-Support-r17 SEQUENCE { | | | |
| nr-on-demand-DL-PRS-InformationSup-r17 | Not checked | Rel-17 onwards | |
| nr-on-demand-DL-PRS-ConfigurationsSup-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| nr-los-nlos-IndicatorSupport-r17 SEQUENCE { | | | |
| type-r17 | Not checked | Rel-17 onwards | |
| granularity-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| scheduledLocationRequestSupported-r17 | Not checked | Rel-17 onwards | |
| nr-dl-prs-AssistanceDataValidity-r17 | Not checked | Rel-17 onwards | |
| multiMeasInSameMeasReport-r17 | Not checked | Rel-17 onwards | |
| mg-ActivationRequest-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_mg_ActivationRequest_DL_AoD |
| posMeasGapSupport-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_posMeasGapSupport_DL_AoD |
| multiLocationEstimateInSameMeasReport-r17 | Not checked | Rel-17 onwards | |
| locationCoordinateTypes-r18 | Not checked | Rel-18 onwards | |
| nr-IntegrityAssistanceSupport-r18 | Not checked | Rel-18 onwards | |
| periodicReportingIntervalMsSupport-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| nr-DL-TDOA-ProvideCapabilities-r16 SEQUENCE { | Present or not present dependent on (pc_UEA_DL_TDOA OR pc_UEB_DL_TDOA) | Rel-16 onwards | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-Mode-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-PRS-Capability-r16 | Set according to Table 9.3.1.1.3.3-7. | | |
| nr-DL-TDOA-MeasurementCapability-r16 | Set according to Table 9.3.1.1.3.3-13. | | |
| nr-DL-PRS-QCL-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-9. | | |
| nr-DL-PRS-ProcessingCapability-r16 | Set according to Table 9.3.1.1.3.3-10. | | |
| additionalPathsReport-r16 | Dependent on UE capabilities | | |
| periodicalReporting-r16 | Dependent on UE capabilities | | |
| ten-ms-unit-ResponseTime-r17 | Not checked | Rel-17 onwards | |
| nr-PosCalcAssistanceSupport-r17 | Not checked | Rel-17 onwards | |
| nr-los-nlos-AssistanceDataSupport-r17 SEQUENCE { | | | |
| type-r17 | Not checked | Rel-17 onwards | |
| granularity-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-Sup-r17 | Not checked | Rel-17 onwards | |
| nr-DL-TDOA-On-Demand-DL-PRS-Support-r17 SEQUENCE { | | | |
| nr-on-demand-DL-PRS-InformationSup-r17 | Not checked | Rel-17 onwards | |
| nr-on-demand-DL-PRS-ConfigurationsSup-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| nr-los-nlos-IndicatorSupport-r17 SEQUENCE { | | | |
| type-r17 | Not checked | Rel-17 onwards | |
| granularity-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| additionalPathsExtSupport-r17 | Not checked | Rel-17 onwards | |
| scheduledLocationRequestSupported-r17 | Not checked | Rel-17 onwards | |
| nr-dl-prs-AssistanceDataValidity-r17 | Not checked | Rel-17 onwards | |
| multiMeasInSameMeasReport-r17 | Not checked | Rel-17 onwards | |
| mg-ActivationRequest-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_mg_ActivationRequest_DL_TDOA |
| posMeasGapSupport-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_posMeasGapSupport_DL_TDOA |
| multiLocationEstimateInSameMeasReport-r17 | Not checked | Rel-17 onwards | |
| locationCoordinateTypes-r18 | Not checked | Rel-18 onwards | |
| periodicAssistanceData-r18 | Not checked | Rel-18 onwards | |
| nr-IntegrityAssistanceSupport-r18 | Not checked | Rel-18 onwards | |
| nr-DL-TDOA-OnDemandPRS-ForBWA-Support-r18 | Not checked | Rel-18 onwards | |
| periodicReportingIntervalMsSupport-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| nr-UL-ProvideCapabilities-r16 SEQUENCE { | Present or not present and value dependent on UE capabilities. | Rel-16 onwards | |
| nr-UL-SRS-Capability-r16 | Set according to Table 9.3.1.1.3.3-11 | | |
| nr-UE-TEG-Capability-r17 | Set according to Table 9.3.1.1.3.3-14 | Rel-17 onwards | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-6: NR-ECID-ProvideCapabilities (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.5.9.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-ECID-ProvideCapabilities-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| nr-ECID-MeasSupported-r16 | Dependent on UE capabilities | | |
| periodicalReporting-r16 | Dependent on UE capabilities | | |
| triggeredReporting-r16 | Dependent on UE capabilities | | |
| ten-ms-unit-ResponseTime-r17 | Not checked | Rel-17 onwards | |
| scheduledLocationRequestSupported-r17 | Not checked | Rel-17 onwards | |
| periodicReportingIntervalMsSupport-r18 | Not checked | Rel-18 onwards | |
| } | | | |

**Table 9.3.1.1.3.3-7: NR-DL-PRS-ResourcesCapability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-ResourcesCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| maxNrOfDL-PRS-ResourceSetPerTrpPerFrequencyLayer-r16 | Dependent on UE capabilities | | |
| maxNrOfTRP-AcrossFreqs-r16 | Dependent on UE capabilities | | |
| maxNrOfPosLayer-r16 | Dependent on UE capabilities | | |
| dl-PRS-ResourcesCapabilityBandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| maxNrOfDL-PRS-ResourcesPerResourceSet-r16 | Dependent on UE capabilities | | |
| maxNrOfDL-PRS-ResourcesPerPositioningFrequencylayer-r16 | Dependent on UE capabilities | | |
| } | | | |
| dl-PRS-ResourcesBandCombinationList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| bandList-r16 | Not checked | | |
| maxNrOfDL-PRS-ResourcesAcrossAllFL-TRP-ResourceSet-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-8: NR-Multi-RTT-MeasurementCapability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.5.12.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-Multi-RTT-MeasurementCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| maxNrOfRx-TX-MeasFR1-r16 | Dependent on UE capabilities | | |
| maxNrOfRx-TX-MeasFR2-r16 | Dependent on UE capabilities | | |
| supportOfRSRP-MeasFR1-r16 | Dependent on UE capabilities | | |
| supportOfRSRP-MeasFR2-r16 | Dependent on UE capabilities | | |
| srs-AssocPRS-MultiLayersFR1-r16 | Dependent on UE capabilities | | |
| srs-AssocPRS-MultiLayersFR2-r16 | Dependent on UE capabilities | | |
| nr-UE-TEG-Capability-r17 | Set according to Table 9.3.1.1.3.3-14 | | |
| multi-RTT-MeasCapabilityBandList-r17 SEQUENCE (SIZE (1..nrMaxBands-r16)) OF { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r17 | Not ckecked | | |
| supportOfDL-PRS-FirstPathRSRP-r17 | Not checked | Rel-17 onwards | |
| dl-PRS-MeasRRC-Inactive-r17 | Not checked | Rel-17 onwards | |
| supportOfDL-PRS-BWA-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_DL_PRS_BWA_RRC_Connected_Multi_RTT |
| supportOfDL-PRS-BWA-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| nr-NTN-MeasAndReport-r18 | Not checked | Rel-18 onwards | |
| nr-DL-PRS-RSCP-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_nr_DL_PRS_RSCP_Reporting RRC_Connected_Multi_RTT |
| nr-DL-PRS-RSCP-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| assocSingleRx-Tx-WithUpToNsampleRSCP-r18 | Not checked | Rel-18 onwards | |
| supportOfSymbolTimeStampForRSCP-r18 | Not checked | Rel-18 onwards | |
| supportOfFinerTimingReportGranularityForPRS-Meas-r18 | Not checked | Rel-18 onwards | |
| supportOfMeasurementsInTimeWindow-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-9: NR-DL-PRS-QCL-ProcessingCapability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-QCL-ProcessingCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| dl-PRS-QCL-ProcessingCapabilityBandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| ssb-FromNeighCellAsQCL-r16 | Dependent on UE capabilities | | |
| prs-FromServNeighCellAsQCL-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-10: NR-DL-PRS-ProcessingCapability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-ProcessingCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| prs-ProcessingCapabilityBandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| supportedBandwidthPRS-r16 | Dependent on UE capabilities | | |
| dl-PRS-BufferType-r16 | Dependent on UE capabilities | | |
| durationOfPRS-Processing-r16 SEQUENCE { | | | |
| durationOfPRS-ProcessingSymbols-r16 | Dependent on UE capabilities | | |
| durationOfPRS-ProcessingSymbolsInEveryTms-r16 | Dependent on UE capabilities | | |
| } | | | |
| maxNumOfDL-PRS-ResProcessedPerSlot-r16 SEQUENCE { | | | |
| scs15-r16 | Dependent on UE capabilities | | |
| scs30-r16 | Dependent on UE capabilities | | |
| scs60-r16 | Dependent on UE capabilities | | |
| scs120-r16 | Dependent on UE capabilities | | |
| } | | | |
| supportedDL-PRS-ProcessingSamples-RRC-CONNECTED-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_supportedDL_PRS_ProcessingSamples_RRC_CONNECTED |
| prs-ProcessingWindowType1A-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_prs_Proc essingWindo wType1A |
| prs-ProcessingWindowType1B-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_prs_Proc essingWindo wType1B |
| prs-ProcessingWindowType2-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_prs_Proc essingWindo wType2 |
| prs-ProcessingCapabilityOutsideMGinPPW-r17 SEQUENCE (SIZE(1..3)) OF { | Size n of SEQUENCE is dependent on UE capabilities | | |
| prsProcessingType-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_prsProcessingType |
| ppw-dl-PRS-BufferType-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_dl_PRS_BufferType |
| ppw-durationOfPRS-Processing1-r17 SEQUENCE { | | | |
| ppw-durationOfPRS-ProcessingSymbolsN-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_durationOfPRS_ProcessingSymbolsN |

| | | | |
|---|---|---|---|
| ppw-durationOfPRS-ProcessingSymbolsT-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_durationOfPRS_ProcessingSymbolsT |
| } | | | |
| ppw-durationOfPRS-Processing2-r17 SEQUENCE { | | | |
| ppw-durationOfPRS-ProcessingSymbolsN2-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_durationOfPRS_ProcessingSymbolsN2 |
| ppw-durationOfPRS-ProcessingSymbolsT2-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_durationOfPRS_ProcessingSymbolsT2 |
| } | | | |
| ppw-maxNumOfDL-PRS-ResProcessedPerSlot-r17 SEQUENCE { | | | |
| scs15-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_maxNumOfDL_PRS_ResProcessedPerSlot_SCS15 |
| scs30-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_maxNumOfDL_PRS_ResProcessedPerSlot_SCS30 |
| scs60-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_maxNumOfDL_PRS_ResProcessedPerSlot_SCS60 |
| scs120-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_maxNumOfDL_PRS_ResProcessedPerSlot_SCS120 |
| } | | | |
| ppw-maxNumOfDL-Bandwidth-r17 | Dependent on UE capabilities | Rel-17 onwards | pc_ppw_maxNumOfDL_Bandwidth_FR1 / pc_ppw_maxNumOfDL_Bandwidth_FR2 |
| } | | | |
| dl-PRS-BufferType-RRC-Inactive-r17 | Not checked | Rel-17 onwards | |
| durationOfPRS-Processing-RRC-Inactive-r17 SEQUENCE { | | | |
| durationOfPRS-ProcessingSymbols-r17 | Not checked | Rel-17 onwards | |
| durationOfPRS-ProcessingSymbolsInEveryTms-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| maxNumOfDL-PRS-ResProcessedPerSlot-RRC-Inactive-r17 SEQUENCE { | | | |
| scs15-r17 | Not checked | Rel-17 onwards | |
| scs30-r17 | Not checked | Rel-17 onwards | |
| scs60-r17 | Not checked | Rel-17 onwards | |
| scs120-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| supportedLowerRxBeamSweepingFactor-FR2-r17 | Not checked | Rel-17 onwards | |
| supportedDL-PRS-ProcessingSamples-RRC-Inactive-r17 | Not checked | Rel-17 onwards | |
| prs-MeasurementWithoutMG-r17 | Not checked | Rel-17 onwards | |

| | | | |
|---|---|---|---|
| supportOfDL-PRS-BWA-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_DL_PRS_BWA_RRC_Connected_Multi_RTT |
| supportOfDL-PRS-BWA-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| nr-NTN-MeasAndReport-r18 | Not checked | Rel-18 onwards | |
| nr-DL-PRS-RSCP-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_nr_DL_PRS_RSCP_Reporting RRC_Connected_Multi_RTT |
| nr-DL-PRS-RSCP-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| assocSingleRx-Tx-WithUpToNsampleRSCP-r18 | Not checked | Rel-18 onwards | |
| supportOfSymbolTimeStampForRSCP-r18 | Not checked | Rel-18 onwards | |
| supportOfFinerTimingReportGranularityForPRS-Meas-r18 | Not checked | Rel-18 onwards | |
| supportOfMeasurementsInTimeWindow-r18 | Not checked | Rel-18 onwards | |
| maxNumOfOneSymbolPRS-ResProcessedPerSlot-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| maxNumOfOneSymbolPRS-ResProcessedPerSlot-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| ppw-maxNumOfOneSymbolPRS-ResProcessedPerSlot-r18 | Not checked | Rel-18 onwards | |
| prs-BWA-TwoContiguousIntrabandInMG-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| prs-BWA-ThreeContiguousIntrabandInMG-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| prs-BWA-TwoContiguousIntraband-RRC-IdleAndInactive-r18 | Not checked | Rel-18 onwards | |
| prs-BWA-ThreeContiguousIntraband-RRC-IdleAndInactive-r18 | Not checked | Rel-18 onwards | |
| reducedNumOfSampleInMeasurementWithPRS-BWA-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| reducedNumOfSampleInMeasurementWithPRS-BWA-RRC-IdleAndInactive-r18 | Not checked | Rel-18 onwards | |
| dl-PRS-MeasurementWithRxFH-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| dl-PRS-MeasurementWithRxFH-RRC-Idle-r18 | Not checked | Rel-18 onwards | |
| reducedNumOfSampleForMeasurementWithFH-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| reducedNumOfSampleForMeasurementWithFH-RRC-IdleAndInactive-r18 | Not checked | Rel-18 onwards | |
| supportOfPRS-BWA-WithTwoPFL-Combination-r18 | Not checked | Rel-18 onwards | |
| dl-PRS-MeasurementWithRxFH-RRC-Connected-r18 SEQUENCE { | | | |
| maximumPRS-BandwidthAcrossAllHopsFR1-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_maximumPRS_BandwidthAcrossAllHopsFR1_value |
| maximumPRS-BandwidthAcrossAllHopsFR2-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_maximumPRS_BandwidthAcrossAllHopsFR2_value |
| maximumFH-Hops-r18 | Not checked | Rel-18 onwards | |
| processingDuration-r18 SEQUENCE { | | | |
| processingPRS-SymbolsDurationN3-r18 | Not checked | Rel-18 onwards | |
| processingDurationT3-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| rf-RxRetuneTimeFR1-r18 | Not checked | Rel-18 onwards | |
| rf-RxRetuneTimeFR2-r18 | Not checked | Rel-18 onwards | |
| numOfOverlappingPRB-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| maxSupportedFreqLayers-r16 | Dependent on UE capabilities | | |
| simulLTE-NR-PRS-r16 | Dependent on UE capabilities | | |
| } | | | |

**Table 9.3.1.1.3.3-11: NR-UL-SRS-Capability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-UL-SRS-Capability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| srs-CapabilityBandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| olpc-SRS-Pos-r16 SEQUENCE { | | | |
| olpc-SRS-PosBasedOnPRS-Serving-r16 | Dependent on UE capabilities | | |
| olpc-SRS-PosBasedOnSSB-Neigh-r16 | Dependent on UE capabilities | | |
| olpc-SRS-PosBasedOnPRS-Neigh-r16 | Dependent on UE capabilities | | |
| maxNumberPathLossEstimatePerServing-r16 | Dependent on UE capabilities | | |
| } | | | |
| spatialRelationsSRS-Pos-r16 SEQUENCE { | | | |
| spatialRelation-SRS-PosBasedOnSSB-Serving-r16 | Dependent on UE capabilities | | |
| spatialRelation-SRS-PosBasedOnCSI-RS-Serving-r16 | Dependent on UE capabilities | | |
| spatialRelation-SRS-PosBasedOnPRS-Serving-r16 | Dependent on UE capabilities | | |
| spatialRelation-SRS-PosBasedOnSRS-r16 | Dependent on UE capabilities | | |
| spatialRelation-SRS-PosBasedOnSSB-Neigh-r16 | Dependent on UE capabilities | | |
| spatialRelation-SRS-PosBasedOnPRS-Neigh-r16 | Dependent on UE capabilities | | |
| } | | | |
| posSRS-RRC-Inactive-InInitialUL-BWP-r17 SEQUENCE { | | | |
| maxNumOfSRSposResourceSets-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicAndSemiPersistentSRSposResources-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicAndSemiPersistentSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicSRSposResources-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| dummy1 | Not checked | Rel-17 onwards | |
| dummy2 | Not checked | Rel-17 onwards | |
| } | | | |
| posSRS-RRC-Inactive-OutsideInitialUL-BWP-r17 SEQUENCE { | | | |
| maxSRSposBandwidthForEachSCS-withinCC-FR1-r17 | Not checked | Rel-17 onwards | |
| maxSRSposBandwidthForEachSCS-withinCC-FR2-r17 | Not checked | Rel-17 onwards | |
| maxNumOfSRSposResourceSets-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicSRSposResources-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| differentNumerologyBetweenSRSposAndInitialBWP-r17 | Not checked | Rel-17 onwards | |
| srsPosWithoutRestrictionOnBWP-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicAndSemiPersistentSRSposResources-r17 | Not checked | Rel-17 onwards | |
| maxNumOfPeriodicAndSemiPersistentSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| differentCenterFreqBetweenSRSposAndInitialBWP-r17 | Not checked | Rel-17 onwards | |
| maxNumOfSemiPersistentSRSposResources-r17 | Not checked | Rel-17 onwards | |

| | | | |
|---|---|---|---|
| maxNumOfSemiPersistentSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| switchingTimeSRS-TX-OtherTX-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| olpc-SRS-PosRRC-Inactive-r17 SEQUENCE { | | | |
| olpc-SRS-PosBasedOnPRS-Serving-r16 | Not checked | Rel-17 onwards | |
| olpc-SRS-PosBasedOnSSB-Neigh-r16 | Not checked | Rel-17 onwards | |
| olpc-SRS-PosBasedOnPRS-Neigh-r16 | Not checked | Rel-17 onwards | |
| maxNumberPathLossEstimatePerServing-r16 | Not checked | Rel-17 onwards | |
| } | | | |
| spatialRelationsSRS-PosRRC-Inactive-r17 SEQUENCE { | | | |
| spatialRelation-SRS-PosBasedOnSSB-Serving-r16 | Not checked | Rel-17 onwards | |
| spatialRelation-SRS-PosBasedOnCSI-RS-Serving-r16 | Not checked | Rel-17 onwards | |
| spatialRelation-SRS-PosBasedOnPRS-Serving-r16 | Not checked | Rel-17 onwards | |
| spatialRelation-SRS-PosBasedOnSRS-r16 | Not checked | Rel-17 onwards | |
| spatialRelation-SRS-PosBasedOnSSB-Neigh-r16 | Not checked | Rel-17 onwards | |
| spatialRelation-SRS-PosBasedOnPRS-Neigh-r16 | Not checked | Rel-17 onwards | |
| } | | | |
| posSRS-SP-RRC-Inactive-InInitialUL-BWP-r17 SEQUENCE { | | | |
| maxNumOfSemiPersistentSRSposResources-r17 | Not checked | Rel-17 onwards | |
| maxNumOfSemiPersistentSRSposResourcesPerSlot-r17 | Not checked | Rel-17 onwards | |
| } | | | |
| posSRS-Preconfigured-RRC-InactiveInitialUL-BWP-r18 | Not checked | Rel-18 onwards | |
| posSRS-Preconfigured-RRC-InactiveOutsideInitialUL-BWP-r18 | Not checked | Rel-18 onwards | |
| posSRS-ValidityAreaRRC-InactiveInitialUL-BWP-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_posSRS_ValidityAreaRRC_InactiveInitialUL_BWP |
| posSRS-ValidityAreaRRC-InactiveOutsideInitialUL-BWP-r18 | Not checked | Rel-18 onwards | |
| posSRS-TxFH-RRC-Connected-r18 | Not checked | Rel-18 onwards | |
| posSRS-TxFH-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| posSRS-TxFH-WithTimeWindow-r18 | Not checked | Rel-18 onwards | |
| posSRS-BWA-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| srs-PosResourceConfigCA-BandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| maxNumberSRS-PosResourceSetsPerBWP-r16 | Dependent on UE capabilities | | |
| maxNumberSRS-PosResourcesPerBWP-r16 | Dependent on UE capabilities | | |
| maxNumberPeriodicSRS-PosResourcesPerBWP-r16 | Dependent on UE capabilities | | |
| maxNumberAP-SRS-PosResourcesPerBWP-r16 | Dependent on UE capabilities | | |
| maxNumberSP-SRS-PosResourcesPerBWP-r16 | Dependent on UE capabilities | | |
| } | Dependent on UE capabilities | | |
| maxNumberSRS-PosPathLossEstimateAllServingCells-r16 | Dependent on UE capabilities | | |
| maxNumberSRS-PosSpatialRelationsAllServingCells-r16 | Dependent on UE capabilities | | |
| } | | | |

**Table 9.3.1.1.3.3-12: NR-DL-AoD-MeasurementCapability-r16 (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.5.11.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-AoD-MeasurementCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| maxDL-PRS-RSRP-MeasurementFR1-r16 | Dependent on UE capabilities | | |
| maxDL-PRS-RSRP-MeasurementFR2-r16 | Dependent on UE capabilities | | |
| dl-AoD-MeasCapabilityBandList-r16 SEQUENCE (SIZE (1..n)) OF SEQUENCE { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r16 | Dependent on UE capabilities | | |
| simul-NR-DL-AoD-DL-TDOA-r16 | Dependent on UE capabilities | | |
| simul-NR-DL-AoD-Multi-RTT-r16 | Dependent on UE capabilities | | |
| maxDL-PRS-FirstPathRSRP-MeasPerTRP-r17 | Not checked | Rel-17 onwards | |
| dl-PRS-MeasRRC-Inactive-r17 | Not checked | Rel-17 onwards | |
| supportOfMeasurementsInTimeWindow-r18 | Not checked | Rel-18 onwards | |
| supportOfPRS-MeasurementRRC-Idle-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| maxDL-PRS-RSRP-MeasurementFR1-v1730 | Not checked | Rel-17 onwards | |
| maxDL-PRS-RSRP-MeasurementFR2-v1730 | Not checked | Rel-17 onwards | |
| } | | | |

**Table 9.3.1.1.3.3-13: NR-DL-TDOA-MeasurementCapability (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.5.10.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-TDOA-MeasurementCapability-r16 ::= SEQUENCE { | | Rel-16 onwards | |
| dl-RSTD-MeasurementPerPairOfTRP-FR1-r16 | Dependent on UE capabilities | | |
| dl-RSTD-MeasurementPerPairOfTRP-FR2-r16 | Dependent on UE capabilities | | |
| supportOfDL-PRS-RSRP-MeasFR1-r16 | Dependent on UE capabilities | | |
| supportOfDL-PRS-RSRP-MeasFR2-r16 | Dependent on UE capabilities | | |
| nr-UE-TEG-Capability-r17 | Set according to Table 9.3.1.1.3.3-14 | | |
| dl-tdoa-MeasCapabilityBandList-r17 SEQUENCE (SIZE (1..nrMaxBands-r16)) OF { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r17 | Not ckecked | | |
| supportOfDL-PRS-FirstPathRSRP-r17 | Not checked | Rel-17 onwards | |
| dl-PRS-MeasRRC-Inactive-r17 | Not checked | Rel-17 onwards | |
| supportOfDL-PRS-BWA-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_DL_PRS_BWA_RRC_Connected_DL_TDOA |
| supportOfDL-PRS-BWA-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| supportOfDL-PRS-BWA-RRC-Idle-r18 | Not checked | Rel-18 onwards | |
| nr-DL-PRS-RSCPD-RRC-Connected-r18 | Dependent on UE capabilities | Rel-18 onwards | pc_nr_DL_PRS_RSCPD_ReportingRRC_Connected_DL_TDOA |
| assocSingleRSTD-WithUpToNsampleRSCPD-r18 | Not checked | Rel-18 onwards | |
| nr-DL-PRS-RSCPD-MeasurementRRC-Idle-r18 | Not checked | Rel-18 onwards | |
| supportOfUE-basedCarrierPhasePositioning-r18 | Not checked | Rel-18 onwards | |
| supportOfSymbolTimeStampForRSCPD-r18 | Not checked | Rel-18 onwards | |
| supportOfFinerTimingReportGranularityForPRS-Meas-r18 | Not checked | Rel-18 onwards | |
| supportOfMeasurementsInTimeWindow-r18 | Not checked | Rel-18 onwards | |
| supportOfPRS-MeasurementRRC-Idle-r18 | Not checked | Rel-18 onwards | |
| nr-DL-PRS-RSCPD-RRC-Inactive-r18 | Not checked | Rel-18 onwards | |
| } | | | |
| } | | | |

**Table 9.3.1.1.3.3-14: NR-UE-TEG-Capability (Table 9.3.1.1.3.3-5)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-UE-TEG-Capability-r17 ::= SEQUENCE { | | Rel-17 onwards | |
| nr-UE-TEG-ID-CapabilityBandList-r17 SEQUENCE (SIZE (1..nrMaxBands-r16)) OF { | Size n of SEQUENCE is dependent on UE capabilities | | |
| freqBandIndicatorNR-r17 | Not checked | | |
| nr-UE-RxTEG-ID-MaxSupport-r17 | Dependent on UE capabilities | | pc_nr_UE_RxTEG_ID_MaxSupport |
| nr-UE-TxTEG-ID-MaxSupport-r17 | Dependent on UE capabilities | | pc_nr_UE_TxTEG_ID_MaxSupport |
| nr-UE-RxTxTEG-ID-MaxSupport-r17 | Dependent on UE capabilities | | pc_nr_UE_RxTxTEG_ID_MaxSupport |
| measureSameDL-PRS-ResourceWithDifferentRxTEGs-r17 | Dependent on UE capabilities | | pc_measureSameDL_PRS_ResourceWithDifferentRxTEGs |
| measureSameDL-PRS-ResourceWithDifferentRxTEGsSimul-r17 | Dependent on UE capabilities | | pc_measureSameDL_PRS_ResourceWithDifferentRxTEGsSimul |
| } | | | |
| } | | | |

## 9.3.1.2 LPP Abort

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

### 9.3.1.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives a LPP Abort message carrying the transaction ID of an on-going procedure }
    then { UE aborts the on-going procedure }
          }
```

### 9.3.1.2.2 Conformance requirements

As defined in clause 7.3.5.1.2.

### 9.3.1.2.3 Test description

#### 9.3.1.2.3.1 Pre-test conditions

System Simulator:

- For Test Configuration B (Table 9.3.1.2.3.2-1):
  - Sub-tests 11, 12, 13, 15, 16, 17, 19: NR Cell 1.

- Sub-test 5: NR Cell 1 and independent LTE Cell 1, as specified in 8.2.2.
- Sub-test 20: NR Cell 1 and NR Cell 2 (only applicable to UE-Based DL-TDOA).
- Sub-test 21: NR Cell 1, NR Cell 2 and NR Cell 3 (only applicable to UE-Based DL-AoD).

- For Test Configuration D (Table 9.3.1.2.3.2-1):
  - Sub-tests 11, 12, 13, 15, 16, 17: LTE Cell 1.
  - Sub-test 5: LTE Cell 1, as specified in 8.2.2.

UE:

- The UE shall begin the tests with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.3.1.2.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
  - Sub-test 5: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.1.2.3.1-1 for the configuration of measurement gaps for OTDOA (LTE).
  - Sub-test 19: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.1.2.3.1-1 for the configuration of measurement gaps for Multi-RTT and then the SS shall execute the steps in Table 9.3.1.2.3.1-2 for the configuration of UL-SRS for Multi-RTT.
  - Sub-test 20: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.1.2.3.1-1 for the configuration of measurement gaps for DL-AoD.
  - Sub-test 21: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.1.2.3.1-1 for the configuration of measurement gaps for DL-TDOA.

**Table 9.3.1.2.3.1-1: Configuration of measurement gaps**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.1-1. | **<--** | RRCReconfiguration | **-** | **-** |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

**Table 9.3.1.2.3.1-2: Configuration of UL-SRS for Multi-RTT**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.2-1. | **<--** | RRCReconfiguration | **-** | **-** |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

- For Test Configuration D (Table 9.3.1.2.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

### 9.3.1.2.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.1.2.3.2-0 below:

**Table 9.3.1.2.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 5 | UE supporting OTDOA (LTE) (Rel-15 onwards) |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 19 | UE supporting Multi-RTT (Rel-16 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

Note that this test case does not include a sub-test for the case where ECID (LTE), NR E-CID or Sensor is supported by the UE as the behaviour required cannot be guaranteed in these cases.

**Table 9.3.1.2.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.5.1 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour is as defined in Table 7.3.5.1.3.2-1.

### 9.3.1.2.3.3 Specific message contents

As defined in clause 7.3.5.1.3.3, with the following exceptions:

Table 9.3.1.2.3.3-0 replaces Table 7.3.5.1.3.3-0, Table 9.3.1.2.3.3-1 replaces Table 7.3.5.1.3.3-1, Table 9.3.1.2.3.3-2 replaces Table 7.3.5.1.3.3-2, Table 9.3.1.2.3.3-2a replaces Table 7.3.5.1.3.3-2c, Table 9.3.1.2.3.3-2b replaces Table 7.3.5.1.3.3-2d, Table 9.3.1.2.3.3-3 replaces Table 7.3.5.1.3.3-5, Table 9.3.1.2.3.3-4 replaces Table 7.3.5.1.3.3-6, Table 9.3.1.2.3.3-5 replaces Table 7.3.5.1.3.3-3 and Table 9.3.1.2.3.3-6 replaces Table 7.3.5.1.3.3-7.

**Table 9.3.1.2.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 00, Table 7.3.5.1.3.2-1)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.5.1.3.3-0 with the following exceptions: | | | |
| UE Positioning Technology | Sub-test 19: 0 0 0 0 0 1 1 0<br>Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1 | Sub-test 19: Multi-RTT<br>Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA | |

**Table 9.3.1.2.3.3-1: *DLInformationTransfer* (steps 0, 0b, 0c, 1 and 2, Table 7.3.5.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.1.2.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.2.3.3-2: DL NAS TRANSPORT (steps 0, 0b, 0c, 1 and 2, Table 7.3.5.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 0:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 0b:** Set according to Table 7.3.5.1.3.3-2b | LPP Acknowledgement | |
| | **Step 0c:** Set according to Table 9.3.1.2.3.3-2a | LPP Provide Assistance Data | |
| | **Step 1:** Set according to Table 9.3.1.2.3.3-5 | LPP Request Location Information | |
| | **Step 2:** Set according to Table 7.3.5.1.3.3-4 | LPP Abort | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.1.2.3.3-2a: LPP Provide Assistance Data (step 0c, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| OTDOA-NeighbourCellInfoList ::= SEQUENCE (SIZE(1)) OF SEQUENCE { | | | Sub-test 5 |
| SEQUENCE (SIZE(18)) OF SEQUENCE { | Sequence contains 18 instances of the following data. | | |
| physCellId | Refer to Sequence data values in Table 9.3.1.2.3.3-2b | | |
| cellGlobalId | For values of cellidentity refer to Sequence data values in Table 9.3.1.2.3.3-2b | | |
| earfcn | Not present | Same as for the reference cell | |
| cpLength | Not present | Same as for the reference cell | |
| prsInfo | Not present | Same as for the reference cell | |
| antennaPortConfig | Not present | Same as for the reference cell | |
| slotNumberOffset | Not present | Same as for the reference cell | |
| prs-SubframeOffset | Not present | Same as for the reference cell | |
| expectedRSTD | Refer to Sequence data values in Table 9.3.1.2.3.3-2b | | |
| expectedRSTD-Uncertainty | Refer to Sequence data values in Table 9.3.1.2.3.3-2b | | |
| earfcn-v9a0 | Not present | Same as for the reference cell | |
| tpId-r14 | Not present | Transmission Points not used | Rel-14 onwards |
| prs-only-tp-r14 | Not present | Not required | Rel-14 onwards |
| cpLengthCRS-r14 | Not present | Not required | Rel-14 onwards |
| sameMBSFNconfigNeighbour-r14 | TRUE | Same as for the reference cell | Rel-14 onwards |
| dlBandwidth-r14 | Not present | Same as for the reference cell and PRS frequency hopping not used | Rel-14 onwards |
| addPRSconfigNeighbour-r14 | Not present | No additional PRS configuration(s) | Rel-14 onwards |
| tdd-config-v1520 | Not present | | Rel-15 onwards |
| } | | | |
| } | | | |

**Table 9.3.1.2.3.3-2b: Sequence data values for 18 instances of sequence for Table 9.3.1.2.3.3-2a**

| Cell | Value physCellId | Value cellidentity (E-UTRAN Cell Identity) | | Value expectedRSTD | Value expectedRSTD-Uncertainty | Comment |
|---|---|---|---|---|---|---|
| | | Value eNB ID | Value Cell Identity | | | |
| Cell 2 | 2 | '0000 0000 0000 0000 0001'B | '0000 0010'B | 8192 | 10 | As defined for Cell 2 in 36.508 [8] |
| Cell 4 | 4 | '0000 0000 0000 0000 0011'B | ‘0000 0100’B | 8192 | 10 | As defined for Cell 4 in 36.508 [8] |
| Dummy cell | 1 | '0000 0000 0000 0000 0001'B | '0000 0001'B | 8253 | 51 | |
| Dummy cell | 3 | '0000 0000 0000 0000 0010'B | '0000 0011'B | 8211 | 51 | |
| Dummy cell | 6 | '0000 0000 0000 0000 0100'B | '0000 0110'B | 8221 | 51 | |
| Dummy cell | 7 | '0000 0000 0000 0000 0110'B | ‘0000 0111’B | 8192 | 51 | |
| Dummy cell | 8 | '0000 0000 0000 0000 0010'B | '0000 1000'B | 8233 | 51 | |
| Dummy cell | 9 | '0000 0000 0000 0000 0100'B | ‘0000 1001’B | 8161 | 51 | |
| Dummy cell | 10 | '0000 0000 0000 0000 0101'B | '0000 1010'B | 8226 | 51 | |
| Dummy cell | 11 | '0000 0000 0000 0000 0110'B | '0000 1011'B | 8232 | 51 | |
| Dummy cell | 16 | '0000 0000 0000 0000 0010'B | '0001 0000'B | 8223 | 51 | |
| Dummy cell | 111 | '0000 0000 0000 0000 1100'B | '0110 1111'B | 8236 | 51 | |
| Dummy cell | 118 | '0000 0000 0000 0000 1111'B | ‘0111 0110’B | 8223 | 51 | |
| Dummy cell | 119 | '0000 0000 0000 0000 1110'B | ‘0111 0111’B | 8221 | 51 | |
| Dummy cell | 120 | '0000 0000 0000 0000 1111'B | ‘0111 1000’B | 8223 | 51 | |
| Dummy cell | 122 | '0000 0000 0000 0000 1010'B | ‘0111 1010’B | 8243 | 51 | |
| Dummy cell | 125 | '0000 0000 0000 0000 1011'B | ‘0111 1101’B | 8253 | 51 | |
| Dummy cell | 126 | '0000 0000 0000 0000 1100'B | ‘0111 1110’B | 8257 | 51 | |

**Table 9.3.1.2.3.3-3: *ULInformationTransfer* (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.1.2.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.1.2.3.3-4: UL NAS TRANSPORT (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.3.1.1.3.3-5 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.1.1.3.2-1) | |

**Table 9.3.1.2.3.3-5: LPP Request Location Information (step 1, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | 0 | | |
| } | | | |
| locationInformationType | locationEstimateRequired | In case of "UE-based" Location method supported by the UE | |
| | locationMeasurementsRequired | In case of "UE-assisted" Location method supported by the UE | |
| time | 10 | | |

**Table 9.3.1.2.3.3-6: LPP Provide Capabilities (step 0a, Table 7.3.5.1.3.2-1)**

| Derivation Path: Table 7.3.5.1.3.3-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table ith the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.2 LPP Transport

### 9.3.2.1 LPP Duplicated Message

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.2.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives a LPP message carrying the same sequence number as that last received for the
         associated location session }
    then { UE discards the LPP message }
}
```

#### 9.3.2.1.2 Conformance requirements

As defined in clause 7.3.2.1.2.

9.3.2.1.3 Test description

9.3.2.1.3.1 Pre-test conditions

System Simulator:

- For Test Configuration B (Table 9.3.2.1.3.2-1): NR Cell 1.
- For Test Configuration D (Table 9.3.2.1.3.2-1): LTE Cell 1.

UE:

-

Preamble:

- For Test Configuration B (Table 9.3.2.1.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.2.1.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

9.3.2.1.3.2 Test procedure sequence

**Table 9.3.2.1.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.2.1 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour is as defined in Table 7.3.2.1.3.2-1.

9.3.2.1.3.3 Specific message contents

As defined in clause 7.3.2.1.3.3, with the following exceptions:

Table 9.3.2.1.3.3-1 replaces Table 7.3.2.1.3.3-1, Table 9.3.2.1.3.3-2 replaces Table 7.3.2.1.3.3-2, Table 9.3.2.1.3.3-3 replaces Table 7.3.2.1.3.3-4, Table 9.3.2.1.3.3-4 replaces Table 7.3.2.1.3.3-5, Table 9.3.2.1.3.3-2a replaces Table 7.3.2.1.3.3-3 and Table 9.3.2.1.3.3-5 replaces Table 7.3.2.1.3.3-6.

**Table 9.3.2.1.3.3-1: *DLInformationTransfer* (steps 1, 2, and 3a, Table 7.3.2.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.1.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.1.3.3-2: DL NAS TRANSPORT (steps 1, 2 and 3a, Table 7.3.2.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1 and 2:** Set according to Table 9.3.2.1.3.3-2a | LPP Request Capabilities | |
| | **Step 3a:** Set according to Table 7.3.2.1.3.3-7 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.2.1.3.3-2a: LPP Request Capabilities (steps 1 and 2, Table 7.3.2.1.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| sequenceNumber | 0 | | |

**Table 9.3.2.1.3.3-3: *ULInformationTransfer* (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.1.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.1.3.3-4: UL NAS TRANSPORT (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.3.2.1.3.3-5 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.2.1.3.2-1) | |

**Table 9.3.2.1.3.3-5: LPP Provide Capabilities (step 3, Table 7.3.2.1.3.2-1)**

| Derivation Path: Table 7.3.2.2.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.2.2.3.3-6 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.3.2.2 LPP Acknowledgment

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.2.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives a LPP message carrying an acknowledgement request indicator  }
    then { UE returns an acknowledgement response }
          }
```

#### 9.3.2.2.2 Conformance requirements

As defined in clause 7.3.2.2.2.

#### 9.3.2.2.3 Test description

##### 9.3.2.2.3.1 Pre-test conditions

System Simulator:

- For Test Configuration B (Table 9.3.2.2.3.2-1): NR Cell 1.
- For Test Configuration D (Table 9.3.2.2.3.2-1): LTE Cell 1.

UE:

-

Preamble:

- For Test Configuration B (Table 9.3.2.2.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.2.2.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

##### 9.3.2.2.3.2 Test procedure sequence

**Table 9.3.2.2.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.2.2 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in Table 7.3.2.2.3.2-1.

9.3.2.2.3.3 Specific message contents

As defined in clause 7.3.2.2.3.3, with the following exceptions:

Table 9.3.2.2.3.3-1 replaces Table 7.3.2.2.3.3-1, Table 9.3.2.2.3.3-2 replaces Table 7.3.2.2.3.3-2, Table 9.3.2.2.3.3-3 replaces Table 7.3.2.2.3.3-4, Table 9.3.2.2.3.3-4 replaces Table 7.3.2.2.3.3-5, Table 9.3.2.2.3.3-2a replaces Table 7.3.2.2.3.3-3 and Table 9.3.2.2.3.3-5 replaces Table 7.3.2.2.3.3-6.

**Table 9.3.2.2.3.3-1: *DLInformationTransfer* (steps 1, and 3, Table 7.3.2.2.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.2.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.2.3.3-2: DL NAS TRANSPORT (steps 1, and 3, Table 7.3.2.2.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 9.3.2.2.3.3-2a | LPP Request Capabilities | |
| | **Step 3:** Set according to Table 7.3.2.2.3.3-8 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.2.2.3.3-2a: LPP Request Capabilities (step 1, Table 7.3.2.2.3.2-1)**

| Derivation Path: Table 8.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-2 with the following exceptions: | | | |
| sequenceNumber | 0 | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |

**Table 9.3.2.2.3.3-3: *ULInformationTransfer* (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.2.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.2.3.3-4: UL NAS TRANSPORT (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.3.2.2.3.3-5 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.2.2.3.2-1) | |

**Table 9.3.2.2.3.3-5: LPP Provide Capabilities (step 2, Table 7.3.2.2.3.2-1)**

| Derivation Path: Table 7.3.2.2.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.2.2.3.3-6 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.2.3 LPP Retransmission

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

### 9.3.2.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection for EPC-NI-LR session existing}
ensure that {
  when { UE does not receive an LPP acknowledgement for an LPP message which requires
         acknowledgement }
    then { UE retransmits the LPP message up to three times. If still unacknowledged after that, the
           UE aborts all LPP activity for the associated session}
}
```

### 9.3.2.3.2 Conformance requirements

As defined in clause 7.3.2.3.2.

#### 9.3.2.3.3 Test description

##### 9.3.2.3.3.1 Pre-test conditions

System Simulator:

- For Test Configuration B (Table 9.3.2.3.3.2-1): NR Cell 1.
- For Test Configuration D (Table 9.3.2.3.3.2-1): LTE Cell 1.

UE:

-

Preamble:

- For Test Configuration B (Table 9.3.2.3.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.2.3.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

##### 9.3.2.3.3.2 Test procedure sequence

**Table 9.3.2.3.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.2.3 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in Table 7.3.2.3.3.2-1.

##### 9.3.2.3.3.3 Specific message contents

As defined in clause 7.3.2.3.3.3, with the following exceptions:

Table 9.3.2.3.3.3-1 replaces Table 7.3.2.3.3.3-1, Table 9.3.2.3.3.3-2 replaces Table 7.3.2.3.3.3-2, Table 9.3.2.3.3.3-3 replaces Table 7.3.2.3.3.3-4, Table 9.3.2.3.3.3-4 replaces Table 7.3.2.3.3.3-5 and Table 9.3.2.3.3.3-5 replaces Table 7.3.2.3.3.3-6.

**Table 9.3.2.3.3.3-1: *DLInformationTransfer* (step 1, Table 7.3.2.3.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.3.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.3.3.3-2: DL NAS TRANSPORT (steps 1 and 2a, Table 7.3.2.3.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 8.4-2 | LPP Request Capabilities | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.2.3.3.3-3: *ULInformationTransfer* (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.2.3.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.3.3.3-4: UL NAS TRANSPORT (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table Table 9.3.2.3.3.3-5 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.2.3.3.2-1) | |

**Table 9.3.2.3.3.3-5: LPP Provide Capabilities (steps 2, 4, 6, and 8, Table 7.3.2.3.3.2-1)**

| Derivation Path: Table 7.3.2.3.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.2.3.3.3-6 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.3 LPP Error Handling

### 9.3.3.1 Void

### 9.3.3.1A Void

### 9.3.3.1B LPP Requested Method not Supported – UE-Assisted

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.3.1B.1 Test Purpose (TP)

(1)

```
with {  a UE supporting at least one of UE-assisted GNSS, UE-assisted OTDOA (LTE) (LPP Rel-15
onwards), UE-assisted ECID (LTE) (Test Configuration D only), UE-assisted WLAN, UE-assisted
Bluetooth, UE-assisted Sensor, UE-assisted MBS, UE-assisted DL-TDOA, UE-assisted DL-AoD, UE-assisted
Multi-RTT or UE-assisted NR E-CID but not all of them }
and with { a NAS signalling connection existing }
ensure that {
  when { UE receives a LPP message requesting at least one location method not supported }
    then { the UE provides location information for the supported methods}
}
```

#### 9.3.3.1B.2 Conformance requirements

As defined in clause 7.3.3.1B.2.

#### 9.3.3.1B.3 Test description

##### 9.3.3.1B.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.3.3.1B.3.2-1):

- If GNSS is supported by the UE: NR Cell 1 and satellite signals, as specified in 8.2.1.
- If OTDOA (LTE) is supported by the UE: NR Cell 1 independent and LTE Cell 1 and LTE Cell 2 as specified in 8.2.2.
- If WLAN is supported by the UE: NR Cell 1 and WLAN signals, as specified in 8.2.5.
- If Bluetooth is supported by the UE: NR Cell 1 and Bluetooth signals, as specified in 8.2.6.
- If Sensor is supported by the UE: NR Cell 1.
- If MBS is supported by the UE: NR Cell 1 and MBS signals, as specified in 8.2.4.
- If DL-TDOA is supported by the UE: NR Cell 1 and NR Cell 2, as specified in 8.2.11.
- If DL-AoD is supported by the UE: NR Cell 1, as specified in 8.2.10.
- If Multi-RTT is supported by the UE: NR Cell 1, as specified in 8.2.9.
- If NR E-CID is supported by the UE: NR Cell 1, as specified in 8.2.12.

For Test Configuration D (Table 9.3.3.1B.3.2-1):

- If GNSS is supported by the UE: LTE Cell 1 and satellite signals, as specified in 8.2.1.

- If OTDOA (LTE) is supported by the UE: LTE Cell 1 and LTE Cell 2, as specified in 8.2.2
- If ECID (LTE) is supported by the UE: LTE Cell 1 and LTE Cell 2, as specified in 8.2.3.
- If WLAN is supported by the UE: LTE Cell 1 and WLAN signals, as specified in 8.2.5.
- If Bluetooth is supported by the UE: LTE Cell 1 and Bluetooth signals, as specified in 8.2.6.
- If Sensor is supported by the UE: LTE Cell 1.
- If MBS is supported by the UE: LTE Cell 1 and MBS signals, as specified in 8.2.4.

UE:

-

Preamble:

- For Test Configuration B (Table 9.3.3.1B.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
  - If OTDOA (LTE) is supported by the UE then after the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.3.1B.3.1-1 for the configuration of measurement gaps for OTDOA (LTE).
  - If DL-TDOA is supported by the UE then after the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.3.1B.3.1-1 for the configuration of measurement gaps for DL-TDOA.
  - If DL-AoD is supported by the UE then after the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.3.1B.3.1-1 for the configuration of measurement gaps for DL-AoD.
  - If Multi-RTT is supported by the UE then after the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.3.1B.3.1-1 for the configuration of measurement gaps for Multi-RTT and then the SS shall execute the steps in Table 9.3.3.1B.3.1-2 for the configuration of UL-SRS for Multi-RTT.

**Table 9.3.3.1B.3.1-1: Configuration of measurement gaps**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.1-1. | **<--** | RRCReconfiguration | **-** | **-** |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

**Table 9.3.3.1B.3.1-2: Configuration of UL-SRS for Multi-RTT**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.2-1. | **<--** | RRCReconfiguration | **-** | **-** |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

- For Test Configuration D (Table 9.3.3.1B.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

9.3.3.1B.3.2 Test procedure sequence

**Table 9.3.3.1B.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.3.1B |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in clause 7.3.3.1B.3.2.

9.3.3.1B.3.3 Specific message contents

As defined in clause 7.3.3.1B.3.3, with the following exceptions:

Table 9.3.3.1B.3.3-1 replaces Table 7.3.3.1B.3.3-1, Table 9.3.3.1B.3.3-2 replaces Table 7.3.3.1B.3.3-2, Table 9.3.3.1B.3.3-2a replaces Table 7.3.3.1B.3.3-3, Table 9.3.3.1B.3.3-3 replaces Table 7.3.3.1B.3.3-5, Table 9.3.3.1B.3.3-4 replaces Table 7.3.3.1B.3.3-6, Table 9.3.3.1B.3.3.5 replaces Table 7.3.3.1B.3.3-4, Table 9.3.3.1B.3.3.6 replaces Table 7.3.3.1B.3.3-6a and Table 9.3.3.1B.3.3-7 replaces Table 7.3.3.1B.3.3-7.

**Table 9.3.3.1B.3.3-1: *DLInformationTransfer* (steps 0, 0b, 1, 2 and 3a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.3.1B.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.3.1B.3.3-2: DL NAS TRANSPORT (steps 0, 0b, 1, 2 and 3a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 0:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 1:** Set according to Table 9.3.3.1B.3.3-2a | LPP Provide Assistance Data | |
| | **Step 2:** Set according to Table 9.3.3.1B.3.3-5 | LPP Request Location Information | |
| | **Steps 0b and 3a:** Set according to Table 7.3.3.1B.3.3-8 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.3.1B.3.3-2a: LPP Provide Assistance data (step 1, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| Initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| a-gnss-ProvideAssistanceData | Present for all supported GNSSs if UE supports UE-assisted A-GNSS. As defined in clause 8.4 | | |
| otdoa-ProvideAssistanceData | Present if UE supports UE-assisted OTDOA (LTE). As defined in clause 8.4 | | |
| sensor-ProvideAssistanceData-r14 | Present if UE supports UE-assisted Sensor. As defined in clause 8.4 | Rel-14 onwards | |
| tbs-ProvideAssistanceData-r14 | Present if UE supports UE-assisted MBS. As defined in clause 8.4 | Rel-14 onwards | |
| wlan-ProvideAssistanceData-r14 | Present if UE supports UE-assisted WLAN. As defined in clause 8.4 | Rel-14 onwards | |
| nr-Multi-RTT-ProvideAssistanceData | Present if UE supports UE-assisted Multi-RTT. As defined in clause 8.4 | Rel-16 onwards | |
| nr-DL-AoD-ProvideAssistanceData-r16 | Present if UE supports UE-assisted DL-AoD. As defined in clause 8.4 | Rel-16 onwards | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | Present if UE supports UE-assisted DL-TDOA. As defined in clause 8.4 | Rel-16 onwards | |

**Table 9.3.3.1B.3.3-3: *ULInformationTransfer* (steps 0a and 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.3.1B.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.3.1B.3.3-4: UL NAS TRANSPORT (steps 0a and 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 0a:** Set according to Table 9.3.3.1B.3.3-6 | LPP Provide Capabilities | |
| | **Step 3:** Set according to Table 9.3.3.1B.3.3-7 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.3.1B.3.2-1) | |

**Table 9.3.3.1B.3.3-5: LPP Request Location Information (step 2, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsRequired | | |
| a-gnss-RequestLocationInformation | Present. As defined in Table 5.4-4 | | |
| gnss-Methods | GNSS-ID-Bitmap: bits 0, 3, 4, 5 = 1 | | |
| otdoa-RequestLocationInformation | Present. As defined in Table 5.4-5 | | |
| ecid-RequestLocationInformation | Present. As defined in Table 5.4-6 | | |
| requestedMeasurements | Test Configuration D: bits 0, 1, 2 = 1 | | |
| tbs-RequestLocationInformation-r13 | Present. As defined in Table 5.4-7 | Rel-13 onwards | |
| sensor-RequestLocationInformation-r13 | Present. As defined in Table 5.4-10 | Rel-13 onwards | |
| wlan-RequestLocationInformation-r13 | Present. As defined in Table 5.4-8 | Rel-13 onwards | |
| bt-RequestLocationInformation-r13 | Present. As defined in Table 5.4-9 | Rel-13 onwards | |
| nr-ECID-RequestLocationInformation-r16 | Present. As defined in Table 8.4-4 | Rel-16 onwards | |
| nr-Multi-RTT-RequestLocationInformation-r16 | Present. As defined in Table 8.4-5 | Rel-16 onwards | |
| nr-DL-AoD-RequestLocationInformation-r16 | Present. As defined in Table 8.4-6 | Rel-16 onwards | |
| nr-DL-TDOA-RequestLocationInformation-r16 | Present. As defined in Table 8.4-7 | Rel-16 onwards | |

**Table 9.3.3.1B.3.3-6: LPP Provide Capabilities (step 0a, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 7.3.3.1B.3.3-6a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.3.1B.3.3-6a with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-ECID-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.3.1B.3.3-7: LPP Provide Location Information (step 3, Table 7.3.3.1B.3.2-1)**

| Derivation Path: Table 7.3.3.1B.3.3-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.3.1B.3.3-7 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | Present if UE supports UE-assisted Multi-RTT. | Rel-16 onwards | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-Multi-RTT-Error-r16 | May be present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present if UE supports UE-assisted DL-AoD. | Rel-16 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-AoD-LocationInformation-r16 | Not present | | |
| nr-DL-AoD-Error-r16 | May be present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present if UE supports UE-assisted DL-TDOA. | Rel-16 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-tdoa-LocationInformation-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | May be present | | |
| } | | | |
| nr-ECID-ProvideLocationInformation-r16 SEQUENCE { | Present if UE supports UE-assisted NR E-CID. | Rel-16 onwards | |
| nr-ECID-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-ECID-Error-r16 | May be present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.4 LPP Positioning Procedures

### 9.3.4.1 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Based

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.4.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
```

```
  when { UE receives assistance data and a location request for UE-based  }
    then { UE sends a PROVIDE LOCATION INFORMATION  message containing a location estimate }
           }
```

#### 9.3.4.1.2 Conformance requirements

As defined in clause 7.3.4.1.2.

#### 9.3.4.1.3 Test description

##### 9.3.4.1.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.3.4.1.3.2-1):NR Cell 1.

- Satellite signals (sub-test case 15): as specified in 8.2.1.
- MBS signals (Sub-test 16): as specified in 8.2.4.
- WLAN signals (Sub-test 17): as specified in 8.2.5.
- Sub-test 20: NR Cell 1 and NR Cell 2, as specified in 8.2.10.
- Sub-test 21: NR Cell 1, NR Cell 2 and NR Cell 3, as specified in 8.2.11.

For Test Configuration D (Table 9.3.4.1.3.2-1): LTE Cell 1.

- Satellite signals (sub-test case 15): as specified in 8.2.1.
- MBS signals (Sub-test 16): as specified in 8.2.4.
- WLAN signals (Sub-test 17): as specified in 8.2.5.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.3.4.1.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.4.1.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

##### 9.3.4.1.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.1.3.2-0 below:

**Table 9.3.4.1.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.3.4.1.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.4.1 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in Table 7.3.4.1.3.2-1.

### 9.3.4.1.3.3 Specific message contents

As defined in clause 7.3.4.1.3.3, with the following exceptions:

Table 9.3.4.1.3.3-0 replaces Table 7.3.4.1.3.3-1, Table 9.3.4.1.3.3-1 replaces Table 7.3.4.1.3.3-2, Table 9.3.4.1.3.3-2 replaces Table 7.3.4.1.3.3-3, Table 9.3.4.1.3.3-2a replaces Table 7.3.4.1.3.3-4, Table 9.3.4.1.3.3-3 replaces Table 7.3.4.1.3.3-6, Table 9.3.4.1.3.3-4 replaces Table 7.3.4.1.3.3-7, Table 9.3.4.1.3.3-2b replaces Table 7.3.4.1.3.3-5, Table 9.3.4.1.3.3-5 replaces Table 7.3.4.1.3.3-7a and Table 9.3.4.1.3.3-6 replaces Table 7.3.4.1.3.3-8.

**Table 9.3.4.1.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.1.3.2-1)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.1.3.3-1 with the following exceptions: | | | |
| UE Positioning Technology | Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1 | Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA | |

**Table 9.3.4.1.3.3-1: *DLInformationTransfer* (steps 1a, 1c, 2, 3 and 4a, Table 7.3.4.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.1.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.1.3.3-2: DL NAS TRANSPORT (steps 1a, 1c, 2, 3 and 4a, Table 7.3.4.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1a:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 2:** Set according to Table 9.3.4.1.3.3-2a | LPP Provide Assistance Data | |
| | **Step 3:** Set according to Table 9.3.4.1.3.3-2b | LPP Request Location Information | |
| | **Steps 1c and 4a:** Set according to Table 7.3.4.1.3.3-9 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.1.3.3-2a: LPP Provide Assistance data (step 2, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 9.3.4.1.3.3-2b: LPP Request Location Information (step 3, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| locationInformationType | locationEstimateRequired | | |

**Table 9.3.4.1.3.3-3: *ULInformationTransfer* (steps 1b and 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.1.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.1.3.3-4: UL NAS TRANSPORT (steps 1b and 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1b:** Set according to Table 9.3.4.1.3.3-5 | LPP Provide Capabilities | |
| | **Step 4:** Set according to Table 9.3.4.1.3.3-6 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.4.1.3.2-1) | |

**Table 9.3.4.1.3.3-5: LPP Provide Capabilities. (step 1b, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 7.3.4.1.3.3-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.1.3.3-7a with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.1.3.3-6: LPP Provide Location Information (step 4, Table 7.3.4.1.3.2-1)**

| Derivation Path: Table 7.3.4.1.3.3-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.1.3.3-8 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-AoD-LocationInformation-r16 | Present. Any value acceptable | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 21 | Rel-16 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-tdoa-LocationInformation-r16 | Present. Any value acceptable | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

### 9.3.4.2 E-SMLC Initiated Assistance Data Delivery followed by Location Information Transfer: UE-Assisted

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.4.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives assistance data and a location request for UE-assisted  }
    then { UE sends a PROVIDE LOCATION INFORMATION  message containing location measurements }
           }
```

#### 9.3.4.2.2 Conformance requirements

As defined in clause 7.3.4.2.2.

#### 9.3.4.2.3 Test description

##### 9.3.4.2.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.3.4.2.3.2-1):

- Sub-tests 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22: NR Cell 1.
- Sub-tests 5 and 7: NR Cell 1 and independent LTE Cell 1 and LTE Cell 2 as specified in 8.2.2.
- Sub-test 6: not applicable.
- Satellite signals (Sub-test 15): as specified in 8.2.1.
- WLAN signals (Sub-tests 11, 17): as specified in 8.2.5.
- MBS signals (Sub-tests 12, 16): as specified in 8.2.4
- Bluetooth signals (Sub-test 13): as specified in 8.2.6.
- Sub-test 21: NR Cell 1 and NR Cell 2, as specified in 8.2.11.

For Test Configuration D (Table 9.3.4.2.3.2-1):

- Sub-tests 11, 12, 13, 14, 15, 16, 17, 18: LTE Cell 1.
- Sub-tests 5 and 7: LTE Cell 1 and LTE Cell 2, as specified in 8.2.2.
- Sub-test 6: LTE Cell 1 and LTE Cell 2, as specified in 8.2.3.
- Satellite signals (Sub-test 15): as specified in 8.2.1.
- WLAN signals (Sub-tests 11, 17): as specified in 8.2.5.
- MBS signals (Sub-tests 12, 16): as specified in 8.2.4

- Bluetooth signals (Sub-test 13): as specified in 8.2.6.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.3.4.2.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

  - Sub-test 5 and 7: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.2.3.1-1 for the configuration of measurement gaps for OTDOA (LTE).

  - Sub-test 19: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.2.3.1-1 for the configuration of measurement gaps for Multi-RTT and then the SS shall execute the steps in Table 9.3.4.2.3.1-2 for the configuration of UL-SRS for Multi-RTT.

  - Sub-test 20: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.2.3.1-1 for the configuration of measurement gaps for DL-AoD.

  - Sub-test 21 After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.2.3.1-1 for the configuration of measurement gaps for DL-TDOA.

**Table 9.3.4.2.3.1-1: Configuration of measurement gaps**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.1-1. | <-- | RRCReconfiguration | - | - |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

**Table 9.3.4.2.3.1-2: Configuration of UL-SRS for Multi-RTT**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.2-1. | <-- | RRCReconfiguration | - | - |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

- For Test Configuration D (Table 9.3.4.2.3.2-1): FFS.

Related PICS/PIXIT Statements:

-

### 9.3.4.2.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.2.3.2-0 below:

**Table 9.3.4.2.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 5 | UE supporting OTDOA (LTE) (Rel-15 onwards) |
| 6 | UE supporting ECID (LTE) (Test Configuration D only) |
| 7 | UE supporting GNSS(1) and OTDOA (LTE) (Rel-15 onwards) |
| 11 | UE supporting WLAN (Rel-13 only) |
| 12 | UE supporting MBS (Rel-13 only) |
| 13 | UE supporting Bluetooth |
| 14 | UE supporting Sensor (Rel-13 only) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 19 | UE supporting Multi-RTT (Rel-16 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| 22 | UE supporting NR E-CID (Rel-16 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.3.4.2.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.4.2 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in clause 7.3.4.2.3.2.

For sub-test 19 or sub-test 20 or sub-test 21, the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE at step 1. Then the stored assistance data in the UE are cleared.

For sub-test 22 NR E-CID, the SS does not send LPP message of type Provide Assistance Data at step 2.

### 9.3.4.2.3.3 Specific message contents

As defined in clause 7.3.4.2.3.3, with the following exceptions:

Table 9.3.4.2.3.3-0 replaces Table 7.3.4.2.3.3-1, Table 9.3.4.2.3.3-1 replaces Table 7.3.4.2.3.3-3, Table 9.3.4.2.3.3-2 replaces Table 7.3.4.2.3.3-4, Table 9.3.4.2.3.3-2a replaces Table 7.3.4.2.3.3-5, Table 9.3.4.2.3.3-3 replaces Table 7.3.4.2.3.3-7, Table 9.3.4.2.3.3-4 replaces Table 7.3.4.2.3.3-8, Table 9.3.4.2.3.3-5 replaces Table 7.3.4.2.3.3-6, Table 9.3.4.2.3.3-6 replaces Table 7.3.4.2.3.3-8a and Table 9.3.4.2.3.3-7 replaces Table 7.3.4.2.3.3-9.

**Table 9.3.4.2.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.2.3.2-1)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.2.3.3-1 with the following exceptions: | | | |
| UE Positioning Technology | Sub-test 19: 0 0 0 0 0 1 1 0<br>Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1 | Sub-test 19: Multi-RTT<br>Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA | |

**Table 9.3.4.2.3.3-1: *DLInformationTransfer* (steps 1b, 1d, 2, 3, 4a2, 4b2 and 4b4, Table 7.3.4.2.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.2.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.2.3.3-2: DL NAS TRANSPORT (steps 1b, 1d, 2, 3, 4a2, 4b2 and 4b4, Table 7.3.4.2.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1b:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 2:** Set according to Table 9.3.4.2.3.3-2a | LPP Provide Assistance Data | |
| | **Step 3:** Set according to Table9.3.4.2.3.3-5 | LPP Request Location Information | |
| | **Steps 1d, 4a2, 4b2 and 4b4:** Set according to Table 7.3.4.2.3.3-10 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.2.3.3-2a: LPP Provide Assistance data (step 2, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 9.3.4.2.3.3-3: *ULInformationTransfer* (steps 1c, 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.2.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.2.3.3-4: UL NAS TRANSPORT (steps 1c, 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1c:** Set according to Table 9.3.4.2.3.3-6 | LPP Provide Capabilities | |
| | **Steps 4 a1, 4b1 and 4b3:** Set according to Table 9.3.4.2.3.3-7 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.4.2.3.2-1) | |

**Table 9.3.4.2.3.3-5: LPP Request Location Information (step 3, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsRequired | | |
| } | | | |
| ecid-RequestLocationInformation SEQUENCE { | | | |
| requestedMeasurements | Test Configuration D: bits 0, 1, 2 = 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.2.3.3-6: LPP Provide Capabilities. (step 1c, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 7.3.4.2.3.3-8a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.2.3.3-8a with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.2.3.3-7: LPP Provide Location Information (steps 4 a1, 4b1 and 4b3, Table 7.3.4.2.3.2-1)**

| Derivation Path: Table 7.3.4.2.3.3-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.2.3.3-9 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| nr-ECID-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 22. Not present for sub-tests 19, 20 and 21(Note1). | Rel-16 onwards | |
| nr-ECID-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-ECID-Error-r16 | May be present | | |
| } | | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 19. Not present for sub-tests 20, 21 and 22(Note1). | Rel-16 onwards | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-Multi-RTT-Error-r16 | May be present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 20. Not present for sub-tests 19, 21 and 22(Note1). | Rel-16 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-DL-AoD-Error-r16 | May be present | | |
| nr-dl-AoD-LocationInformation-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 21. Not present for sub-tests 19, 20 and 22(Note1). | Rel-16 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-tdoa-LocationInformation-r16 | Not present | | |
| nr-DL-TDOA-Error-r16 | May be present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| Note 1: In addition, a-gnss-ProvideLocationInformation, otdoa-ProvideLocationInformation, ecid-ProvideLocationInformation, sensor-ProvideLocationInformation-r13, tbs-ProvideLocationInformation-r13, wlan-ProvideLocationInformation-r13 and bt-ProvideLocationInformation-r13 shall also be not present. | | | |

## 9.3.4.3 E-SMLC Initiated Position Measurement without assistance data: UE-Based

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

#### 9.3.4.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and receives a location request for UE-based and the UE
         requires assistance data in order to fulfil the location request }
    then {    UE sends a REQUEST ASSISTANCE DATA message followed by a PROVIDE LOCATION INFORMATION
         message containing a location estimate }
           }
```

#### 9.3.4.3.2 Conformance requirements

As defined in clause 7.3.4.3.2.

#### 9.3.4.3.3 Test description

##### 9.3.4.3.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.3.4.3.3.2-1): NR Cell 1.

- Satellite signals (sub-test 15): as specified in 8.2.1.
- MBS signals (sub-test 16): as specified in 8.2.4.
- WLAN signals (sub-test 17): as specified in 8.2.5.
- Sub-test 20: NR Cell 1 and NR Cell 2, as specified in 8.2.10.
- Sub-test 21: NR Cell 1, NR Cell 2 and NR Cell 3, as specified in 8.2.11.

For Test Configuration D (Table 9.3.4.3.3.2-1): LTE Cell 1.

- Satellite signals (sub-test 15): as specified in 8.2.1.
- MBS signals (sub-test 16): as specified in 8.2.4.
- WLAN signals (sub-test 17): as specified in 8.2.5.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.3.4.3.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- For Test Configuration D (Table 9.3.4.3.3.2-1): FFS.

Related PICS/PIXIT Statements:

Method of triggering an LPP Request Assistance Data message.

##### 9.3.4.3.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.3.3.2-0 below:

**Table 9.3.4.3.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.3.4.3.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.4.3 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in Table 7.3.4.3.3.2-1.

### 9.3.4.3.3.3 Specific message contents

As defined in clause 7.3.4.3.3.3, with the following exceptions:

Table 9.3.4.3.3.3-0 replaces Table 7.3.4.3.3.3-1, Table 9.3.4.3.3.3-1 replaces Table 7.3.4.3.3.3-2, Table 9.3.4.3.3.3-2 replaces Table 7.3.4.3.3.3-3, Table 9.3.4.3.3.3-2a replaces Table 7.3.4.3.3.3-9, Table 9.3.4.3.3.3-3 replaces Table 7.3.4.3.3.3-6, Table 9.3.4.3.3.3-4 replaces Table 7.3.4.3.3.3-7, Table 9.3.4.3.3.3-5 replaces Table 7.3.4.3.3.3-7a, Table 9.3.4.3.3.3-6 replaces Table 7.3.4.3.3.3-8 and Table 9.3.4.3.3.3-7 replaces Table 7.3.4.3.3.3-10.

**Table 9.3.4.3.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.3.3.2-1)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.3.3.3-1 with the following exceptions: | | | |
| UE Positioning Technology | Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1 | Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA | |

**Table 9.3.4.3.3.3-1: *DLInformationTransfer* (steps 1a, 1c, 2, 4 and 5a, Table 7.3.4.3.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.3.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.3.3.3-2: DL NAS TRANSPORT (steps 1a, 1c, 2, 4 and 5a, Table 7.3.4.3.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1a:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 2:** Set according to Table 9.3.4.3.3.3-2b | LPP Request Location Information | |
| | **Step 4:** Set according to Table 9.3.4.3.3.3-2a | LPP Provide Assistance Data | |
| | **Steps 1c and 5a:** Set according to Table 7.3.4.3.3.3-11 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.3.3.3-2a: LPP Provide Assistance Data (step 4, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Assistance Data message in step 3 Table 7.3.4.3.3.2-1 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | Present if acknowledgement field is included by the UE at step 3, Table 7.3.4.3.3.2-1. | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value as the sequenceNumber in step 3, Table 7.3.4.3.3.2-1 | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| a-gnss-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to TS 37.571-5 [12]. | | |
| sensor-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 8.4.1.5. | Rel-14 onwards | |
| tbs-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 8.4.1.3. | Rel-14 onwards | |
| wlan-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 8.4.1.4. | Rel-14 onwards | |
| nr-DL-AoD-ProvideAssistanceData-r16 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 8.4.1.7. | Rel-16 onwards | |
| nr-DL-TDOA-ProvideAssistanceData-r16 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.3.3.2-1 which are available according to subclause 8.4.1.8. | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.3.3.3-2b: LPP Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 5.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 5.4-3 with the following exceptions: | | | |
| locationInformationType | locationEstimateRequired | | |
| a-gnss-RequestLocationInformation | Set according to Table 7.3.4.3.3.3-5 | | Sub-test 15 |
| sensor-RequestLocationInformation-r14 | Set according to Table 7.3.4.3.3.3-5B | Rel-14 onwards | Sub-test 18 |
| tbs-RequestLocationInformation-r13 | Set according to Table 7.3.4.3.3.3-5A | Rel-13 onwards | Sub-test 16 |
| wlan-RequestLocationInformation-r14 | Set according to Table 7.3.4.3.3.3-5C | Rel-14 onwards | Sub-test 17 |
| nr-DL-AoD-RequestLocationInformation-r16 | As defined in Table 9.3.4.3.3.3-2c | Rel-16 onwards | Sub-test 20 |
| nr-DL-TDOA-RequestLocationInformation-r16 | As defined in Table 9.3.4.3.3.3-2d | Rel-16 onwards | Sub-test 21 |

**Table 9.3.4.3.3.3-2c: NR DL-AoD Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 8.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-6 with the following exceptions: | | | |
| nr-AssistanceAvailability-r16 | TRUE | | |

**Table 9.3.4.3.3.3-2d: NR DL-TDOA Request Location Information (step 2, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 8.4-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-7 with the following exceptions: | | | |
| nr-AssistanceAvailability-r16 | TRUE | | |

**Table 9.3.4.3.3.3-3: *ULInformationTransfer* (steps 1b, 3 and 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.3.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.3.3.3-4: UL NAS TRANSPORT (steps 1b, 3 and 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1b:** Set according to Table 9.3.4.3.3.3-5 | LPP Provide Capabilities | |
| | **Step 3:** Set according to Table 9.3.4.3.3.3-6 | LPP Request Assistance Data | |
| | **Step 5:** Set according to Table 9.3.4.3.3.3-7 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.4.3.3.2-1) | |

**Table 9.3.4.3.3.3-5: LPP Provide Capabilities. (step 1b, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 7.3.4.3.3.3-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.3.3.3-7a with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1a, Table 7.3.4.3.3.2-1 | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-DL-AoD-RequestCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-RequestCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.3.3.3-6: LPP Request Assistance Data (step 3, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 7.3.4.3.3.3-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.3.3.3-8 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| nr-DL-AoD-RequestAssistanceData-r16 | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-TDOA-RequestAssistanceData-r16 | Present for sub-test 21 | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.3.3.3-7: LPP Provide Location Information (step 5, Table 7.3.4.3.3.2-1)**

| Derivation Path: Table 7.3.4.3.3.3-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.3.3.3-10 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-TDOA-ProvideLocationInformation-r16 | Present for sub-test 21 | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.4.4 E-SMLC Initiated Position Measurement without assistance data: UE-Assisted

Editor's note: Test configuration D is incomplete:

- The corresponding attach procedure for NG-RAN E-UTRA has not yet been defined.
- The message contents need to be revised for Test Configuration D.

### 9.3.4.4.1 Test Purpose (TP)

```
with { a NAS signalling connection existing }
ensure that {
  when {    UE has no assistance data stored and receives a location request for UE-assisted and the
            UE requires assistance data in order to fulfil the location request }
    then {  UE sends a REQUEST ASSISTANCE DATA message followed by a PROVIDE LOCATION INFORMATION
            message containing location measurements }
            }
```

### 9.3.4.4.2 Conformance requirements

As defined in clause 7.3.4.4.2.

### 9.3.4.4.3 Test description

#### 9.3.4.4.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.3.4.4.3.2-1):

- Sub-tests 15, 16, 17, 18, 19, 20: NR Cell 1.
- Sub-tests 5 and 7: NR Cell 1 and independent LTE Cell 1 and LTE Cell 2 as specified in 8.2.2.
- Satellite signals (sub-test 15): as specified in 8.2.1.
- MBS signals (sub-test 16): as specified in 8.2.4.
- WLAN signals (sub-test 17): as specified in 8.2.5.

- Sub-test 21: NR Cell 1 and NR Cell 2, as specified in 8.2.11.

For Test Configuration D (Table 9.3.4.4.3.2-1):

- Sub-tests 15, 16, 17, 18: LTE Cell 1.
- Sub-tests 5 and 7: LTE Cell 1 and LTE Cell 2, as specified in 8.2.2.
- Satellite signals (sub-test 15): as specified in 8.2.1.
- MBS signals (sub-test 16): as specified in 8.2.4.
- WLAN signals (sub-test 17): as specified in 8.2.5.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.3.4.4.3.2-1): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
  - Sub-test 5 and 7: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.4.3.1-1 for the configuration of measurement gaps for OTDOA (LTE).
  - Sub-test 19: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.4.3.1-1 for the configuration of measurement gaps for Muti-RTT and then the SS shall execute the steps in Table 9.3.4.4.3.1-2 for the configuration of UL-SRS for Muti-RTT.
  - Sub-test 20: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.4.3.1-1 for the configuration of measurement gaps for DL-AoD.
  - Sub-test 21: After the UE is in state 3N-A, the SS shall execute the steps in Table 9.3.4.4.3.1-1 for the configuration of measurement gaps for DL-TDOA.

**Table 9.3.4.4.3.1-1: Configuration of measurement gaps**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.1-1. | <-- | RRCReconfiguration | - | - |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

**Table 9.3.4.4.3.1-2: Configuration of UL-SRS for Multi-RTT**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an RRCReconfiguration message as in Table 8.3.2-1. | <-- | RRCReconfiguration | - | - |
| 2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |

- For Test Configuration D (Table 9.3.4.4.3.2-1): FFS.

Related PICS/PIXIT Statements:

Method of triggering an LPP Request Assistance Data message.

#### 9.3.4.4.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.4.3.2-0 below:

**Table 9.3.4.4.3.2-0: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 5 | UE supporting OTDOA (LTE) (Rel-15 onwards) |
| 7 | UE supporting GNSS(1) and OTDOA (LTE) (Rel-15 onwards) |
| 15 | UE supporting GNSS(1) |
| 16 | UE supporting MBS (Rel-14 onwards) |
| 17 | UE supporting WLAN (Rel-14 onwards) |
| 18 | UE supporting Sensor (Rel-14 onwards) |
| 19 | UE supporting Multi-RTT (Rel-16 onwards) |
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.3.4.4.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| A | EN-DC | Functionality is tested by test case 7.3.4.4 |
| B | NG-RAN NR | |
| C | NE-DC | Functionality is tested by test configuration B |
| D | NG-RAN E-UTRA | |
| E | NGEN-DC | Functionality is tested by test configuration D |

Main behaviour as defined in Table 7.3.4.4.3.2-1.

For sub-test 19 or sub-test 20 or sub-test 21, the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE at step 1. Then the stored assistance data in the UE are cleared.

#### 9.3.4.4.3.3 Specific message contents

As defined in clause 7.3.4.4.3.3, with the following exceptions:

Table 9.3.4.4.3.3-0 replaces Table 7.3.4.4.3.3-1, Table 9.3.4.4.3.3-1 replaces Table 7.3.4.4.3.3-3, Table 9.3.4.4.3.3-2 replaces Table 7.3.4.4.3.3-4,Table 9.3.4.4.3.3-2a replaces Table 7.3.4.4.3.3-10, Table 9.3.4.4.3.3-3 replaces Table 7.3.4.4.3.3-7, Table 9.3.4.4.3.3-4 replaces Table 7.3.4.4.3.3-8, Table 9.3.4.4.3.3-2b replaces Table 7.3.4.4.3.3-5, Table 9.3.4.4.3.3-5 replaces Table 7.3.4.4.3.3-8a, Table 9.3.4.4.3.3-6 replaces Table 7.3.4.4.3.3-9 and Table 9.3.4.4.3.3-7 replaces Table 7.3.4.4.3.3-11.

**Table 9.3.4.4.3.3-0: RESET UE POSITIONING STORED INFORMATION (step 1, Table 7.3.4.4.3.2-1)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.4.3.3-1 with the following exceptions: | | | |
| UE Positioning Technology | Sub-test 19: 0 0 0 0 0 1 1 0<br>Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1 | Sub-test 19: Multi-RTT<br>Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA | |

**Table 9.3.4.4.3.3-1: *DLInformationTransfer* (steps 1b, 1d, 2, 4, 4b, 5a2, 5b2 and 5b4, Table 7.3.4.4.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.4.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.4.3.3-2: DL NAS TRANSPORT (steps 1b, 1d, 2, 4, 4b, 5a2, 5b2 and 5b4, Table 7.3.4.4.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1b:** Set according to Table 8.4-2 | LPP Request Capabilities | |
| | **Step 2:** Set according to Table 9.3.4.4.3.3-2b | LPP Request Location Information | |
| | **Steps 4 and 4b:** Set according to Table 9.3.4.4.3.3-2a | LPP Provide Assistance Data | |
| | **Steps 1d, 5a2, 5b2 and 5b4:** Set according to Table 7.3.4.4.3.3-12 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.4.3.3-2a: LPP Provide Assistance Data (steps 4 and 4b, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Assistance Data message in step 3 or 4a Table 7.3.4.4.3.2-1. | |

| | | | |
|---|---|---|---|
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | Present if acknowledgement field is included by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1. | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value as the sequenceNumber in step 3 or 4a, Table 7.3.4.4.3.2-1. | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| a-gnss-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1 which are available according to TS 37.571-5 [12]. | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then the SS sends two seperate LPP Provide Assistance Data messages in steps 4 and 4b each containing the relevant assistance data. | Sub-tests 7, 15 |
| otdoa-ProvideAssistanceData | The SS provides the assistance data requested by the UE at step 3 or 4a, Table 7.3.4.4.3.2-1 according to subclause 8.4.1.2. | For sub-test 7, in case the UE sends two separate LPP Request Assistance Data messages in steps 3 and 4a then the SS sends two seperate LPP Provide Assistance Data messages in steps 4 and 4b each containing the relevant assistance data. | Sub-tests 5,7 |

| | | | |
|---|---|---|---|
| sensor-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.5. | Release 14 onwards | Sub-test 18 |
| tbs-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.3. | Release 14 onwards | Sub-test 16 |
| wlan-ProvideAssistanceData-r14 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.4. | Release 14 onwards | Sub-test 17 |
| nr-Multi-RTT-ProvideAssistanceData-r16 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.6. | Release 16 onwards | Sub-test 19 |
| nr-DL-AoD-ProvideAssistanceData-r16 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.7. | Release 16 onwards | Sub-test 20 |
| nr-DL-TDOA-ProvideAssistanceData-r16 | The SS provides the assistance data requested by the UE at step 3, Table 7.3.4.4.3.2-1 which are available according to subclause 8.4.1.8. | Release 16 onwards | Sub-test 21 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.4.3.3-2b: LPP Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 7.3.4.4.3.3-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.4.3.3-5 with the following exceptions: | | | |
| nr-Multi-RTT-RequestLocationInformation-r16 | As defined in Table 9.3.4.4.3.3-2c | Rel-16 onwards | Sub-test 19 |
| nr-DL-AoD-RequestLocationInformation-r16 | As defined in Table 9.3.4.4.3.3-2d | Rel-16 onwards | Sub-test 20 |
| nr-DL-TDOA-RequestLocationInformation-r16 | As defined in Table 9.3.4.4.3.3-2e | Rel-16 onwards | Sub-test 21 |

**Table 9.3.4.4.3.3-2c: NR Multi-RTT Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 8.4-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-5 with the following exceptions: | | | |
| nr-AssistanceAvailability-r16 | TRUE | | |

**Table 9.3.4.4.3.3-2d: NR DL-AoD Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 8.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-6 with the following exceptions: | | | |
| nr-AssistanceAvailability-r16 | TRUE | | |

**Table 9.3.4.4.3.3-2e: NR DL-TDOA Request Location Information (step 2, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 8.4-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-7 with the following exceptions: | | | |
| nr-AssistanceAvailability-r16 | TRUE | | |

**Table 9.3.4.4.3.3-3: *ULInformationTransfer* (steps 1c, 3, 4a, 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.4.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.4.3.3-4: UL NAS TRANSPORT (steps 1c, 3, 4a, 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1c:** Set according to Table 9.3.4.4.3.3-5 | LPP Provide Capabilities | |
| | **Steps 3 and 4a:** Set according to Table 9.3.4.4.3.3-6 | LPP Request Assistance Data | |
| | **Steps 5a1, 5b1 and 5b3:** Set according to Table 9.3.4.4.3.3-7 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 7.3.4.4.3.2-1) | |

**Table 9.3.4.4.3.3-5: LPP Provide Capabilities. (step 1c, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 7.3.4.4.3.3-8a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.4.3.3-8a with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| otdoa-ProvideCapabilities | Present or not present dependent on pc_OTDOA_onNR | | |
| ecid-ProvideCapabilities | Present or not present dependent on pc_ECID_onNR | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.4.3.3-6: LPP Request Assistance Data (steps 3 and 4a, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 7.3.4.4.3.3-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.4.3.3-9 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| nr-Multi-RTT-RequestAssistanceData-r16 | Present for sub-test 19 | Rel-16 onwards | |
| nr-DL-AoD-RequestAssistanceData-r16 | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-TDOA-RequestAssistanceData-r16 | Present for sub-test 21 | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.4.3.3-7: LPP Provide Location Information (steps 5a1, 5b1 and 5b3, Table 7.3.4.4.3.2-1)**

| Derivation Path: Table 7.3.4.4.3.3-11 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 7.3.4.4.3.3-11 with the following exceptions: | | | |

| | | | |
|---|---|---|---|
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 | Present for sub-test 19 | Rel-16 onwards | |
| nr-DL-AoD-ProvideLocationInformation-r16 | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-TDOA-ProvideLocationInformation-r16 | Present for sub-test 21 | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.4.5 LPP carrier phase positioning procedure

### 9.3.4.5.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives assistance data and a LPP REQUEST LOCATION INFORMATION message indicating the
UE to provide DL carrier phase measurement}
    then { UE sends a PROVIDE LOCATION INFORMATION message containing DL RSCPD/RSCP measurements
together with the DL RSTD/UE Rx–Tx time difference measurements }
}
```

### 9.3.4.5.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.214 clause 5.1.6.5.2, TS 38.305 clauses 8.10 and clause 8.12. Unless otherwise stated these are Rel-18 requirements.

[TS 38.214, clause 5.1.6.5.2]

For DL UE positioning measurement reporting in higher layer parameter *NR-DL-TDOA-SignalMeasurementInformation*, the UE may be configured to report the DL Reference Signal Carrier Phase Difference (RSCPD) [7, TS 38.215] measurement along with the DL RSTD measurement. When the UE reports RSCPD measurements, the reference *nr-DL-PRS-ReferenceInfo* is the same as the one reported, for the RSTD measurements. For DL UE positioning measurement reporting in higher layer parameter *NR-Multi-RTT-SignalMeasurementInformation*, the UE may be configured to report the DL Reference Signal Carrier Phase (RSCP) measurement [7, TS 38,215] along with the UE Rx-Tx time difference measurement. When the UE reports DL RSCPD measurement(s) along with DL RSTD measurement(s) or DL RSCP measurement(s) along with UE Rx-Tx time difference measurement(s), the DL RSCPD and/or DL RSCP measurement(s) should be measured from a single DL PRS positioning frequency layer. For a UE in RRC_CONNECTED state, DL RSCP/RSCPD measurements are measured within the configured measurement gap.

The UE is expected to obtain each DL RSCP or DL RSCPD measurement with $N_{sample} = 1$ as defined in [11, TS 38.133]. If the UE reports a DL RSTD measurement with $N_{sample}$ = 2 or 4 samples as defined in [11, TS 38.133], up to $N_{sample}$ DL RSCPD measurements can be reported associated with the DL RSTD measurement. If the UE reports a UE Rx-Tx time difference measurement with $N_{sample}$ = 2 or 4 samples as defined in [11, TS 38.133], up to $N_{sample}$ DL RSCP measurements can be reported associated with the UE Rx-Tx time difference measurement. Each DL RSCP or DL RSCPD measurement has its own timestamp.

[TS 38.305, clause 8.10]

In the Multi-RTT positioning method, the UE position is estimated based on measurements performed at both, UE and TRPs. The measurements performed at the UE and TRPs are UE/gNB Rx-Tx time difference measurements (and optionally DL-PRS-RSRP, DL-PRS-RSRPP, UL-SRS-RSRP, UL-SRS-RSRPP, and/or DL-RSCP/UL-RSCP) of DL-PRS and UL-SRS, which are used by an LMF to determine the RTTs.

…

The information that may be signalled from UE to the LMF is listed in Table 8.10.2.2-1. The individual UE measurements are defined in TS 38.215 [37].

**Table 8.10.2.2-1: Measurement results that may be transferred from UE to the LMF**

| Information |
|---|
| PCI, GCI, and PRS ID, ARFCN, PRS resource ID, PRS resource set ID for each measurement |
| DL-PRS-RSRP measurement |
| UE Rx-Tx time difference measurement |
| DL-RSCP measurement[NOTE1, NOTE2] |
| Time stamp of the measurement |
| Quality for each measurement |
| TA offset used by UE |
| UE Rx TEG IDs, UE Tx TEG IDs, and UE RxTx TEG IDs associated with UE Rx-Tx time difference measurements |
| LOS/NLOS information for UE measurements |
| DL-PRS-RSRPP measurement |
| The association of UE Tx TEG ID and SRS |
| Indication that DL-PRS bandwidth aggregation has been used for UE Rx-Tx time difference measurement |
| Indication that the reported measurements are based on receiving single or multiple hops of DL-PRS |
| UE Rx – Tx time difference subframe offset |
| DL timing drift |
| NOTE 1: The DL-RSCP measurement may be reported along with the UE Rx-Tx time difference measurement.<br>NOTE 2: The DL-RSCP is measured from a single DL PRS positioning frequency layer. |

[TS 38.305, clause 8.12]

In the DL-TDOA positioning method, the UE position is estimated based on DL RSTD (and optionally DL-PRS-RSRP and/or DL-PRS-RSRPP and/or DL-RSCPD) measurements taken at the UE of downlink radio signals from multiple NR TRPs, along with knowledge of the geographical coordinates of the TRPs and their relative downlink timing.

The UE while connected to a gNB may require measurement gaps to perform the DL-TDOA measurements from NR TRPs. The UE may request measurement gaps from a gNB using the procedure described in clause 7.4.1.1. The UE may also request to activate pre-configured measurement gaps as described in clause 7.7.2.

The specific positioning techniques used to estimate the UE's location from this information are beyond the scope of this specification.

…

The information that may be signalled from UE to the LMF is listed in Table 8.12.2.2.0-1. The individual UE measurements are defined in TS 38.215 [37].

**Table 8.12.2.2.0-1: Measurement results that may be transferred from UE to the LMF**

| Information | UE-assisted | UE-based |
|---|---|---|
| Latitude/Longitude/Altitude, together with uncertainty shape | No | Yes |
| PCI, GCI, ARFCN, PRS resource ID, PRS resource set ID and PRS ID for each measurement | Yes | No |
| DL RSTD measurement | Yes | No |
| DL-PRS-RSRP measurement | Yes | No |
| DL-RSCPD measurement[NOTE 1, NOTE 2] | Yes | No |
| Time stamp of the measurements | Yes | No |
| Time stamp of location estimate | No | Yes |
| Quality for each measurement | Yes | No |
| UE Rx TEG IDs for DL RSTD measurements | Yes | No |
| DL-PRS-RSRPP measurement | Yes | No |
| LOS/NLOS information for UE measurements | Yes | No |
| Indication that DL-PRS bandwidth aggregation has been used for DL RSTD measurement | Yes | No |
| Indication that the reported measurements are based on receiving single or multiple hops of DL-PRS. | Yes | No |
| Protection Level, optionally together with achievable Target Integrity Risk | No | Yes |
| NOTE 1: The DL-RSCPD measurement may be reported along with the DL RSTD measurement.<br>NOTE 2: The DL-RSCPD is measured from a single DL PRS positioning frequency layer. | | |

### 9.3.4.5.3 Test description

#### 9.3.4.5.3.1 Pre-test conditions

System Simulator:

- Sub-test 30: NR Cell 1 as specified in 8.2.9 and system information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 32: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11 and system information combination NR-4 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

#### 9.3.4.5.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.5.3.2-1 below:

**Table 9.3.4.5.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 30 | UE supporting Multi-RTT (Rel-18 onwards) |
| 32 | UE supporting DL-TDOA (Rel-18 onwards) |

**Table 9.3.4.5.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.3.4.5.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS sends an RRCReconfiguration message. | <-- | RRCReconfiguration | | |
| 6 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | | |
| 7 | The SS sends an LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 8 | Check: Does the UE send an LPP message of type Provide Location Information including DL RSCPD/RSCP measurements together with the DL RSTD/UE Rx–Tx time difference measurements? | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 9 | IF the UE LPP message at step 8 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

9.3.4.5.3.3 Specific message contents

**Table 9.3.4.5.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 30: 0 0 0 0 0 1 1 0<br>Sub-test 32: 0 0 0 0 0 1 1 1 | Sub-test 30: Multi-RTT<br>Sub-test 32: DL-TDOA | |

**Table 9.3.4.5.3.3-2: DLInformationTransfer (step 1, step 3, step 4, step 7 and step 9, Table 9.3.4.5.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.5.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.5.3.3-3: DL NAS TRANSPORT (DLInformationTransfer, Table 9.3.4.5.3.3-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3 and 9:** Set according to Table 9.3.4.5.3.3-4 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 9.3.4.5.3.3-5 | LPP Provide Assistance Data | |
| | **Step 7:** Set according to Table 9.3.4.5.3.3-6 | LPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.5.3.3-4: LPP Acknowledgement (DLInformationTransfer, Table 9.3.4.5.3.3-2)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the | |

| | | | |
|---|---|---|---|
| | | sequenceNumber field in steps 2 or 8, Table 9.3.4.5.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.3.4.5.3.3-5: LPP Provide Assistance data (DLInformationTransfer, Table 9.3.4.5.3.3-2)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 9.3.4.5.3.3-6: LPP Request Location Information (DLInformationTransfer, Table 9.3.4.5.3.3-2)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| } | | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 30 |
| nr-RequestedMeasurements-r16 | bit 2 = 1 (dl-PRS-RSCP-Request-r18) | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 32 |
| nr-RequestedMeasurements-r16 | bit 2 = 1 (dl-PRS-RSCPD-Request-r18) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.5.3.3-7: RRCReconfiguration (step 5, Table 9.3.4.5.3.2-3)**

| Derivation Path: TS 38.508-1, table 4.6.1-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| measConfig | MeasConfig as defined in Table 8.3.1-2 | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension SEQUENCE { | | | Sub-test 30 |
| masterCellGroup | CellGroupConfig as defined in Table 8.3.2-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.5.3.3-8: ULInformationTransfer (step 2 and step 10, Table 9.3.4.5.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.5.3.3-9 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.5.3.3-9: UL NAS TRANSPORT (ULInformationTransfer, Table 9.3.4.5.3.3-8)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 9.3.4.5.3.3-10 | LPP Provide Capabilities | |
| | **Step 10:** Set according to Table 9.3.4.5.3.3-11 | LPP Provide Location Information | |

| | | | |
|---|---|---|---|
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.3.4.5.3.2-3) | |

**Table 9.3.4.5.3.3-10: LPP Provide Capabilities (ULInformationTransfer, Table 9.3.4.5.3.3-8)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.3.4.5.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.5.3.3-11: LPP Provide Location Information (ULInformationTransfer, Table 9.3.4.5.3.3-8)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 7, Table 9.3.4.5.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 30 | | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Present. Any integer between (0..3599) | | |

| | | | |
|---|---|---|---|
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 32 | | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 1 entry | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Present. Any integer between (0..3599) | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.3.4.6 LPP frequency hopping positioning procedure for Redcap UEs

#### 9.3.4.6.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives assistance data and a LPP REQUEST LOCATION INFORMATION message indicating the
UE to use DL-PRS receiver frequency hopping for performing positioning measurements }
    then { UE sends a PROVIDE LOCATION INFORMATION message containing a location estimate and
indicates that the reported measurement is based on receiving single or multiple hops of DL-PRS }
}
```

#### 9.3.4.6.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.214 clauses 5.1.6.5.1, and TS 38.305 clauses 8.10.2.2 and 8.10.3.1.3.1. Unless otherwise stated these are Rel-18 requirements.

[TS 38.214, clause 5.1.6.5.1]

The reduced capability UE may be configured to measure and report, subject to UE capability, via *nr-DL-PRS-RxHoppingRequest* the DL RSTD, DL PRS-RSRP, DL PRS-RSRPP, or UE Rx-Tx time difference using receiver frequency hopping for a DL PRS resource, with a requested bandwidth of all hops that may be greater than the maximum reduced capability UE bandwidth. The reduced capability UE performing receiver frequency hopping may report via *nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx* one measurement associated with one received frequency hop or one measurement based on multiple hops of the DL PRS. The reduced capability UE may report whether the measurement is associated with one received frequency hop or multiple frequency hops of the DL PRS. In RRC_CONNECTED mode, the reduced capability UE is expected to use a single instance of a configured measurement gap to receive all hops of the DL PRS using receiver frequency hopping.

[TS 38.305, clause 8.10.2.2]

The information that may be signalled from UE to the LMF is listed in Table 8.10.2.2-1. The individual UE measurements are defined in TS 38.215 [37].

**Table 8.10.2.2-1: Measurement results that may be transferred from UE to the LMF**

| Information |
|---|
| PCI, GCI, and PRS ID, ARFCN, PRS resource ID, PRS resource set ID for each measurement |
| DL-PRS-RSRP measurement |
| UE Rx-Tx time difference measurement |
| DL-RSCP measurement[NOTE1, NOTE2] |
| Time stamp of the measurement |
| Quality for each measurement |
| TA offset used by UE |
| UE Rx TEG IDs, UE Tx TEG IDs, and UE RxTx TEG IDs associated with UE Rx-Tx time difference measurements |
| LOS/NLOS information for UE measurements |
| DL-PRS-RSRPP measurement |
| The association of UE Tx TEG ID and SRS |
| Indication that DL-PRS bandwidth aggregation has been used for UE Rx-Tx time difference measurement |
| Indication that the reported measurements are based on receiving single or multiple hops of DL-PRS |
| UE Rx – Tx time difference subframe offset |
| DL timing drift |
| NOTE 1: The DL-RSCP measurement may be reported along with the UE Rx-Tx time difference measurement.<br>NOTE 2: The DL-RSCP is measured from a single DL PRS positioning frequency layer. |

[TS 38.305, clause 8.10.3.1.3.1]

Figure 8.10.3.1.3.1-1 shows the Location Information Transfer operations for the Multi-RTT positioning method when the procedure is initiated by the LMF.




**Figure 8.10.3.1.3.1-1: LMF-initiated Location Information Transfer Procedure**

(1) The LMF sends an LPP Request Location Information message to the UE. This request includes indication of Multi-RTT measurements requested, including any needed measurement configuration information, and required response time.
The LPP Request Location Information message may include one or more time windows during which the target device is requested to perform the Multi-RTT measurements on indicated DL-PRS Resource Sets.
The LPP Request Location Information message may include a request to perform joint UE Rx-Tx time difference measurement(s) across two or three DL-PRS positioning frequency layers.
The LPP Request Location Information message may include a request to perform the Multi-RTT measurements using receiver frequency hopping for a DL PRS resource within a configured measurement gap.

NOTE: The LMF may provide the same configurations of time windows to two devices e.g., target UE and PRU, for simultaneous measurements by the two devices.

(2) The UE obtains Multi-RTT measurements as requested in step 1. The UE then sends an LPP Provide Location Information message to the LMF, before the Response Time provided in step (1) elapsed, and includes the obtained Multi-RTT measurements. If the UE is unable to perform the requested measurements, or the Response Time elapsed before any of the requested measurements were obtained, the UE returns any information that can be provided in an LPP message of type Provide Location Information which includes a cause indication for the not provided location information.

### 9.3.4.6.3 Test description

#### 9.3.4.6.3.1 Pre-test conditions

System Simulator:

- Sub-test 30: NR Cell 1 as specified in 8.2.9 and system information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 31: NR Cell 1 and NR Cell 2 as specified in 8.2.10 and system information combination NR-2 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 32: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11 and system information combination NR-4 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

#### 9.3.4.6.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.6.3.2-1 below:

**Table 9.3.4.6.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 30 | UE supporting Multi-RTT (Rel-18 onwards) |
| 31 | UE supporting DL-AoD (Rel-18 onwards) |
| 32 | UE supporting DL-TDOA (Rel-18 onwards) |

**Table 9.3.4.6.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.3.4.6.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS sends an RRCReconfiguration message. | <-- | RRCReconfiguration | | |
| 6 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | | |
| 7 | The SS sends an LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 8 | Check: Does the UE send an LPP message of type Provide Location Information including measurements results with frequency hopping indication? | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 9 | IF the UE LPP message at step 8 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

9.3.4.6.3.3 Specific message contents

**Table 9.3.4.6.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 30: 0 0 0 0 0 1 1 0<br>Sub-test 31: 0 0 0 0 1 0 0 0<br>Sub-test 32: 0 0 0 0 0 1 1 1 | Sub-test 30: Multi-RTT<br>Sub-test 31: DL-AoD<br>Sub-test 32: DL-TDOA | |

**Table 9.3.4.6.3.3-2: DLInformationTransfer (step 1, step 3, step 4, step 7 and step 9, Table 9.3.4.6.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.6.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-3: DL NAS TRANSPORT (DLInformationTransfer, Table 9.3.4.6.3.3-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3 and 9:** Set according to Table 9.3.4.6.3.3-4 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 9.3.4.6.3.3-5 | LPP Provide Assistance Data | |
| | **Step 7:** Set according to Table 9.3.4.6.3.3-7 | LPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.6.3.3-4: LPP Acknowledgement (DLInformationTransfer, Table 9.3.4.6.3.3-2)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in steps 2 or 8, Table 9.3.4.6.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.3.4.6.3.3-5: LPP Provide Assistance data (DLInformationTransfer, Table 9.3.4.6.3.3-2)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | Sub-test 30 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.3.4.6.3.3-6 | | |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | Sub-test 31 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.3.4.6.3.3-6 | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | Sub-test 32 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.3.4.6.3.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-6: NR-DL-PRS-AssistanceData (LPP Provide Assistance data, Table 9.3.4.6.3.3-5)**

| Derivation Path: Table 8.4.1.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4.1.6-1 with the following exceptions: | | | |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 268 PRBs | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | In case of sub-test 32 UE-based DL-TDOA method supported by the UE as defined in clause 8.2.11 |
| nr-DL-PRS-PositioningFrequencyLayer-r16 SEQUENCE { | | | |
| dl-PRS-ResourceBandwidth-r16 | 61 | 268 PRBs | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-7: LPP Request Location Information (DLInformationTransfer, Table 9.3.4.6.3.3-2)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| } | | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 30 |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | Set as the maximumPRS-BandwidthAcrossAllHopsFR1-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR1 |

| | | | |
|---|---|---|---|
| fr2 | Set as the maximumPRS-BandwidthAcrossAllHopsFR2-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR2 |
| } | | | |
| } | | | |
| } | | | |
| nr-DL-AoD-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 31 |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | Set as the maximumPRS-BandwidthAcrossAllHopsFR1-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR1 |
| fr2 | Set as the maximumPRS-BandwidthAcrossAllHopsFR2-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR2 |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 32 |
| nr-DL-PRS-RxHoppingRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-RxHoppingTotalBandwidth-r18 CHOICE { | | | |
| fr1 | Set as the maximumPRS-BandwidthAcrossAllHopsFR1-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR1 |
| fr2 | Set as the maximumPRS-BandwidthAcrossAllHopsFR2-r18 value reported by UE | | pc_maximumPRS_BandwidthAcrossAllHopsFR2 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-8: RRCReconfiguration (step 5, Table 9.3.4.6.3.2-3)**

| Derivation Path: TS 38.508-1, table 4.6.1-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| measConfig | MeasConfig as defined in Table 8.3.1-2 | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension SEQUENCE { | | | Sub-test 30 |
| masterCellGroup | CellGroupConfig as defined in Table 8.3.2-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-9: ULInformationTransfer (step 2 and step 8, Table 9.3.4.6.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.6.3.3-10 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-10: UL NAS TRANSPORT (ULInformationTransfer, Table 9.3.4.6.3.3-9)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 9.3.4.6.3.3-11 | LPP Provide Capabilities | |
| | **Step 10:** Set according to Table 9.3.4.6.3.3-12 | LPP Provide Location Information | |

| | | | |
|---|---|---|---|
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.3.4.6.3.2-3) | |

**Table 9.3.4.6.3.3-11: LPP Provide Capabilities (ULInformationTransfer, Table 9.3.4.6.3.3-9)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.3.4.6.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.6.3.3-12: LPP Provide Location Information (ULInformationTransfer, Table 9.3.4.6.3.3-9)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 7, Table 9.3.4.6.3.3-7 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | Present for sub-test 30 |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Present. Any value acceptable | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | | | Present for sub-test 31 |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-DL-AoD-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-AoD-MeasElement-r16 { | 2 entries | | |
| NR-DL-AoD-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Present. Any value acceptable | | |
| nr-DL-PRS-RxBeamIndex-r16 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurements-r16 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-DL-AoD-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Present. Any value acceptable | | |
| } | | | |
| NR-DL-AoD-MeasElement-r16[2] | | entry 2 NR-DL-AoD-MeasElement-r16[2] contains the same message contents as NR-DL-AoD-MeasElement-r16[1]. | |
| } | | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | Present for sub-test 32 |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 2 entries | | |

| | | | |
|---|---|---|---|
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | Not checked | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Present. Any value acceptable | | |
| } | | | |
| NR-DL-TDOA-MeasElement-r16[2] | | entry 2 NR-DL-TDOA-MeasElement-r16[2] contains the same message contents as NR-DL-TDOA-MeasElement-r16[1]. | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.3.4.7 PRS aggregation positioning procedure

### 9.3.4.7.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives assistance data and a LPP REQUEST LOCATION INFORMATION message indicating the
UE is requested to perform joint measurement across aggregated PFLs }
    then { UE sends a PROVIDE LOCATION INFORMATION message containing a location estimate and
indicating the measurement is based on aggregated DL-PRS resources across PFLs }
}
```

### 9.3.4.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.214 clauses 5.1.6.5.3. Unless otherwise stated these are Rel-18 requirements.

[TS 38.214, clause 5.1.6.5.3]

When the UE is expected to perform aggregated measurements for bandwidth aggregation across DL PRS positioning frequency layers, the UE expects to be configured with linkage information, via higher layer parameter *nr-DL-PRS-AggregationInfo*, between DL PRS resource sets across DL PRS positioning frequency layers. For the linked DL PRS resource sets, the UE is expected to be configured with the same values of *dl-PRS-QCL-Info*, *dl-PRS-Periodicity-and-ResourceSetSlotOffset*, *dl-PRS-NumSymbols*, *dl-PRS-ResourceTimeGap*, *dl-PRS-ResourceRepetitionFactor*, *dl-PRS-ResourceSymbolOffset*, *dl-PRS-MutingBitRepetitionFactor*, *dl-PRS-SubcarrierSpacing*, *dl-PRS-CyclicPrefix*, *dl-PRS-CombSizeN*, *dl-PRS-ResourcePower*, *NR-MutingPattern*, and *NR-DL-PRS-SFN0-Offset*, and the UE is expected to be configured with DL PRS resources that maintain uniformly spaced DL PRS RE pattern within a symbol across aggregated DL PRS positioning frequency layers. The UE assumes that DL PRS resources across the linked DL PRS resource sets which satisfy the above conditions are linked for bandwidth aggregation, and the UE may assume phase continuity on the DL PRS resources on same symbol(s); otherwise, the UE does not assume that PRS resources from the linked DL PRS resource sets are linked for bandwidth aggregation.

The UE may be indicated by the network that aggregated DL PRS resource set(s) can be used as the reference for the aggregated DL RSTD, DL PRS-RSRP, DL PRS-RSRPP, and UE Rx-Tx time difference measurements.

The UE may be configured to measure and report, subject to UE capability, up to 4 aggregated DL RSTD measurement(s) per pair of *dl-PRS-ID*, from a different pair of aggregated DL PRS resources across two or three DL PRS positioning frequency layers. The UE may report up to 4 RSTD measurements associated with different aggregated DL PRS resources per UE Rx TEG per *dl-PRS-ID*.

The UE may be configured to measure and report, subject to UE capability, up to 4 aggregated UE Rx-Tx time difference measurement(s) from aggregated DL PRS resources across two or three DL PRS positioning frequency layers. The UE may report up to 4 UE Rx-Tx time difference measurements associated with different aggregated DL PRS resources per UE RxTx TEG per *dl-PRS-ID*.

The UE may be requested via higher layer parameter *nr-DL-PRS-JointMeasurementRequestedPFL-List* to perform the aggregated DL RSTD measurement(s) or the aggregated UE Rx-Tx time difference measurement(s) across two or three DL PRS positioning frequency layers.

The UE may report via higher layer parameter *nr-RSTD-BasedOnAggregatedResources* or *nr-UE-RxTxTimeDiffBasedOnAggregatedResources* in a measurement report whether the aggregated DL RSTD measurement(s) or the aggregated UE Rx-Tx time difference measurement(s) is performed. If any aggregated measurement is performed, the two or three DL PRS positioning frequency layers to be used may also be reported by reporting PRS resource set IDs.

If the UE reports a DL PRS-RSRP or a DL PRS-RSRPP with aggregated DL RSTD measurement(s) or aggregated UE Rx-Tx time difference measurement(s), the DL PRS-RSRP or the DL PRS-RSRPP correspond to the aggregated DL PRS resources across two or three DL PRS positioning frequency layers.

For PRS resources on multiple DL PRS positioning frequency layers (PFLs) linked for aggregation, the channel over which a symbol on one PFL for PRS transmission is conveyed can be inferred from the channel over which the same symbol of another PFL or the aggregated PFL is conveyed.

### 9.3.4.7.3 Test description

#### 9.3.4.7.3.1 Pre-test conditions

System Simulator:

- Sub-test 30: NR Cell 1 as specified in 8.2.9 and system information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 32: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11 and system information combination NR-4 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

#### 9.3.4.7.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.3.4.7.3.2-1 below:

**Table 9.3.4.7.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 30 | UE supporting Multi-RTT (Rel-18 onwards) |
| 32 | UE supporting DL-TDOA (Rel-18 onwards) |

**Table 9.3.4.7.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.3.4.7.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS sends an RRCReconfiguration message. | <-- | RRCReconfiguration | | |
| 6 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | | |
| 7 | The SS sends an LPP message of type Request Location Information including a request for performing joint measurement across aggregated PFLs. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 8 | Check: Does the UE send an LPP message of type Provide Location Information including measurements results with indicating the measurement is based on aggregated DL-PRS resources? | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 9 | IF the UE LPP message at step 8 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

9.3.4.7.3.3 Specific message contents

**Table 9.3.4.7.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 30: 0 0 0 0 0 1 1 0<br>Sub-test 32: 0 0 0 0 0 1 1 1 | Sub-test 30: Multi-RTT<br>Sub-test 32: DL-TDOA | |

**Table 9.3.4.7.3.3-2: DLInformationTransfer (step 1, step 3, step 4, step 7 and step 9, Table 9.3.4.7.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.7.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-3: DL NAS TRANSPORT (DLInformationTransfer, Table 9.3.4.7.3.3-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3 and 9:** Set according to Table 9.3.4.7.3.3-4 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 9.3.4.7.3.3-5 | LPP Provide Assistance Data | |
| | **Step 7:** Set according to Table 9.3.4.7.3.3-9 | LPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.3.4.7.3.3-4: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.3.4.7.3.3-3)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in steps 2 or 8, Table 9.3.4.7.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.3.4.7.3.3-5: LPP Provide Assistance data (DL NAS TRANSPORT, Table 9.3.4.7.3.3-3)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | Sub-test 30 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.3.4.7.3.3-6 | | |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | Sub-test 32 |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.3.4.7.3.3-6 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-6: NR-DL-PRS-AssistanceData (LPP Provide Assistance data, Table 9.3.4.7.3.3-5)**

| Derivation Path: Table 8.4.1.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4.1.6-1 with the following exceptions: | | | |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 1 | entry 1 | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 2 | entry 2 | sub-test 32 |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 3 | entry 3 | sub-test 32 |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 3 entries | | |

| | | | |
|---|---|---|---|
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 1 | entry 1 | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 2 | entry 2 | sub-test 32 |
| NR-DL-PRS-AssistanceDataPerTRP-r16[3] | As specified in Table 9.3.4.7.3.3-7 with condition NR cell 3 | entry 3 | sub-test 32 |
| } | | | |
| } | | | |
| } | | | |
| nr-DL-PRS-AggregationInfo-r18 SEQUENCE (SIZE (1..nrMaxNumPRS-BandWidthAggregation-r18)) OF NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18 { | 3 entries | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[1] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 1 | |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 0 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[2] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 2 | sub-test 32 |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 1 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| NR-linkedDL-PRS-ResourceSetID-PRS-AggregationList-r18[3] SEQUENCE (SIZE (2..3)) OF NR-DL-PRS-AggregationElement-r18 { | 2 entries | entry 3 | sub-test 32 |
| NR-DL-PRS-AggregationElement-r18[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 0 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| NR-DL-PRS-AggregationElement-r18[2] SEQUENCE { | | entry 2 | |
| nr-DL-PRS-FrequencyLayerIndex-r18 | 1 | | |
| nr-DL-PRS-TRP-Index-r18 | 2 | | |
| nr-DL-PRS-ResourceSetIndex-r18 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-7: NR-DL-PRS-AssistanceDataPerTRP-r16 (NR-DL-PRS-AssistanceData, Table 9.3.4.7.3.3-6)**

| Derivation Path: 37.355 clause 6.4.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NR-DL-PRS-AssistanceDataPerTRP-r16 ::= SEQUENCE { | | | |
| dl-PRS-ID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| | 2 | | NR Cell 3 |
| nr-PhysCellID-r16 | As defined in TS 38.508-1 table 4.4.2-2, Physical layer cell Identity for SIG for NR Cell 1/2/3 | | |
| nr-CellGlobalID-r16 SEQUENCE { | | | |
| mcc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 1/2/3 | | |
| mnc-r15 | As defined TS 38.508-1 table 4.4.2-3 for NR Cell 1/2/3 | | |
| nr-cellidentity-r15 | Set to NR Cell 1/2/3 Identifier defined in TS 38.508-1 Table 4.4.2-2 | | |
| } | | | |
| nr-ARFCN-r16 | ARFCN-ValueNR for NR Cell 1/2/3 frequency | | |
| nr-DL-PRS-SFN0-Offset-r16 SEQUENCE { | | | |
| sfn-Offset-r16 | 0 | | |
| integerSubframeOffset-r16 | 0 | | |
| } | | | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 4 | About 1 µs | |
| nr-DL-PRS-Info-r16 | NR-DL-PRS-Info-r16 as specified in Table 8.4.1.6-3 | | |
| prs-OnlyTP-r16 | Not present | | |
| nr-DL-PRS-ExpectedAoD-or-AoA-r17 | Not present | | |
| } | | | |

**Table 9.3.4.7.3.3-8: NR-DL-PRS-Info (NR-DL-PRS-AssistanceDataPerTRP-r16, Table 9.3.4.7.3.3-7)**

| Derivation Path: Table 8.4.1.6-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4.1.6-3 with the following exceptions: | | | |
| NR-DL-PRS-Info-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-ResourceSetList-r16 SEQUENCE (SIZE (1..nrMaxSetsPerTrp-r16)) OF NR-DL-PRS-ResourceSet-r16 { | 1 entry | | |
| NR-DL-PRS-ResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ResourceList-r16 SEQUENCE (SIZE (1..nrMaxResourcesPerSet-r16)) OF NR-DL-PRS-Resource-r16 { | 1 entry | | |
| NR-DL-PRS-Resource-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ResourceID-r16 | 0 | | NR-DL-PRS-AssistanceDataPerFreq-r16[1] |
| | 1 | | NR-DL-PRS-AssistanceDataPerFreq-r16[2] |
| dl-PRS-SequenceID-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| | 2 | | NR Cell 3 |
| dl-PRS-CombSizeN-AndReOffset-r16 CHOICE { | | | |
| n2-r16 | 0 | | NR Cell 1 |
| | 1 | | NR Cell 2 |
| | 0 | | NR Cell 3 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-9: LPP Request Location Information (DLInformationTransfer, Table 9.3.4.7.3.3-2)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation-r9 SEQUENCE { | | | |
| commonIEsRequestLocationInformation SEQUENCE { | | | |
| locationInformationType | locationMeasurementsPreferred | | |
| } | | | |
| nr-Multi-RTT-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 30 |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |
| nr-DL-TDOA-RequestLocationInformation-r16 SEQUENCE { | | | Sub-test 32 |
| nr-DL-PRS-JointMeasurementRequest-r18 SEQUENCE { | | | |
| nr-DL-PRS-JointMeasurementRequestedPFL-List-r18 SEQUENCE (SIZE (2..3)) OF INTEGER (0..nrMaxFreqLayers-1-r16) { | | | |
| INTEGER[1] | 0 | | |
| INTEGER[2] | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-10: RRCReconfiguration (step 5, Table 9.3.4.7.3.2-3)**

| Derivation Path: TS 38.508-1, table 4.6.1-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | | | |
| measConfig | MeasConfig as defined in Table 8.3.1-2 | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension SEQUENCE { | | | Sub-test 30 |
| masterCellGroup | CellGroupConfig as defined in Table 8.3.2-2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-11: ULInformationTransfer (step 2 and step 8, Table 9.3.4.7.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.3.4.7.3.3-12 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-12: UL NAS TRANSPORT (ULInformationTransfer, Table 9.3.4.7.3.3-11)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 9.3.4.7.3.3-13 | LPP Provide Capabilities | |
| | **Step 10:** Set according to Table 9.3.4.7.3.3-14 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.3.4.7.3.2-3) | |

**Table 9.3.4.7.3.3-13: LPP Provide Capabilities (ULInformationTransfer, Table 9.3.4.7.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.3.4.7.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.7.3.3-14: LPP Provide Location Information (ULInformationTransfer, Table 9.3.4.7.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 7, Table 9.3.4.7.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | | | Present for sub-test 30 |
| nr-Multi-RTT-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| nr-Multi-RTT-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-Multi-RTT-MeasElement-r16 { | 1 entry | | |
| NR-Multi-RTT-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-UE-RxTxTimeDiff-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-RxTx-TEG-Info-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-Multi-RTT-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCP-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |

| | | | |
|---|---|---|---|
| nr-RSCP-AddSampleMeasurements-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| nr-NTN-UE-RxTxMeasurements-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-NTA-Offset-r16 | Not checked | | |
| nr-SRS-TxTEG-Set-r17 | Not present | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-TxTEG-TimingErrorMargin-r17 | Not present | | |
| nr-UE-RxTxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | | | Present for sub-test 32 |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | | | |
| dl-PRS-ReferenceInfo-r16 | Not checked | | |
| nr-DL-TDOA-MeasList-r16 SEQUENCE (SIZE(1..nrMaxTRPs-r16)) OF NR-DL-TDOA-MeasElement-r16 { | 1 entry | | |
| NR-DL-TDOA-MeasElement-r16[1] SEQUENCE { | | entry 1 | |
| dl-PRS-ID-r16 | Not checked | | |
| nr-PhysCellID-r16 | Not checked | | |
| nr-CellGlobalID-r16 | Not checked | | |
| nr-ARFCN-r16 | Not checked | | |
| nr-DL-PRS-ResourceID-r16 | Not checked | | |
| nr-DL-PRS-ResourceSetID-r16 | Not checked | | |
| nr-TimeStamp-r16 | Not checked | | |
| nr-RSTD-r16 | Present. Any value acceptable | | |
| nr-AdditionalPathList-r16 | Not checked | | |
| nr-TimingQuality-r16 | Not checked | | |
| nr-DL-PRS-RSRP-Result-r16 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurements-r16 | Not checked | | |
| nr-UE-Rx-TEG-ID-r17 | Not checked | | |
| nr-DL-PRS-FirstPathRSRP-Result-r17 | Not checked | | |
| nr-los-nlos-Indicator-r17 | Not checked | | |
| nr-AdditionalPathListExt-r17 | Not checked | | |
| nr-DL-TDOA-AdditionalMeasurementsExt-r17 | Not checked | | |
| nr-MeasBasedOnAggregatedResources-r18 | true | | |
| nr-AggregatedDL-PRS-ResourceInfo-List-r18 | Not checked | | |
| nr-RSCPD-r18 | Not checked | | |
| nr-PhaseQuality-r18 | Not checked | | |
| nr-RSCPD-AddMeasurementSamples-r18 | Not checked | | |
| nr-ReportDL-PRS-MeasBasedOnSingleOrMultiHopRx-r18 | Not checked | | |
| } | | | |
| } | | | |
| nr-UE-RxTEG-TimingErrorMargin-r17 | Not present | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

# 9.4 RRC Protocol Procedures

## 9.4.1 PosSIB broadcasting followed by location information transfer

### 9.4.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and receives the positioning assistance data via posSIBs
and then UE receives a location request from LMF}
    then { the UE sends a PROVIDE LOCATION INFORMATION message containing a location estimate }
}
```

### 9.4.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 37.355 clauses 7.1, 7.2, and TS 38.331 clause 5.2.2.3.5. Unless otherwise stated these are Rel-16 requirements.

[TS 37.355, clause 7.1]

Broadcast of positioning assistance data is supported via Positioning System Information Blocks (posSIBs) as specified in TS 36.331 [12] or TS 38.331 [35]. The posSIBs are carried in RRC System Information (SI) messages (TS 36.331 [12] or TS 38.331 [35]).

For LTE RRC System Information (SI), a single *SystemInformationBlockPos* IE is defined in TS 36.331 [12] which is carried in IE *PosSystemInformation-r15-IEs* specified in TS 36.331 [12]. The mapping of positioning SIB type (*posSibType*) to assistance data carried in *SystemInformationBlockPos* is specified in clause 7.2.

For NR RRC System Information (SI), a single *SIBpos* IE is defined in TS 38.331 [35] which is carried in IE *PosSystemInformation-r16-IEs* specified in TS 38.331 [35]. The mapping of positioning SIB type (*posSibType*) to assistance data carried in *SIBpos* is specified in clause 7.2.

[TS 37.355, clause 7.2]

The supported *posSibType*'s are specified in Table 7.2-1. The GNSS Common and Generic Assistance Data IEs are defined in clause 6.5.2.2. The OTDOA Assistance Data IEs and NR DL-TDOA/DL-AoD Assistance Data IEs are defined in clause 7.4.2. The Barometric Assistance Data IEs are defined in clause 6.5.5.8. The TBS (based on MBS signals) Assistance Data IEs are defined in clause 6.5.4.8.

**Table 7.2-1: Mapping of posSibType to assistanceDataElement**

| | ***posSibType*** | ***assistanceDataElement*** |
|---|---|---|
| GNSS Common Assistance Data (clause 6.5.2.2) | *posSibType1-1* | *GNSS-ReferenceTime* |
| | *posSibType1-2* | *GNSS-ReferenceLocation* |
| | *posSibType1-3* | *GNSS-IonosphericModel* |
| | *posSibType1-4* | *GNSS-EarthOrientationParameters* |
| | *posSibType1-5* | *GNSS-RTK-ReferenceStationInfo* |
| | *posSibType1-6* | *GNSS-RTK-CommonObservationInfo* |
| | *posSibType1-7* | *GNSS-RTK-AuxiliaryStationData* |
| | *posSibType1-8* | *GNSS-SSR-CorrectionPoints* |

| | | |
|---|---|---|
| GNSS Generic Assistance Data (clause 6.5.2.2) | *posSibType2-1* | *GNSS-TimeModelList* |
| | *posSibType2-2* | *GNSS-DifferentialCorrections* |
| | *posSibType2-3* | *GNSS-NavigationModel* |
| | *posSibType2-4* | *GNSS-RealTimeIntegrity* |
| | *posSibType2-5* | *GNSS-DataBitAssistance* |
| | *posSibType2-6* | *GNSS-AcquisitionAssistance* |
| | *posSibType2-7* | *GNSS-Almanac* |
| | *posSibType2-8* | *GNSS-UTC-Model* |
| | *posSibType2-9* | *GNSS-AuxiliaryInformation* |
| | *posSibType2-10* | *BDS-DifferentialCorrections* |
| | *posSibType2-11* | *BDS-GridModelParameter* |
| | *posSibType2-12* | *GNSS-RTK-Observations* |
| | *posSibType2-13* | *GLO-RTK-BiasInformation* |
| | *posSibType2-14* | *GNSS-RTK-MAC-CorrectionDifferences* |
| | *posSibType2-15* | *GNSS-RTK-Residuals* |
| | *posSibType2-16* | *GNSS-RTK-FKP-Gradients* |
| | *posSibType2-17* | *GNSS-SSR-OrbitCorrections* |
| | *posSibType2-18* | *GNSS-SSR-ClockCorrections* |
| | *posSibType2-19* | *GNSS-SSR-CodeBias* |
| | *posSibType2-20* | *GNSS-SSR-URA* |
| | *posSibType2-21* | *GNSS-SSR-PhaseBias* |
| | *posSibType2-22* | *GNSS-SSR-STEC-Correction* |
| | *posSibType2-23* | *GNSS-SSR-GriddedCorrection* |
| | *posSibType2-24* | *NavIC-DifferentialCorrections* |
| | *posSibType2-25* | *NavIC-GridModelParameter* |
| OTDOA Assistance Data (clause 7.4.2) | *posSibType3-1* | *OTDOA-UE-Assisted* |
| Barometric Assistance Data (clause 6.5.5.8) | *posSibType4-1* | *Sensor-AssistanceDataList* |
| TBS Assistance Data (clause 6.5.4.8) | *posSibType5-1* | *TBS-AssistanceDataList* |
| NR DL-TDOA/DL-AoD Assistance Data (clauses 6.4.3, 7.4.2) | *posSibType6-1* | *NR-DL-PRS-AssistanceData* |
| | *posSibType6-2* | *NR-UEB-TRP-LocationData* |
| | *posSibType6-3* | *NR-UEB-TRP-RTD-Info* |

[TS 38.331, clause 5.2.2.3.5]

The UE shall:

1> if the UE is in RRC_CONNECTED with an active BWP not configured with common search space with the field *searchSpaceOtherSystemInformation* and the UE has not stored a valid version of a SIB or posSIB, in accordance with sub-clause 5.2.2.2.1, of one or several required SIB(s) or posSIB(s) in accordance with sub-clause 5.2.2.1:

2> for the SI message(s) that, according to the *si-SchedulingInfo* or *posSI-SchedulingInfo* in the stored SIB1, contain at least one required SIB or requested posSIB:

3> if *onDemandSIB-Request* is configured and timer T350 is not running:

4> initiate transmission of the *DedicatedSIBRequest* message in accordance with 5.2.2.3.6;

4> start timer T350 with the timer value set to the *onDemandSIB-RequestProhibitTimer*;

1> else if the UE is in RRC_CONNECTED with an active BWP configured with common search space with the field *searchSpaceOtherSystemInformation* and the UE has not stored a valid version of a SIB or posSIB, in accordance with sub-clause 5.2.2.2.1, of one or several required SIB(s) or posSIB(s) in accordance with sub-clause 5.2.2.1:

2> for the SI message(s) that, according to the *si-SchedulingInfo* in the stored SIB1, contain at least one required SIB and for which *si-BroadcastStatus* is set to *broadcasting*:

3> acquire the SI message(s) as defined in sub-clause 5.2.2.3.2;

2> for the SI message(s) that, according to the *si-SchedulingInfo* in the stored SIB1, contain at least one required SIB and for which *si-BroadcastStatus* is set to *notBroadcasting*:

3> if *onDemandSIB-Request* is configured and timer T350 is not running:

4> initiate transmission of the *DedicatedSIBRequest* message in accordance with 5.2.2.3.6;

4> start timer T350 with the timer value set to the *onDemandSIB-RequestProhibitTimer*;

4> acquire the requested SI message(s) corresponding to the requested SIB(s) as defined in sub-clause 5.2.2.3.2.

2> for the SI message(s) that, according to the *posSI-SchedulingInfo* in the stored SIB1, contain at least one requested posSIB and for which *posSI-BroadcastStatus* is set to *broadcasting*:

3> acquire the SI message(s) as defined in sub-clause 5.2.2.3.2;

2> for the SI message(s) that, according to the *posSI-SchedulingInfo* in the stored SIB1, contain at least one requested posSIB and for which *posSI-BroadcastStatus* is set to *notBroadcasting*:

3> if *onDemandSIB-Request* is configured and timer T350 is not running:

4> initiate transmission of the *DedicatedSIBRequest* message in accordance with 5.2.2.3.6;

4> start timer T350 with the timer value set to the *onDemandSIB-RequestProhibitTimer*;

4> acquire the requested SI message(s) corresponding to the requested posSIB(s) as defined in sub-clause 5.2.2.3.2.

NOTE: UE may include on demand request for SIB and/or posSIB(s) in the same *DedicatedSIBRequest* message.

### 9.4.1.3 Test description

#### 9.4.1.3.1 Pre-test conditions

System Simulator:

For Test Configuration B (Table 9.4.1.3.2-2): NR Cell 1 and system information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

- Satellite signals (sub-test case 25): as specified in 8.2.1.
- MBS signals (Sub-test 23): as specified in 8.2.4.
- Sub-test 20: NR Cell 1 and NR Cell 2 as specified in 8.2.10 and system information combination NR-2 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 21: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11 and system information combination NR-4 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- For Test Configuration B (Table 9.4.1.3.2-2): The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.
- The UE is switched-off.

Related PICS/PIXIT Statements:

-

#### 9.4.1.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.4.1.3.2-1 below:

**Table 9.4.1.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| 23 | UE supporting MBS(2) (Rel-16 onwards) |
| 24 | UE supporting Sensor (Rel-16 onwards) |
| 25 | UE supporting GNSS(1) (Rel-15 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.4.1.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.1.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The UE is switched on. | | | - | - |
| 2 | System information that includes the needed posSystemInformation is broadcasted.<br>Note: The sysinfo combination is NR-16 for Sub-test 20, NR-17 for Sub-test 21, NR-15 for Sub-test 23, Sub-test 24 and Sub-test 25. | <-- | RRC: SYSTEM INFORMATION (BCCH) | - | - |
| 3-10 | Steps 1 to 8 of the NR RRC_CONNECTED procedure in TS 38.508-1 [3] Table 4.5.4.2-3 are executed. | - | - | | |
| - | EXCEPTION: Steps 11a1 to 11a2 describe behaviour that depends on the UE capability; the "lower case letter" identifies a step sequence that takes place if a capability is supported. | | | | |
| 11a1 | IF<br>sub-test 20 or 21<br>THEN<br>The SS sends an RRCReconfiguration message as in Table 8.3.1-1 to configure the measurement gap. | <-- | RRC RRCReconfiguration | - | - |
| 11a2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |
| 12 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 13 | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 14 | IF<br>the UE LPP message at step 13 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 15 | The SS sends a LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 16 | The UE sends a LPP message of type Provide Location Information including location estimate. | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 17 | IF<br>the UE LPP message at step 16 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

## 9.4.1.3.3 Specific message contents

**Table 9.4.1.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 20: 0 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 0 1 1 1<br>Sub-test 23: 0 0 0 0 0 0 0 1 0<br>Sub-test 24: 0 0 0 0 0 0 1 0 1<br>Sub-test 25: 0 0 0 0 0 0 0 0 0 | Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA<br>Sub-test 23: MBS<br>Sub-test 24: Sensor<br>Sub-test 25: GNSS | |

**Table 9.4.1.3.3-2: SIB1 (step 2, Table 9.4.1.3.2-3)**

| Derivation Path: TS 38.508-1, Table 4.6.1-28 with condition posSIB. |
|---|

**Table 9.4.1.3.3-3: PosSI-SchedulingInfo (SIB1, Table 9.4.1.3.3-2)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PosSI-SchedulingInfo-r16 ::= SEQUENCE { | | | |
| posSchedulingInfoList-r16 SEQUENCE (SIZE (1..maxSI-Message)) OF PosSchedulingInfo-r16 { | 2 entries | The size of PosSchedulingInfo-r16 is 1 for Sub-tests 20, 21, 23 and 24.<br>If Sub-test 25 and UE supports X GNSS systems, the size of PosSchedulingInfo-r16 is (X+1) and the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | |
| PosSchedulingInfo-r16[1] SEQUENCE { | | entry 1 | |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 3 entries for Sub-test 25, 1 entry for Sub-tests 20, 21, 23 and 24 | PosSibType-n = posSibType1-1, posSibType1-2, posSibType1-3 | Sub-test 25 |
| | | PosSibType-n = posSibType4-1 | Sub-test 24 |
| | | PosSibType-n = posSibType5-1 | Sub-test 23 |
| | | PosSibType-n = posSibType6-1 | Sub-test 20 or Sub-test 21 |

| | | | |
|---|---|---|---|
| PosSIB-Type-r16[n] SEQUENCE { | | entry n | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Not present | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-n | The posSibType-r16 is depending on the sub–test cases and entry number n | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| PosSchedulingInfo-r16[2] SEQUENCE { | | entry 2<br>If the UE supports X GNSS systems, the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | Sub-test 25 |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 6 entries for Sub-test 25 | posSibType2-1, posSibType2-3, posSibType2-6, posSibType2-7, posSibType2-8, posSibType2-9 | |
| PosSIB-Type-r16[m] SEQUENCE { | | entry m | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Set depending on the GNSS supported by the UE | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-m, | The posSibType-r16 is depending on the entry number m | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.1.3.3-4: SystemInformation – Sub-tests 20, 21, 23 and 24 (step 2, Table 9.4.1.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB. |

**Table 9.4.1.3.3-4A: SystemInformation – Sub-test 25, message 1 (step 2, Table 9.4.1.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB and posSib1-x. |

**Table 9.4.1.3.3-4B: SystemInformation – Sub-test 25, message 2 and subsequent (step 2, Table 9.4.1.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB and posSib2-x. |

**Table 9.4.1.3.3-5: PosSystemInformation-r16-IEs (step 2, Table 9.4.1.3.2-3)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PosSystemInformation-r16-IEs ::= SEQUENCE { | | | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 3 entries | | Sub-test 25 posSib1-x |
| posSib1-1-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 1 | |
| posSib1-2-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 2 | |
| posSib1-3-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 3 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 6 entries | | Sub-test 25, posSib2-x |
| posSib2-1-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 1 | |
| posSib2-3-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 2 | |
| posSib2-6-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 3 | |
| posSib2-7-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 4 | |
| posSib2-8-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 5 | |
| posSib2-9-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 6 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 24 |
| posSib4-1-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 1 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 23 |
| posSib5-1-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 1 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 20, Sub-test 21 |
| posSib6-1-r16 | SIBpos as defined in Table 9.4.1.3.3-6 | entry 1 | |
| } | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.4.1.3.3-6: SIBpos (PosSystemInformation-r16-IEs, Table 9.4.1.3.3-5)**

| Derivation Path: TS 38.501-1, Table 4.6.2a-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SIBpos-r16 ::= SEQUENCE { | | | |
| assistanceDataSIB-Element-r16 | OCTET STRING containing AssistanceDataSIBelement | | |
| } | | | |

**Table 9.4.1.3.3-7: AssistanceDataSIBelement**

| Derivation Path: TS 37.355, clause 7.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AssistanceDataSIBelement-r15 ::= SEQUENCE { | | | |
| valueTag-r15 | Not present | | |
| expirationTime-r15 | Not present | | |
| cipheringKeyData-r15 | Not present | | |
| segmentationInfo-r15 | Not present | | Not present when the posSIB is not segmented |
| segmentationInfo-r15 SEQUENCE { | | | Present when the posSIB is segmented |
| segmentationOption-r15 | octet-string-seg | | |
| assistanceDataSegmentType-r15 | notLastSegment | | Used when posSIB is segmented and the segment is not the last segment |
| | lastSegment | | Used when posSIB is not segmented, or posSIB is segmented and the segment is the last segment |
| assistanceDataSegmentNumber-r15 | index of the segmented posSIB | | |
| } | | | |
| assistanceDataElement-r15 | OCTET STRING containing LPP IEs (GNSS-ReferenceTime, GNSS-ReferenceLocation, GNSS-IonosphericModel,GNSS-TimeModelList, GNSS-NavigationModel,<br><br>GNSS-AcquisitionAssistance, GNSS-Almanac, GNSS-UTC-Model, GNSS-AuxiliaryInformation ) as defined in TS 37.571-5 [12] and mapped according to the content of posSibType-r16 in Table 9.4.1.3.3-3. | | Sub-test 25 |
| | OCTET STRING containing sensor-AssistanceDataList as defined in Table 8.4-1 | | Sub-test 24 |
| | OCTET STRING containing tbs-AssistanceDataList as defined in Table 8.4-1 | | Sub-test 23 |
| | OCTET STRING containing NR-DL-PRS-AssistanceData as defined in Table 8.4.1.6-1. | | Sub-test 20, Sub-test 21 |
| } | | | |

**Table 9.4.1.3.3-8: DLInformationTransfer (step 12, step 14, step 15 and step 17, Table 9.4.1.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.1.3.3-9 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.1.3.3-9: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.1.3.2-8)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 12:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 15:** Set according to Table 9.3.4.1.3.3-10 | LPP Request Location Information | |
| | **Steps 14 and 17:** Set according to Table 9.4.1.3.3-11 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.1.3.3-10: LPP Request Location Information (DL NAS TRANSPORT, Table 9.4.1.3.2-9)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsPreferred | | |

**Table 9.4.1.3.3-11: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.1.3.2-9)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same | |

| | | | |
|---|---|---|---|
| | | value of the sequenceNumber field in step 13 or 16, Table 9.4.1.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.1.3.3-12: ULInformationTransfer (steps 13 and 16, Table 9.4.1.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.1.3.3-13 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.1.3.3-13: UL NAS TRANSPORT (ULInformationTransfer, Table 9.4.1.3.3-12)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 13:** Set according to Table 9.4.1.3.3-14 | LPP Provide Capabilities | |
| | **Step 16:** Set according to Table 9.4.1.3.3-15 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 12 Table 9.4.1.3.2-3) | |

**Table 9.4.1.3.3-14: LPP Provide Capabilities (UL NAS TRANSPORT, Table 9.4.1.3.3-13)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 12, Table 9.4.1.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-16 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.1.3.3-15: LPP Provide Location Information (UL NAS TRANSPORT, Table 9.4.1.3.3-13)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 15, Table 9.4.1.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | Present for sub-test 25 | | |
| gnss-SignalMeasurementInformation SEQUENCE { | Present. | | |
| measurementReferenceTime | Any value acceptable | | |
| gnss-MeasurementList | Any value acceptable | | |
| } | | | |
| gnss-LocationInformation SEQUENCE { | May be present | | |
| measurementReferenceTime | Any value acceptable | | |
| agnss-List | Any value acceptable | | |
| } | | | |
| gnss-Error | Not present | | |
| } | | | |
| sensor-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 24 | Rel-16 onwards | |
| sensor-MeasurementInformation-r13 | Present. Any value acceptable | | |
| sensor-Error-r13 | Not present | | |
| sensor-MotionInformation-r15 | May be present | | |
| } | | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 23 | Rel-16 onwards | |
| tbs-MeasurementInformation-r13 | Present. Any value acceptable | | |
| tbs-Error-r13 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 SEQUENCE { | Present. | | |
| nr-DL-AoD-MeasList-r16 | Any value acceptable | | |
| } | | | |
| nr-dl-AoD-LocationInformation-r16 | May be present | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |

| | | | |
|---|---|---|---|
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 21 | Rel-16 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 SEQUENCE { | Present. | | |
| dl-PRS-ReferenceInfo-r16 | Any value acceptable | | |
| nr-DL-TDOA-MeasList-r16 | Any value acceptable | | |
| } | | | |
| nr-dl-tdoa-LocationInformation-r16 | May be present | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.2 PosSIB broadcasting followed by location information transfer / Positioning SI messages offset

### 9.4.2.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and receives the positioning assistance data via posSIBs
and offsetToSI-Used-r16 is present and then UE receives a location request from LMF}
    then { the UE acquires the SI message at the correct position and then sends a PROVIDE LOCATION
INFORMATION message containing a location estimate }
}
```

### 9.4.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.331 clauses 5.2.2.3.2. Unless otherwise stated these are Rel-16 requirements.

[TS 38.331, clause 5.2.2.3.2]

For SI message acquisition PDCCH monitoring occasion(s) are determined according to *searchSpaceOtherSystemInformation*. If *searchSpaceOtherSystemInformation* is set to zero, PDCCH monitoring occasions for SI message reception in SI-window are same as PDCCH monitoring occasions for *SIB1* where the mapping between PDCCH monitoring occasions and SSBs is specified in TS 38.213[13]. If *searchSpaceOtherSystemInformation* is not set to zero, PDCCH monitoring occasions for SI message are determined based on search space indicated by *searchSpaceOtherSystemInformation*. PDCCH monitoring occasions for SI message which are not overlapping with UL symbols (determined according to *tdd-UL-DL-ConfigurationCommon*) are sequentially numbered from one in the SI window. The $[x \times N+K]^{th}$ PDCCH monitoring occasion (s) for SI message in SI-window corresponds to the $K^{th}$ transmitted SSB, where x = 0, 1, ...X-1, K = 1, 2, …N, N is the number of actual transmitted SSBs determined according to *ssb-PositionsInBurst* in *SIB1* and X is equal to CEIL(number of PDCCH monitoring occasions in SI-window/N). The actual transmitted SSBs are sequentially numbered from one in ascending order of their SSB indexes. The UE assumes that, in the SI window, PDCCH for an SI message is transmitted in at least one PDCCH monitoring occasion corresponding to each transmitted SSB and thus the selection of SSB for the reception SI messages is up to UE implementation.

When acquiring an SI message, the UE shall:

1> determine the start of the SI-window for the concerned SI message as follows:

2> if the concerned SI message is configured in the *schedulingInfoList*:

3> for the concerned SI message, determine the number *n* which corresponds to the order of entry in the list of SI messages configured by *schedulingInfoList* in *si-SchedulingInfo* in *SIB1*;

3> determine the integer value $x = (n – 1) \times w$, where $w$ is the *si-WindowLength*;

3> the SI-window starts at the slot #$a$, where $a = x$ mod N, in the radio frame for which SFN mod $T$ = FLOOR($x$/N), where $T$ is the *si-Periodicity* of the concerned SI message and N is the number of slots in a radio frame as specified in TS 38.213 [13];

2> else if the concerned SI message is configured in the *posSchedulingInfoList* and *offsetToSI-Used* is not configured:

3> create a concatenated list of SI messages by appending the *posSchedulingInfoList* in *posSI-SchedulingInfo* in *SIB1* to *schedulingInfoList* in *si-SchedulingInfo* in *SIB1*;

3> for the concerned SI message, determine the number $n$ which corresponds to the order of entry in the concatenated list;

3> determine the integer value $x = (n – 1) \times w$, where $w$ is the *si-WindowLength*;

3> the SI-window starts at the slot #$a$, where $a = x$ mod N, in the radio frame for which SFN mod $T$ = FLOOR($x$/N), where $T$ is the *posSI-Periodicity* of the concerned SI message and N is the number of slots in a radio frame as specified in TS 38.213 [13];

2> else if the concerned SI message is configured by the *posSchedulingInfoList* and *offsetToSI-Used* is configured:

3> determine the number $m$ which corresponds to the number of SI messages with an associated *si-Periodicity* of 8 radio frames (80 ms), configured by *schedulingInfoList* in *SIB1*;

3> for the concerned SI message, determine the number $n$ which corresponds to the order of entry in the list of SI messages configured by *posSchedulingInfoList* in *SIB1*;

3> determine the integer value $x = m \times w + (n – 1) \times w$, where $w$ is the *si-WindowLength;*

3> the SI-window starts at the slot #$a$, where $a = x$ mod N, in the radio frame for which SFN mod $T$ = FLOOR($x$/N) +8, where $T$ is the *posSI-Periodicity* of the concerned SI message and N is the number of slots in a radio frame as specified in TS 38.213 [13];.

### 9.4.2.3 Test description

#### 9.4.2.3.1 Pre-test conditions

Same as in clause 9.4.1.3.1.

#### 9.4.2.3.2 Test procedure sequence

Same as in clause 9.4.1.3.2.

#### 9.4.2.3.3 Specific message contents

Same as 9.4.1.3.3 except that Table 9.4.2.3.3-1 replaces Table 9.4.1.3.3-3.

**Table 9.4.2.3.3-1: PosSI-SchedulingInfo (SIB1, Table 9.4.1.3.3-2)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PosSI-SchedulingInfo-r16 ::= SEQUENCE { | | | |
| posSchedulingInfoList-r16 SEQUENCE (SIZE (1..maxSI-Message)) OF PosSchedulingInfo-r16 { | 2 entries | The size of PosSchedulingInfo-r16 is 1 for Sub-tests 20, 21, 23 and 24.<br>If Sub-test 25 and UE supports X GNSS systems, the size of PosSchedulingInfo-r16 is (X+1) and the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | |
| PosSchedulingInfo-r16[1] SEQUENCE { | | entry 1 | |
| offsetToSI-Used-r16 | true | | |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 3 entries for Sub-test 25, 1 entry for Sub-tests 20, 21, 23 and 24 | PosSibType-n = posSibType1-1, posSibType1-2, posSibType1-3 | Sub-test 25 |
| | | PosSibType-n = posSibType4-1 | Sub-test 24 |
| | | PosSibType-n = posSibType5-1 | Sub-test 23 |
| | | PosSibType-n = posSibType6-1 | Sub-test 20 or Sub-test 21 |

| | | | |
|---|---|---|---|
| PosSIB-Type-r16[n] SEQUENCE { | | entry n | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Not present | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-n | The posSibType-r16 is depending on the sub–test cases and entry number n | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| PosSchedulingInfo-r16[2] SEQUENCE { | | entry 2<br>If the UE supports X GNSS systems, the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | Sub-test 25 |
| offsetToSI-Used-r16 | true | | |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 6 entries for Sub-test 25 | posSibType2-1, posSibType2-3, posSibType2-6, posSibType2-7, posSibType2-8, posSibType2-9 | |
| PosSIB-Type-r16[m] SEQUENCE { | | entry m | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Set depending on the GNSS supported by the UE | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-m, | The posSibType-r16 is depending on the entry number m | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.3 On-demand PosSIB followed by location information transfer / RRC_connected state

### 9.4.3.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and receives a REQUEST LOCATION INFORMATION message }
    then { the UE sends the DedicatedSIBRequest message indicating the requested posSIB }
}
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
```

```
  when { UE receives the positioning assistance data via posSIBs after sending the onDemandSIB-
Request message }
    then { the UE sends a PROVIDE LOCATION INFORMATION message containing a location estimate }
}
```

### 9.4.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.331 clauses 5.2.2.1, 5.2.2.3.5, 5.2.2.3.6 and TS 37.355 clause7.2. Unless otherwise stated these are Rel-16 requirements.

[TS 38.331, clause 5.2.2.1]

The UE applies the SI acquisition procedure to acquire the AS, NAS- and positioning assistance data information. The procedure applies to UEs in RRC_IDLE, in RRC_INACTIVE and in RRC_CONNECTED.

[TS 38.331, clause 5.2.2.3.5]

The UE shall:

1> if the UE is in RRC_CONNECTED with an active BWP not configured with common search space with the field *searchSpaceOtherSystemInformation* and the UE has not stored a valid version of a SIB or posSIB, in accordance with sub-clause 5.2.2.2.1, of one or several required SIB(s) or posSIB(s) in accordance with sub-clause 5.2.2.1:

2> for the SI message(s) that, according to the *si-SchedulingInfo* or *posSI-SchedulingInfo* in the stored SIB1, contain at least one required SIB or requested posSIB:

3> if *onDemandSIB-Request* is configured and timer T350 is not running:

4> initiate transmission of the *DedicatedSIBRequest* message in accordance with 5.2.2.3.6;

4> start timer T350 with the timer value set to the *onDemandSIB-RequestProhibitTimer*;

1> else if the UE is in RRC_CONNECTED with an active BWP configured with common search space with the field *searchSpaceOtherSystemInformation* and the UE has not stored a valid version of a SIB or posSIB, in accordance with sub-clause 5.2.2.2.1, of one or several required SIB(s) or posSIB(s) in accordance with sub-clause 5.2.2.1:

…

2> for the SI message(s) that, according to the *posSI-SchedulingInfo* in the stored SIB1, contain at least one requested posSIB and for which *posSI-BroadcastStatus* is set to *broadcasting*:

3> acquire the SI message(s) as defined in sub-clause 5.2.2.3.2;

2> for the SI message(s) that, according to the *posSI-SchedulingInfo* in the stored SIB1, contain at least one requested posSIB and for which *posSI-BroadcastStatus* is set to *notBroadcasting*:

3> if *onDemandSIB-Request* is configured and timer T350 is not running:

4> initiate transmission of the *DedicatedSIBRequest* message in accordance with 5.2.2.3.6;

4> start timer T350 with the timer value set to the *onDemandSIB-RequestProhibitTimer*;

4> acquire the requested SI message(s) corresponding to the requested posSIB(s) as defined in sub-clause 5.2.2.3.2.

[TS 38.331, clause 5.2.2.3.6]

The UE shall set the contents of *DedicatedSIBRequest* message as follows:

1> if the procedure is triggered to request the required SIB(s):

2> include *requestedSIB-List* in the *onDemandSIB-RequestList* to indicate the requested SIB(s);

1> if the procedure is triggered to request the required posSIB(s):

2> include *requestedPosSIB-List* in the *onDemandSIB-RequestList* to indicate the requested posSIB(s).

The UE shall submit the *DedicatedSIBRequest* message to lower layers for transmission.

[TS 37.355, clause 7.2]

The supported *posSibType*'s are specified in Table 7.2-1. The GNSS Common and Generic Assistance Data IEs are defined in clause 6.5.2.2. The OTDOA Assistance Data IEs and NR DL-TDOA/DL-AoD Assistance Data IEs are defined in clause 7.4.2. The Barometric Assistance Data IEs are defined in clause 6.5.5.8. The TBS (based on MBS signals) Assistance Data IEs are defined in clause 6.5.4.8.

**Table 7.2-1: Mapping of posSibType to assistanceDataElement**

| | ***posSibType*** | ***assistanceDataElement*** |
|---|---|---|
| GNSS Common Assistance Data (clause 6.5.2.2) | *posSibType1-1* | *GNSS-ReferenceTime* |
| | *posSibType1-2* | *GNSS-ReferenceLocation* |
| | *posSibType1-3* | *GNSS-IonosphericModel* |
| | *posSibType1-4* | *GNSS-EarthOrientationParameters* |
| | *posSibType1-5* | *GNSS-RTK-ReferenceStationInfo* |
| | *posSibType1-6* | *GNSS-RTK-CommonObservationInfo* |
| | *posSibType1-7* | *GNSS-RTK-AuxiliaryStationData* |
| | *posSibType1-8* | *GNSS-SSR-CorrectionPoints* |
| GNSS Generic Assistance Data (clause 6.5.2.2) | *posSibType2-1* | *GNSS-TimeModelList* |
| | *posSibType2-2* | *GNSS-DifferentialCorrections* |
| | *posSibType2-3* | *GNSS-NavigationModel* |
| | *posSibType2-4* | *GNSS-RealTimeIntegrity* |
| | *posSibType2-5* | *GNSS-DataBitAssistance* |
| | *posSibType2-6* | *GNSS-AcquisitionAssistance* |
| | *posSibType2-7* | *GNSS-Almanac* |
| | *posSibType2-8* | *GNSS-UTC-Model* |
| | *posSibType2-9* | *GNSS-AuxiliaryInformation* |
| | *posSibType2-10* | *BDS-DifferentialCorrections* |
| | *posSibType2-11* | *BDS-GridModelParameter* |
| | *posSibType2-12* | *GNSS-RTK-Observations* |
| | *posSibType2-13* | *GLO-RTK-BiasInformation* |
| | *posSibType2-14* | *GNSS-RTK-MAC-CorrectionDifferences* |
| | *posSibType2-15* | *GNSS-RTK-Residuals* |
| | *posSibType2-16* | *GNSS-RTK-FKP-Gradients* |
| | *posSibType2-17* | *GNSS-SSR-OrbitCorrections* |
| | *posSibType2-18* | *GNSS-SSR-ClockCorrections* |
| | *posSibType2-19* | *GNSS-SSR-CodeBias* |
| | *posSibType2-20* | *GNSS-SSR-URA* |
| | *posSibType2-21* | *GNSS-SSR-PhaseBias* |
| | *posSibType2-22* | *GNSS-SSR-STEC-Correction* |
| | *posSibType2-23* | *GNSS-SSR-GriddedCorrection* |
| | *posSibType2-24* | *NavIC-DifferentialCorrections* |
| | *posSibType2-25* | *NavIC-GridModelParameter* |
| OTDOA Assistance Data (clause 7.4.2) | *posSibType3-1* | *OTDOA-UE-Assisted* |
| Barometric Assistance Data (clause 6.5.5.8) | *posSibType4-1* | *Sensor-AssistanceDataList* |
| TBS Assistance Data (clause 6.5.4.8) | *posSibType5-1* | *TBS-AssistanceDataList* |
| NR DL-TDOA/DL-AoD Assistance Data (clauses 6.4.3, 7.4.2) | *posSibType6-1* | *NR-DL-PRS-AssistanceData* |
| | *posSibType6-2* | *NR-UEB-TRP-LocationData* |
| | *posSibType6-3* | *NR-UEB-TRP-RTD-Info* |

### 9.4.3.3 Test description

#### 9.4.3.3.1 Pre-test conditions

System Simulator:

- Sub-test 23, sub-test 24 and sub-test 25: NR Cell 1 and system information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Satellite signals (sub-test case 25): as specified in 8.2.1.
- MBS signals (Sub-test 23): as specified in 8.2.4.
- Sub-test 20: NR Cell 1 and NR Cell 2 as specified in 8.2.10 and system information combination NR-2 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.
- Sub-test 21: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11 and system information combination NR-4 as defined in TS 38.508-1 [30] clause 4.4.3.1.2..

UE:

The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

#### 9.4.3.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a Sub-Test Case Number as defined in Table 9.4.3.3.2-1 below:

**Table 9.4.3.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 20 | UE supporting DL-AoD (Rel-16 onwards) |
| 21 | UE supporting DL-TDOA (Rel-16 onwards) |
| 23 | UE supporting MBS(2) (Rel-16 onwards) |
| 24 | UE supporting Sensor (Rel-16 onwards) |
| 25 | UE supporting GNSS(1) (Rel-15 onwards) |
| NOTE 1: The GNSS combination of GPS, GLONASS, Galileo, BDS supported by the UE | |

**Table 9.4.3.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.3.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| - | EXCEPTION: Steps 1a1 to 1a2 describe behaviour that depends on the UE capability; the "lower case letter" identifies a step sequence that takes place if a capability is supported. | | | | |
| 1a1 | IF<br>sub-test 20 or 21<br>THEN<br>The SS sends an RRCReconfiguration message as in Table 8.3.1-1 to configure the measurement gap. | <-- | RRC RRCReconfiguration | - | - |
| 1a2 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |
| 2 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer*<br>(LPP REQUEST CAPABILITIES) | - | - |
| 3 | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer*<br>(LPP PROVIDE CAPABILITIES) | - | - |
| 4 | IF<br>the UE LPP message at step 3 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |
| 5 | The SS sends a LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer*<br>(LPP REQUEST LOCATION INFORMATION) | - | - |
| 6 | Check: Does the UE send the DedicatedSIBRequest message indicating the requested posSIBs within 5 S? | --> | RRC: DedicatedSIBRequest | 1 | P |
| - | EXCEPTION: In parallel to events described in step 6 steps 7a and 7b may take place. | | | | |
| 7a | The UE sends the LPP message of type REQUEST ASSISTANCE DATA message | --> | *ULInformationTransfer*<br>(LPP REQUEST ASSISTANCE DATA) | | |
| 7b | The SS does not respond to the LPP REQUEST ASSISTANCE DATA message. | | | | |
| 8 | The requested posSIBs that includes the needed posSystemInformation are broadcasted.<br>NOTE: The sysinfo combination is NR-16 for Sub-test 20, NR-17 for Sub-test 21, NR-15 for Sub-test 23, Sub-test 24 and Sub-test 25. | <-- | RRC: SYSTEM INFORMATION (BCCH) | | |
| 9 | Check: Does the UE send an LPP message of type Provide Location Information including location estimate? | --> | *ULInformationTransfer*<br>(LPP PROVIDE LOCATION INFORMATION) | 2 | P |
| 10 | IF<br>the UE LPP message at step 9 includes an acknowledgment request<br>THEN<br>SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer*<br>(LPP ACKNOWLEDGEMENT) | - | - |

## 9.4.3.3.3 Specific message contents

**Table 9.4.3.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| UE Positioning Technology | Sub-test 20: 0 0 0 0 1 0 0 0<br>Sub-test 21: 0 0 0 0 0 1 1 1<br>Sub-test 23: 0 0 0 0 0 0 1 0<br>Sub-test 24: 0 0 0 0 0 1 0 1<br>Sub-test 25: 0 0 0 0 0 0 0 0 | Sub-test 20: DL-AoD<br>Sub-test 21: DL-TDOA<br>Sub-test 23: MBS<br>Sub-test 24: Sensor<br>Sub-test 25: GNSS | |

**Table 9.4.3.3.3-2: SIB1 (Preamble)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-28 with condition posSIB. |

**Table 9.4.3.3.3-3: PosSI-SchedulingInfo (SIB1, Table 9.4.3.3.3-2)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PosSI-SchedulingInfo-r16 ::= SEQUENCE { | | | |
| posSchedulingInfoList-r16 SEQUENCE (SIZE (1..maxSI-Message)) OF PosSchedulingInfo-r16 { | 2 entries | The size of PosSchedulingInfo-r16 is 1 for Sub-tests 20, 21, 23 and 24.<br>If Sub-test 25 and UE supports X GNSS systems, the size of PosSchedulingInfo-r16 is (X+1) and the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | |
| PosSchedulingInfo-r16[1] SEQUENCE { | | entry 1 | |
| offsetToSI-Used-r16 | Not present | | |
| posSI-Periodicity-r16 | rf32 | | |
| posSI-BroadcastStatus-r16 | notBroadcasting | | |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 3 entries for Sub-test 25, 1 entry for Sub-tests 20, 21, 23 and 24 | PosSibType-n = posSibType1-1, posSibType1-2, posSibType1-3 | Sub-test 25 |
| | | PosSibType-n = posSibType4-1 | Sub-test 24 |
| | | PosSibType-n = posSibType5-1 | Sub-test 23 |
| | | PosSibType-n = posSibType6-1 | Sub-test 20 or Sub-test 21 |

| | | | |
|---|---|---|---|
| PosSIB-Type-r16[n] SEQUENCE { | | entry n | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Not present | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-n | The posSibType-r16 is depending on the sub–test cases and entry number n | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| PosSchedulingInfo-r16[2] SEQUENCE { | | entry 2<br>If the UE supports X GNSS systems, the PosSchedulingInfo-r16 needs to be broadcasted X times, once for each different GNSS. | Sub-test 25 |
| offsetToSI-Used-r16 | Not present | | |
| posSI-Periodicity-r16 | rf32 | | |
| posSI-BroadcastStatus-r16 | notBroadcasting | | |
| posSIB-MappingInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF PosSIB-Type-r16 { | 6 entries for Sub-test 25 | posSibType2-1, posSibType2-3, posSibType2-6, posSibType2-7, posSibType2-8, posSibType2-9 | |
| PosSIB-Type-r16[m] SEQUENCE { | | entry m | |
| encrypted-r16 | Not present | | |
| gnss-id-r16 | Set depending on the GNSS supported by the UE | | |
| sbas-id-r16 | Not present | | |
| posSibType-r16 | PosSibType-m, | The posSibType-r16 is depending on the entry number m | |
| areaScope-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-4: RRCReconfiguration (Preamble, step 7 as defined in TS 38.508-1 [30] Table 4.5.4.2-3)**

| Derivation Path: TS 38.331 [6], clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| onDemandSIB-Request-r16 CHOICE SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| onDemandSIB-RequestProhibitTimer-r16 | s5 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-5: DLInformationTransfer (step 2, step 4, step 5 and step 10, Table 9.4.3.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.3.3.3-6 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-6: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.3.3.3-4)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 5:** Set according to Table 9.4.3.3.3-7 | LPP Request Location Information | |
| | **Steps 4 and 10:** Set according to Table 9.4.3.3.3-8 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.3.3.3-7: LPP Request Location Information (DL NAS TRANSPORT, Table 9.4.3.3.3-5)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsPreferred | | |

**Table 9.4.3.3.3-8: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.3.3.3-5)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 3 or 9, Table 9.4.3.3.2- | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.3.3.3-9: DedicatedSIBRequest (step 6, Table 9.4.3.3.2-3)**

| Derivation Path: TS 38.508-1, Table 4.6.1-2A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DedicatedSIBRequest-r16 ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| dedicatedSIBRequest-r16 SEQUENCE { | | | |
| onDemandSIB-RequestList-r16 SEQUENCE { | | | |
| requestedSIB-List-r16 | Not checked | | |
| requestedPosSIB-List-r16 SEQUENCE (SIZE (1..maxOnDemandPosSIB-r16)) OF PosSIB-ReqInfo-r16 { | 1 entry for Sub-tests 20, 21, 23 and 24; 9 entries for Sub-test 25 | The size of PosSIB-ReqInfo-r16 is 1 for Sub-tests 20, 21, 23 and 24.<br>If Sub-test 25 , the size of PosSIB-ReqInfo-r16 is dependent on the number of posSIB the UE needed in the positioning procedures.. | |
| PosSIB-ReqInfo-r16[n] SEQUENCE { | | entry n | |
| gnss-id-r16 | Not present | | Sub-test 25 posSibType 2-x |
| | Set depending on the GNSS supported by the UE | | Sub-test 25 posSibType 2-x |
| sbas-id-r16 | Not checked | | |
| posSibType-r16 | PosSibType-n | Present. Any posSibType for GNSS systems acceptable.<br>PosSibType-n = posSibType1-1, posSibType1-2, posSibType1-3, posSibType2-1, posSibType2-3, posSibType2-6, posSibType2-7, posSibType2-8, posSibType2-9 | Sub-test 25 |
| | | PosSibType-n = posSibType4-1 | Sub-test 24 |
| | | PosSibType-n = posSibType5-1 | Sub-test 23 |
| | | PosSibType-n = posSibType6-1 | Sub-test 20 or Sub-test 21 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-10: ULInformationTransfer (steps 3, 7a and 9, Table 9.4.3.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.3.3.3-11 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-11: UL NAS TRANSPORT (ULInformationTransfer, Table 9.4.3.3.3-10)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 3:** Set according to Table 9.4.3.3.3-12 | LPP Provide Capabilities | |
| | **Step 7a:** Set according to Table 9.4.3.3.3-13 | LPP Request Assistance Data | |
| | **Step 9:** Set according to Table 9.4.3.3.3-14 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 2 Table Table 9.4.3.3.2-3) | |

**Table 9.4.3.3.3-12: LPP Provide Capabilities (UL NAS TRANSPORT, Table 9.4.3.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 12, Table 9.4.1.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | Rel-14 onwards | |
| a-gnss-ProvideCapabilities | Dependent on UE capabilities | | |
| sensor-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-16 onwards | |
| tbs-ProvideCapabilities-r13 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-13: LPP Request Assistance Data (UL NAS TRANSPORT, Table 9.4.3.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | targetDevice | | |
| transactionNumber | (0..255) | | |
| } | | | |
| endTransaction | FALSE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |

| | | | |
|---|---|---|---|
| requestAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| requestAssistanceData-r9 SEQUENCE { | | | |
| commonIEsRequestAssistanceData | Present or not present | | |
| a-gnss-RequestAssistanceData | Present for sub-test 25 | Rel-15 onwards | |
| otdoa-RequestAssistanceData | Not present | | |
| epdu-RequestAssistanceData | Not present | | |
| sensor-RequestAssistanceData-r14 | Present for sub-test 24 | Rel-16 onwards | |
| tbs-RequestAssistanceData-r14 | Present for sub-test 23 | Rel-16 onwards | |
| wlan-RequestAssistanceData-r14 | Not present | | |
| nr-Multi-RTT-RequestAssistanceData-r16 | Not present | | |
| nr-DL-AoD-RequestAssistanceData-r16 | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-TDOA-RequestAssistanceData-r16 | Present for sub-test 21 | Rel-16 onwards | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.3.3.3-14: LPP Provide Location Information (UL NAS TRANSPORT, Table 9.4.3.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 15, Table 9.4.1.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| a-gnss-ProvideLocationInformation SEQUENCE { | Present for sub-test 25 | | |
| gnss-SignalMeasurementInformation | Present. Any value acceptable | | |
| gnss-LocationInformation | May be present | | |
| gnss-Error | Not present | | |
| } | | | |
| sensor-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 24 | Rel-16 onwards | |
| sensor-MeasurementInformation-r13 | Present. Any value acceptable | | |
| sensor-Error-r13 | Not present | | |
| sensor-MotionInformation-r15 | May be present | | |
| } | | | |
| tbs-ProvideLocationInformation-r13 SEQUENCE { | Present for sub-test 23 | Rel-16 onwards | |
| tbs-MeasurementInformation-r13 | Present. Any value acceptable | | |
| tbs-Error-r13 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 20 | Rel-16 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 21 | Rel-16 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

**Table 9.4.3.3.3-15: SystemInformation – Sub-tests 20, 21, 23 and 24 (step 8, Table 9.4.3.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB. |

**Table 9.4.3.3.3-16: SystemInformation – Sub-test 25, message 1 (step 8, Table 9.4.3.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB and posSib1-x. |

**Table 9.4.3.3.3-17: SystemInformation – Sub-test 25, message 2 (step 8, Table 9.4.3.3.2-3)**

| |
|---|
| Derivation Path: TS 38.508-1, Table 4.6.1-29 with condition posSIB and posSib2-x. |

**Table 9.4.3.3.3-18: PosSystemInformation-r16-IEs (step 8, Table 9.4.3.3.2-3)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PosSystemInformation-r16-IEs ::= SEQUENCE { | | | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 3 entries | | Sub-test 25 posSib1-x |
| posSib1-1-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 1 | |
| posSib1-2-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 2 | |
| posSib1-3-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 3 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 6 entries | | Sub-test 25, posSib2-x |
| posSib2-1-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 1 | |
| posSib2-3-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 2 | |
| posSib2-6-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 3 | |
| posSib2-7-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 4 | |
| posSib2-8-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 5 | |
| posSib2-9-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 6 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 24 |
| posSib4-1-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 1 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 23 |
| posSib5-1-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 1 | |
| posSIB-TypeAndInfo-r16 SEQUENCE (SIZE (1..maxSIB)) OF CHOICE { | 1 entry | | Sub-test 20, Sub-test 21 |
| posSib6-1-r16 | SIBpos as defined in Table 9.4.3.3.3-19 | entry 1 | |
| } | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.4.3.3.3-19: SIBpos (PosSystemInformation-r16-IEs, Table 9.4.3.3.3-18)**

| Derivation Path: TS 38.508-1, Table 4.6.2a-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SIBpos-r16 ::= SEQUENCE { | | | |
| assistanceDataSIB-Element-r16 | OCTET STRING containing AssistanceDataSIBelement | | |
| } | | | |

**Table 9.4.3.3.3-20: AssistanceDataSIBelement**

| Derivation Path: TS 37.355, clause 7.4.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AssistanceDataSIBelement-r15 ::= SEQUENCE { | | | |
| valueTag-r15 | Not present | | |
| expirationTime-r15 | Not present | | |
| cipheringKeyData-r15 | Not present | | |
| segmentationInfo-r15 | Not present | | Not present when the posSIB is not segmented |
| segmentationInfo-r15 SEQUENCE { | | | Present when the posSIB is segmented |
| segmentationOption-r15 | octet-string-seg | | |
| assistanceDataSegmentType-r15 | notLastSegment | | Used when posSIB is segmented and the segment is not the last segment |
| | lastSegment | | Used when posSIB is not segmented, or posSIB is segmented and the segment is the last segment |
| assistanceDataSegmentNumber-r15 | index of the segmented posSIB | | |
| } | | | |
| assistanceDataElement-r15 | OCTET STRING containing LPP IEs (GNSS-ReferenceTime, GNSS-ReferenceLocation, GNSS-IonosphericModel,GNSS-TimeModelList, GNSS-NavigationModel,<br><br>GNSS-AcquisitionAssistance, GNSS-Almanac, GNSS-UTC-Model, GNSS-AuxiliaryInformation ) as defined in TS 37.571-5 [12] and mapped according to the content of posSibType-r16 in Table 9.4.1.3.3-3. | | Sub-test 25 |
| | OCTET STRING containing sensor-AssistanceDataList as defined in Table 8.4-1 | | Sub-test 24 |
| | OCTET STRING containing tbs-AssistanceDataList as defined in Table 8.4-1 | | Sub-test 23 |

| | | | |
|---|---|---|---|
| | OCTET STRING containing NR-DL-PRS-AssistanceData as defined in Table 8.4.1.6-1. | | Sub-test 20, Sub-test 21 |
| } | | | |

## 9.4.4 Pre-configured Measurement Gap Procedures

### 9.4.4.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and pre-configured positioning measurement gap was
configured with associated ID via RRCReconfiguration message}
    then { UE sends a RRCReconfigurationComplete message to confirm the reception of pre-configured
measurement gap configuration }
}
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE is configured with pre-configured measurement gap and the pre-configured measurement gap
is sufficient for the location measurement and UE receives a location request message}
    then { UE sends a measurement gap activation request using UL MAC CE to activate the positioning
measurement gap}
}
```

(3)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives Positioning Measurement Gap Activation command containing an ID to activate the
associated measurement gap }
    then { UE activates the positioning measurement gap and sends a PROVIDE LOCATION INFORMATION
message containing a location estimate }
}
```

(4)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE stops performing location measurements and the preconfigured positioning measurement gap
is activated}
    then { UE sends a Measurement Gap Deactivation Request to deactivate all the activated
measurement gaps}
}
```

### 9.4.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.305 clause 7.7.2, TS 38.331 clause 5.5.6.2, TS 38.321 clauses 5.18.20, 5.25, 6.1.3.40 and 6.1.3.41. Unless otherwise stated these are Rel-17 requirements.

[TS 38.305, clause 7.7.2]

The pre-configured measurement gap procedure is used by the network to provide measurement gap for NR DL-PRS measurements. The gNB may activate/deactivate the pre-configurated measurement gap upon receiving the request from a UE or LMF.

…

Figure 7.7.2-1 shows the general positioning procedure for Pre-configured Measurement Gap.

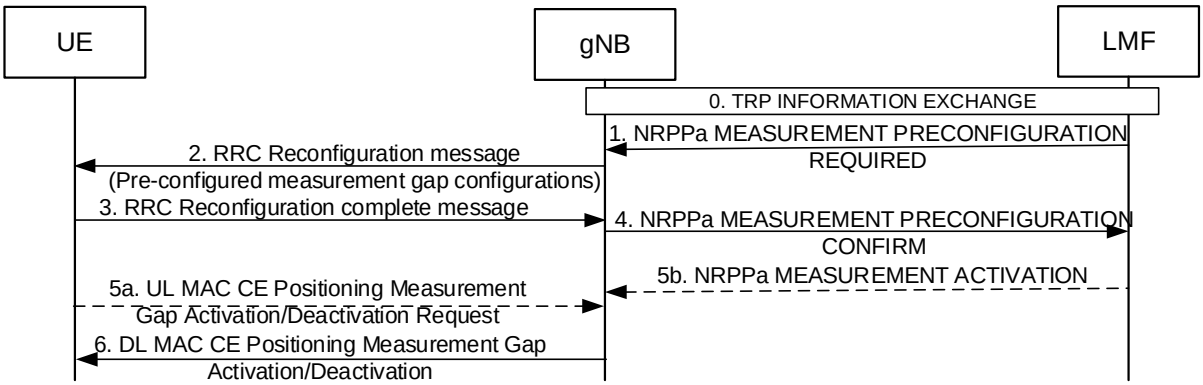



**Figure 7.7.2-1: Pre-configured measurement gap configuration procedure**

0. LMF obtains the TRP information required for positioning services from the gNBs.

1. The LMF provides the PRS information of the neighbour TRPs to the serving gNB and requests the serving gNBs to pre-configure measurement gap via NRPPa MEASUREMENT PRECONFIGURATION REQUIRED message.

2. Based on the assistance information from the LMF and the UE capability, the serving gNB provides pre-configured measurement gap configuration(s) with associated ID(s) to the UE by sending RRC Reconfiguration message specified in TS 38.331 [14].

3. The UE sends RRC Reconfiguration complete message to the gNB to confirm the reception of pre-configured measurement gap configuration(s).

4. The gNB sends the confirmation message to the LMF to indicate the success of the pre-configuration via NRPPa MEASUREMENT PRECONFIGURATION CONFIRM message.

5a. If the UE requires measurement gaps for performing the requested location measurements, and the triggering condition for UL MAC CE as specified in TS 38.331 [14] is met, the UE sends UL MAC CE Positioning Measurement Gap Activation/Deactivation Request to the gNB and indicates the requested measurement gap configuration based on the ID configured in step 1.

5b. LMF may send the NRPPa MEASUREMENT ACTIVATION message to request for measurement gap activation.

6. Based on the request from the UE in step 5a or the request from the LMF in step 5b, the gNB may send DL MAC CE Positioning Measurement Gap Activation/Deactivation containing an ID to activate/deactivate the associated measurement gap.

[TS 38.331, clause 5.5.6.2]

The UE shall:

1> if and only if upper layers indicate to start performing location measurements towards E-UTRA or NR or start subframe and slot timing detection towards E-UTRA, and the UE requires measurement gaps for these operations while measurement gaps are either not configured or not sufficient:

2> if preconfigured measurement gaps for positioning are configured and the UE considers that at least one of the preconfigured measurement gaps for positioning is sufficient for the location measurement when activated:

3> trigger the lower layers to initiate the measurement gap activation request using UL MAC CE as specified in TS 38.321 [3];

2> else:

3> initiate the procedure to indicate start as specified in clause 5.5.6.3;

NOTE 1: The UE verifies the measurement gap situation only upon receiving the indication from upper layers. If at this point in time sufficient gaps are available, the UE does not initiate the procedure. Unless it receives a new indication from upper layers, the UE is only allowed to further repeat the procedure in the same PCell once per frequency of the target RAT if the provided measurement gaps are insufficient.

NOTE 1a: When indication is received from upper layers for performing location measurement and there is pre-configured measurement gap configured (not preconfigured measurement gap for positioning), the UE considers this preconfigured measurement gap to be not sufficient if the measurement gap is not considered to be always activated according to clause 9.1.7.2 of TS 38.133 [14].

1> if and only if upper layers indicate to stop performing location measurements towards E-UTRA or NR or stop subframe and slot timing detection towards E-UTRA:

2> if there is no activated preconfigured measurement gap for positioning:

3> if there is previously triggered UL MAC CE transmission for the measurement gap activation for positioning:

4> indicate to the lower layers to cancel the triggered UL MAC CE transmission for the measurement gap activation as specified in TS 38.321 [3];

3> else:

4> initiate the procedure to indicate stop as specified in 5.5.6.3.

2> else if there is activated preconfigured measurement gap for positioning:

3> trigger the lower layers to deactivate all the activated measurement gap(s) for positioning as specified in TS 38.321 [3].

NOTE 2: The UE may initiate the procedure to indicate stop even if it did not previously initiate the procedure to indicate start.

[TS 38.321, clause 5.18.20]

If the UE is configured with pre-configured measurement gaps, the network may send DL MAC CE for Positioning Measurement Gap Activation/Deactivation Command to the UE as in clause 6.1.3.41. For the activated measurement gap, the UE shall follow the specified UE behaviour in clause 5.14.

Upon the reception of the MAC CE for Positioning Measurement Gap Activation/Deactivation Command, the MAC entity shall:

1> if the Measurement Gap Activation/Deactivation Command MAC CE indicates the deactivation of a pre-configured positioning measurement gap:

2> deactivate the positioning measurement gap.

1> else if the Positioning Measurement Gap Activation/Deactivation Command MAC CE indicates the activation of a pre-configured measurement gap:

2> activate the positioning measurement gap and perform the procedure specified in clause 5.14.

[TS 38.321, clause 5.25]

If the UE is configured with pre-configured measurement gap, the UE may request the network to activate or deactivate the Positioning measurement gap with UL MAC CE for Positioning Measurement Gap Activation/Deactivation Request in clause 6.1.3.40.

The MAC entity shall, when triggered by the upper layer to send Positioning Measurement Gap Activation/Deactivation Request, cancel the triggered Positioning Measurement Gap Activation/Deactivation Request, if any and trigger another Positioning Measurement Gap Activation/Deactivation Request according to the upper layer's request.

The MAC entity shall,

1>if Positioning Measurement Gap Activation/Deactivation Request MAC CE has been triggered, and not cancelled:

2> if indication from upper layer has been received that the triggered Positioning Measurement Gap Activation/Deactivation Request MAC CE should be cancelled; or

2> if the pre-configured measurement gap indicated in the Positioning Measurement Gap Activation/Deactivation Request MAC CE has already been activated/deactivated according to clause 5.18.20:

3> cancel the triggered Positioning Measurement Gap Activation/Deactivation Request MAC CE.

2> if UL-SCH resources are available for a new transmission and these UL-SCH resources can accommodate the Positioning Measurement Gap Activation/Deactivation Request MAC CE plus its subheader as a result of logical channel prioritization:

3> instruct the Multiplexing and Assembly procedure to generate the Positioning Measurement Gap Activation/Deactivation Request MAC CE according to the upper layer's request;

3> cancel triggered Positioning Measurement Gap Activation/Deactivation Request MAC CE.

2> else:

3> trigger a Scheduling Request for Positioning Measurement Gap Activation/Deactivation Request MAC CE.

[TS 38.321, clause 6.1.3.40]

The Positioning Measurement Gap Activation/deactivation request MAC CE is identified by MAC subheader with eLCID as specified in Table 6.2.1-2b.

It has a fixed 8-bit size defined as follows (Figure 6.1.3.40-1):

- Positioning MG ID: This field indicates the identifier for the pre-configured positioning measurement gap. The length of the field is 4 bits;
- A/D: This field indicates the activation or deactivation of the Positioning Measurement Gap. The field is set to 1 to indicate activation, otherwise it indicates deactivation. The length of the field is 1 bit;
- R: Reserved bit, set to 0.

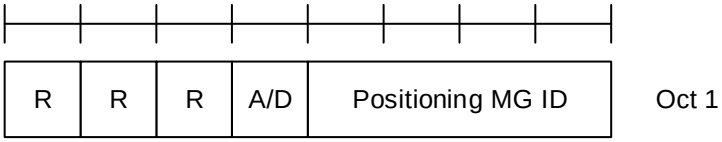



**Figure 6.1.3.40-1: Positioning Measurement Gap Activation/Deactivation Request MAC CE**

[TS 38.321, clause 6.1.3.41]

The Positioning Measurement Gap Activation/Deactivation Command MAC CE is identified by MAC subheader with eLCID as specified in Table 6.2.1-1b.

It has a fixed 8-bit size defined as follows (Figure 6.1.3.41-1):

- Positioning MG ID: This field indicates the identifier for the preconfigured positioning measurement gap. The length of the field is 4 bits;
- A/D: This field indicates the activation or deactivation of the Positioning Measurement Gap. The field is set to 1 to indicate activation, otherwise it indicates deactivation. The length of the field is 1 bit;
- R: Reserved bit, set to 0.

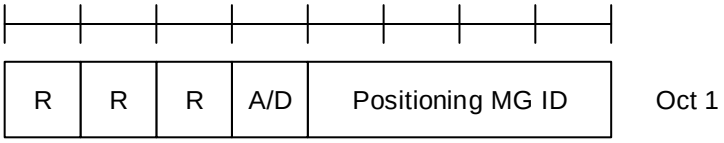



**Figure 6.1.3.41-1: Positioning Measurement Gap Activation/Deactivation Command MAC CE**

## 9.4.4.3 Test description

### 9.4.4.3.1 Pre-test conditions

System Simulator:

- Sub-test 26: NR Cell 1 as specified in 8.2.9
- Sub-test 27: NR Cell 1 and NR Cell 2 as specified in 8.2.10.
- Sub-test 28: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11.

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

### 9.4.4.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 9.4.4.3.2-1 below:

**Table 9.4.4.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 26 | UE supporting Multi-RTT (Rel-17 onwards) |
| 27 | UE supporting DL-AoD (Rel-17 onwards) |
| 28 | UE supporting DL-TDOA (Rel-17 onwards) |

**Table 9.4.4.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.4.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS sends an RRCReconfiguration message to configure pre-configured positioning measurement gap with associated ID. | <-- | RRCReconfiguration | | |
| 6 | Check: Does the UE send an RRCReconfigurationComplete message to confirm the reception of pre-configured measurement gap configuration? | --> | RRCReconfigurationComplete | 1 | P |
| 7 | The SS sends an LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 8 | Check: Does the UE send a measurement gap activation request using UL MAC CE containing the measurement gap ID to activate the positioning measurement gap? | --> | MAC CE (Positioning Measurement Gap Activation Request) | 2 | P |
| 9 | The SS sends a Positioning Measurement Gap Activation command containing an ID to activate the associated measurement gap | <-- | MAC CE (Positioning Measurement Gap Activation Command) | | |
| - | EXCEPTION: In parallel with the events described in Steps 10 to 11, the step specified in Table 9.4.4.3.2-4 takes place. | | | | |
| 10 | Check: Does the UE send an LPP message of type Provide Location Information including location estimate or location measurements? | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 3 | P |
| 11 | IF the UE LPP message at step 10 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

**Table 9.4.4.3.2-4: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | Check: Does the UE send a Measurement Gap Deactivation Request to deactivate the activated measurement gap? | --> | MAC CE (Positioning Measurement Gap Deactivation Request) | 4 | P |

## 9.4.4.3.3 Specific message contents

**Table 9.4.4.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |

| | | | |
|---|---|---|---|
| UE Positioning Technology | Sub-test 26: 0 0 0 0 0 1 1 0<br>Sub-test 27: 0 0 0 0 1 0 0 0<br>Sub-test 28: 0 0 0 0 0 1 1 1 | Sub-test 26: Multi-RTT<br>Sub-test 27: DL-AoD<br>Sub-test 28: DL-TDOA | |

**Table 9.4.4.3.3-2: DLInformationTransfer (step 1, step 3, step 4, step 7 and step 11, Table 9.4.4.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.4.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.4.3.3-3: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.4.3.3-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3 and 11:** Set according to Table 9.4.4.3.3-4 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 9.4.4.3.3-5 | LPP Provide Assistance Data | |
| | **Step 7:** Set according to Table 9.4.4.3.3-6 | LPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.4.3.3-4: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.4.3.3-3)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |

| | | | |
|---|---|---|---|
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in steps 2 or 10, Table 9.4.4.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.4.3.3-5: LPP Provide Assistance data (DL NAS TRANSPORT, Table 9.4.4.3.3-3)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |

**Table 9.4.4.3.3-6: LPP Request Location Information (DL NAS TRANSPORT, Table 9.4.4.3.3-3)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsPreferred | | |

**Table 9.4.4.3.3-7: RRCReconfiguration (step 5, Table 9.4.4.3.2-3)**

| Derivation Path: TS 38.331 [6], clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| radioBearerConfig | Not present | | |
| secondaryCellGroup | Not present | | |
| measConfig SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList | Not present | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList | Not present | | |
| s-MeasureConfig | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig SEQUENCE { | | | |
| gapFR2 | Not present | | |
| gapFR1 | Not present | | |
| gapUE | Not present | | |
| gapToAddModList-r17 | Not present | | |
| gapToReleaseList-r17 | Not present | | |
| posMeasGapPreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPreConfigPosGapId-r17)) OF PosGapConfig-r17 { | 2 entries | | |
| PosGapConfig-r17[1] SEQUENCE { | | entry 1 | |
| measPosPreConfigGapId-r17 | 1 | | |
| gapOffset-r17 | 0 | | |
| mgl-r17 | ms6 | | |
| mgrp-r17 | ms40 | | |
| mgta-r17 | ms0 | | |
| gapType-r17 | perFR1 | | |
| } | | | |
| PosGapConfig-r17[2] SEQUENCE { | | entry 2 | |
| measPosPreConfigGapId-r17 | 2 | | |
| gapOffset-r17 | 0 | | |
| mgl-r17 | ms5dot5 | | |
| mgrp-r17 | ms40 | | |
| mgta-r17 | ms0 | | |
| gapType-r17 | perFR2 | | |
| } | | | |
| } | | | |
| posMeasGapPreConfigToReleaseList-r17 | Not present | | |
| } | | | |
| measGapSharingConfig | Not present | | |
| interFrequencyConfig-NoGap-r16 | Not present | | |
| } | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.4.3.3-8: ULInformationTransfer (step 2 and step 10, Table 9.4.4.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.4.3.3-9 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.4.3.3-9: UL NAS TRANSPORT (ULInformationTransfer, Table 9.4.4.3.3-8)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 9.4.4.3.3-10 | LPP Provide Capabilities | |
| | **Step 10:** Set according to Table 9.4.4.3.3-11 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.4.4.3.2-3) | |

**Table 9.4.4.3.3-10: LPP Provide Capabilities (UL NAS TRANSPORT, Table 9.4.4.3.3-9)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.4.4.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.4.3.3-11: LPP Provide Location Information (UL NAS TRANSPORT, Table 9.4.4.3.3-8)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 7, Table 9.4.4.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 26 | Rel-17 onwards | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 27 | Rel-17 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-AoD-LocationInformation-r16 | Not checked | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 28 | Rel-17 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.5 Pre-configured PRS processing window procedures

### 9.4.5.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE has no assistance data stored and PRS processing window was configured with associated
ID via RRCReconfiguration message}
    then { the UE sends a RRCReconfigurationComplete message to confirm the reception of pre-
configured PRS processing window configuration }
}
```

(2)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE was configured with PRS processing window and receives PPW Activation Command and then
UE receives a location request from LMF }
    then { the UE sends a PROVIDE LOCATION INFORMATION message containing a location estimate }
}
```

### 9.4.5.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.305 clause 7.8, TS 38.321 clauses 5.18.21, 5.24 and 6.1.3.42, TS 38.214 clause 5.1.6.5 and TS 38.133 clause 9.9.1.2. Unless otherwise stated these are Rel-17 requirements.

[TS 38.305, clause 7.8]

The pre-configured PRS processing window procedure is used by the network to provide PRS processing window for NR DL-PRS measurements to the UE without measurement gap. The gNB may activate the pre-configurated PRS processing window upon receiving the request from LMF.

…

Figure 7.8.2-1 shows the general positioning procedure for Pre-configured PRS processing window.

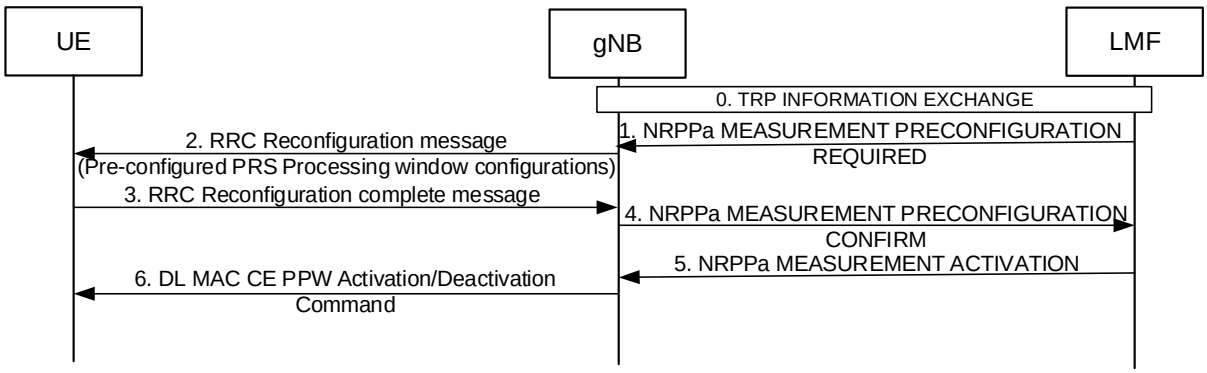



**Figure 7.8.2-1: Pre-configured PRS processing window configuration procedure**

0. LMF obtains the TRP information required for positioning services from the gNBs.

1. The LMF provides the PRS information of the neighbour TRPs to the serving gNB and requests the serving gNBs to pre-configure PRS processing window configuration(s) via NRPPa MEASUREMENT PRECONFIGURATION REQUIRED message.

2. Based on the assistance information from the LMF and the UE capability, the serving gNB provides pre-configured PRS processing window configuration(s) with associated ID(s) to the UE by sending RRC Reconfiguration message specified in TS 38.331 [14].

3. The UE sends RRC Reconfiguration complete message to the gNB to confirm the reception of pre-configured PRS processing window configuration(s).

4. The gNB sends the confirmation message to the LMF to indicate the success of the pre-configuration via NRPPa MEASUREMENT PRECONFIGURATION CONFIRM message.

5. The LMF sends the NRPPa MEASUREMENT ACTIVATION message to request the gNB to (de)activate the preconfigured PRS processing window.

6. Based on the request from the LMF in step 5, the gNB sends DL MAC CE PPW Activation/Deactivation Command containing an ID to activate/deactivate the associated PRS processing window.

[TS 38.321, clause 5.18.21]

If the UE is configured with pre-configured PPW, the network may send DL MAC CE for PPW Activation/Deactivation Command to the UE as in clause 6.1.3.42. For the activated PPW, the UE shall follow the specified UE behaviour in clause 5.24.

Upon activation of DL BWP, the PPW(s) configured for that BWP are considered deactivated. Upon reconfiguration of PPW(s) of the active DL BWP, all the PPW(s) for that BWP are considered deactivated.

Upon the reception of the MAC CE for PPW Activation/Deactivation Command, the MAC entity shall:

1> if the DL MAC CE for PPW Activation/Deactivation Command indicates the deactivation of a pre-configured PPW:

2> deactivate the PPW.

1> else if the DL MAC CE for PPW Activation/Deactivation Command indicates the activation of a pre-configured PPW:

2> activate the PPW according to the procedure specified in clause 5.24.

[TS 38.321, clause 6.1.3.42]

The PPW Activation/Deactivation Command MAC CE is identified by MAC subheader with eLCID as specified in Table 6.2.1-1b.

It has variable size defined as follows (Figure 6.1.3.42-1):

- numEntry: This field indicates the number of entries N-1 in the MAC CE. 00 indicates that N equals to 2; 01 indicates that N equals to 3 and so on. The length of the field is 2 bits;

- Serving Cell ID: This field indicates the identity of the Serving Cell for which the MAC CE applies. The length of the field is 5 bits;

- PPW ID: This field indicates the index of the PPW configured on active DL BWP of the Serving Cell identified by the above Serving Cell ID. Index 0 corresponds to the first entry within the list of the PPW configuration by the increasing order of *PPW-ID* in TS 38.331 [5] in this BWP, index 1 corresponds to the second entry in the list and so on. The length of the field is 2 bits;

- A/D: This field indicates the activation or deactivation of the PPW. The field is set to 1 to indicate activation, otherwise it indicates deactivation. The length of the field is 1 bit;

- R: Reserved bit, set to 0.

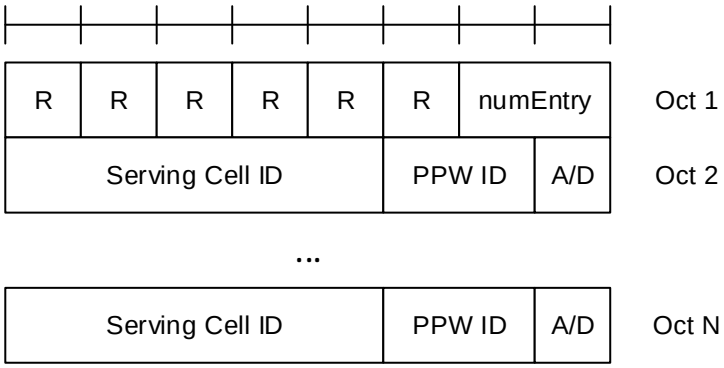



**Figure 6.1.3.42-1: PPW Activation/Deactivation Command MAC CE**

[TS 38.321, clause 5.24]

When PPW is activated and PRS has higher priority than DL channel and signals, for the affected symbols within the PPW according to clause 5.1.6.5 in TS 38.214 [7], the MAC entity shall:

1> if the *ra-ResponseWindow* or the *ra-ContentionResolutionTimer* or the *msgB-ResponseWindow* is running:

2> monitor the PDCCH as specified in clauses 5.1.4 and 5.1.5.

1> else:

2> not receive DL-SCH;

2> not receive PDCCH.

[TS 38.214, clause 5.1.6.5]

The UE is expected to measure the DL PRS outside the measurement gap, subject to UE capability, if the DL PRS is inside the active DL BWP and has the same numerology as the active DL BWP and is within the DL PRS processing window indicated by higher layer parameter *DL-PPW-PreConfig*. The UE is not expected to measure the DL PRS outside the measurement gap if the expected received timing difference between the DL PRS from the non-serving cell and that from the serving cell, determined by the higher layer parameters *nr-DL-PRS-ExpectedRSTD* and *nr-DL-PRS-ExpectedRSTD-Uncertainty*, is larger than maximum Rx timing difference provided by UE capability. For receiving the DL PRS outside the measurement gap and within the DL PRS processing window, the UE determines the DL PRS priority as indicated by higher layer parameter *priority* subject to UE capability or as implied by UE capability:

- with value *'st1'* where the DL PRS is higher priority than all the DL signal/channels except SSB, or
- with value *'st2'* where the DL PRS is lower priority than PDCCH and the PDSCH scheduled by DCI formats 1_1 or 1_2 with the priority indicator field in the corresponding DCI format set to 1, and is higher priority than other DL signals/channels except SSB, or
- with value *'st3'* where the DL PRS is lower priority than all the DL signals/channels except SSB.

Inside one *DL-PPW-PreConfig* the UE is only expected to measure a single DL PRS positioning frequency layer.

When the UE is expected to measure the DL PRS outside the measurement gap in a configured PRS processing window with *type1A* and if the DL PRS is determined to be higher priority than the DL signals and channels inside the PRS processing window, those DL signals and channels are not expected to be measured by the UE. When the UE is expected to measure the DL PRS outside the measurement gap in a configured PRS processing window with *type1B* and if the DL PRS is determined to be higher priority than the DL signals and channels inside the PRS processing window, those DL signals and channels in the same band as the DL PRS are not expected to be measured by the UE. When the UE is expected to measure the DL PRS outside the measurement gap in a configured PRS processing window with *type2* if the DL PRS is determined to be higher priority than the DL signals and channels inside the PRS processing window, those DL signals and channels from the impacted serving cells are not expected to be measured by the UE on the overlapped symbols with the DL PRS, where impacted serving cells refer to the serving cell on which the *DL-PPW-PreConfig* is configured for a frequency range 1 band, and all the serving cells in the same band as the DL PRS for a frequency range 2 band.

…

For a UE configured with PRS Processing Window(s), when the UE receives an activation/deactivation command, as described in clause 6.1.3.42 of [10, TS 38.321], for a PRS processing window activation, and when the UE would transmit a PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the command, the corresponding actions in [10, TS 38.321] and the UE assumptions shall be applied starting from the first slot that is after slot $n + 3N_{slot}^{subframe,\mu}$ where $\mu$ is the SCS configuration for the PUCCH. The UE is not expected to be indicated with more than 4 activated PRS processing windows by higher layer parameter *DL-PPW-PreConfig* across all active DL BWPs and is not expected to be indicated with the activated PRS processing windows *DL-PPW-PreConfig* that overlap in time.

[TS 38.133, clause 9.9.1.2]

The requirements for RSTD, PRS-RSRP, UE Rx-Tx time difference, and PRS-RSRPP measurement without measurement gaps specified in clauses 9.9.2.7, 9.9.3.6, 9.9.4.6 and 9.9.6.6 shall apply provided that:

UE is configured with PPW,

No active BWP switching occurs during PPW,

PRS is within PPW and does not overlap with other signals/channels of higher priority,

- for PPW type 1A/1B, the PPW does not overlap with any symbol for SSB-based RLM/BFD/CBD/L1-RSRP/L1-SINR measurement on any CC or for SSB based RRM measurement on any MOs that are measured outside measurement gaps,
- for PPW type 2, PRS does not overlap with any symbol for SSB-based RLM/BFD/CBD/L1-RSRP/L1-SINR measurement on any CC or for SSB based RRM measurement on any MOs that are measured outside measurement gaps,

max $|\Delta T| \leq$ THR, where

$\Delta T$ is the time difference between the start of a slot containing PRS from the neighbour cell/TRP and the start of the closest slot from the serving cell;

the range of $\Delta T$ is determined by the expected RSTD and expected RSTD uncertainty in the assistance data;

THR is the threshold as reported in UE capability *prs-MeasurementWithoutMG-r17*.

SCS of PRS within PPW and SCS of DL active BWP are the same.

## 9.4.5.3 Test description

### 9.4.5.3.1 Pre-test conditions

System Simulator:

- Sub-test 26: NR Cell 1 as specified in 8.2.9
- Sub-test 27: NR Cell 1 and NR Cell 2 as specified in 8.2.10.
- Sub-test 28: NR Cell 1, NR Cell 2 and NR Cell 3 as specified in 8.2.11.

UE:

- The UE shall begin the test with no assistance data stored.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.
- Then the SS sends the RESET UE POSITIONING STORED INFORMATION message to the UE to clear the stored assistance data in the UE.

Related PICS/PIXIT Statements:

-

### 9.4.5.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 9.4.5.3.2-1 below:

**Table 9.4.5.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 26 | UE supporting Multi-RTT (Rel-17 onwards) |
| 27 | UE supporting DL-AoD (Rel-17 onwards) |
| 28 | UE supporting DL-TDOA (Rel-17 onwards) |

**Table 9.4.5.3.2-2: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.5.3.2-3: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS sends an RRCReconfiguration message to configure PRS processing window with associated ID. | <-- | RRCReconfiguration | | |
| 6 | Check: Does the UE send an RRCReconfigurationComplete message to confirm the reception of pre-configured PRS processing window configuration? | --> | RRCReconfigurationComplete | 1 | P |
| 7 | The SS sends a PPW Activation Command containing a PPW ID to activate the associated PRS processing window. | <-- | MAC CE (PPW Activation Command) | | |
| 8 | The SS sends an LPP message of type Request Location Information including a request for location estimate and location measurements. | <-- | *DLInformationTransfer* (LPP REQUEST LOCATION INFORMATION) | - | - |
| 9 | Check: Does the UE send an LPP message of type Provide Location Information including location estimate or location measurements? | --> | *ULInformationTransfer* (LPP PROVIDE LOCATION INFORMATION) | 2 | P |
| 10 | IF the UE LPP message at step 9 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |

9.4.5.3.3 Specific message contents

**Table 9.4.5.3.3-1: RESET UE POSITIONING STORED INFORMATION (Preamble)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | Sub-test 26: 0 0 0 0 0 1 1 0<br>Sub-test 27: 0 0 0 0 1 0 0 0<br>Sub-test 28: 0 0 0 0 0 1 1 1 | Sub-test 26: Multi-RTT<br>Sub-test 27: DL-AoD<br>Sub-test 28: DL-TDOA | |

**Table 9.4.5.3.3-2: DLInformationTransfer (step 1, step 3, step 4, step 8 and step 10, Table 9.4.5.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.5.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-3: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.5.3.3-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3 and 10:** Set according to Table 9.4.5.3.3-4 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 9.4.5.3.3-5 | LPP Provide Assistance Data | |
| | **Step 8:** Set according to Table 9.4.5.3.3-7 | LPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.5.3.3-4: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.5.3.3-3)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in steps 2 or 9, Table 9.4.5.3.2-3 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.5.3.3-5: LPP Provide Assistance data (DL NAS TRANSPORT, Table 9.4.5.3.3-3)**

| Derivation Path: Table 8.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-1 with the following exceptions: | | | |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData-r9 SEQUENCE { | | | |
| nr-Multi-RTT-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.4.5.3.3-6 | | Sub-test 26 |
| } | | | |
| nr-DL-AoD-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.4.5.3.3-6 | | Sub-test 27 |
| } | | | |
| nr-DL-TDOA-ProvideAssistanceData-r16 SEQUENCE { | | | |
| nr-DL-PRS-AssistanceData-r16 | As defined in Table 9.4.5.3.3-6 | | Sub-test 28 |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-6: NR-DL-PRS-AssistanceData**

| Derivation Path: Table 8.4.1.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4.1.6-1 with the following exceptions: | | | |
| NR-DL-PRS-AssistanceData-r16 ::= SEQUENCE { | | | |
| nr-DL-PRS-AssistanceDataList-r16 SEQUENCE (SIZE (1..nrMaxFreqLayers-r16)) OF NR-DL-PRS-AssistanceDataPerFreq-r16 { | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 2 entries | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[2] SEQUENCE { | | entry 2 | Sub-test 27, Sub-test 28 |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| NR-DL-PRS-AssistanceDataPerFreq-r16[2] SEQUENCE { | | entry 2 | Sub-test 28 |
| nr-DL-PRS-AssistanceDataPerFreq-r16 SEQUENCE (SIZE (1..nrMaxTRPsPerFreq-r16)) OF NR-DL-PRS-AssistanceDataPerTRP-r16{ | 1 entry | | |
| NR-DL-PRS-AssistanceDataPerTRP-r16[1] SEQUENCE { | | entry 1 | |
| nr-DL-PRS-ExpectedRSTD-r16 | 0 | | |
| nr-DL-PRS-ExpectedRSTD-Uncertainty-r16 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-7: LPP Request Location Information (DL NAS TRANSPORT, Table 9.4.5.3.3-3)**

| Derivation Path: Table 8.4-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-3 with the following exceptions: | | | |
| locationInformationType | locationMeasurementsPreferred | | |

**Table 9.4.5.3.3-8: RRCReconfiguration (step 5, Table 9.4.5.3.2-3)**

| Derivation Path: TS 38.508-1 [30], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | OCTET STRING (CONTAINING CellGroupConfig) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-9: *CellGroupConfig* (RRCReconfiguration, Table 9.4.5.3.3-8)**

| Derivation Path: 38.508-1 [30], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| spCellConfig SEQUENCE { | | | |
| spCellConfigDedicated SEQUENCE { | | | |
| initialDownlinkBWP SEQUENCE { | | | |
| dl-PPW-PreConfigToAddModList-r17 SEQUENCE (SIZE (1..maxNrofPPW-Config-r17)) OF DL-PPW-PreConfig-r17 { | 1 entry | | |
| DL-PPW-PreConfig-r17[1] SEQUENCE { | | entry 1 | |
| dl-PPW-ID-r17 | 0 | | |
| dl-PPW-PeriodicityAndStartSlot-r17 | The periodicity is 160ms and the starting slot offset is 11 ms for any SCS configuration | | |
| length-r17 | The length of DL-PRS processing window is 11 ms for any SCS configuration | | |
| type-r17 | type1A or type1B or type2 | Set depending on UE capabilities. Set to any supported type when UE supporting multiple PPW types | |
| priority-r17 | st1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-10: ULInformationTransfer (step 2 and step 9, Table 9.4.5.3.2-3)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.5.3.3-11 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-11: UL NAS TRANSPORT (ULInformationTransfer, Table 9.4.5.3.3-10)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 9.4.5.3.3-12 | LPP Provide Capabilities | |
| | **Step 9:** Set according to Table 9.4.5.3.3-13 | LPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.4.5.3.2-3) | |

**Table 9.4.5.3.3-12: LPP Provide Capabilities (UL NAS TRANSPORT, Table 9.4.5.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.4.5.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| commonIEsProvideCapabilities | Dependent on UE capabilities | | |
| nr-Multi-RTT-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-AoD-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-DL-TDOA-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.5.3.3-13: LPP Provide Location Information (UL NAS TRANSPORT, Table 9.4.5.3.3-11)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in LPP Request Location Information message in step 8, Table 9.4.5.3.2-3 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation-r9 SEQUENCE { | | | |
| commonIEsProvideLocationInformation | May be present. Any value acceptable | | |
| nr-Multi-RTT-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 26 | Rel-17 onwards | |
| nr-Multi-RTT-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-Multi-RTT-Error-r16 | Not present | | |
| nr-Multi-RTT-SignalMeasurementInstances-r17 | Not present | | |
| } | | | |
| nr-DL-AoD-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 27 | Rel-17 onwards | |
| nr-DL-AoD-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-AoD-LocationInformation-r16 | Not checked | | |
| nr-DL-AoD-Error-r16 | Not present | | |
| nr-DL-AoD-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-AoD-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| nr-DL-TDOA-ProvideLocationInformation-r16 SEQUENCE { | Present for sub-test 28 | Rel-17 onwards | |
| nr-DL-TDOA-SignalMeasurementInformation-r16 | Present. Any value acceptable | | |
| nr-dl-tdoa-LocationInformation-r16 | Not checked | | |
| nr-DL-TDOA-Error-r16 | Not present | | |
| nr-DL-TDOA-SignalMeasurementInstances-r17 | Not present | | |
| nr-DL-TDOA-LocationInformationInstances-r17 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.6 UE Positioning Assistance Information

### 9.4.6.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives RRCReconfiguration message requesting the association information of UL SRS
resources for positioning with Tx TEGs }
    then { the UE sends a UEPositioningAssistanceInfo message including TxTEG information }
}
```

### 9.4.6.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.305 clause 7.4.1.2 and TS 38.331 clause 5.7.14. Unless otherwise stated these are Rel-17 requirements.

[TS 38.305, clause 7.4.1.2]

The UE Positioning Assistance Information procedure is used by UE to report the UE Positioning Assistance Information for UL-TDOA. The UE reports the association between UL-SRS resources for positioning and the UE Tx TEG ID.

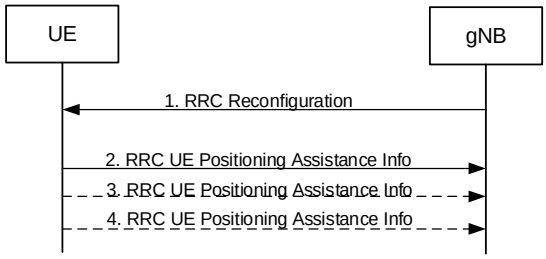



**Figure 7.4.1.2-1: RRC procedure for UE TxTEG**

**Precondition:** The serving gNB of a UE has received a NRPPa message from an LMF requesting the TxTEG of the UE for NR UL-TDOA positioning.

1. The serving gNB may send a RRC Reconfiguration message to the UE, requesting the UE to provide the association information of UL SRS resources for positioning with Tx TEGs to the serving gNB if the UE supports UE Tx TEG reporting. Based on the request from the LMF, the RRC Reconfiguration message from the serving gNB to the UE indicates the UE should provide either a single report or a periodic report of UE TxTEG association to the serving gNB.

2. When the UE receives the request via RRC Reconfiguration message, the UE sends a UE Positioning Assistance Info message to the serving gNB to report the UE TxTEG information, including all the changes of the UE TxTEG during the report period if the UE is required to report UE Tx TEG periodically. The UE will report all the UE TxTEG at the time when the RRC Reconfiguration message is received if the UE is only required to report the one-shot UE TxTEG information.

[TS 38.331, clause 5.7.14]

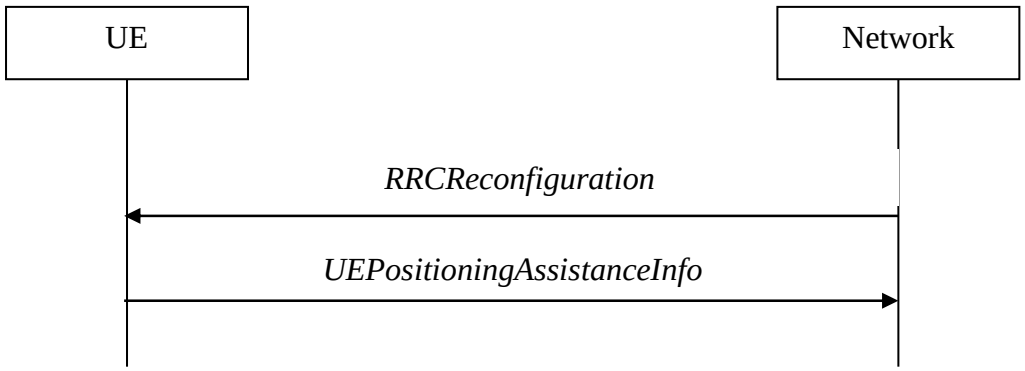



**Figure 5.7.14.1-1: UE Positioning Assistance Information procedure**

The UE Positioning Assistance Information procedure is used by UE to report the UE Positioning Assistance Information. The UE reports the association between UL-SRS resources for positioning and the UE Tx TEG ID as defined in TS 38.305 [73].

[TS 38.331, clause 5.7.14.2]

A UE capable of providing the association between UL SRS Resource for positioning and UE Tx TEG ID in RRC_CONNECTED may initiate the procedure upon being configured to provide this association information.

Upon initiation of the procedure, the UE shall:

1> initiate transmission of the *UEPositioningAssistanceInfo* message in accordance with 5.7.14.3 to provide the association.

[TS 38.331, clause 5.7.14.3]

The UE shall set the contents of the *UEPositioningAssistanceInfo* message as follows:

1> if *ue-TxTEG-RequestUL-TDOA-Config* in *RRCReconfiguration* message is configured with *periodicReporting*:

2> for all the association changes store *ue-TxTEG-Association* corresponding to each *ue-TxTEG-ID* with *nr-TimeStamp*;

2> include the results in *ue-TxTEG-AssociationList* in the *UEPositioningAssistanceInfo* message on expiry of each configured period;

2> optionally include one *ue-TxTEG-TimingErrorMarginValue* for each *UEPositioningAssistanceInfo* message;

1> else if *ue-TxTEG-RequestUL-TDOA-Config* in *RRCReconfiguration* message is configured with *oneShot*:

2> identify the *ue-TxTEG-Association* corresponding to each *ue-TxTEG-ID* with *nr-TimeStamp*;

2> include the results in *ue-TxTEG-AssociationList* in the *UEPositioningAssistanceInfo* message only one time;

2> optionally include one *ue-TxTEG-TimingErrorMarginValue* for each *UEPositioningAssistanceInfo* message.

The UE shall submit the *UEPositioningAssistanceInfo* message to lower layers for transmission.

### 9.4.6.3 Test description

#### 9.4.6.3.1 Pre-test conditions

System Simulator:

- Sub-test 29: NR Cell 1

UE:

-

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

-

### 9.4.6.3.2 Test procedure sequence

**Table 9.4.6.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.6.3.2-2: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends a LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an RRCReconfiguration message requesting the association information of UL SRS resources for positioning with Tx TEGs | <-- | RRCReconfiguration | | |
| 5 | Check: Does the UE send a UEPositioningAssistanceInfo message including TxTEG information? | --> | UEPositioningAssistanceInfo | 1 | P |

### 9.4.6.3.3 Specific message contents

**Table 9.4.6.3.3-1: DLInformationTransfer (step 1 and step 3, Table 9.4.6.3.2-2)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.6.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.6.3.3-2: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.6.3.3-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Step 3** Set according to Table 9.4.6.3.3-3 | LPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.6.3.3-3: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.6.3.3-2)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 1, Table 9.4.6.3.2-2 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.6.3.3-4: RRCReconfiguration (Step 4, Table 9.4.6.3.2-2)**

| Derivation Path: TS 38.508-1 [30], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| ue-TxTEG-RequestUL-TDOA-Config-r17 CHOICE SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| UE-TxTEG-RequestUL-TDOA-Config-r17 CHOICE { | | | |
| oneShot-r17 | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.6.3.3-5: ULInformationTransfer (step 2, Table 9.4.6.3.2-2)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.6.3.3-6 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.6.3.3-6: UL NAS TRANSPORT (ULInformationTransfer, Table 9.4.6.3.3-5)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:**<br>Set according to Table 9.4.6.3.3-7 | LPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.4.6.3.3-2) | |

**Table 9.4.6.3.3-7: LPP Provide Capabilities (UL NAS TRANSPORT, Table 9.4.6.3.3-6)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID SEQUENCE { | | | |
| initiator | locationServer | | |
| transactionNumber | (0..255) | Contains the same value as the corresponding field in the LPP Request Capabilities message in step 1, Table 9.4.6.3.2-2 | |
| } | | | |
| endTransaction | TRUE | | |
| sequenceNumber | (0..255) | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| lpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| provideCapabilities-r9 SEQUENCE { | | | |
| nr-UL-ProvideCapabilities-r16 | Dependent on UE capabilities | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.6.3.3-8: UEPositioningAssistanceInfo (step 5, Table 9.4.6.3.2-2)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UEPositioningAssistanceInfo-r17 ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| uePositioningAssistanceInfo-r17 SEQUENCE { | | | |
| ue-TxTEG-AssociationList-r17 SEQUENCE (SIZE (1..maxNrOfTxTEGReport-r17)) OF UE-TxTEG-Association-r17 { | n entries | The size of UE-TxTEG-Association-r17 is depending on UE capabilities. | |
| UE-TxTEG-Association-r17[n] SEQUENCE { | | | |
| ue-TxTEG-ID-r17 | Present. Depending on UE capabilities. | | |
| nr-TimeStamp-r17 SEQUENCE { | | | |
| nr-SFN-r17 | (0..1023) | | |
| nr-Slot-r17 | Present. Any value acceptable | | |
| } | | | |
| associatedSRS-PosResourceIdList-r17 | Present. Any value acceptable | | |
| servCellId-r17 | Not checked | | |
| } | | | |
| } | | | |
| lateNonCriticalExtension { } | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| ue-TxTEG-TimingErrorMarginValue-r17 | Not checked | | |
| nonCriticalExtension | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.4.7 LPHAP / SRS configuration enhancements / srs-PosRRC-InactiveValidityAreaNonPreConfig

### 9.4.7.1 Test Purpose (TP)

(1)

```
with { UE in NR RRC_INACTIVE state and there is an on-going positioning SRS transmission }
ensure that {
  when { UE performs cell reselection on NR Cell 2 which is not included in the same srs-
PosConfigValidityArea as NR Cell 1 }
    then { UE transmits an RRCResumeRequest message on NR Cell 2 with resumeCause set to srs-
PosConfigOrActivationReq }
}
```

### 9.4.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.321 clauses 5.26.1, and TS 38.331 clauses 5.3.8.3 and 5.3.13. Unless otherwise stated these are Rel-18 requirements.

[TS 38.321, clause 5.26.1]

Periodic and semi-persistent Positioning SRS with or without positioning SRS bandwidth aggregation can be configured for Positioning SRS transmission in RRC_INACTIVE.

SRS for positioning Tx frequency hopping as in clause 5.32 can also be configured for Positioning SRS transmission in RRC_INACTIVE.

The MAC entity shall,

1> if the TA of the configured Positioning SRS is valid according to clause 5.26.2, and the conditions for positioning SRS transmission in clause 7.3.1 of TS 38.213 [6] and clause 6.2.1.4 of TS 38.214 [7] are satisfied:

2> if the UE is configured with UTW and the *UplinkTimeWindowTimer* is running according to clause 5.32; or

2> if the UE is not configured with UTW:

3> instruct to the lower layer according to TS 38.214 [7] to transmit Periodic or Semi-Persistent Positioning SRS that is activated according to clause 5.18.17 or clause 5.18.37.

[TS 38.331, clause 5.3.8.3]

The UE shall:

…

1> if the *RRCRelease* includes *suspendConfig*:

…

2> if *srs-PosRRC-InactiveValidityAreaNonPreConfig* is set to *setup*:

3> apply the SRS for positioning configuration in RRC_INACTIVE and instruct MAC to start the *inactivePosSRS-ValidityAreaTAT*;

2> else if *srs-PosRRC-InactiveValidityAreaNonPreConfig* is set to *release*:

3> release *srs-PosRRC-InactiveValidityAreaNonPreConfig*, if available;

[TS 38.331, clause 5.3.13.2]

Upon initiation of the procedure, the UE shall:

…

1> else if *srs-PosRRC-InactiveValidityAreaPreConfigList* or *srs-PosRRC-InactiveValidityAreaNonPreConfig* is configured:

2> if the resumption of the RRC connection is triggered due to cell reselection as specified in clause 5.3.13.6; or

2> if the resumption of the RRC connection is triggered due to the need for SRS for positioning configuration and no stored *srs-PosRRC-InactiveValidityAreaPreConfigList* for the camped cell exists:

3> if an emergency service is ongoing:

4> select '2' as the Access Category;

4> set the *resumeCause* to *emergency*;

3> else:

4> select '8' as the Access Category;

4> set the *resumeCause* to *srs-PosConfigOrActivationReq*;

[TS 38.331, clause 5.3.13.4]

1> if *srs-PosRRC-InactiveValidityAreaNonPreConfig* is configured; or

1> if field *srs-PosRRC-InactiveValidityAreaPreConfigList* is configured and if the cell is not listed in *srs-PosConfigValidityArea*:

2> instruct the MAC entity to stop *inactivePosSRS-ValidityAreaTAT*, if it is running;

[TS 38.331, clause 5.3.13.6]

The UE shall:

…

1> else if cell reselection occurs when *srs-PosRRC-InactiveValidityAreaPreConfigList* or *srs-PosRRC-InactiveValidityAreaNonPreConfig* is configured and if there is an on-going SRS for positioning transmission procedure in RRC_INACTIVE:

2> if the selected cell is not included in the *srs-PosConfigValidityArea*:

3> indicate to the lower layer to stop *inactivePosSRS-ValidityAreaTAT*;

3> initiate RRC connection resume procedure in 5.3.13.2;

2> else if the cell is included in the *srs-PosConfigValidityArea*:

3> if the selected cell and the previously camped cell are in the same *srs-PosConfigValidityArea*:

4> if *autonomousTA-AdjustmentEnabled* is configured and if the Timing Advance validation requirements specified in clause 5.6.6.3 of TS 38.133 [14] is met:

5> indicate to the lower layer to update Timing Advance and stored RSRP;

4> instruct lower layers to continue transmitting SRS if Timing Advance validation condition as specified in TS 38.321 [3] are satisfied;

3> if the selected cell and previously camped cell are in the different *srs-PosConfigValidityArea*:

4> initiate RRC connection resume procedure in 5.3.13.2;

4> indicate to the lower layer to stop *inactivePosSRS-ValidityAreaTAT*.

## 9.4.7.3 Test description

### 9.4.7.3.1 Pre-test conditions

System Simulator:

- Sub-test 30: NR Cell 1 and NR Cell 2 as specified in 8.2.9 and system information combination NR-2 as defined in TS 38.508-1 [30] clause 4.4.3.1.2.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

-

### 9.4.7.3.2 Test procedure sequence

Table 9.4.7.3.2-2/3 illustrates the downlink power levels and other changing parameters to be applied for the cells at various time instants of the test execution. Row marked "T0" denotes the initial conditions after preamble, while columns marked "T1" is to be applied subsequently. The exact instants on which these values shall be applied are described in the texts in this clause.

**Table 9.4.7.3.2-1: Test Configuration**

| Test Configuration | Network Deployment Type | Test Implementation |
|---|---|---|
| B | NG-RAN NR | |

**Table 9.4.7.3.2-2: Time instances of cell power level and parameter changes for FR1**

| | Parameter | Unit | NR Cell 1 | NR Cell 2 | Remark |
|---|---|---|---|---|---|
| **T0** | SS/PBCH SSS EPRE | dBm/ SCS | -88 | Off | Only NR Cell 1 is on |
| **T1** | SS/PBCH SSS EPRE | dBm/ SCS | -94 | -88 | NR Cell 2 becomes the highest ranked cell. |
| NOTE 1: Power level "Off" is defined in TS 38.508-1 [30] Table 6.2.2.1-3. | | | | | |

**Table 9.4.7.3.2-3 Time instances of cell power level and parameter changes for FR2**

| | Parameter | Unit | NR Cell 1 | NR Cell 2 | Remark |
|---|---|---|---|---|---|
| **T0** | SS/PBCH SSS EPRE | dBm/ SCS | -82 | Off | Only NR Cell 1 is on |
| **T1** | SS/PBCH SSS EPRE | dBm/ SCS | -91 | -82 | NR Cell 2 becomes the highest ranked cell. |
| NOTE 1: Power level "Off" is defined in TS 38.508-1 [30] Table 6.2.2.1-3. | | | | | |

**Table 9.4.7.3.2-4: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an LPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (LPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an LPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (LPP PROVIDE CAPABILITIES) | - | - |
| 3 | IF the UE LPP message at step 2 includes an acknowledgment request THEN SS sends a LPP Acknowledgement response. | <-- | *DLInformationTransfer* (LPP ACKNOWLEDGEMENT) | - | - |
| 4 | The SS sends an LPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (LPP PROVIDE ASSISTANCE DATA) | - | - |
| 5 | The SS transmits an RRCReconfiguration message with the DRX configuration. | <-- | RRCReconfiguration | - | - |
| 6 | The UE transmits RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |
| 7 | The SS transmits an RRCRelease message with srs-PosRRC-InactiveValidityAreaNonPreConfig configuration and transit the UE to RRC_INACTIVE state | <-- | RRCRelease | - | - |
| 8 | The SS changes NR Cell 1 and NR Cell 2 power level according to the row "T1" in table 9.4.7.3.2-2/3. | - | - | - | - |
| 9 | Check: Does the UE transmit an *RRCResumeRequest* message on NR Cell 2 with resumeCause to srs-PosConfigOrActivationReq? | --> | RRCResumeRequest | 1 | P |
| 10 | The SS transmits an RRCRelease message. | <-- | RRCRelease | - | - |

9.4.7.3.3 Specific message contents

**Table 9.4.7.3.3-1: DLInformationTransfer (step 1, step 3 and step 4, Table 9.4.7.3.2-4)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.4.7.3.3-2. | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.7.3.3-2: DL NAS TRANSPORT (DLInformationTransfer, Table 9.4.7.3.3-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 0011 | LTE Positioning Protocol (LPP) message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.4-2 | LPP Request Capabilities. | |
| | **Steps 3:** Set according to Table 9.4.7.3.3-3 | LPP Acknowledgement | |
| | **Step 4:** Set according to Table 8.4-1 | LPP Provide Assistance Data | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.4.7.3.3-3: LPP Acknowledgement (DL NAS TRANSPORT, Table 9.4.7.3.3-2)**

| Derivation Path: 37.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| LPP-Message ::= SEQUENCE { | | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in steps 2, | |

| | | | |
|---|---|---|---|
| | | Table 9.4.7.3.2-4 | |
| } | | | |
| lpp-MessageBody | Not present. | | |
| } | | | |

**Table 9.4.7.3.3-4: RRCReconfiguration (step 5, Table 9.4.7.3.2-4)**

| Derivation Path: TS 38.508-1 [30], table 4.6.1-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| rrcReconfiguration SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| masterCellGroup | CellGroupConfig | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.7.3.3-5: CellGroupConfig (RRCReconfiguration, Table 9.4.7.3.3-4)**

| Derivation Path: 38.508-1 [30], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| mac-CellGroupConfig | As defined in 38.508-1 [30] Table 4.6.3-68 with Condition DRX. | | |
| } | | | |

**Table 9.4.7.3.3-6: RRCRelease (step 7, Table 9.4.7.3.2-4)**

| Derivation Path: TS 38.508-1 [30], table 4.6.1-16 with condition NR_RRC_INACTIVE | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCRelease ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier | | |
| criticalExtensions CHOICE { | | | |
| rrcRelease SEQUENCE { | | | |
| suspendConfig SEQUENCE { | | | |
| srs-PosRRC-InactiveEnhanced-r18 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-InactiveEnhanced-r18 | OCTET STRING containing SRS-PosRRC-InactiveEnhancedConfig-r18 as defined in Table 9.4.7.3.3-7 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4.7.3.3-7: SRS-PosRRC-InactiveEnhancedConfig-r18 (RRCRelease, Table 9.4.7.3.3-6)**

| Derivation Path: TS 38.331 [33], clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SRS-PosRRC-InactiveEnhancedConfig-r18 ::= SEQUENCE { | | | |
| srs-PosRRC-InactiveAggBW-ConfigList-r18 | Not present | | |
| srs-PosTx-Hopping-r18 | Not present | | |
| srs-PosRRC-InactiveValidityAreaPreConfigList-r18 | Not present | | |
| srs-PosRRC-InactiveValidityAreaNonPreConfig-r18 CHOICE { | | | |
| setup SEQUENCE { | | | |
| SRS-PosRRC-InactiveValidityAreaConfig-r18 SEQUENCE { | | | |
| srs-PosConfigValidityArea-r18 SEQUENCE (SIZE(1..maxNrOfCellsInVA-r18)) OF CellIdentity { | | | |
| CellIdentity[1] | Same as NR Cell 1 | | |
| } | | | |
| srs-PosConfigNUL-r18 SEQUENCE { | | | |
| srs-PosResourceSetToReleaseList-r17 | Not present | | |
| srs-PosResourceSetToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResourceSets-r16)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceSet-r16[1] SEQUENCE { | | entry 1 | |
| srs-PosResourceSetId-r16 | 0 | | |
| srs-PosResourceIdList-r16 SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet)) OF SRS-PosResourceId-r16 { | 1 entry | | |
| SRS-PosResourceId-r16[1] | 0 | entry 1 | |
| } | | | |
| resourceType-r16 CHOICE { | | | |
| periodic-r16 SEQUENCE { | | | |
| } | | | |
| } | | | |
| alpha-r16 | Not present | | |
| p0-r16 | -100 | | |
| pathlossReferenceRS-Pos-r16 CHOICE { | | | |
| ssb-IndexServing-r16 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosResourceToReleaseList-r17 | Not present | | |
| srs-PosResourceToAddModList-r17 SEQUENCE (SIZE(1..maxNrofSRS-PosResources-r16)) OF SRS-PosResource-r16 { | 1 entry | | |
| SRS-PosResource-r16[1] | As defined in Table 8.3.2-4 | entry 1 | |
| } | | | |
| } | | | |
| srs-PosConfigSUL-r18 | Not present | | |
| bwp-NUL-r18 | Not present | | |
| bwp-SUL-r18 | Not present | | |
| areaValidityTA-Config-r18 SEQUENCE { | | | |
| inactivePosSRS-ValidityAreaTAT-r18 | infinity | | |
| inactivePosSRS-ValidityAreaRSRP-r18 | dB34 | | |
| autonomousTA-AdjustmentEnabled-r18 | Not present | | |
| } | | | |
| srs-PosConfigValidityAreaExt-v1830 | Not present | | |
| srs-PosRRC-InactiveAggBW-AdditionalCarriersPerVA-r18 | Not present | | |
| srs-PosRRC-InactiveAggBW-ConfigListPerVA-r18 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| srs-PosRRC-InactiveAggBW-AdditionalCarriers-r18 | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 9.4.7.3.3-8: RRCResumeRequest (step 9, Table 9.4.7.3.2-4)**

| Derivation Path: TS 38.508-1 [30], Table 4.6.1-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCResumeRequest ::= SEQUENCE { | | | |
| rrcResumeRequest SEQUENCE { | | | |
| resumeCause | srs-PosConfigOrActivationReq-v1800 | | |
| } | | | |
| } | | | |

# 9.5 SLPP Procedures

## 9.5.1 SLPP Common Procedures

### 9.5.1.1 Position Capability Transfer between LMF and UE

#### 9.5.1.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives an SLPP message of type REQUEST CAPABILITIES }
    then { UE sends an SLPP message of type PROVIDE CAPABILITIES with the correct supported
capabilities }
}
```

#### 9.5.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 5.1.2 and clause 5.1.5. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 5.1.2]

The Capability Transfer procedure is shown in Figure 5.1.2-1.

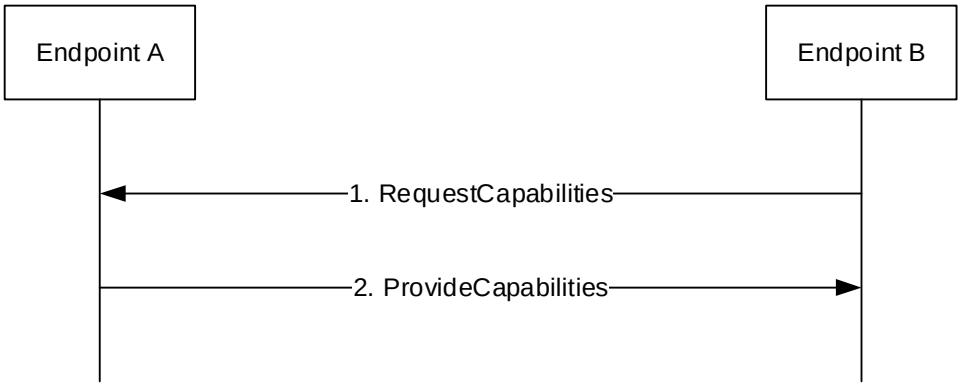



**Figure 5.1.2-1: SLPP Capability Transfer procedure**

1. Endpoint B sends a *RequestCapabilities* message to Endpoint A. Endpoint B may indicate the types of capability requested.

2. Endpoint A responds with a *ProvideCapabilities* message to Endpoint B. The capabilities shall correspond to the capability types specified in step 1. This message shall include the field *endTransaction* set to TRUE.

[TS 38.355, clause 5.1.5]

Upon receiving a *RequestCapabilities* message, Endpoint A shall generate a *ProvideCapabilities* message as a response.

Endpoint A shall:

1> for each positioning method for which a request for capabilities is included in the message:

2> if Endpoint A supports this positioning method:

3> include the capabilities of Endpoint A for that supported positioning method in the response message;

1> set the field *sessionID* in the response message to the same value as the field *sessionID* in the received message if received;

1> set the field *transactionID* in the response message to the same value as the field *transactionID* in the received message;

1> deliver the response message to lower layers for transmission.

### 9.5.1.1.3 Test description

#### 9.5.1.1.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

-

#### 9.5.1.1.3.2 Test procedure sequence

**Table 9.5.1.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an SLPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | 1 | P |
| 3 | IF the UE SLPP message at step 2 includes an acknowledgment request THEN SS sends an SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |

9.5.1.1.3.3 Specific message contents

**Table 9.5.1.1.3.3-1: DL InformationTransfer (steps 1 and 3, Table 9.5.1.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.1.1.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-2: DL NAS TRANSPORT (steps 1 and 3, Table 9.5.1.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.5-1 | SLPP Request Capabilities | |
| | **Step 3:** Set according to Table 9.5.1.1.3.3-3 | SLPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.1.1.3.3-3: SLPP Acknowledgement (step 3, Table 9.5.1.1.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 9.5.1.1.3.2-1. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.1.1.3.3-4: ULInformationTransfer (step 2, Table 9.5.1.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.1.1.3.3-5 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-5: UL NAS TRANSPORT (step 2, Table 9.5.1.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.5.1.1.3.3-6 | SLPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.5.1.1.3.2-1) | |

**Table 9.5.1.1.3.3-6: SLPP Provide Capabilities (step 2, Table 9.5.1.1.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | Present | Contains the same value of the transactionID field in step 1, Table 9.5.1.1.3.2-1. | |
| endTransaction | True | | |
| sequenceNumber | (0..255) | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | Present, or not present | | |
| ackRequested | True | | |
| ackIndicator | (0..255) | | |
| } | | | |
| slpp-MessageBody CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideCapabilities SEQUENCE { | | | |
| commonIEsProvideCapabilities | Not checked | | |
| commonSL-PRS-MethodsIEsProvideCapabilities | Present or not present dependent on UE capabilities. Set according to Table 9.5.1.1.3.3-7. | OCTET STRING containing CommonSL-PRS-MethodsIEsProvide Capabilities | |
| sl-AoA-ProvideCapabilities | Present or not present dependent on (pc_Target_UEB_SL_AoA OR pc_Target_UEA_SL_AoA) Set according to Table 9.5.1.1.3.3-8. | OCTET STRING containing SL-AoA-ProvideCapabilities | |
| sl-RTT-ProvideCapabilities | Present or not present dependent on (pc_Target_UEB_SL_RTT OR pc_Target_UEA_SL_RTT) Set according to Table 9.5.1.1.3.3-9. | OCTET STRING containing SL-RTT-ProvideCapabilities | |
| sl-TDOA-ProvideCapabilities | Present or not present dependent on (pc_Target_UEB_SL_TDOA OR pc_Target_UEA_SL_TDOA). Set according to Table 9.5.1.1.3.3-10. | OCTET STRING containing SL-TDOA-ProvideCapabilities | |
| sl-TOA-ProvideCapabilities | Present or not present dependent on (pc_Target_UEB_SL_TOA OR pc_Target_UEA_SL_TOA) Set according to Table 9.5.1.1.3.3-11. | OCTET STRING containing SL-TOA-ProvideCapabilities | |
| lateNonCriticalExtension | Not checked | | |
| nonCriticalExtensio | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not checked | | |
| } | | | |

**Table 9.5.1.1.3.3-7: CommonSL-PRS-MethodsIEsProvideCapabilities (Table 9.5.1.1.3.3-6)**

| Derivation Path: 38.355 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CommonSL-PRS-MethodsIEsProvideCapabilities ::= SEQUENCE { | | | |
| sl-PRS-CommonProcCapabilityPerUE | Not checked | | |
| sl-PRS-CapabilityBandList SEQUENCE (SIZE (1..nrMaxBands)) OF SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| freqBandIndicatorNR | Present. Any value acceptable. | | |
| sl-PRS-CommonProcCapabilityPerBand SEQUENCE { | | | |
| maxSL-PRS-Bandwidth | Dependent on UE capabilities | | |
| maxNumOfActiveSL-PRS-ResourcesInOneSlot | Dependent on UE capabilities | | |
| maxNumOfSlotsWithActiveSL-PRS-Resources | Dependent on UE capabilities | | |
| minTimeAfterEndofSlotCarryActiveSL-PRS-Resources | Dependent on UE capabilities | | |
| } | | | |
| sl-PositioningARP-LocationProvision | Not checked | | |
| dummy1 | Not checked | | |
| dummy2 | Not checked | | |
| sl-PRS-RxInSharedResourcePool | Not checked | | |
| sl-PRS-RxInDedicatedResourcePool | Not checked | | |
| sl-PRS-TxInSharedResourcePool | Not checked | | |
| sl-PRS-TxScheme1InDedicatedResourcePool | Not checked | | |
| sl-PRS-TxScheme2InDedicatedResourcePool | Not checked | | |
| sl-PRS-RSRP-Meas | Not checked | | |
| sl-PRS-RSRPP-Meas | Not checked | | |
| sl-PRS-TDM-Multiplexing | Not checked | | |
| sl-PRS-RxCombMultiplexing | Not checked | | |
| sl-PRS-AdditionalPathsReport | Not checked | | |
| sl-PRS-LOS-NLOS-Indication | Not checked | | |
| sl-PRS-RxForBandWithSL-CA | Not checked | | |
| sl-PRS-TxForBandWithSL-CA | Not checked | | |
| sl-PRS-ReportTxARP-ID | Not checked | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-8: SL-AoA-ProvideCapabilities (Table 9.5.1.1.3.3-6)**

| Derivation Path: 38.355 clause 6.7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-AoA-ProvideCapabilities ::= SEQUENCE { | | | |
| positioningModes | Dependent on UE capabilities | | |
| tenMsUnitResponseTime | Not checked | | |
| periodicalReporting | Not checked | | |
| scheduledLocationRequestSupported | Not checked | | |
| locationCoordinateTypes | Not checked | | |
| velocityTypes | Not checked | | |
| sl-AoA-CapabilityBandList SEQUENCE (SIZE (1..nrMaxBands)) SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| sl-AoA-Meas | Dependent on UE capabilities | | |
| dummy | Not checked | | |
| measurementsForMultipleARP-IDs-Rx | Not checked | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-9: SL-RTT-ProvideCapabilities (Table 9.5.1.1.3.3-6)**

| Derivation Path: 38.355 clause 6.8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-ProvideCapabilities ::= SEQUENCE { | | | |
| positioningModes | Dependent on UE capabilities | | |
| tenMsUnitResponseTime | Not checked | | |
| periodicalReporting | Not checked | | |
| scheduledLocationRequestSupported | Not checked | | |
| locationCoordinateTypes | Not checked | | |
| velocityTypes | Not checked | | |
| sl-RTT-CapabilityBandList SEQUENCE (SIZE (1..nrMaxBands)) SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| sl-PRS-RxTxTimeDiffWithoutTxTimeStamp | Dependent on UE capabilities | | |
| sl-PRS-RxTxTimeDiffWithTxTimeStamp SEQUENCE { | Present or not Present dependent on UE capabilities | | |
| numOfMeasForSameSL-PRS | Not checked | | |
| maxMeasReportingForDiffSL-PRS | Not checked | | |
| } | | | |
| dummy | Not checked | | |
| measurementsForMultipleARP-IDs-Rx | Not checked | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-10: SL-TDOA-ProvideCapabilities (Table 9.5.1.1.3.3-6)**

| Derivation Path: 38.355 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TDOA-ProvideCapabilities ::= SEQUENCE { | | | |
| positioningModes | Dependent on UE capabilities | | |
| tenMsUnitResponseTime | Not checked | | |
| periodicalReporting | Not checked | | |
| scheduledLocationRequestSupported | Not checked | | |
| locationCoordinateTypes | Not checked | | |
| velocityTypes | Not checked | | |
| sl-TDOA-CapabilityBandList SEQUENCE (SIZE (1..nrMaxBands)) SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| sl-PRS-RSTD-Meas | Dependent on UE capabilities | | |
| dummy | Not checked | | |
| measurementsForMultipleARP-IDs-Rx | Not checked | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.3.3-11: SL-TOA-ProvideCapabilities (Table 9.5.1.1.3.3-6)**

| Derivation Path: 38.355 clause 6.10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TOA-ProvideCapabilities ::= SEQUENCE { | | | |
| positioningModes | Dependent on UE capabilities | | |
| tenMsUnitResponseTime | Not checked | | |
| periodicalReporting | Not checked | | |
| scheduledLocationRequestSupported | Not checked | | |
| locationCoordinateTypes | Not checked | | |
| velocityTypes | Not checked | | |
| sl-TOA-CapabilityBandList SEQUENCE (SIZE (1..nrMaxBands)) SEQUENCE { | | Size n of SEQUENCE is dependent on UE capabilities | |
| sl-RTOA-Meas | Dependent on UE capabilities | | |
| dummy | Not checked | | |
| measurementsForMultipleARP-IDs-Rx | Not checked | | |
| } | | | |
| } | | | |

## 9.5.1.2 Position Capability Transfer between UEs (Target UE is in coverage)

### 9.5.1.2.1 Test Purpose (TP)

(1)

```
with { Target UE is in coverage and having established PC5 RRC connection with peer UE on unicast
sidelink }
ensure that {
  when { UE receives an SLPP message of type REQUEST CAPABILITIES }
    then { UE sends an SLPP message of type PROVIDE CAPABILITIES with the correct supported
capabilities }
}
```

### 9.5.1.2.2 Conformance requirements

As defined in clause 9.5.1.1.2.

#### 9.5.1.2.3 Test description

##### 9.5.1.2.3.1 Pre-test conditions

System Simulator:

SS-NW

- NR Cell 1.
- System information combination NR-14 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 3N-B RRC_CONNECTED_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

-

9.5.1.2.3.2 Test procedure sequence

**Table 9.5.1.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The NR-SS-UE 1 sends an SLPP message of type Request Capabilities. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 2 | The UE sends an SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | SLPP PROVIDE CAPABILITIES | 1 | P |
| 3 | IF<br>the UE SLPP message at step 2 includes an acknowledgment request<br>THEN<br>NR-SS-UE 1 sends an SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |

9.5.1.2.3.3 Specific message contents

**Table 9.5.1.2.3.3-1: SLPP REQUEST CAPABILITIES (step 1, Table 9.5.1.2.3.2-1)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. |
|---|

**Table 9.5.1.2.3.3-2: SLPP PROVIDE CAPABILITIES (step 2, Table 9.5.1.2.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sessionID | Present | Contains the same value of the transactionID field in step 1, Table 9.5.1.2.3.2-1. | |
| } | | | |

**Table 9.5.1.2.3.3-3: SLPP Acknowledgement (step 3, Table 9.5.1.2.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | '00 00 01 00'H | Application Layer ID in initiating UE side | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Present | Contains the same value of the transactionID field in step 1, Table 9.5.1.2.3.2-1. | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 9.5.1.2.3.2-1. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

## 9.5.1.3 Position Capability Transfer between UEs (Target UE is out of coverage)

### 9.5.1.3.1 Test Purpose (TP)

(1)

```
with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives an SLPP message of type REQUEST CAPABILITIES }
    then { UE sends an SLPP message of type PROVIDE CAPABILITIES with the correct supported
capabilities }
}
```

### 9.5.1.3.2 Conformance requirements

As defined in clause 9.5.1.1.2.

### 9.5.1.3.3 Test description

#### 9.5.1.3.3.1 Pre-test conditions

System Simulator:

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

-

#### 9.5.1.3.3.2 Test procedure sequence

As defined in clause 9.5.1.2.3.2.

#### 9.5.1.3.3.3 Specific message contents

As defined in clause 9.5.1.2.3.3.

### 9.5.1.4 SLPP Abort between LMF and UE

#### 9.5.1.4.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives a SLPP Abort message carrying the transaction ID of an on-going procedure }
    then { UE aborts the on-going procedure }
}
```

#### 9.5.1.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 5.5.1, 5.5.2 and clause 5.5.3. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 5.5.1]

The purpose of the abort procedure is to allow Endpoints to abort an ongoing procedure due to some unexpected event (e.g., cancellation of a location request by an LCS client). It can also be used to stop an ongoing procedure (e.g., periodic location reporting from an Endpoint).

[TS 38.355, clause 5.5.2]

Figure 5.5.2-1 shows the Abort procedure.

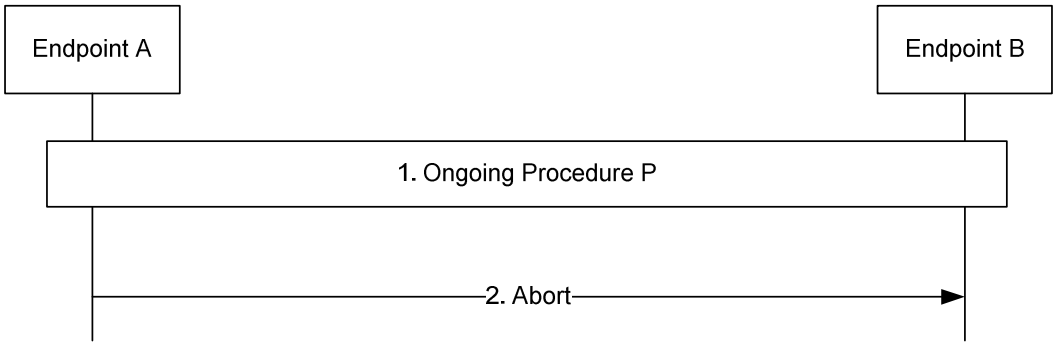



**Figure 5.5.2-1: SLPP Abort procedure**

1. A procedure P is ongoing between endpoints A and B.
2. Endpoint A determines that the procedure must be aborted and sends an *Abort* message to Endpoint B carrying the field *sessionID* (if PC5-U is used as transport layer) and the field *transactionID* for procedure P. Endpoint B aborts procedure P.

[TS 38.355, clause 5.5.3]

Upon receiving an *Abort* message, Endpoint shall:

1> abort any ongoing procedure associated with the field *sessionID* and the field *transactionID* indicated in the message.

#### 9.5.1.4.3 Test description

##### 9.5.1.4.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- Sub-test 33: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.14.
- Sub-test 34: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.15.
- Sub-test 35: NR-SS-UE1, NR-SS-UE2 and NR-SS-UE3 (if required) as specified in clause 8.2.16.
- Sub-test 36: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.17.
- All NR-SS-UEs use GNSS as the synchronization reference source and operate as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

- FFS

#### 9.5.1.4.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 9.5.1.4.3.2-1 below:

**Table 9.5.1.4.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 33 | UE supporting SL-RTT (Rel-18 onwards) |
| 34 | UE supporting SL-AoA (Rel-18 onwards) |
| 35 | UE supporting SL-TDOA (Rel-18 onwards) |
| 36 | UE supporting SL-TOA (Rel-18 onwards) |

**Table 9.5.1.4.3.2-2: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 2 | The SS sends an RRCReconfiguration message with SL-PRS resource pool configuration as defined in Table 8.3.3-1. | <-- | RRCReconfiguration | - | - |
| 3 | The UE sends an RRCReconfigurationComplete message. | --> | RRCReconfigurationComplete | - | - |
| 4 | The SS sends a SLPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 5 | The UE sends a SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 6 | IF the UE SLPP message at step 5 includes an acknowledgment request THEN SS sends a SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |
| 7 | The SS sends a SLPP message of type Provide Assistance Data. | <-- | *DLInformationTransfer* (SLPP PROVIDE ASSISTANCE DATA) | - | - |
| 8 | The SS sends a SLPP message of type Request Location Information including a transaction ID. | <-- | *DLInformationTransfer* (SLPP REQUEST LOCATION INFORMATION) | - | - |
| 9 | Immediately after step 8, the SS sends a SLPP message of type Abort using the same transaction ID chosen in step 8. | <-- | *DLInformationTransfer* (SLPP ABORT) | - | - |
| 10 | The SS waits for 10 seconds to ensure the UE does not send a SLPP message of type Provide Location Information with the same transaction ID as in step 8. | | | 1 | P |

#### 9.5.1.4.3.3 Specific message contents

**Table 9.5.1.4.3.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | FFS | FFS | |

**Table 9.5.1.4.3.3-2: DL InformationTransfer (steps 4, 6, 7, 8 and 9, Table 9.5.1.4.3.2-2)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.1.4.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.4.3.3-3: DL NAS TRANSPORT (steps 4, 6, 7, 8 and 9, Table 9.5.1.4.3.2-2)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 4:** Set according to Table 8.5-1 | SLPP Request Capabilities | |
| | **Step 6:** Set according to Table Table 9.5.1.4.3.3-4 | SLPP Acknowledgement | |
| | **Step 7:** Set according to Table 8.5-2 | SLPP Provide Assistance Data | |
| | **Step 8:** Set according to Table 8.5-3 | SLPP Request Location Information | |
| | **Step 9:** Set according to Table Table 9.5.1.4.3.3-5 | SLPP Abort | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.1.4.3.3-4: SLPP Acknowledgement (step 6, Table 9.5.1.4.3.2-2)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 5, Table 9.5.1.4.3.2-2. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.1.4.3.3-5: SLPP Abort (step 9, Table 9.5.1.4.3.2-2)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 8, Table 9.5.1.4.3.2-2. | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| abort SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| abort SEQUENCE { | | | |
| commonIEsAbort SEQUENCE { | | | |
| abortCause | undefined | | |
| } | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.1.4.3.3-6: ULInformationTransfer (step 5, Table 9.5.1.4.3.2-2)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.1.1.3.3-5 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.4.3.3-7: UL NAS TRANSPORT (step 5, Table 9.5.1.4.3.2-2)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.5.1.1.3.3-6 | SLPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 4 Table 9.5.1.4.3.2-2) | |

## 9.5.1.5 SLPP Abort between UEs (Target UE is in coverage)

### 9.5.1.5.1 Test Purpose (TP)

(1)

```
with { Target UE is in coverage and having established PC5 RRC connection with peer UE on unicast
sidelink }
ensure that {
  when { UE receives a SLPP Abort message carrying the transaction ID of an on-going procedure }
    then { UE aborts the on-going procedure }
}
```

### 9.5.1.5.2 Conformance requirements

As defined in clause 9.5.1.4.2.

#### 9.5.1.5.3 Test description

##### 9.5.1.5.3.1 Pre-test conditions

System Simulator:

SS-NW

- NR Cell 1.
- System information combination NR-36 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- Sub-test 33: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.14.
- Sub-test 34: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.15.
- Sub-test 35: NR-SS-UE1, NR-SS-UE2 and NR-SS-UE3 (if required) as specified in clause 8.2.16.
- Sub-test 36: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.17.
- All NR-SS-UEs use GNSS as the synchronization reference source and operate as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 1N-B RRC_IDLE_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

- FFS

##### 9.5.1.5.3.2 Test procedure sequence

This test case includes sub-test cases dependent on the positioning method(s) supported by the UE. Each sub-test case is identified by a sub-test case number as defined in Table 9.5.1.5.3.2-1 below:

**Table 9.5.1.5.3.2-1: Sub-test case numbers**

| Sub-Test Case Number | Supported Positioning Methods |
|---|---|
| 33 | UE supporting SL-RTT (Rel-18 onwards) |
| 34 | UE supporting SL-AoA (Rel-18 onwards) |
| 35 | UE supporting SL-TDOA (Rel-18 onwards) |
| 36 | UE supporting SL-TOA (Rel-18 onwards) |

**Table 9.5.1.5.3.2-2: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The NR-SS-UE1 sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 2 | The NR-SS-UE1 sends a SLPP message of type Request Capabilities. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 3 | The UE sends a SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 4 | IF<br>the UE SLPP message at step 3 includes an acknowledgment request<br>THEN<br>NR-SS-UE1 sends a SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |
| 5 | The NR-SS-UE1 sends a SLPP message of type Provide Assistance Data. | <-- | SLPP PROVIDE ASSISTANCE DATA | - | - |
| 6 | The NR-SS-UE1 sends a SLPP message of type Request Location Information including a transaction ID. | <-- | SLPP REQUEST LOCATION INFORMATION | - | - |
| 7 | Immediately after step 6, the NR-SS-UE1 sends a SLPP message of type Abort using the same transaction ID chosen in step 6. | <-- | SLPP ABORT | - | - |
| 8 | The NR-SS-UE1 waits for 10 seconds to ensure the UE does not send a SLPP message of type Provide Location Information with the same transaction ID as in step 6. | | | 1 | P |

9.5.1.5.3.3 Specific message contents

**Table 9.5.1.5.3.3-1: RESET UE POSITIONING STORED INFORMATION (step 1, Table 9.5.1.5.3.2-2)**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | FFS | FFS | |

**Table 9.5.1.5.3.3-2: SLPP REQUEST CAPABILITIES (step 2, Table 9.5.1.5.3.2-2)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. |
|---|

**Table 9.5.1.5.3.3-3: SLPP PROVIDE CAPABILITIES (step 3, Table 9.5.1.5.3.2-2)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table 9.5.1.5.3.2-2. | |
| } | | | |

**Table 9.5.1.5.3.3-4: SLPP Acknowledgement (step 4, Table 9.5.1.5.3.2-2)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table 9.5.1.5.3.2-2. | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 3, Table 9.5.1.5.3.2-2. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.1.5.3.3-5: SLPP PROVIDE ASSISTANCE DATA (step 5, Table 9.5.1.5.3.2-2)**

| Derivation Path: Table 8.5-2 with UE-only mode condition. |
|---|

**Table 9.5.1.5.3.3-6: SLPP REQUEST LOCATION INFORMATION (step 6, Table 9.5.1.5.3.2-2)**

| Derivation Path: Table 8.5-3 with UE-only mode condition. |
|---|

**Table 9.5.1.5.3.3-7: SLPP Abort (step 7, Table 9.5.1.5.3.2-2)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 6, Table 9.5.1.5.3.2-2. | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table 9.5.1.5.3.2-2. | |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| abort SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| abort SEQUENCE { | | | |
| commonIEsAbort SEQUENCE { | | | |
| abortCause | undefined | | |
| } | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.1.5.3.3-8: SLPP PROVIDE CAPABILITIES (step 3, Table Table 9.5.1.5.3.2-2)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table 9.5.1.5.3.2-2. | |
| } | | | |

## 9.5.1.6 SLPP Abort between UEs (Target UE is out of coverage)

### 9.5.1.6.1 Test Purpose (TP)

(1)

```
with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives a SLPP Abort message carrying the transaction ID of an on-going procedure }
    then { UE aborts the on-going procedure }
}
```

#### 9.5.1.6.2 Conformance requirements

As defined in clause 9.5.1.4.2.

#### 9.5.1.6.3 Test description

##### 9.5.1.6.3.1 Pre-test conditions

System Simulator:

NR-SS-UE

- Sub-test 33: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.14.
- Sub-test 34: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.15.
- Sub-test 35: NR-SS-UE1, NR-SS-UE2 and NR-SS-UE3 (if required) as specified in clause 8.2.16.
- Sub-test 36: NR-SS-UE1 and NR-SS-UE2 (if required) as specified in clause 8.2.17.
- All NR-SS-UEs use GNSS as the synchronization reference source and operate as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.

**Table 9.5.1.6.3.1-1: UE/ USIM configuration**

| USIM field | Priority | Value | Access Technology Identifier |
|---|---|---|---|
| $EF_{UST}$ | | Service n°119 (V2X) supported | |
| $EF_{VST}$ | | As per TS 38.508-1 [30] clause 4.8.3.3.3 | |
| $EF_{V2XP_PC5}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |

- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4 except for SL-PreconfigurationNR with SLPP condition.

**Table 9.5.1.6.3.1-2: UE / USIM configuration**

| USIM field | Priority | Value | Access Technology Identifier |
|---|---|---|---|
| $EF_{UST}$ | | Service n°139 5G ProSe is "available" | |
| $EF_{5G_PROSE_ST}$ | | Service n°1, n°2 and n°4 are "available" | |
| $EF_{5G_PROSE_DD}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |
| $EF_{5G_PROSE_DC}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |
| $EF_{5G_PROSE_RU}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |

- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

- FFS

#### 9.5.1.6.3.2 Test procedure sequence

As defined in clause 9.5.1.5.3.2.

#### 9.5.1.6.3.3 Specific message contents

As defined in clause 9.5.1.6.3.3.

## 9.5.2 SLPP Transport

### 9.5.2.1 SLPP Duplicated Message between LMF and UE

#### 9.5.2.1.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives an SLPP message carrying the same sequence number as that last received for the
associated location session }
    then { UE discards the SLPP message }
}
```

#### 9.5.2.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 4.3.2. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 4.3.2]

A sender shall include a sequence number in all SLPP messages sent for a particular location session. The sequence number shall be distinct for different SLPP messages sent by the same endpoint for the same endpoint in the same location session (e.g., may start at zero in the first SLPP message and increase monotonically in each succeeding SLPP message). Sequence numbers used in the messages transmitted from different endpoints or for different endpoints are independent (e.g., can be the same).

A receiver shall record the most recent received sequence number for each pair of endpoints of each location session. If a message is received carrying the same sequence number as that last received for the same pair of endpoints and the associated location session, it shall be discarded. Otherwise (i.e., if the sequence number is different or the sequence number is same but for different pair of endpoints), the message shall be processed.

Sending and receiving sequence numbers shall be deleted in a server when the associated location session is terminated and shall be deleted in the UE(s) when there has been no activity for a particular location session for 10 minutes.

#### 9.5.2.1.3 Test description

##### 9.5.2.1.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

-

##### 9.5.2.1.3.2 Test procedure sequence

**Table 9.5.2.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an SLPP message of type Request Capabilities including a sequence number. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 2 | Immediately after step 1, the SS sends the same SLPP message as in step 1. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 3 | The UE sends an SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 4 | IF the UE SLPP message at step 3 includes an acknowledgment request THEN SS sends an SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |
| 5 | The SS waits for 10 seconds to ensure the UE does not send another SLPP message of type Provide Capabilities with the same transaction ID as received in step 1 or 2. | | | 1 | P |

9.5.2.1.3.3 Specific message contents

**Table 9.5.2.1.3.3-1: DLInformationTransfer (steps 1, 2 and 4, Table 9.5.2.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.1.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.1.3.3-2: DL NAS TRANSPORT (steps 1, 2 and 4, Table 9.5.2.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Steps 1 and 2:** Set according to Table 9.5.2.1.3.3-3 | SLPP Request Capabilities | |
| | **Step 4:** Set according to Table 9.5.2.1.3.3-4 | SLPP Acknowledgement | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.2.1.3.3-3: SLPP Request Capabilities (steps 1 and 2, Table 9.5.2.1.3.2-1)**

| Derivation Path: Table 8.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.5-1 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sequenceNumber | 0 | | |
| } | | | |

**Table 9.5.2.1.3.3-4: SLPP Acknowledgement (step 4, Table 9.5.1.1.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 3, Table 9.5.2.1.3.2-1. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.2.1.3.3-5: ULInformationTransfer (step 3, Table 9.5.2.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.1.3.3-6 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.1.3.3-6: UL NAS TRANSPORT (step 3, Table 9.5.2.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table 9.5.1.1.3.3-6 | SLPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.5.2.1.3.2-1) | |

## 9.5.2.2 SLPP Duplicated Message between UEs (Target UE is in coverage)

### 9.5.2.2.1 Test Purpose (TP)

(1)

```
with { Target UE is in coverage and having established PC5 RRC connection with peer UE on unicast
sidelink }
ensure that {
  when { UE receives an SLPP message carrying the same sequence number as that last received for the
associated location session}
    then { UE discards the SLPP message }
}
```

### 9.5.2.2.2 Conformance requirements

As defined in clause 9.5.2.1.2.

### 9.5.2.2.3 Test description

#### 9.5.2.2.3.1 Pre-test conditions

System Simulator:

SS-NW

- NR Cell 1.
- System information combination NR-14 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.

- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 3N-B RRC_CONNECTED_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

-

### 9.5.2.2.3.2 Test procedure sequence

**Table 9.5.2.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The NR-SS-UE 1 sends an SLPP message of type Request Capabilities including a sequence number. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 2 | Immediately after step 1, the NR-SS-UE 1 sends the same SLPP message as in step 1. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 3 | The UE sends an SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 4 | IF<br>the UE SLPP message at step 3 includes an acknowledgment request<br>THEN<br>SS sends an SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |
| 5 | The SS waits for 10 minutes to ensure the UE does not send another SLPP message of type Provide Capabilities with the same transaction ID as received in step 1 or 2. | | | 1 | P |

9.5.2.2.3.3 Specific message contents

**Table 9.5.2.2.3.3-1: SLPP REQUEST CAPABILITIES (steps 1 and 2, Table 9.5.2.2.3.2-1)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sequenceNumber | 0 | | |
| } | | | |

**Table 9.5.2.2.3.3-2: SLPP PROVIDE CAPABILITIES (step 3, Table 9.5.2.2.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sessionID | Present | Contains the same value of the transactionID field in step 1, Table 9.5.2.2.3.2-1. | |
| } | | | |

**Table 9.5.2.2.3.3-3: SLPP Acknowledgement (step 4, Table 9.5.2.2.3.2-1)**

| Derivation Path: Table 9.5.2.1.3.3-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.2.1.3.3-4 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | '00 00 01 00'H | Application Layer ID in initiating UE side | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Present | Contains the same value of the transactionID field in step 1, Table 9.5.2.2.3.2-1. | |
| } | | | |

## 9.5.2.3 SLPP Duplicated Message between UEs (Target UE is out of coverage)

### 9.5.2.3.1 Test Purpose (TP)

(1)

```
with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives an SLPP message carrying the same sequence number as that last received for the
associated location session}
    then { UE discards the SLPP message }
}
```

### 9.5.2.3.2 Conformance requirements

As defined in clause 9.5.2.1.2.

#### 9.5.2.3.3 Test description

##### 9.5.2.3.3.1 Pre-test conditions

System Simulator:

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

-

##### 9.5.2.3.3.2 Test procedure sequence

As defined in clause 9.5.2.2.3.2.

##### 9.5.2.3.3.3 Specific message contents

As defined in clause 9.5.2.2.3.3.

### 9.5.2.4 SLPP Acknowledgment between LMF and UE

#### 9.5.2.4.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE receives an SLPP message carrying an acknowledgement request indicator }
    then { UE returns an SLPP acknowledgement response }
```

}

#### 9.5.2.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 4.3.3. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 4.3.3]

Each SLPP message may carry an acknowledgement request and/or an acknowledgement indicator. An SLPP message including an acknowledgement request (i.e., that include the field *ackRequested* set to TRUE) shall also include a sequence number. Upon reception of an SLPP message which includes the field *ackRequested* set to TRUE, a receiver returns an SLPP message with an acknowledgement response (i.e., that includes the field *ackIndicator* set to the same sequence number of the message being acknowledged). An acknowledgement response may contain no SLPP message body (in which case only the sequence number being acknowledged is significant); alternatively, the acknowledgement may be sent in an SLPP message along with an SLPP message body. An acknowledgement is returned for each received SLPP message that requested an acknowledgement including any duplicate(s). Once a sender receives an acknowledgement for an SLPP message, and provided any included sequence number is matching, it is permitted to send the next SLPP message. No message reordering is needed at the receiver since this stop-and-wait method of sending ensures that messages normally arrive in the correct order.

When an SLPP message is transported via a NAS SL-MO-LR request, the message does not request an acknowledgement.

#### 9.5.2.4.3 Test description

##### 9.5.2.4.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

-

9.5.2.4.3.2 Test procedure sequence

**Table 9.5.2.4.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an SLPP message of type Request Capabilities including a request for acknowledgement. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 2 Option 1 | Option 1: The UE sends an acknowledgement along with an SLPP message of type Provide Capabilities. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES, incl. acknowledgement response) | 1 | P |
| 2 Option 2 | Option 2: The UE sends an SLPP Acknowledgement response, followed by an SLPP message of type Provide Capabilities. | --><br>--> | *ULInformationTransfer* (SLPP ACKNOWLEDGEMENT)<br>*ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | 1 | P |
| 3 | IF the UE SLPP message at step 2 includes an acknowledgment request THEN SS sends an SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |

9.5.2.4.3.3 Specific message contents

**Table 9.5.2.4.3.3-1: DLInformationTransfer (steps 1 and 3, Table 9.5.2.4.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.4.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.4.3.3-2: DL NAS TRANSPORT (steps 1 and 3, Table 9.5.2.4.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 9.5.2.4.3.3-3 | SLPP Request Capabilities | |
| | **Step 3:** Set according to Table 9.5.2.4.3.3-4 | SLPP Acknowledgement | |

| | | | |
|---|---|---|---|
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.2.4.3.3-3: SLPP REQUEST CAPABILITIES (step 1, Table 9.5.2.4.3.2-1)**

| Derivation Path: Table 8.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.5-1 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sequenceNumber | 0 | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | True | | |
| ackIndicator | (0..255) | | |
| } | | | |
| } | | | |

**Table 9.5.2.4.3.3-4: SLPP Acknowledgement (step 3, Table 9.5.2.4.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 9.5.2.4.3.2-1. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.2.4.3.3-5: ULInformationTransfer (step 2, Table 9.5.2.4.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.4.3.3-6 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.4.3.3-6: UL NAS TRANSPORT (step 2, Table 9.5.2.4.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:**<br>Set according to Table Table 9.5.2.4.3.3-7 | SLPP Provide Capabilities | |
| | **Step 2 (Option 2)** :<br>Set according to Table Table 9.5.2.4.3.3-8 | SLPP Acknowledgement | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.5.2.4.3.2-1) | |

**Table 9.5.2.4.3.3-7: SLPP Provide Capabilities (step 2, Table 9.5.2.4.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 1, Table 9.5.2.4.3.2-1. | |
| endTransaction | True | | |
| sequenceNumber | (0..255) | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | Present, or not present.<br>Present for Option 1. | | |
| ackRequested | TRUE or FALSE | | |
| ackIndicator | 0 (Option 1)<br>Not present (Option 2) | | |
| } | | | |
| } | | | |

**Table 9.5.2.4.3.3-8: SLPP Acknowledgement (step 2 – Option 2, Table 9.5.2.4.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | Not present | | |
| endTransaction | False | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | 0 | | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

## 9.5.2.5 SLPP Acknowledgment between UEs (Target UE is in coverage)

### 9.5.2.5.1 Test Purpose (TP)

(1)

```
with { Target UE is in coverage and having established PC5 RRC connection with peer UE on unicast
sidelink }
ensure that {
  when { UE receives an SLPP message carrying an acknowledgement request indicator }
    then { UE returns an SLPP acknowledgement response }
            }
```

### 9.5.2.5.2 Conformance requirements

As defined in clause 9.5.2.4.2.

### 9.5.2.5.3 Test description

#### 9.5.2.5.3.1 Pre-test conditions

System Simulator:

SS-NW

- NR Cell 1.
- System information combination NR-14 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.

- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.

- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.

- UE is synchronised on GNSS.

Preamble:

- The UE is in state 3N-B RRC_CONNECTED_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:

  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;

- for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.Related PICS/PIXIT Statements:

-

9.5.2.5.3.2 Test procedure sequence

**Table 9.5.2.5.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The NR-SS-UE 1 sends an SLPP message of type Request Capabilities including a request for acknowledgement. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 2<br>Option 1 | Option 1:<br>The UE sends an acknowledgement along with an SLPP message of type Provide Capabilities. | --> | SLPP PROVIDE CAPABILITIES, incl. acknowledgement response | 1 | P |
| 2<br>Option 2 | Option 2:<br>The UE sends an SLPP Acknowledgement response, followed by an SLPP message of type Provide Capabilities. | --><br>--> | SLPP ACKNOWLEDGEMENT<br>SLPP PROVIDE CAPABILITIES | 1 | P |
| 3 | IF<br>the UE SLPP message at step 2 includes an acknowledgment request<br>THEN<br>NR-SS-UE 1 sends an SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |

9.5.2.5.3.3 Specific message contents

**Table 9.5.2.5.3.3-1: SLPP REQUEST CAPABILITIES (step 1, Table 9.5.2.5.3.2-1)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.5-1 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sequenceNumber | 0 | | |
| sessionID | Present. Any value acceptable. | OCTET STRING (SIZE (6)) | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | True | | |
| ackIndicator | (0..255) | | |
| } | | | |
| } | | | |

**Table 9.5.2.5.3.3-2: SLPP Provide Capabilities (step 2, Table 9.5.2.5.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 1, Table 9.5.2.5.3.2-1. | |
| endTransaction | True | | |
| sequenceNumber | (0..255) | | |
| sessionID | Present. | Contains the same value of the sessionID field in step 1, Table 9.5.2.5.3.2-1. | |
| acknowledgement SEQUENCE { | Present, or not present. Present for Option 1. | | |
| ackRequested | TRUE or FALSE | | |
| ackIndicator | 0 (Option 1)<br>Not present (Option 2) | | |
| } | | | |
| } | | | |

**Table 9.5.2.5.3.3-3: SLPP Acknowledgement (step 2 – Option 2, Table 9.5.2.5.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | Not present | | |
| endTransaction | False | | |
| sequenceNumber | Not present | | |
| sessionID | Present. | Contains the same value of the sessionID field in step 1, Table 9.5.2.5.3.2-1. | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | 0 | | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.2.5.3.3-4: SLPP Acknowledgement (step 3, Table 9.5.2.5.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | '00 00 01 00'H | Application Layer ID in initiating UE side | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Present. Any value acceptable. | OCTET STRING (SIZE (6)) | |
| sessionID | Present. | Contains the same value of the sessionID field in step 2, Table 9.5.2.5.3.2-1. | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in step 2, Table 9.5.2.5.3.2-1. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

## 9.5.2.6 SLPP Acknowledgment between UEs (Target UE is out of coverage)

### 9.5.2.6.1 Test Purpose (TP)

(1)

```
with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives an SLPP message carrying an acknowledgement request indicator }
    then { UE returns an SLPP acknowledgement response }
          }
```

### 9.5.2.6.2 Conformance requirements

As defined in clause 9.5.2.4.2.

### 9.5.2.6.3 Test description

#### 9.5.2.6.3.1 Pre-test conditions

System Simulator:

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.

- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

-

#### 9.5.2.6.3.2 Test procedure sequence

As defined in clause 9.5.2.5.3.2.

#### 9.5.2.6.3.3 Specific message contents

As defined in clause 9.5.2.5.3.3.

### 9.5.2.7 SLPP Retransmission between LMF and UE

#### 9.5.2.7.1 Test Purpose (TP)

(1)

```
with { a NAS signalling connection existing }
ensure that {
  when { UE does not receive an SLPP acknowledgement for an SLPP message which requires
acknowledgement }
    then { UE retransmits the SLPP message up to three times. If still unacknowledged after that,
the UE aborts all SLPP activity for the associated session}
}
```

#### 9.5.2.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 4.3.4.1 and clause 4.3.4.2. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 4.3.4.1]

This capability builds on the acknowledgement and duplicate detection capabilities. When an SLPP message which requires acknowledgement is sent and not acknowledged, it is resent by the sender following a timeout period up to three times. If still unacknowledged after that, the sender aborts all SLPP activity for this Endpoint. The timeout period is determined by the sender implementation but shall not be less than a minimum value of 250 ms.

[TS 38.355, clause 4.3.4.2]

Figure 4.3.4.2-1 shows the procedure related to retransmission when combined with acknowledgement and duplicate detection.

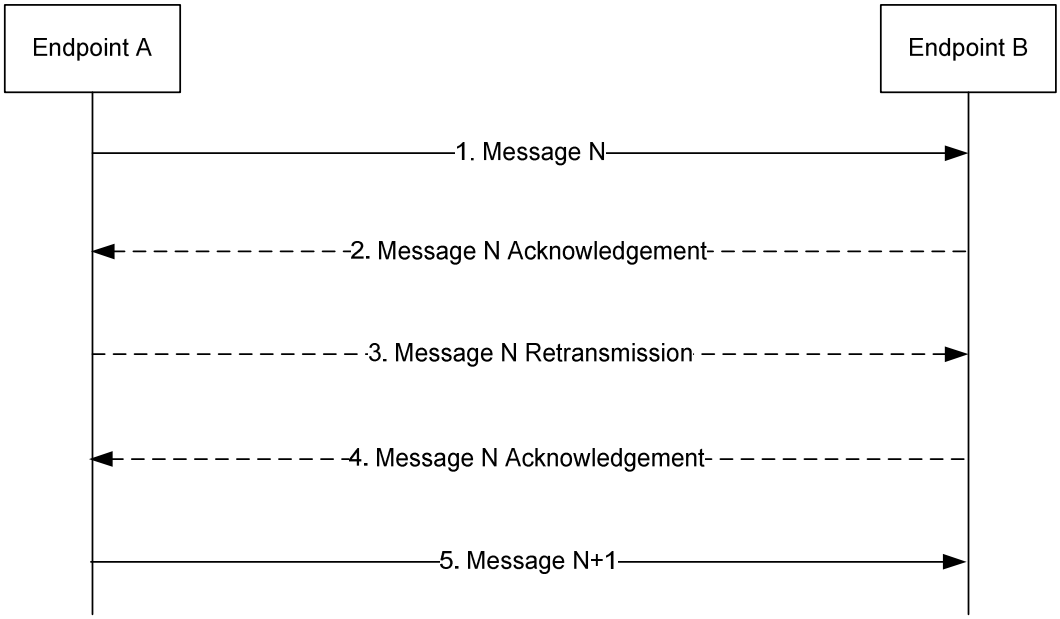



**Figure 4.3.4.2-1: SLPP Retransmission procedure**

1. Endpoint A sends an SLPP message *N* to Endpoint B for a particular location session and includes a request for acknowledgement along with a sequence number.
2. If SLPP message *N* is received and Endpoint B is able to decode the *ackRequested* value and sequence number (regardless of whether the message body can be correctly decoded), Endpoint B shall return an acknowledgement for message *N*. If the acknowledgement is received by Endpoint A (such that the acknowledged message can be identified and sequence numbers are matching), Endpoint A skips steps 3 and 4.
3. If the acknowledgement in step 2 is not received after a timeout period, Endpoint A shall retransmit SLPP message *N* and shall include the same sequence number as in step 1.
4. If SLPP message *N* in step 3 is received and Endpoint B is able to decode the *ackRequested* value and sequence number (regardless of whether the message body can be correctly decoded and whether or not the message is considered a duplicate), Endpoint B shall return an acknowledgement. Steps 3 may be repeated one or more times if the acknowledgement in step 4 is not received after a timeout period by Endpoint A. If the acknowledgement in step 4 is still not received after sending three retransmissions, Endpoint A shall abort all procedures and activity associated with SLPP support for this Endpoint B.
5. Once an acknowledgement in step 2 or step 4 is received, Endpoint A sends the next SLPP message *N+1* for the location session to Endpoint B when this message is available.

### 9.5.2.7.3 Test description

#### 9.5.2.7.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

- UE supporting sending of acknowledgement request in SLPP Provide Capabilities message.

9.5.2.7.3.2 Test procedure sequence

**Table 9.5.2.7.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an SLPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 2 | The UE sends an SLPP message of type Provide Capabilities including a request for acknowledgement along with a sequence number. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 3 | SS does not send an acknowledgement | | | - | - |
| 4 | After an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | 1 | P |
| 5 | SS does not send an acknowledgement | | | - | - |
| 6 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 7 | SS does not send an acknowledgement | | | - | - |
| 8 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 9 | SS does not send an acknowledgement | | | | |
| 10 | UE aborts all procedures and activity associated with SLPP support for the location session.<br>SS waits for 10 seconds to ensure the UE does not send another SLPP message. | | | 1 | P |

9.5.2.7.3.3 Specific message contents

**Table 9.5.2.7.3.3-1: DLInformationTransfer (step 1, Table 9.5.2.7.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.7.3.3-2 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.7.3.3-2: DL NAS TRANSPORT (step 1, Table 9.5.2.7.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 1:** Set according to Table 8.5-1 | SLPP Request Capabilities | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.2.7.3.3-3: ULInformationTransfer (steps 2, 4, 6 and 8, Table 9.5.2.7.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.2.7.3.3-4 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.2.7.3.3-4: UL NAS TRANSPORT (steps 2, 4, 6 and 8, Table 9.5.2.7.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | Set according to Table Table 9.5.2.7.3.3-5 | SLPP Provide Capabilities | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 1 Table 9.5.2.7.3.2-1) | |

**Table 9.5.2.7.3.3-5: SLPP Provide Capabilities (steps 2, 4, 6 and 8, Table 9.5.2.7.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 1, Table 9.5.2.7.3.2-1. | |
| endTransaction | True | | |
| sequenceNumber | (0..255) | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| } | | | |

## 9.5.2.8 SLPP Retransmission between UEs (Target UE is in coverage)

### 9.5.2.8.1 Test Purpose (TP)

(1)

```
with { Target UE is in coverage and having established PC5 RRC connection with peer UE on unicast
sidelink }
ensure that {
  when { UE does not receive an SLPP acknowledgement for an SLPP message which requires
acknowledgement }
    then { UE retransmits the SLPP message up to three times. If still unacknowledged after that,
the UE aborts all SLPP activity for the associated session}
```

}

#### 9.5.2.8.2 Conformance requirements

As defined in clause 9.5.2.7.2.

#### 9.5.2.8.3 Test description

##### 9.5.2.8.3.1 Pre-test conditions

SS-NW

- NR Cell 1.
- System information combination NR-14 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 3N-B RRC_CONNECTED_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
- for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.Related PICS/PIXIT Statements:
- UE supporting sending of acknowledgement request in SLPP Provide Capabilities message.

9.5.2.8.3.2 Test procedure sequence

**Table 9.5.2.8.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| 1 | The SS sends an SLPP message of type Request Capabilities. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 2 | The UE sends an SLPP message of type Provide Capabilities including a request for acknowledgement along with a sequence number. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 3 | SS does not send an acknowledgement | | | - | - |
| 4 | After an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | SLPP PROVIDE CAPABILITIES | 1 | P |
| 5 | SS does not send an acknowledgement | | | - | - |
| 6 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 7 | SS does not send an acknowledgement | | | - | - |
| 8 | The UE either proceeds directly to step 10 or after an implementation specific timeout period, the UE retransmits the SLPP message from step 2 and includes the same sequence number as in step 2. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 9 | SS does not send an acknowledgement | | | | |
| 10 | UE aborts all procedures and activity associated with SLPP support for the location session.<br>SS waits for 10 seconds to ensure the UE does not send another SLPP message. | | | 1 | P |

9.5.2.8.3.3 Specific message contents

**Table 9.5.2.8.3.3-1: SLPP REQUEST CAPABILITIES (step 1, Table 9.5.2.8.3.2-1)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. |
|---|

**Table 9.5.2.8.3.3-2: SLPP Provide Capabilities (steps 2, 4, 6 and 8, Table 9.5.2.8.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Application Layer ID in target UE side | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 1, Table 9.5.2.8.3.2-1. | |
| endTransaction | True | | |
| sequenceNumber | (0..255) | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | TRUE | | |
| ackIndicator | Not present | | |
| } | | | |
| } | | | |

## 9.5.2.9 SLPP Retransmission between UEs (Target UE is out of coverage)

### 9.5.2.9.1 Test Purpose (TP)

(1)

```
with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE does not receive an SLPP acknowledgement for an SLPP message which requires
acknowledgement }
    then { UE retransmits the SLPP message up to three times. If still unacknowledged after that,
the UE aborts all SLPP activity for the associated session}
}
```

### 9.5.2.9.2 Conformance requirements

As defined in clause 9.5.2.7.2.

### 9.5.2.9.3 Test description

#### 9.5.2.9.3.1 Pre-test conditions

NR-SS-UE

- NR-SS-UE1 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- NR-SS-UE1 uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

- UE supporting sending of acknowledgement request in SLPP Provide Capabilities message.

#### 9.5.2.9.3.2 Test procedure sequence

As defined in clause 9.5.2.8.3.2.

#### 9.5.2.9.3.3 Specific message contents

As defined in clause 9.5.2.8.3.3.

## 9.5.3 SLPP Error Handling

### 9.5.3.1 SLPP Requested Method not Supported between LMF and UE - UE-Assisted

#### 9.5.3.1.1 Test Purpose (TP)

(1)

```
with { a UE supporting at least one of target UE-assisted SL-RTT, target UE-assisted SL-AOA, target
UE-assisted SL-TDOA or target UE-assisted SL-TOA but not all of them }
and with { a NAS signalling connection existing }
ensure that {
  when { UE receives a SLPP message requesting at least one positioning method not supported }
    then { the UE provides location information for the supported methods}
}
```

#### 9.5.3.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TS 38.355 clauses 5.3.5 and clause 5.4.3. Unless otherwise stated these are Rel-18 requirements.

[TS 38.355, clause 5.3.5]

Upon receiving a *RequestLocationInformation* message, Endpoint A shall:

1> if the requested information is compatible with Endpoint A's capabilities and configuration:

2> include the requested information in a *ProvideLocationInformation* message;

2> set the field *sessionID* in the response message to the same value as the field *sessionID* in the received message if received;

2> set the field *transactionID* in the response to the same value as the field *transactionID* in the received message;

2> deliver the *ProvideLocationInformation* message to lower layers for transmission.

1> else if one or more positioning methods are included that Endpoint A does not support:

2> continue to process the message as if it contained only information for the supported positioning methods;

2> handle the signaling content of the unsupported positioning methods by SLPP error detection as in 5.4.3.

[TS 38.355, clause 5.4.3]

Upon receiving any SLPP message, the receiving endpoint shall attempt to decode the message and verify the presence of any errors and:

…

1> if the message type is an SLPP *RequestAssistanceData* or *RequestLocationInformation* and some or all of the requested information is not supported:

2> return any information that can be provided in a normal response, which includes indications on other information that is not supported.

#### 9.5.3.1.3 Test description

##### 9.5.3.1.3.1 Pre-test conditions

System Simulator:

- NR Cell 1.
- System information combination NR-1 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1, NR-SS-UE2 (if required) and NR-SS-UE3 (if required) as specified in clause 8.2 operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- All NR-SS-UEs uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- None.

Preamble:

- The UE is in state 3N-A as defined in TS 38.508-1 [30], subclause 4.4A on NR Cell 1.

Related PICS/PIXIT Statements:

- FFS

9.5.3.1.3.2 Test procedure sequence

**Table 9.5.3.1.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| **1** | The SS sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 2 | The SS sends a SLPP message of type Request Capabilities. | <-- | *DLInformationTransfer* (SLPP REQUEST CAPABILITIES) | - | - |
| 3 | The UE sends a SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | *ULInformationTransfer* (SLPP PROVIDE CAPABILITIES) | - | - |
| 4 | IF the UE SLPP message at step 3 includes an acknowledgment request THEN SS sends a SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |
| 5 | The SS sends a SLPP message of type Provide Assistance Data containing the data for all supported positioning methods. | <-- | *DLInformationTransfer* (SLPP PROVIDE ASSISTANCE DATA) | - | - |
| 6 | The SS sends a SLPP message of type Request Location Information including all specified positioning methods. | <-- | *DLInformationTransfer* (SLPP REQUEST LOCATION INFORMATION) | - | - |
| 7 | The UE sends a SLPP message of type Provide Location Information including information for the supported method(s). | --> | *ULInformationTransfer* (SLPP PROVIDE LOCATION INFORMATION) | 1 | P |
| 8 | IF the UE SLPP message at step 7 includes an acknowledgment request THEN SS sends a SLPP Acknowledgement response. | <-- | *DLInformationTransfer* (SLPP ACKNOWLEDGEMENT) | - | - |

9.5.3.1.3.3 Specific message contents

**Table 9.5.3.1.3.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | FFS | FFS | |

**Table 9.5.3.1.3.3-2: DL InformationTransfer (steps 2, 4, 5, 6 and 8, Table 9.5.3.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DLInformationTransfer ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | | | |
| criticalExtensions CHOICE { | | | |
| dlInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.3.1.3.3-3 | DL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.1.3.3-3: DL NAS TRANSPORT (steps 2, 4, 5, 6 and 8, Table 9.5.3.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.11.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | Downlink generic NAS transport | |
| DL NAS TRANSPORT message identity | 01101000 | DL NAS transport | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 2:** Set according to Table 8.5-1 | SLPP Request Capabilities | |
| | **Steps 4 and 8:** Set according to Table Table 9.5.3.1.3.3-4 | SLPP Acknowledgement | |
| | **Step 5:** Set according to Table 9.5.3.1.3.3-5 | SLPP Provide Assistance Data | |
| | **Step 6:** Set according to Table 9.5.3.1.3.3-6 | SLPP Request Location Information | |
| Additional information | Present | Routing Identifier/Correlation ID | |

**Table 9.5.3.1.3.3-4: SLPP Acknowledgement (steps 4 and 8, Table 9.5.3.1.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in 1 or 7, Table 9.5.1.4.3.2-2. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.3.1.3.3-5: SLPP PROVIDE ASSISTANCE DATA (step 5, Table 9.5.3.1.3.2-1)**

| Derivation Path: Table 8.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.5-2 with the following exceptions: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| sl-AoA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-AoA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_AoA |
| sl-RTT-ProvideAssistanceData | Present if UE supports target UE-assisted SL-RTT. As defined in clause 8.5.1 | | pc_Target_UEA_SL_RTT |
| sl-TDOA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-TDOA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_TDOA |
| sl-TOA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-TOA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_TOA |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.1.3.3-6: SLPP REQUEST LOCATION INFORMATION (step 6, Table 9.5.3.1.3.2-1)**

| Derivation Path: Table 8.5-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-2 with the following exceptions: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| sl-AoA-RequestLocationInformation | Present. | OCTET STRING containing SL-AoA-RequestLocationInformation as defined in Table 8.5-4 | |
| sl-RTT-RequestLocationInformation | Present | OCTET STRING containing SL-RTT-RequestLocationInformation as defined in Table 8.5-5 | |
| sl-TDOA-RequestLocationInformation | Present | OCTET STRING containing SL-TDOA-RequestLocationInformation as defined in Table 8.5-6 | |
| sl-TOA-RequestLocationInformation | Present | OCTET STRING containing SL-TOA-RequestLocationInformation as defined in Table 8.5-7 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.1.3.3-7: ULInformationTransfer (steps 3 and 7, Table 9.5.3.1.3.2-1)**

| Derivation Path: 38.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULInformationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| ulInformationTransfer SEQUENCE { | | | |
| dedicatedNAS-Message OCTET STRING | Set according to Table 9.5.3.1.3.3-8 | UL NAS TRANSPORT | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.1.3.3-8: UL NAS TRANSPORT (steps 3 and 7, Table 9.5.3.1.3.2-1)**

| Derivation Path: 24.501 Table 8.2.10.1.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Extended Protocol discriminator | 01111110 | 5GS mobility management messages | |
| Security header type | 0000 | Plain 5GS NAS message | |
| Spare half octet | 0000 | | |
| UL NAS TRANSPORT message identity | 01100111 | UL NAS TRANSPORT | |
| Payload container type | 1100 | SLPP message container | |
| Spare half octet | 0000 | | |
| Payload container | **Step 3:** Set according to Table 9.5.1.1.3.3-6 | SLPP Provide Capabilities | |
| | **Step 7:** Set according to Table 9.5.3.1.3.3-9 | SLPP Provide Location Information | |
| Additional information | Present | The UE includes the Routing Identifier received in the Additional Information IE of the DOWNLINK GENERIC NAS TRANSPORT message (step 2 or 6 Table 9.5.1.4.3.2-2) | |

**Table 9.5.3.1.3.3-9: SLPP PROVIDE LOCATION INFORMATION (step 7, Table 9.5.3.1.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | (0..255) | | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not checked | | |
| commonSL-PRS-MethodsIEsProvideLocationInformation | Not checked | | |
| sl-AoA-ProvideLocationInformation | Present if UE supports target UE-assisted SL-AoA | OCTET STRING containing SL-AoA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-10 | pc_Target_UEA_SL_AoA |
| sl-RTT-ProvideLocationInformation SEQUENCE { | Present if UE supports target UE-assisted SL-RTT | OCTET STRING containing SL-RTT-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-11 | pc_Target_UEA_SL_RTT |
| sl-TDOA-ProvideLocationInformation SEQUENCE { | Present if UE supports target UE-assisted SL-TDOA | OCTET STRING containing SL-TDOA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-12 | pc_Target_UEA_SL_TDOA |
| sl-TOA-ProvideLocationInformation SEQUENCE { | Present if UE supports target UE-assisted SL-TOA | OCTET STRING containing SL-TOA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-13 | pc_Target_UEA_SL_TOA |
| lateNonCriticalExtension | Not checked | | |
| nonCriticalExtension | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not checked | | |
| } | | | |

**Table 9.5.3.1.3.3-10: SL-AoA-ProvideLocationInformation**

| Derivation Path: 38.355 clause 6.7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-AoA-ProvideLocationInformation ::= SEQUENCE { | | | |
| sl-AoA-SignalMeasurementInformation | Present. Any value acceptable | | |
| sl-AoA-Error | May be present | | |
| } | | | |

**Table 9.5.3.1.3.3-11: SL-RTT-ProvideLocationInformation**

| Derivation Path: 38.355 clause 6.8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-RTT-ProvideLocationInformation ::= SEQUENCE { | | | |
| sl-RTT-SignalMeasurementInformation | Present. Any value acceptable | | |
| sl-RTT-Error | May be present | | |
| } | | | |

**Table 9.5.3.1.3.3-12: SL-TDOA-ProvideLocationInformation**

| Derivation Path: 38.355 clause 6.9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TDOA-ProvideLocationInformation ::= SEQUENCE { | | | |
| sl-RSTD-ReferenceUE-Info | Present. Any value acceptable | | |
| sl-TDOA-SignalMeasurementInformation | Present. Any value acceptable | | |
| sl-TDOA-Error | May be present | | |
| } | | | |

**Table 9.5.3.1.3.3-13: SL-TOA-ProvideLocationInformation**

| Derivation Path: 38.355 clause 6.10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SL-TOA-ProvideLocationInformation ::= SEQUENCE { | | | |
| sl-TOA-SignalMeasurementInformation | Present. Any value acceptable | | |
| sl-TOA-Error | May be present | | |
| } | | | |

## 9.5.3.2 SLPP Requested Method not Supported between UEs (Target UE is in coverage) - UE-Assisted

### 9.5.3.2.1 Test Purpose (TP)

(1)

```
with { a UE supporting at least one of target UE-assisted SL-RTT, target UE-assisted SL-AOA, target
UE-assisted SL-TDOA or target UE-assisted SL-TOA but not all of them }
and with { Target UE is in coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives a SLPP message requesting at least one positioning method not supported }
    then { the UE provides location information for the supported methods}
}
```

### 9.5.3.2.2 Conformance requirements

As defined in clause 9.5.3.1.2.

### 9.5.3.2.3 Test description

#### 9.5.3.2.3.1 Pre-test conditions

System Simulator:

SS-NW

- NR Cell 1.
- System information combination NR-36 as defined in TS 38.508-1 [30] clause 4.4.3.1.3 is used in NR cell.

NR-SS-UE

- NR-SS-UE1, NR-SS-UE2 (if required) and NR-SS-UE3 (if required) operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- All NR-SS-UEs uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.
- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4.
- UE is synchronised on GNSS.

Preamble:

- The UE is in state 1N-B RRC_IDLE_with_SL and Test Mode (On) with UE test loop mode E as defined in TS 38.508-1 [30], subclause 4.4A using generic procedure parameter Sidelink (On), test mode (On) and GNSS Sync (On) as defined in TS 38.508-1 [30], subclause 4.5.1, using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

- FFS

9.5.3.2.3.2 Test procedure sequence

**Table 9.5.3.2.3.2-1: Main behaviour**

| St | Procedure | Message Sequence | | TP | Verdict |
|---|---|---|---|---|---|
| | | U - S | Message | | |
| **1** | The NR-SS-UE1 sends a RESET UE POSITIONING STORED INFORMATION message. | <-- | RESET UE POSITIONING STORED INFORMATION | - | - |
| 2 | The NR-SS-UE1 sends a SLPP message of type Request Capabilities. | <-- | SLPP REQUEST CAPABILITIES | - | - |
| 3 | The UE sends a SLPP message of type Provide Capabilities including the UE positioning capabilities. | --> | SLPP PROVIDE CAPABILITIES | - | - |
| 4 | IF the UE SLPP message at step 3 includes an acknowledgment request THEN NR-SS-UE1 sends a SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |
| 5 | The NR-SS-UE1 sends a SLPP message of type Provide Assistance Data containing the data for all supported positioning methods. | <-- | SLPP PROVIDE ASSISTANCE DATA | - | - |
| 6 | The NR-SS-UE1 sends a SLPP message of type Request Location Information including all specified positioning methods. | <-- | SLPP REQUEST LOCATION INFORMATION | - | - |
| 7 | The UE sends a SLPP message of type Provide Location Information including information for the supported method(s). | --> | SLPP PROVIDE LOCATION INFORMATION | 1 | P |
| 8 | IF the UE SLPP message at step 7 includes an acknowledgment request THEN NR-SS-UE1 sends a SLPP Acknowledgement response. | <-- | SLPP ACKNOWLEDGEMENT | - | - |

9.5.3.2.3.3 Specific message contents

**Table 9.5.3.2.3.3-1: RESET UE POSITIONING STORED INFORMATION**

| Derivation Path: 38.509 clause 6.6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UE Positioning Technology | FFS | FFS | |

**Table 9.5.3.2.3.3-2: SLPP REQUEST CAPABILITIES (step 2, Table 9.5.3.2.3.2-1)**

| Derivation Path: Table 8.5-1 with UE-only mode condition. |
|---|

**Table 9.5.3.2.3.3-3: SLPP PROVIDE CAPABILITIES (step 3, Table 9.5.3.2.3.2-1)**

| Derivation Path: Table 9.5.1.1.3.3-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 9.5.1.1.3.3-6 with the following exception: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table9.5.3.2.3.1-2. | |
| } | | | |

**Table 9.5.3.2.3.3-4: SLPP Acknowledgement (steps 4 and 8, Table 9.5.3.2.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Not present | | |
| transactionID | Not present | | |
| endTransaction | TRUE | | |
| sequenceNumber | Not present | | |
| sessionID | Not present | | |
| acknowledgement SEQUENCE { | | | |
| ackRequested | FALSE | | |
| ackIndicator | (0..255) | Contains the same value of the sequenceNumber field in 3 or 7, Table 9.5.1.4.3.2-2. | |
| } | | | |
| slpp-MessageBody | Not present | | |
| nonCriticalExtension | Not present | | |
| } | | | |

**Table 9.5.3.2.3.3-5: SLPP PROVIDE ASSISTANCE DATA (step 5, Table 9.5.3.2.3.2-1)**

| Derivation Path: Table 8.5-2 with UE-only mode condition. | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.5-2 with the following exceptions: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideAssistanceData SEQUENCE { | | | |
| sl-AoA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-AoA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_AoA |
| sl-RTT-ProvideAssistanceData | Present if UE supports target UE-assisted SL-RTT. As defined in clause 8.5.1 | | pc_Target_UEA_SL_RTT |
| sl-TDOA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-TDOA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_TDOA |
| sl-TOA-ProvideAssistanceData | Present if UE supports target UE-assisted SL-TOA. As defined in clause 8.5.1 | | pc_Target_UEA_SL_TOA |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.2.3.3-6: SLPP REQUEST LOCATION INFORMATION (step 6, Table 9.5.3.2.3.2-1)**

| Derivation Path: Table 8.5-3 with UE-only mode condition. | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| As defined in Table 8.4-2 with the following exceptions: | | | |
| SLPP-Message ::= SEQUENCE { | | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| requestLocationInformation SEQUENCE { | | | |
| sl-AoA-RequestLocationInformation | Present. | OCTET STRING containing SL-AoA-RequestLocationInformation as defined in Table 8.5-4 | |
| sl-RTT-RequestLocationInformation | Present | OCTET STRING containing SL-RTT-RequestLocationInformation as defined in Table 8.5-5 | |
| sl-TDOA-RequestLocationInformation | Present | OCTET STRING containing SL-TDOA-RequestLocationInformation as defined in Table 8.5-6 | |
| sl-TOA-RequestLocationInformation | Present | OCTET STRING containing SL-TOA-RequestLocationInformation as defined in Table 8.5-7 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.3.2.3.3-7: SLPP PROVIDE LOCATION INFORMATION (step 7, Table 9.5.3.2.3.2-1)**

| Derivation Path: 38.355 clause 6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SLPP-Message ::= SEQUENCE { | | | |
| applicationLayerID | Present. Any value acceptable | | |
| transactionID | (0..255) | Contains the same value of the transactionID field in step 6, Table 9.5.3.2.3.2-1. | |
| endTransaction | FALSE | | |
| sequenceNumber | Not present | | |
| sessionID | Present | Contains the same value of the transactionID field in step 2, Table 9.5.3.2.3.2-1. | |
| acknowledgement | Not present | | |
| slpp-MessageBody CHOICE { | | | |
| c1 CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| provideLocationInformation SEQUENCE { | | | |
| commonIEsProvideLocationInformation | Not checked | | |
| commonSL-PRS-MethodsIEsProvideLocationInformation | Not checked | | |
| sl-AoA-ProvideLocationInformation | Present if UE supports target UE-assisted SL-AoA | OCTET STRING containing SL-AoA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-10 | pc_Target_UEA_SL_AoA |
| sl-RTT-ProvideLocationInformation | Present if UE supports target UE-assisted SL-RTT | OCTET STRING containing SL-RTT-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-11 | pc_Target_UEA_SL_RTT |
| sl-TDOA-ProvideLocationInformation | Present if UE supports target UE-assisted SL-TDOA | OCTET STRING containing SL-TDOA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-12 | pc_Target_UEA_SL_TDOA |
| sl-TOA-ProvideLocationInformation | Present if UE supports target UE-assisted SL-TOA | OCTET STRING containing SL-TOA-ProvideLocationInformation as defined in Table 9.5.3.1.3.3-13 | pc_Target_UEA_SL_TOA |
| lateNonCriticalExtension | Not checked | | |
| nonCriticalExtension | Not checked | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| nonCriticalExtension | Not checked | | |
| } | | | |

## 9.5.3.3 SLPP Requested Method not Supported between UEs (Target UE is out of coverage) - UE-Assisted

### 9.5.3.3.1 Test Purpose (TP)

(1)

```
with { a UE supporting at least one of target UE-assisted SL-RTT, target UE-assisted SL-AOA, target
UE-assisted SL-TDOA or target UE-assisted SL-TOA but not all of them }
and with { Target UE is out of coverage and having established PC5 RRC connection with peer UE on
unicast sidelink }
ensure that {
  when { UE receives a SLPP message requesting at least one positioning method not supported }
    then { the UE provides location information for the supported methods}
}
```

### 9.5.3.3.2 Conformance requirements

As defined in clause 9.5.3.1.2.

### 9.5.3.3.3 Test description

#### 9.5.3.3.3.1 Pre-test conditions

System Simulator:

NR-SS-UE

- NR-SS-UE1, NR-SS-UE2 (if required) and NR-SS-UE3 (if required) operating as NR sidelink communication device on the resources (i.e. the frequency included in pre-configuration) that UE is expected to use for transmission and reception via PC5 interface.
- All NR-SS-UEs uses GNSS as the synchronization reference source.

GNSS simulator

- The GNSS simulator is started and configured for Scenario #1.

UE:

- UE is authorised to perform NR sidelink communication.
- For V2X capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.3.

**Table 9.5.3.3.3.1-1: UE/ USIM configuration**

| USIM field | Priority | Value | Access Technology Identifier |
|---|---|---|---|
| $EF_{UST}$ | | Service n°119 (V2X) supported | |
| $EF_{VST}$ | | As per TS 38.508-1 [30] clause 4.8.3.3.3 | |
| $EF_{V2XP_PC5}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |

- For 5G ProSe capable UE, the UE is equipped with a USIM containing default values as per TS 38.508-1 [30] clause 4.8.3.3.4 except for SL-PreconfigurationNR with SLPP condition.

**Table 9.5.3.3.3.1-2: UE / USIM configuration**

| USIM field | Priority | Value | Access Technology Identifier |
|---|---|---|---|
| $EF_{UST}$ | | Service n°139 5G ProSe is "available" | |
| $EF_{5G_PROSE_ST}$ | | Service n°1, n°2 and n°4 are "available" | |
| $EF_{5G_PROSE_DD}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |
| $EF_{5G_PROSE_DC}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |
| $EF_{5G_PROSE_RU}$ | | SL-PreconfigurationNR field as defined in TS 38.508-1 [30], table 4.10.1-1 with SLPP condition | |

- UE is synchronised on GNSS.

Preamble:

- The UE is in state 4-A and Test Mode (*On*), Test Loop Function (*Off*) as defined in TS 38.508-1 [30], Table 4.5.7.2-1 using generic procedure parameter Sidelink (*On*),Cast Type (*Unicast*), UE initiating unicast mode NR sidelink communication, GNSS Sync (*On*), using the following procedures:
  - for V2X capable UE, using UE initiated unicast mode NR sidelink communication procedure in subclause 4.9.22;
  - for 5G ProSe capable UE, using UE initiated unicast mode ProSe Direct communication procedure in subclause 4.9.39.

Related PICS/PIXIT Statements:

- FFS

### 9.5.3.3.3.2 Test procedure sequence

As defined in clause 9.5.3.2.3.2.

### 9.5.3.3.3.3 Specific message contents

As defined in clause 9.5.3.2.3.3.

# Annex A (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment** | **Old** | **New** |
| 36.571-2 | | | | | | | |
| 2010-08 | R5#48 | R5-104119 | | | Initial skeleton proposal | | 0.0.0 |
| | R5#48 | R5-104741 | | | Merge of documents R5-104119, R5-104120, R5-104121, R5-104122, together with small editorial modifications | 0.0.0 | 0.0.1 |
| 2011-02 | R5#50 | R5-110250 | | | Various corrections based on LPP v9.3.0 (R5-106431) | 0.0.1 | |
| | | | | | New test cases: LPP Reliable Transport (R5-106433) | | |
| | | | | | New test cases: CS fallback (R5-106698) | | 0.1.0 |
| 2011-05 | R5#51 | R5-112388 | | | Various corrections (R5-110251) | 0.1.0 | |
| | | | | | OTDOA default conditions (R5-110252) | | 0.2.0 |
| 2011-08 | R5#52 | R5-113770 | | | Small corrections to 36.571-2 baseline text | 0.2.0 | |
| | | R5-113771 | | | Addition of LPP abort test case | | |
| | | R5-113147 | | | Addition of Position Capability Transfer test case | | |
| | | R5-113140 | | | Addition of Notification test cases | | |
| | | R5-113769 | | | Addition of UE Network Capability test case | | |
| | | R5-113847 | | | Addition of LPP Error handling test cases | | 1.0.0 |
| 37.571-2 | | | | | | | |
| 2011-11 | R5#53 | R5-115249 | | | Creation of 37.571-2 based on 36.571-2 v1.0.0 and 34.123-1 v9.6.0 | | 1.0.0 |
| | | R5-115250 | | | Default conditions for ECID signalling test cases in 37.571-2 baseline text | | |
| | | R5-115251 | | | Various corrections to the 37.571-2 baseline text | | |
| | | R5-115252 | | | Default system information for UTRAN A-GNSS tests in 37.571-2 baseline text | | 2.0.0 |
| 2011-12 | RAN#54 | - | - | - | Moved to Rel-9 with editorial changes only. | 2.0.0 | 9.0.0 |
| 2012-03 | RAN#55 | R5-120358 | 0001 | - | Addition of missing test case 7.3.4.1 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120359 | 0002 | - | Addition of missing test case 7.3.4.2 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120360 | 0003 | - | Addition of missing test case 7.3.4.3 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120361 | 0004 | - | Addition of missing test case 7.3.4.4 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120362 | 0005 | - | Editorial corrections to 37.571-2 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120363 | 0006 | - | Completion of Test Case 7.3.1.1 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120364 | 0007 | - | Removal of FFS for Sub-test -7 | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120395 | 0008 | - | Addition of RESET command to EPC MO-LR tests | 9.0.0 | 9.1.0 |
| 2012-03 | RAN#55 | R5-120725 | 0009 | - | Correction of MO-LR CS fallback test cases 7.4.1 | 9.0.0 | 9.1.0 |
| 2012-06 | RAN#56 | R5-121134 | 0010 | - | Clarification to cell synchronization for OTDOA | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | R5-121148 | 0011 | - | Completion of Test Case 7.3.1.1 | 9.1.0 | 9.2.0 |
| 2012-06 | RAN#56 | - | - | - | Upgrade to v10.0.0 with no change. | 9.2.0 | 10.0.0 |
| 2012-09 | RAN#57 | R5-123067 | 0012 | - | Removal of FFS for Sub-test-7 | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123068 | 0013 | - | Correction of references to clauses in 37.571-5 | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123070 | 0014 | - | Editorial Corrections | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123071 | 0015 | - | Correction of MO-LR CS fallback test cases 7.4.1 | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123072 | 0016 | - | Correction to UE Network Capability Test Procedure | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123073 | 0017 | - | Correction to Register and Facility message type content | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123074 | 0018 | - | Addition of RESET command to EPC MO-LR tests | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123698 | 0019 | - | Small corrections to default E-UTRAN message contents | 10.0.0 | 10.1.0 |
| 2013-03 | RAN#59 | R5-130112 | 0021 | - | Correction to LPP Request Location Information Message Content for TC 7.3.4.4 | 10.1.0 | 10.2.0 |
| 2013-03 | RAN#59 | R5-130593 | 0022 | - | Correction of applicability for TC 7.3.2.3 | 10.1.0 | 10.2.0 |
| 2013-06 | RAN#60 | R5-131099 | 0023 | - | Clarification of IE values | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131100 | 0024 | - | Correction of behaviour in 7.3.2.3 | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131101 | 0025 | - | Simplification of test set-up for OTDOA and ECID tests | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131306 | 0026 | - | Correction to LTE UE Positioning test cases | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131326 | 0027 | - | Correction to available GNSS assistance data elements | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131877 | 0028 | - | Clarification of Sub-Test Case Numbers Tables | 10.2.0 | 10.3.0 |
| 2013-06 | RAN#60 | R5-131878 | 0029 | - | New test case for inter-frequency RSTD measurement indication procedure | 10.2.0 | 10.3.0 |
| 2013-09 | RAN#61 | R5-133175 | 0030 | - | Clarifications to 7.3.3.1 | 10.3.0 | 10.4.0 |
| 2013-12 | RAN#62 | R5-134907 | 0031 | - | Addition of missing IEs from otdoa-ProvideCapabilities in 7.3.1.1 | 10.4.0 | 10.5.0 |
| 2013-12 | RAN#62 | R5-134908 | 0032 | - | Change Applicability of test 7.3.5.1 | 10.4.0 | 10.5.0 |
| 2013-12 | RAN#62 | R5-134909 | 0033 | - | Addition of Capability exchange in various clause 7 tests | 10.4.0 | 10.5.0 |
| 2013-12 | RAN#62 | R5-134910 | 0034 | - | Clarification of Provide Capabilities content for test 7.2.2.2 | 10.4.0 | 10.5.0 |
| 2014-03 | RAN#63 | R5-140133 | 0035 | - | Add Assistance Data delivery to test 7.3.5.1 | 10.5.0 | 10.6.0 |
| 2014-03 | RAN#63 | R5-140606 | 0036 | - | Correction to LTE UE Positioning test case 7.2.1.2 | 10.5.0 | 10.6.0 |
| 2014-03 | RAN#63 | R5-140608 | 0037 | - | Correction to LTE UE Positioning test case 7.2.1.3 | 10.5.0 | 10.6.0 |
| 2014-03 | RAN#63 | R5-140793 | 0038 | - | Correction to message content for inter-frequency RSTD measurement indication test case | 10.5.0 | 10.6.0 |
| 2014-06 | RAN#64 | R5-142251 | 0039 | - | Clarification of use of satellite simulator | 10.6.0 | 10.7.0 |
| 2014-06 | RAN#64 | R5-142886 | 0040 | - | Correction to EUTRA UE Positioning test cases 7.3.4.x | 10.6.0 | 10.7.0 |
| 2014-09 | RAN#65 | R5-144194 | 0041 | - | Clarification for configuration of cell 1 in OTDOA tests | 10.7.0 | 10.8.0 |
| 2014-09 | RAN#65 | R5-144195 | 0042 | - | RESET Positioning Information in LPP Abort Procedures | 10.7.0 | 10.8.0 |
| 2014-09 | RAN#65 | R5-144237 | 0043 | - | Adding extra neighbour cells to 7.3.5.1 | 10.7.0 | 10.8.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment | Old | New |
| 2014-09 | RAN#65 | R5-144626 | 0044 | - | Correct OTDOA and ECID Elements in 7.3.4.2 and 7.3.4.4 | 10.7.0 | 10.8.0 |
| 2014-09 | RAN#65 | R5-144702 | 0045 | - | Correction to test case 7.5.1, Table 7.5.1.3.2-1: Main behaviour | 10.7.0 | 10.8.0 |
| 2014-09 | RAN#65 | R5-144703 | 0046 | - | Updates OTDOA Neighbour Cell Info List | 10.7.0 | 10.8.0 |
| 2014-12 | RAN#66 | R5-145137 | 0047 | - | Clarification to OTDOA Assistance Data | 10.8.0 | 10.9.0 |
| 2014-12 | RAN#66 | R5-145347 | 0048 | - | Addition of Galileo in test 7.3.3 | 10.8.0 | 10.9.0 |
| 2014-12 | RAN#66 | R5-145736 | 0049 | - | Correction to OTDOA related default message contents in LPP common procedure for Position Capability Transfer | 10.8.0 | 10.9.0 |
| 2014-12 | RAN#66 | - | - | - | Raised to v 11.0.0 with no change | 10.9.0 | 11.0.0 |
| 2014-12 | RAN#66 | R5-145737 | 0050 | - | Addition of Beidou | 11.0.0 | 12.0.0 |
| 2015-03 | RAN#67 | R5-150741 | 0051 | - | Abbreviation Corrections for BDS in 37.571-2 | 12.0.0 | 12.1.0 |
| 2015-06 | RAN#68 | R5-151109 | 0054 | - | IMS settings for LTE Positioning test cases | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-151981 | 0053 | 1 | Correction of prs-ConfigurationIndex for TDD | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-151982 | 0055 | 1 | LPP updates and corrections | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-151983 | 0056 | 1 | Update of default GNSS Assistance Data Elements | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-151984 | 0057 | 1 | Correction to EUTRA UE Positioning test case 7.3.3.1 | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-152147 | 0058 | 1 | Correction to EUTRA UE Positioning test cases covering the UE-assisted case | 12.1.0 | 12.2.0 |
| 2015-06 | RAN#68 | R5-152149 | 0052 | 2 | Addition of "early fix" to A-GNSS tests | 12.1.0 | 12.2.0 |
| 2015-09 | RAN#69 | R5-153110 | 0059 | - | Correction to GANSS Multi-frequency Measurement Requested IE | 12.2.0 | 12.3.0 |
| 2015-09 | RAN#69 | R5-153111 | 0060 | - | Corrections to MEASUREMENT CONTROL messages | 12.2.0 | 12.3.0 |
| 2015-09 | RAN#69 | R5-153153 | 0061 | - | Missing " earlyFixReport-r12 " in Table 7.2.2.2.3.3-13 | 12.2.0 | 12.3.0 |
| 2015-09 | RAN#69 | - | - | - | update of the "non-specific references" in section 2 according to the approved R5-153582 and an action point on ETSI MCC | 12.2.0 | 12.3.0 |
| 2015-12 | RAN#70 | R5-155101 | 0064 | - | Correction to use of Modernized GPS with BDS | 12.3.0 | 12.4.0 |
| 2016-03 | RAN#71 | R5-160354 | 0067 | - | Addition of extra call flow to test case 7.3.4.4 | 12.4.0 | 12.5.0 |
| 2016-09 | RAN#73 | R5-165993 | 0069 | 1 | Add missing references to GPS and Galileo and A-GPS and A-Galileo | 12.5.0 | 12.6.0 |
| 2016-09 | RAN#73 | R5-165996 | 0070 | 1 | Addition of Indoor Positioning Protocol Conformance Testing (MBS) | 12.6.0 | 13.0.0 |
| 2016-12 | RAN#74 | R5-168065 | 0071 | - | Change of applicability of ECID tests for TDD | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168461 | 0073 | - | Missing Satellite signal sub-test case reference | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168463 | 0075 | - | Incorrect Procedure Step referenced for Main behaviour Table | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169094 | 0074 | 1 | Correction in Table 7.3.4.4.3.3-11 for sub test 6 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169100 | 0072 | 1 | Addition of TC 7.3.3.1A | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169101 | 0076 | 1 | Add WLAN signalling sub-test and references for Indoor Positioning | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169102 | 0077 | 1 | Add BT signalling sub-test and references for Indoor Positioning | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169103 | 0078 | 1 | Add Sensor signalling sub-test and references for Indoor Positioning | 13.0.0 | 13.1.0 |
| 2017-03 | RAN#75 | R5-170736 | 0079 | - | Remove Bluetooth Abbreviations and Add Missing References for WLAN | 13.1.0 | 13.2.0 |
| 2017-03 | RAN#75 | - | - | - | Administrative release upgrade to match the release of 3GPP TS 37.571-1 which was upgraded at RAN#74 to Rel-14 due to Rel-14 relevant CR(s) | 13.2.0 | 14.0.0 |
| 2017-06 | RAN#76 | R5-172963 | 0082 | 1 | Merge GNSS sub-tests into one sub-test | 14.0.0 | 14.1.0 |
| 2017-06 | RAN#76 | R5-172964 | 0084 | 1 | Correction to tbs-ProvideCapabilities in several tables | 14.0.0 | 14.1.0 |
| 2017-06 | RAN#76 | R5-172967 | 0083 | 1 | Introduction of MBS Assistance Data Signalling Sub-test 16 | 14.0.0 | 14.1.0 |
| 2017-09 | RAN#77 | R5-173685 | 0086 | - | Editorial changes for Release 14 alignment with core specification | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-173922 | 0089 | - | Clarifications to test 7.2.2.1 for UEs that support more than one positioning technology | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-174060 | 0091 | - | Editorial change to remove referenced text for ‘signals switched off or not present’ in Section 7 | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-174582 | 0087 | 1 | Correction to Inter-frequency RSTD measurement indication test cases 7.5.1 to support Band > 64 | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-174583 | 0090 | 1 | Editorial change to correct typos and missing Information Elements in tables | 14.1.0 | 14.2.0 |
| 2017-09 | RAN#77 | R5-174585 | 0088 | 1 | Editorial change to correct IEs | 14.1.0 | 14.2.0 |
| 2017-12 | RAN#78 | R5-176088 | 0092 | - | Correction to LPP ProvideAssistanceData for UE Based MBS | 14.2.0 | 14.3.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment | Old | New |
| 2017-12 | RAN#78 | R5-176569 | 0093 | - | Addition of Rel-14 WLAN Positioning Protocol Tests and Sub-Tests | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | R5-176570 | 0094 | - | Addition of Rel-14 Sensor Positioning Protocol Tests and Sub-Tests | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | R5-176574 | 0095 | - | Correction for UE Assisted TBS Request Location Information Table | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | R5-176576 | 0096 | - | Deletion of Duplicated IEs in Table 5.4.1.3-2 | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | R5-177039 | 0097 | 1 | Editorial change to correct typos, grammar, descriptive text in table content, missing references and formatting issues | 14.2.0 | 14.3.0 |
| 2017-12 | RAN#78 | - | - | - | Administrative release upgrade to match the release of 3GPP TS 37.571-1 which was upgraded at RAN#78 to Rel-15 due to Rel-15 relevant CR(s) | 14.3.0 | 15.0.0 |
| 2018-03 | RAN#79 | R5-181111 | 0102 | - | Update Main Behaviour positioning method | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181112 | 0103 | - | Missing Release 14 Information Element for otdoa | 15.0.0 | 15.1.0 |
| 2018-03 | RAN#79 | R5-181113 | 0104 | - | Update Simulated Environment conditions for MBS, WLAN, and Bluetooth | 15.0.0 | 15.1.0 |
| 2018-06 | RAN#80 | R5-182957 | 0106 | - | Correction to UE Positioning TCs 7.2.2.1 + 7.2.2.2 | 15.1.0 | 15.2.0 |
| 2018-06 | RAN#80 | R5-183169 | 0105 | 1 | Addition of Rel-14 LPP message segmentation IEs into Capabilities messages | 15.1.0 | 15.2.0 |
| 2018-09 | RAN#81 | R5-184037 | 0108 | - | Addition of PICS for support of LPP message segmentation in test 7.3.1.1 | 15.2.0 | 15.3.0 |
| 2018-12 | RAN#82 | R5-186618 | 0109 | - | Addition of NR background information | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-186695 | 0110 | - | Addition of Rel-12 missing IEs to LPP message contents | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-186696 | 0111 | - | Addition of Rel-15 missing IEs to LPP message contents | 15.3.0 | 15.4.0 |
| 2018-12 | RAN#82 | R5-187726 | 0112 | 1 | Positioning NSA Protocol tests - LPP Procedures | 15.3.0 | 15.4.0 |
| 2019-03 | RAN#83 | R5-192382 | 0113 | 1 | Addition NR SA positioning tests and removal of NSA | 15.4.0 | 15.5.0 |
| 2019-03 | RAN#83 | - | - | - | Administrative release upgrade to match the release of TS 37.571-1 which was upgraded at RAN#83 to Rel-16 due to a Rel-16 relevant CR | 15.5.0 | 16.0.0 |
| 2019-06 | RAN#84 | R5-195201 | 0119 | - | Addition of information for test environments for NR | 16.0.0 | 16.1.0 |
| 2019-06 | RAN#84 | R5-195203 | 0118 | 1 | Clean Up of NR Positioning Test Cases | 16.0.0 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197164 | 0124 | 1 | Correction to Testcases 7.2.2.2, 7.3.4.2 and 7.3.4.4 | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197166 | 0120 | 1 | Addition of sub-test information for NR tests | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197167 | 0121 | 1 | Alignment of NR terminology | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197168 | 0122 | 1 | Complete positioning protocol tests for NR | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197169 | 0123 | 1 | Add default conditions for FR2 positioning tests | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197170 | 0125 | 1 | Clean Up of NR Positioning Test Cases | 16.1.0 | 16.2.0 |
| 2019-09 | RAN#85 | R5-197175 | 0126 | - | Missing Values from Request Assistance Data Table | 16.1.0 | 16.2.0 |
| 2019-12 | RAN#86 | R5-198966 | 0127 | 1 | Additional test coverage for multi-frequency GNSS test cases | 16.2.0 | 16.3.0 |
| 2019-12 | RAN#86 | R5-198967 | 0128 | 1 | Update to protocol positioning tests | 16.2.0 | 16.3.0 |
| 2020-03 | RAN#87 | R5-201010 | 0131 | 1 | ECID test cases deleted for NR Test Configuration B up to and including LPP Rel-15 | 16.3.0 | 16.4.0 |
| 2020-03 | RAN#87 | R5-201012 | 0129 | 1 | Editorial changes to TS 37.571-X titles to remove references to individual RATs | 16.3.0 | 16.4.0 |
| 2020-06 | RAN#88 | R5-203511 | 0133 | - | Correction of WLAN Assistance Data Element referenced clause | 16.4.0 | 16.5.0 |
| 2020-06 | RAN#88 | R5-204489 | 0132 | 1 | Adding measurement gaps for OTDOA configuration | 16.4.0 | 16.5.0 |
| 2020-06 | RAN#88 | R5-204532 | 0134 | 1 | Addition of OTDOA information for pre-test conditions for test 9.3.3.1B | 16.4.0 | 16.5.0 |
| 2020-12 | RAN#90 | R5-205211 | 0136 | - | Addition of miscellaneous Release 16 fields to tables in test case 7.3.1.1 | 16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | R5-205667 | 0139 | - | Clarification of configuration of measurement gaps for OTDOA in NR tests | 16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206424 | 0137 | 1 | Addition of BDS B1C Signal test contents in TS 37.571-2 | 16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206442 | 0135 | 1 | Deletion of tests 7.3.3.1, 7.3.3.1A, 9.3.3.1 and 9.3.3.1A | 16.5.0 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206443 | 0138 | 1 | Corrections to LPP Provide Assistance Data for NR test cases in clause 9 | 16.5.0 | 16.6.0 |
| 2021-03 | RAN#91 | R5-210258 | 0140 | - | Definition of values for epdu fields | 16.6.0 | 16.7.0 |
| 2021-03 | RAN#91 | R5-210260 | 0141 | - | Corrections and clarifications to default MBS, WLAN and Sensor assistance data in clause 5.4.1 | 16.6.0 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211298 | 0142 | - | Corrections for support of multiple GPS signals | 16.6.0 | 16.7.0 |
| 2021-06 | RAN#92 | R5-213638 | 0144 | 1 | Addition of NR Rel 16 positioning methods information into protocol conformance test cases for NR | 16.7.0 | 16.8.0 |
| 2021-06 | RAN#92 | R5-213639 | 0145 | 1 | Addition of NR Rel 16 positioning methods into default conditions for NR | 16.7.0 | 16.8.0 |
| 2021-09 | RAN#93 | R5-215154 | 0149 | - | Clarifications for OTDOA (LTE) test cases for NR | 16.8.0 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215616 | 0151 | - | Clarification text on LCS Sub-Test Cases | 16.8.0 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216322 | 0147 | 1 | Correction to NR positioning method information in Position Capability Transfer test case | 16.8.0 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216323 | 0148 | 1 | Addition of assistance data information elements for Multi-RTT, DL-AoD and DL-TDOA positioning methods | 16.8.0 | 16.9.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment | Old | New |
| 2021-12 | RAN#94 | R5-217127 | 0155 | - | Correction to pre-test conditions for NR DL-TDOA UE-Based test cases | 16.9.0 | 16.10.0 |
| 2021-12 | RAN#94 | R5-217936 | 0152 | 1 | Addition of RESET UE POSITIONING STORED INFORMATION message contents | 16.9.0 | 16.10.0 |
| 2021-12 | RAN#94 | R5-217937 | 0153 | 1 | Correction of the test steps for Multi-RTT sub-tests | 16.9.0 | 16.10.0 |
| 2021-12 | RAN#94 | R5-217938 | 0154 | 1 | Correction of the conditions for Multi-RTT and DI-AoD assistance data elements | 16.9.0 | 16.10.0 |
| 2022-03 | RAN#95 | R5-221592 | 0156 | 1 | Addition of TC 9.4.1 PosSIB broadcasting followed by location information transfer | 16.10.0 | 16.11.0 |
| 2022-03 | RAN#95 | R5-221593 | 0157 | 1 | Correction of the assistance data elements for NR positioning support | 16.10.0 | 16.11.0 |
| 2022-06 | RAN#96 | R5-223388 | 0158 | 1 | Correction of TC 9.4.1 PosSIB broadcasting followed by location information transfer | 16.11.0 | 16.12.0 |
| 2022-06 | RAN#96 | R5-223389 | 0159 | 1 | Addition of TC 9.4.2 PosSIB broadcasting followed by location information transfer / Positioning SI messages offset | 16.11.0 | 16.12.0 |
| 2022-06 | RAN#96 | R5-223390 | 0160 | 1 | Addition of TC 7.5.2 PosSIB broadcasting followed by location information transfer | 16.11.0 | 16.12.0 |
| 2022-09 | RAN#97 | R5-224406 | 0162 | - | Removal of test case 7.5.2 | 16.12.0 | 16.13.0 |
| 2022-09 | RAN#97 | R5-224407 | 0163 | - | Correction of posSIB broadcasting test case | 16.12.0 | 16.13.0 |
| 2022-09 | RAN#97 | R5-225313 | 0161 | 1 | Correction to LPP Provide Capabilities template for subclause 9 test cases | 16.12.0 | 16.13.0 |
| 2022-09 | RAN#97 | R5-225314 | 0164 | 1 | Addition of on-demand PosSIB followed by location information transfer in RRC_connected state | 16.12.0 | 16.13.0 |
| 2022-12 | RAN#98 | R5-226459 | 0165 | | Correction of on-demand posSIB test case in RRC_connected | 16.13.0 | 16.14.0 |
| 2022-12 | RAN#98 | R5-226513 | 0166 | | Correction to MeasGapConfig | 16.13.0 | 16.14.0 |
| 2022-12 | RAN#98 | R5-226515 | 0168 | | Correction to NR-DL-PRS-Info | 16.13.0 | 16.14.0 |
| 2022-12 | RAN#98 | R5-227526 | 0167 | 1 | Correction to NR DL-PRS Assistance Data | 16.13.0 | 16.14.0 |
| 2022-12 | RAN#98 | R5-227527 | 0169 | 1 | Include measObject as part of the measConfig message | 16.13.0 | 16.14.0 |
| 2023-03 | RAN#99 | R5-231022 | 0171 | - | Correction to NR-DL-PRS-Info parameters | 16.14.0 | 16.15.0 |
| 2023-03 | RAN#99 | R5-230335 | 0170 | - | Introduction of BDS B2a and B3I signal test contents in TS 37.571-2 | 16.15.0 | 17.0.0 |
| 2023-06 | RAN#100 | R5-233395 | 0173 | 1 | Introduction of R17 Positioning Enhancements default test conditions in TS 37.571-2 | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233396 | 0174 | 1 | Addition of new positioning test case for pre-configured measurement gap procedures | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233397 | 0175 | 1 | Addition of new positioning test case for pre-configured PRS processing window procedures | 17.0.0 | 17.1.0 |
| 2023-06 | RAN#100 | R5-233398 | 0176 | 1 | Addition of new positioning test case for UE positioning assistance information procedures | 17.0.0 | 17.1.0 |
| 2023-09 | RAN#101 | R5-234217 | 0179 | - | Correction of test case 9.4.6 UE Positioning Assistance Information | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234218 | 0180 | - | Correction of test case 9.4.5 Pre-configured PRS processing window procedures | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-234219 | 0181 | - | Correction of test case 9.4.4 Pre-configured Measurement Gap Procedures | 17.1.0 | 17.2.0 |
| 2023-09 | RAN#101 | R5-235341 | 0178 | 1 | Addition of Rel-17 missing IEs to LPP position capability transfer test case | 17.1.0 | 17.2.0 |
| 2024-03 | RAN#103 | R5-240513 | 0182 | - | Correction to NR Physical Cell IDs for NR-PRS assistance data | 17.2.0 | 17.3.0 |
| 2024-06 | RAN#104 | R5-243569 | 0183 | 1 | Correction to LPP positioning test case 9.3.4.2 | 17.3.0 | 17.4.0 |
| 2024-12 | RAN#106 | R5-246428 | 0184 | - | Addition of default test condition for Rel-18 positioning work item | 17.4.0 | 18.0.0 |
| 2024-12 | RAN#106 | R5-246429 | 0185 | - | Addition of new LPP carrier phase positioning test case | 17.4.0 | 18.0.0 |
| 2024-12 | RAN#106 | R5-246430 | 0186 | - | Addition of new LPP frequency hopping test case for redcap UE | 17.4.0 | 18.0.0 |
| 2025-03 | RAN#107 | R5-250587 | 0187 | - | Addition of new PRS aggregation positioning procedure test case | 18.0.0 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251373 | 0188 | 1 | Addition of new LPHAP SRS configuration enhancements test case | 18.0.0 | 18.1.0 |
| 2025-06 | RAN#108 | R5-252271 | 0189 | - | Correction message contents of 9.3.4.7 PRS aggregation positioning procedure test case | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253162 | 0190 | 1 | Expanded 9.3.1.1 Position Capability Transfer test case message contents | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253163 | 0191 | 1 | Addition of sidelink positioning test condition to 37.571-2 | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253164 | 0192 | 1 | Addition of new SLPP position capability transfer procedure test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253165 | 0193 | 1 | Addition of new SLPP duplicated message transfer procedure test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253166 | 0194 | 1 | Addition of new SLPP acknowledgment procedure test cases | 18.1.0 | 18.2.0 |
| 2025-06 | RAN#108 | R5-253167 | 0195 | 1 | Addition of new SLPP retransmission procedure test cases | 18.1.0 | 18.2.0 |
| 2025-09 | RAN#109 | R5-255009 | 0196 | 1 | Addition of SLPP default message contents and test condition | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255010 | 0197 | 1 | Addition of new SLPP abort procedure test cases | 18.2.0 | 18.3.0 |
| 2025-09 | RAN#109 | R5-255011 | 0198 | 1 | Addition of new SLPP error handling procedure test cases | 18.2.0 | 18.3.0 |

# History

| Document history | | |
|---|---|---|
| V18.0.0 | February 2025 | Publication |
| V18.1.0 | April 2025 | Publication |
| V18.2.0 | July 2025 | Publication |
| V18.3.0 | October 2025 | Publication |
| | | |